Quasi-Maximum Likelihood Estimation for a Genuinely Unbalanced Dynamic Network Panel Data Model

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Abstract

This paper develops quasi-maximum likelihood estimation for genuinely unbalanced dynamic network panel data models. We first study a baseline one-way fixed effect specification and then extend the framework to two-way fixed effects. The model accommodates contemporaneous and lagged network spillovers, temporal dependence, and a listing effect that operates when a unit first appears in the panel. For the baseline specification, we establish consistency and asymptotic normality of the QMLE, characterize the incidental-parameter bias, and propose a bias-corrected estimator. We further provide a consistent estimator of the asymptotic covariance matrix, to enable feasible inference. Under the two-way fixed effect extension, we derive the corresponding asymptotic distribution, bias correction, and covariance matrix estimator, accounting for the additional bias component induced by the time fixed effects. Monte Carlo simulations examine the finite sample performance of the proposed estimators and inference procedure. An empirical application to Airbnb listings from New Zealand and New York City reveals region-specific patterns in spatial and temporal price transmission, highlighting the importance of modeling genuine unbalancedness in dynamic network settings.

Keywords: Dynamic Network Panel Data Model; Quasi-maximum Likelihood Estimation; Spatial Dynamic Panel Data Model; Unbalanced Panel Data Model

1 Introduction

Panel data are widely used in social science and economics because they simultaneously capture individual heterogeneity and temporal dynamics. Compared with cross-sectional or time-series data, panel data provide richer information for analyzing both long-term relationships and short-term adjustments among variables. When units are spatially or network-connected—such as regions in an economic system, firms in a financial market, or individuals in a social platform—their outcomes often exhibit interdependence across both space and time. This has motivated the development of spatial dynamic panel data models, which integrate network interactions with temporal evolution. These models have been applied to a wide range of contexts, such as regional economics (Amidi et al., 2020), environmental systems (Chang et al., 2021), financial markets (Billé and Caporin, 2022), and social networks (Han et al., 2021).

Unbalanced dynamic network panel data are commonly observed in real-world contexts such as regional rental markets, financial stock markets, and social networks. For instance, in stock markets, firms may enter or exit over time, as illustrated in Figure 1. The figure shows the historical retention status of CSI 300 and S&P 500 constituent stocks and indicates that a substantial share of constituents is replaced approximately

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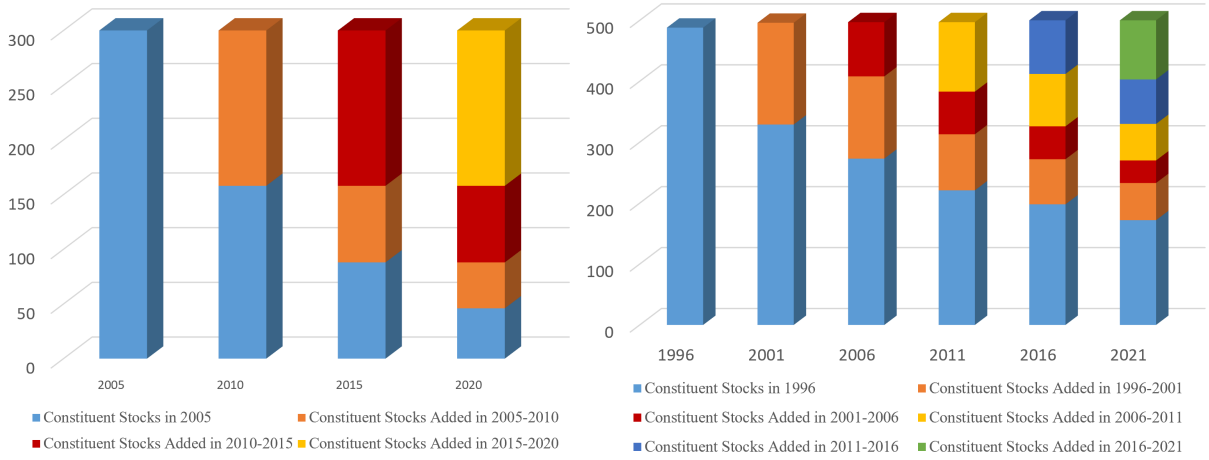
every five years. Critically, these firms are interconnected through business partnerships or ownership ties, giving rise to a complex and evolving network structure. Together, the continuous turnover of units and the network links define an unbalanced dynamic network panel. Similarly, in social network studies, individuals exhibit heterogeneous temporal activity, and relationships evolve as users join or leave the network, which also constitutes a prototypical example of an unbalanced dynamic network panel.



(a) CSI 300 (b) S&P 500

Figure 1: Historical retention status of CSI 300 and S&P 500 constituent stocks

Despite their empirical prevalence, theoretical work on dynamic spatial or network panels has overwhelmingly assumed balanced panels. Kapoor et al. (2007) study the estimation of static spatial panel data models. Yu et al. (2008) and Lee and Yu (2010) propose quasi-maximum likelihood estimation (QMLE) for spatial dynamic panel data models. Lee and Yu (2014) consider the generalized method of moments (GMM) estimation for spatial dynamic panel data. Lee and Yu (2012) and Qu et al. (2017) extend the previous spatial dynamic panel data to scenarios with time-varying or endogenous spatial weight matrices. Jin et al. (2020) propose a least squares estimators based on the eigendecomposition of a spatial weight matrix method for general spatial dynamic panel data models. On this basis, Hou et al. (2024) extend this framework to directed networks and develop a variable selection approach for high-dimensional spatial dynamic panel data models. Another related model is the network autoregressive model. See Zhu et al. (2017) and Zhu et al. (2019). While these studies collectively enrich the methodological toolkit for dynamic spatial and network panel models, they all focus on balanced panel data settings.

In the field of unbalanced panel data, existing studies have focused on specific subtopics but lack integration with spatial dynamic structures. Honoré (1992), Hu (2002), and Honoré and Hu (2004) address censored unbalanced panel data models, while Wooldridge (1995) and Kyriazidou (1997) investigate unbalanced panels with sample selection bias. Su et al. (2025) develop estimation and inference methods for unbalanced panels with interactive fixed effects, and Sasaki and Xin (2017) study dynamic panel data with irregular time intervals. For spatial panel data, Meng and Yang (2026a) examine a static genuinely unbalanced spatial panel data model, while Meng and Yang (2026b) focus on static spatial panel data models with missing observations. None provide a unified framework for estimation and inference in unbalanced dynamic network panels.

This paper bridges this gap. We propose a quasi-maximum likelihood estimator (QMLE) for an unbalanced dynamic network panel model that captures network interactions and temporal dynamics simultaneously, while explicitly accommodating unit entry and exit. We focus on panels that are *genuinely* unbalanced (Meng and Yang, 2026a), **meaning that the unbalancedness is driven by the entry and exit of units rather than by missing observations**. This specification allows us to avoid modeling the missing-data process. Our main contributions and results are as follows.

1. We propose a baseline one-way fixed effect (OWFE) model for genuinely unbalanced dynamic network

panels. The model incorporates contemporaneous network spillovers, lagged network and temporal effects, and a common listing effect in a unit's first active period. For this model, we develop the QMLE, establish consistency and asymptotic normality, characterize the incidental-parameter bias, and derive a bias-corrected estimator with a consistent asymptotic covariance matrix estimator.

2. Since the size and elements of the network weight matrices vary over time, our setting differs from balanced spatial panel models. We establish new linear–quadratic form results that extend the existing theory from balanced spatial dynamic panels to genuinely unbalanced dynamic network panels. These results may also be useful for future studies of unbalanced network panel data.

3. We further extend the baseline framework to a two-way fixed effect (TWFE) specification. Under this extension, the QMLE remains consistent, while the incidental-parameter bias contains both the OWFE bias component and an additional component induced by the time fixed effects. We derive the corresponding asymptotic distribution, bias correction, and consistent asymptotic covariance matrix estimator.

We examine the finite sample performance of the estimator through Monte Carlo simulations. Furthermore, we apply our model to study an empirical Airbnb listing dataset. We compare the estimated results across different regions and spatial scales to analyze the similarities and differences in the dynamic pricing behavior of Airbnb listings across cities.

The rest of the paper is organized as follows. Section 2 introduces the baseline OWFE model and the concentrated likelihood. Section 3 develops the asymptotic theory, bias correction, and covariance estimation for the baseline QMLE. Section 4 extends the framework to TWFE. Section 5 reports Monte Carlo simulations, with additional TWFE results provided in the appendix. Section 6 presents the empirical application, and Section 7 concludes. All proofs are collected in the Supplementary Appendices.

Notation

We define ι_N as the N-dimensional column vector of ones, and I_N as the N-dimensional identity matrix. For any matrix B, B_{ij} represents the (i,j)-th element of the matrix B. $B_{i,\cdot}$ and $B_{\cdot,j}$ are the i-th row and j-th column of B, respectively. Denote $\|B\|_1 \equiv \sup_{1\leqslant j\leqslant N}\sum_{i=1}^N |B_{ij}|$ and $\|B\|_\infty \equiv \sup_{1\leqslant i\leqslant N}\sum_{j=1}^N |B_{ij}|$ as the one-norm and the infinity-norm for matrix B, respectively. For a vector C, let $\|C\|_1 = \sum_{i=1}^N |C_i|$, $\|C\|_\infty = \sup_i |C_i|$, and $\|C\|_2 = (\sum_{i=1}^N C_i^2)^{1/2}$ denote the ℓ_1, infinity, and Euclidean norms of C, respectively. Let $\mathrm{tr}(B)$ denote the trace of matrix B and $|B|$ denote the determinant of B. Define $\sup_t B_t$ as the element-wise supremum of the matrix sequence $\{B_t\}$, which means that $(\sup_t B_t)_{ij} = \sup_t B_{t,ij}$ for $i,j = 1,\cdots,N$. Also, $\mathrm{abs}(B)$ represents the element-wise absolute value of the matrix B, i.e., $[\mathrm{abs}(B)]_{ij} = |B_{ij}|$. For any square matrix B, B^- denotes the strictly lower triangular part of B, obtained by setting the diagonal and upper triangular entries to zero. The notation $\mathrm{diag}(b_1,\cdots,b_N)$ represents a diagonal matrix with diagonal entries $b_1, b_2, \cdots, b_N$. Moreover, $\mathrm{blkdiag}(B_1,\cdots,B_T)$ forms a block-diagonal matrix by the given submatrices. For brevity, the transposes of symmetrically positioned submatrices in a symmetric matrix are sometimes denoted by $*$. Var and Cov represent the variance and covariance of random variables, respectively.

2 The Model and Concentrated Likelihood Function

2.1 The Model

Consider a genuinely unbalanced dynamic network panel, where nodes are fully observed during each period (i.e., there are no missing data), and nodes that are not listed do not have any impact on the whole network. Suppose that we have observations for a network of N nodes over T time periods. However, for the i-th node,

we only have a total of T_i consecutive observations[1]. In the t-th time period, observed nodes may drop out of the network, and new nodes may join the network. We can only observe N_t nodes in period t and let $\mathbb{I}_t = \{i_1, i_2, \cdots, i_{N_t}\}$ denote the set of indices of nodes observed. Their corresponding attributes are (y_{it}, x'_{it}) for each $i \in \mathbb{I}_t$, where y_{it} is a scalar dependent variable and x_{it} is a k-dimensional vector of regressors. The total sample size is $n = \sum_{t=1}^{T} N_t$.

Then we can establish the following model: for $t = 1, 2, \cdots, T$ and $i \in \mathbb{I}_t$,

$$y_{it} = \rho_0 \sum_{j \in \mathbb{I}_t} w_{ij,t} y_{jt} + d_{i,t-1}(\lambda_0 \sum_{j \in \mathbb{I}_{t-1}} m_{ij,t} y_{j,t-1} + \nu_0 y_{i,t-1}) + d_{it}(1 - d_{i,t-1})\gamma_0 + x'_{it}\beta_0 + \alpha_{i0} + v_{it}, \quad (1)$$

where $w_{ij,t}$ and $m_{ij,t}$ are predefined network weights, and d_{it} is an indicator that equals 1 if y_{it} is observed and 0 otherwise. γ_0 and α_{i0} are unknown listing and individual fixed effects, respectively, where the listing effect only takes effect at the first appearance of each individual, and the effect is identical for each individual. The above model assumes that y_{it} is not only affected by its own covariates and the current network structure, but also by the fixed listing effect if node i has just been listed or by its own lagged outcome and the lagged network state if node i was already present. Since we only model the nodes that appear at $t = 0$ or later and cannot observe the data for $t < 0$, to simplify the analysis, we ignore the impact of nodes that existed only before $t = 0$.

Denote $Y_t = (y_{i_1 t}, \cdots, y_{i_{N_t} t})'$ as an N_t-dimensional column vector, $X_t = (x_{i_1 t}, \cdots, x_{i_{N_t} t})'$ as an $N_t \times k$ matrix, and $V_t = (v_{i_1 t}, \cdots, v_{i_{N_t} t})'$ as an N_t-dimensional column vector, where $i_1, \cdots, i_{N_t} \in \mathbb{I}_t$. Define $\alpha_0 = (\alpha_{10}, \alpha_{20}, \cdots, \alpha_{N0})'$. The $N_t \times N$ matrix D_t is obtained by removing the rows corresponding to unobserved nodes during period t from I_N. Then the above model can be written in the following matrix form: for $t = 1, 2, \cdots, T$,

$$Y_t = \rho_0 W_t Y_t + \lambda_0 D_t (I_N - F_t) D'_t M_t Y_{t-1} + \nu_0 D_t D'_{t-1} Y_{t-1} + \gamma_0 D_t F_t \iota_N + X_t \beta_0 + D_t \alpha_0 + V_t. \quad (2)$$

In the above model, W_t and M_t are $N_t \times N_t$ and $N_t \times N_{t-1}$ network weight matrices, respectively. The elements of W_t and M_t are defined as:

$$(W_t)_{k,l} \equiv w_{i_k i_l, t}, \quad \text{for } k, l = 1, 2, \cdots, N_t \text{ and } i_k, i_l \in \mathbb{I}_t,$$

$$(M_t)_{k,l} \equiv m_{i_k j_l, t}, \quad \text{for } k = 1, 2, \cdots, N_t, l = 1, 2, \cdots, N_{t-1}, i_k \in \mathbb{I}_t \text{ and } j_l \in \mathbb{I}_{t-1}.$$

The N-dimensional diagonal matrix $F_t \equiv D'_t D_t (I_N - D'_{t-1} D_{t-1}) = \mathrm{diag}(d_{1t}(1 - d_{1,t-1}), \cdots, d_{N,t}(1 - d_{N,t-1}))$ is the listing index matrix. $F_t \iota_N$ is sparser than other covariates because, over all T periods, fewer than N individuals can newly join the network.

Let $S_t \equiv I_{N_t} - \rho_0 W_t$, which is assumed to be invertible, and define $A_t \equiv D_t[\lambda_0 (I_N - F_t) D'_t M_t + \nu_0 D'_{t-1}] S_{t-1}^{-1}$. Assume that the following infinite summation is well-defined. Then repeated substitution of model (2) gives

$$Y_t = \sum_{h=0}^{\infty} S_t^{-1} A_t^{(h)} [D_{t-h}(\gamma_0 F_{t-h} \iota_N + \alpha_0) + X_{t-h}\beta_0 + V_{t-h}] \equiv \mathcal{E}_t + \mathcal{X}_t \beta_0 + \mathcal{V}_t, \quad (3)$$

where $A_t^{(h)} \equiv A_t A_{t-1} \cdots A_{t-h+1}$ for $h = 1, 2, \cdots$ and $A_t^{(0)} \equiv I_{N_t}$, $\mathcal{E}_t \equiv \sum_{h=0}^{\infty} S_t^{-1} A_t^{(h)} D_{t-h}(\gamma_0 F_{t-h} \iota_N + \alpha_0)$, $\mathcal{X}_t \equiv \sum_{h=0}^{\infty} S_t^{-1} A_t^{(h)} X_{t-h}$ and $\mathcal{V}_t \equiv \sum_{h=0}^{\infty} S_t^{-1} A_t^{(h)} V_{t-h}$. The above decomposition can be viewed as an infinite moving average representation of Y_t. We establish the law of large numbers and the central limit theorem based on this decomposition in the lemmas in Supplementary Appendix A.

[1] We assume that each individual enters the market only once. This condition can be generalized to allow the same individual to enter and exit multiple times.

2.2 Quasi-Maximum Likelihood Method

Define $Z_t \equiv (D_t(I_N - F_t)D_t'M_tY_{t-1}, D_tD_{t-1}'Y_{t-1}, D_tF_t\iota_N, X_t)$, $\delta_0 \equiv (\lambda_0, \nu_0, \gamma_0, \beta_0')'$. Then model (2) can be simplified to

$$Y_t = \rho_0 W_t Y_t + Z_t\delta_0 + D_t\alpha_0 + V_t. \tag{4}$$

Stacking the matrices and vectors in model (4) for $t = 1, 2, \cdots, T$, we have

$$\mathbb{Y} = \rho_0 \mathbb{W}\mathbb{Y} + \mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{V}, \tag{5}$$

where $\mathbb{Y} = (Y_1', \cdots, Y_T')'$, $\mathbb{W} = \text{blkdiag}(W_1, \cdots, W_T)$, $\mathbb{Z} = (Z_1', \cdots, Z_T')'$, $\mathbb{D} = (D_1', \cdots, D_T')'$ and $\mathbb{V} = (V_1', \cdots, V_T')'$. Note that $\mathbb{D}'\mathbb{D} = \sum_{t=1}^T D_t'D_t = \sum_{t=1}^T \text{diag}(d_{1t}, d_{2t}, \cdots, d_{N,t}) = \text{diag}(T_1, T_2, \cdots, T_N) \equiv \mathcal{T}$. Assume v_{it} are i.i.d. random variables with mean 0 and variance σ_0^2. Denote the parameters of interest as $\theta = (\rho, \delta', \sigma^2)'$ and let $\zeta = (\rho, \delta')$. Then the log-likelihood function of model (5) is

$$L_n(\theta, \alpha) = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\sigma^2 + \ln|\mathbb{S}(\rho)| - \frac{1}{2\sigma^2}\mathbb{V}'(\zeta, \alpha)\mathbb{V}(\zeta, \alpha), \tag{6}$$

where $\mathbb{S}(\rho) \equiv I_n - \rho\mathbb{W}$ and $\mathbb{V}(\zeta, \alpha) \equiv \mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta - \mathbb{D}\alpha$. Given ζ, $L_n(\theta, \alpha)$ is maximized at $\hat{\alpha}(\zeta) = (\mathbb{D}'\mathbb{D})^{-1}\mathbb{D}'(\mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta)$. Substituting $\hat{\alpha}$ into (6), we obtain our estimator $\hat{\theta}$ by maximizing the following concentrated log-likelihood function

$$L_n^c(\theta) = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\sigma^2 + \ln|\mathbb{S}(\rho)| - \frac{1}{2\sigma^2}\widetilde{\mathbb{V}}'(\zeta)\widetilde{\mathbb{V}}(\zeta), \tag{7}$$

where $\widetilde{\mathbb{V}}(\zeta) \equiv \mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta - \mathbb{D}\hat{\alpha}(\zeta) = \mathbb{Q}(\mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta)$ and $\mathbb{Q} \equiv I_n - \mathbb{P}$, $\mathbb{P} \equiv \mathbb{D}(\mathbb{D}'\mathbb{D})^{-1}\mathbb{D}'$. Moreover, the corresponding expected value function of $L_n^c(\theta)$ is

$$Q_n(\theta) \equiv E[L_n^c(\theta)] = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\sigma^2 + \ln|\mathbb{S}(\rho)| - \frac{1}{2\sigma^2}E\left[\widetilde{\mathbb{V}}'(\zeta)\widetilde{\mathbb{V}}(\zeta)\right]. \tag{8}$$

We derive the asymptotic properties of the estimators based on $L_n^c(\theta)$ and $Q_n(\theta)$. In practice, we further concentrate out δ and σ^2 to accelerate computation, that is, we optimize

$$L_n^{cc}(\rho) \equiv -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\hat{\sigma}^2 + \ln|\mathbb{S}(\rho)| - \frac{n}{2}$$

instead, where $\hat{\delta}(\rho) = (\mathbb{Z}'\mathbb{Q}\mathbb{Z})^{-1}\mathbb{Z}'\mathbb{Q}\mathbb{S}(\rho)\mathbb{Y}$, and $\hat{\sigma}^2(\rho) = \frac{1}{n}\mathbb{Y}'\mathbb{S}(\rho)'[\mathbb{Q} - \mathbb{Q}\mathbb{Z}(\mathbb{Z}'\mathbb{Q}\mathbb{Z})^{-1}\mathbb{Z}'\mathbb{Q}]\mathbb{S}(\rho)\mathbb{Y}$.

3 Asymptotic properties of QMLE

3.1 Consistency of QMLE

In the proof of consistency, we first establish the consistency of all parameters except γ, and then prove the consistency of γ. Define $\mathbb{Z}^{(-\gamma)}$ as the matrix obtained by removing the third column from $\mathbb{Z}$, and let $\mathbb{Z}^{(\gamma)}$ be the third column of $\mathbb{Z}$. The following assumptions are needed for consistency.

Assumption 1 *The unbalancedness of the model is nonstochastic, which means that d_{it} and D_t are nonstochastic for all i and t. There is a lower bound on the number of observations for all individuals, i.e., $T_i \geqslant c_T T$ for all i, where $c_T > 0$ is a constant that does not vary with N or T. Additionally, $N_0 < c_N N$, where N_0 denotes the number of units observed at $t = 0$ and $0 < c_N < 1$ is a constant that does not vary with N or T.*

Assumption 2 *The error terms v_{it} are i.i.d. across i and t with mean 0, variance σ_0^2, third moment μ_3 and fourth moment μ_4. $E|v_{it}|^{4+2\eta} < \infty$ for some $\eta > 0$.*

Assumption 3 $\sup_{\rho\in\Lambda}\sup_{N,T}\sup_t\|\rho W_t\|_\infty < 1$, *where* Λ *is the parameter space of* ρ. *Moreover,* Λ *is compact and* ρ_0 *lies in the interior of* Λ.

Assumption 4 *The elements of* x_{it} *and* α_{i0} *are nonstochastic and bounded uniformly in* i, t, N *and* T. $\mathbb{Z}'\mathbb{Q}\mathbb{Z}$ *is nonsingular. Also,* $\lim_{n\to\infty}\frac{1}{n}E(\mathbb{Z}^{(-\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(-\gamma)})$ *exists and is nonsingular, where* $n=\sum_{t=1}^T N_t$ *is the total sample size.*

Assumption 5 *The sequences of matrices* $\{W_t\}$, $\{M_t\}$ *and* $\{S_t^{-1}\}$ *are nonstochastic and uniformly bounded in row and column sum norms (for short, UB)* [2] *uniformly for* t. *Moreover,*

$$\max(\|\sup_t \operatorname{abs}(D_t'S_t^{-1}D_t)\|_\infty, \|\sup_t \operatorname{abs}(D_t'W_tD_t)\|_\infty, \|\sup_t \operatorname{abs}(D_t'M_tD_{t-1})\|_\infty) < \infty.$$

Assumption 6 $\sum_{h=1}^\infty \operatorname{abs}\left(D_t'A_t^{(h)}D_{t-h}\right)$ *and* $\sum_{h=t}^\infty \operatorname{abs}\left(D_h'A_h^{(h-t)}D_t\right)$ *are UB uniformly for* t.

Assumption 7 N *is a nondecreasing function of* T, *and* N *will go to infinity as* T *goes to infinity.*

Assumption 1 regulates the unbalancedness of the panel, where we assume the unbalancedness to be nonstochastic. We will leave the more complicated case where d_{it} is stochastic for future research. A lower bound on T_i is imposed to ensure the consistency of the estimator for α, and an upper bound on N_0 is imposed to guarantee the consistency of the estimator for γ, as it requires sufficiently many observed listing effects. If this condition is not satisfied, a similar asymptotic analysis can still be conducted, but the convergence rate of $\hat{\gamma}$ will change. Assumption 2 provides regularity assumptions for v_{it}. Similar assumptions are adopted in Lee (2004) and Yu et al. (2008). Assumption 3 is a standard assumption for spatial (or network) panel data models. It implies that $S_t(\rho)^{-1}$ exists and

$$\sup_{\rho\in\Lambda}\sup_{N,T}\sup_t\|S_t(\rho)^{-1}\|_\infty < \infty, \tag{9}$$

which is proved in equation (S.28) in the supplementary materials of Wu et al. (2025). Although the assumption that ρ_0 lies in the interior of the parameter space is in fact imposed for simplicity and not required for consistency, we state the conditions jointly for notational convenience. Assumption 4 rules out multicollinearity, ensuring that δ is estimable. We consider fixed design in this paper as in Lee (2004, 2007), Yu et al. (2008) and Lee and Yu (2014), among others. This assumption can be relaxed to allow for stochastic exogenous variables by using the near-epoch dependence theory in Jenish and Prucha (2012) or the functional dependence theory in Wu et al. (2025). The proof is similar to the proof under fixed design. Assumption 5 is a standard condition in the literature of spatial and network econometrics. See Lee (2004, 2007), Yu et al. (2008) and Lee and Yu (2014). The assumption that $\sup_{N,T}\sup_t\|S_t^{-1}\|_\infty < \infty$ can in fact be ensured by equation (9). Moreover, Assumption 5 implies that $\{S_t(\rho)^{-1}\}$ is UB uniformly in a neighborhood of ρ_0, from Lemma A.3 of Lee (2004). Assumption 6 restricts the joint effect of network spillovers and temporal dynamics, which is a common assumption in dynamic network panel data, such as Assumption 6 in Yu et al. (2008), Assumption 7 in Lee and Yu (2014) and Assumption 6 in Lee and Yu (2012). Assumptions 5 and 6 are imposed to establish a law of large numbers and a central limit theorem for certain quadratic forms of the network lag spillover term M_tY_{t-1} and the temporal lag term Y_{t-1}. Assumption 7 means that we consider a large-N and large-T setting, in which the QMLE is consistent. For short dynamic network panel data models, other estimation methods, such as GMM, are required.

Denote $\mathbb{S}_0 \equiv \mathbb{S}(\rho_0)$, $\mathbb{G}(\rho) \equiv \mathbb{W}\mathbb{S}(\rho)^{-1}$ and $\mathbb{G}_0 \equiv \mathbb{W}\mathbb{S}_0^{-1}$. Define $\mathcal{H}^{(-\gamma)} \equiv \frac{1}{n}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0, \mathbb{Z}^{(-\gamma)})'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0, \mathbb{Z}^{(-\gamma)})$ and $\sigma_n^2 \equiv \frac{\sigma_0^2}{n}tr[(\mathbb{S}(\rho)\mathbb{S}_0^{-1})'\mathbb{S}(\rho)\mathbb{S}_0^{-1}]$. As in Yu et al. (2008), the following assumption is needed to establish the global identification condition required for consistency.

[2] We say a $N\times N$ matrix B is UB if $\sup_N\max\{\|B\|_1, \|B\|_\infty\} < \infty$. We say a sequence of $N_t\times N_t$ matrices $\{B_t\}$ is UB uniformly for t if $\sup_{N,T}\sup_t\max\{\|B_t\|_1, \|B_t\|_\infty\} < \infty$.

Assumption 8 *At least one of the following two conditions is satisfied: (i)* $\lim_{n\to\infty} E\mathcal{H}^{(-\gamma)}$ *exists and is nonsingular. (ii)* $\lim_{n\to\infty}\left(\frac{1}{n}\ln|\sigma_0^2\mathbb{S}_0^{-1}\mathbb{S}_0^{-1\prime}| - \frac{1}{n}\ln|\sigma_n^2\mathbb{S}(\rho)^{-1}\mathbb{S}(\rho)^{-1\prime}|\right) \neq 0$ *for* $\rho \neq \rho_0$.

Theorem 1 *Under Assumptions 1-8,* $\hat{\theta} \xrightarrow{p} \theta_0$.

3.2 Asymptotic Distribution of QMLE

Since we have concentrated out the N individual fixed effects from the log likelihood function, the QMLE is subject to the incidental parameters problem, which induces an asymptotic bias in the estimator (Meng and Yang, 2026a). We derive the asymptotic distribution of $\hat{\theta}$ by performing a Taylor expansion of $\frac{\partial L_n^c(\hat{\theta})}{\partial\theta}$ around θ_0. The first and second order derivatives of $L_n^c(\theta)$ are derived in Supplementary Appendix B. According to (3), denote $Z_t^\circ \equiv \left(D_t(I_N - F_t)D_t'M_t\mathcal{V}_{t-1}, D_tD_{t-1}'\mathcal{V}_{t-1}, \mathbf{0}_{N_t\times(k+1)}\right)$ and $\mathbb{Z}^\circ \equiv (Z_1^{\circ\prime}, \cdots, Z_T^{\circ\prime})'$. We decompose $\mathbb{QZ}$ into $\mathbb{QZ} = \widetilde{\mathbb{Z}}^* - \widetilde{\mathbb{Z}}^\circ$, where $\widetilde{\mathbb{Z}}^\circ \equiv \mathbb{PZ}^\circ$ and $\widetilde{\mathbb{Z}}^* = \mathbb{QZ} + \widetilde{\mathbb{Z}}^\circ$. Note that Z_t° is independent of V_s when $t \leqslant s$, and $\widetilde{\mathbb{Z}}^* = \mathbb{QZ} + \widetilde{\mathbb{Z}}^\circ = \mathbb{Q}(\mathbb{Z} - \mathbb{Z}^\circ) + \mathbb{Z}^\circ$, where $\mathbb{Z} - \mathbb{Z}^\circ$ is nonstochastic. Decompose $\widetilde{\mathbb{Z}}^*$ into N_t-dimensional vectors $\widetilde{Z}_1^*, \cdots, \widetilde{Z}_T^*$ such that $\widetilde{\mathbb{Z}}^* = (\widetilde{Z}_1^{*\prime}, \cdots, \widetilde{Z}_T^{*\prime})'$. Therefore, $\widetilde{Z}_t^*$ is independent of V_s when $t \leqslant s$. But $\widetilde{\mathbb{Z}}^\circ$ is correlated with $\mathbb{V}$, as proved in Lemma 11 in Supplementary Appendix A.

Similarly, we can decompose $\mathbb{QG}_0\mathbb{Z} = \widetilde{\mathbb{G}_0\mathbb{Z}}^* - \widetilde{\mathbb{G}_0\mathbb{Z}}^\circ$, where $\widetilde{\mathbb{G}_0\mathbb{Z}}^\circ \equiv \mathbb{PG}_0\mathbb{Z}^\circ$ and $\widetilde{\mathbb{G}_0\mathbb{Z}}^* = \mathbb{QG}_0\mathbb{Z} + \widetilde{\mathbb{G}_0\mathbb{Z}}^\circ$. Further, decompose $\widetilde{\mathbb{G}_0\mathbb{Z}}^*$ into N_t-dimensional vectors $\widetilde{GZ}_1^*, \cdots, \widetilde{GZ}_T^*$ such that $\widetilde{\mathbb{G}_0\mathbb{Z}}^* = (\widetilde{GZ}_1^{*\prime}, \cdots, \widetilde{GZ}_T^{*\prime})'$. Similarly, $\widetilde{GZ}_t^*$ is independent of V_s when $t \leqslant s$, whereas $\widetilde{\mathbb{G}_0\mathbb{Z}}^\circ$ is correlated with $\mathbb{V}$.

Let $\Gamma = \text{blkdiag}(I_3, \sqrt{T}, I_{k+1})$ and $\widetilde{\Gamma} = \text{blkdiag}(I_2, \sqrt{T}, I_k)$. Then we can rewrite $\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^c(\theta_0)}{\partial\theta} = \frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta} - \Delta_n$, where

$$\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta} = \begin{pmatrix} \frac{1}{\sqrt{n}\sigma_0^2}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{QV} + \frac{1}{\sqrt{n}\sigma_0^2}\mathbb{V}'\mathbb{G}_0\mathbb{V} + \frac{1}{\sqrt{n}\sigma_0^2}(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)'\mathbb{V} - \frac{1}{\sqrt{n}}\text{tr}(\mathbb{G}_0) \\ \frac{1}{\sqrt{n}\sigma_0^2}\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{*\prime}\mathbb{V} \\ -\frac{n}{2\sqrt{n}\sigma_0^2} + \frac{1}{2\sqrt{n}\sigma_0^4}\mathbb{V}'\mathbb{V} \end{pmatrix},$$

and

$$\Delta_n = \begin{pmatrix} \frac{1}{\sqrt{n}\sigma_0^2}\mathbb{V}'\mathbb{G}_0'\mathbb{PV} + \frac{1}{\sqrt{n}\sigma_0^2}(\widetilde{\mathbb{G}_0\mathbb{Z}}^\circ\delta_0)'\mathbb{V} \\ \frac{1}{\sqrt{n}\sigma_0^2}\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{\circ\prime}\mathbb{V} \\ \frac{1}{2\sqrt{n}\sigma_0^4}\mathbb{V}'\mathbb{PV} \end{pmatrix}.$$

This decomposes $\frac{\partial L_n^c(\theta_0)}{\partial\theta}$ into two components: a mean zero term $\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}$ and a nonzero mean term Δ_n. Δ_n can be regarded as the source of asymptotic bias arising from the estimation of fixed effects. As derived in Supplementary Appendix B.2,

$$E\left(\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}\cdot\frac{1}{\sqrt{n}}\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta'}\Gamma\right) = \Gamma\Sigma_{\theta_0,n}\Gamma + \Gamma\Omega_{\theta_0,n}\Gamma + O\left(\frac{1}{T}\right).$$

Denote $\Sigma_{\theta_0} \equiv \lim_{n\to\infty}\Gamma\Sigma_{\theta_0,n}\Gamma$, $\Omega_{\theta_0} \equiv \lim_{n\to\infty}\Gamma\Omega_{\theta_0,n}\Gamma$, then

$$\lim_{n\to\infty} E\left(\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}\cdot\frac{1}{\sqrt{n}}\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta'}\Gamma\right) = \Sigma_{\theta_0} + \Omega_{\theta_0}. \tag{10}$$

The asymptotic distribution of $\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}$ can be derived by applying the central limit theorem for martingale difference sequences, as proved in Lemma 18 in Supplementary Appendix A. Furthermore, as derived in

Supplementary Appendix B.4, we have $\Delta_n = \sqrt{\frac{N}{c_u T}}\varphi_n(\theta_0) + O_p\left(\frac{1}{\sqrt{T}}\right)$, where $c_u \equiv \frac{n}{NT}$, $\varphi_n(\theta_0) = O(1)$ and $\varphi_n(\theta)$ is defined in equation (B.6) in Supplementary Appendix B.4.

The following assumption ensures the nonsingularity of the information matrix Σ_{θ_0}.

Assumption 9 *At least one of the following two conditions is satisfied: (i)* $\lim_{n\to\infty} E\mathcal{H}^{(-\gamma)}$ *exists and is nonsingular. (ii)* $\lim_{n\to\infty} \frac{1}{n}\left(\mathrm{tr}(\mathbb{G}_0^2 + \mathbb{G}_0'\mathbb{G}_0) - \frac{2}{n}\left[\mathrm{tr}(\mathbb{G}_0)\right]^2\right) \neq 0$.

Theorem 2 *Under Assumptions 1-9,*

$$\sqrt{n}\Gamma^{-1}\left[\hat{\theta} - \left(\theta_0 - \frac{\Sigma_{\theta_0,n}^{-1}\varphi_n(\theta_0)}{c_u T}\right)\right] + O_p\left(\max\left\{\frac{1}{\sqrt{T}}, \sqrt{\frac{N}{T^3}}\right\}\right) \xrightarrow{d} N\left(0, \Sigma_{\theta_0}^{-1}(\Sigma_{\theta_0} + \Omega_{\theta_0})\Sigma_{\theta_0}^{-1}\right), \quad (11)$$

where $\Sigma_{\theta_0,n}^{-1}\varphi_n(\theta_0)$ *is O(1).*

If error terms v_{it} are normal, $\Omega_{\theta_0} = 0$ and the asymptotic variance of equation (11) becomes $\Sigma_{\theta_0}^{-1}$. Theorem 2 shows that $\hat{\theta}$ exhibits an asymptotic bias $-\Sigma_{\theta_0,n}^{-1}\varphi_n(\theta_0)/(c_u T)$. It also shows that $\hat{\theta}^{(-\gamma)}$ is $\sqrt{n}$-consistent and $\hat{\gamma}$ is $\sqrt{c_u N}$-consistent. When $N/T \to 0$, $\hat{\theta}$ is consistent and asymptotically centered normal, which means $\sqrt{n}\Gamma^{-1}\left(\hat{\theta} - \theta_0\right) \xrightarrow{d} N\left(0, \Sigma_{\theta_0}^{-1}(\Sigma_{\theta_0} + \Omega_{\theta_0})\Sigma_{\theta_0}^{-1}\right)$.

3.3 Bias Correction

According to (11), $\hat{\theta}$ exhibits the bias $-\Sigma_{\theta_0,n}^{-1}\varphi_n(\theta_0)/(c_u T)$. Thus a bias-corrected counterpart of $\hat{\theta}$ takes the form

$$\hat{\theta}^1 = \hat{\theta} + \frac{\hat{B}_n}{c_u T},$$

where $\hat{B}_n = \left[\left(-\frac{1}{n}\frac{\partial^2 L_n^c(\theta)}{\partial\theta\partial\theta'}\right)^{-1}\varphi_n(\theta)\right]\Big|_{\theta=\hat{\theta}}$.

Assumption 10 $\sum_{h=0}^{T-t} D_t' A_t^{(h)}(\theta) D_{t-h}$ *and* $\frac{\partial}{\partial\theta}\sum_{h=0}^{T-t} D_t' A_t^{(h)}(\theta) D_{t-h}$ *are UB uniformly in a neighborhood of* θ_0 *over* t.

Theorem 3 *Under Assumptions 1-10, if* $N/T^3 \to 0$,

$$\sqrt{n}\Gamma^{-1}\left(\hat{\theta}^1 - \theta_0\right) \xrightarrow{d} N\left(0, \Sigma_{\theta_0}^{-1}(\Sigma_{\theta_0} + \Omega_{\theta_0})\Sigma_{\theta_0}^{-1}\right).$$

Hence, the estimator after bias correction is asymptotically normal and centered around θ_0 as long as T grows faster than $N^{\frac{1}{3}}$.

3.4 Estimation of the Asymptotic Covariance Matrix

To make the asymptotic distribution in Theorem 3 operational, we construct a plug-in estimator of the asymptotic covariance matrix. The population covariance matrix has the sandwich form $\Sigma_{\theta_0}^{-1}(\Sigma_{\theta_0} + \Omega_{\theta_0})\Sigma_{\theta_0}^{-1}$, where Σ_{θ_0} is the limiting information matrix and Ω_{θ_0} is the additional non-Gaussian correction term. Let $\widehat{\Sigma}$ and $\widehat{\Omega}$ denote the plug-in estimators of the normalized matrices $\Gamma\Sigma_{\theta_0,n}\Gamma$ and $\Gamma\Omega_{\theta_0,n}\Gamma$, respectively, constructed from the bias-corrected estimator and the corresponding residual moments. The explicit formulas are reported in Supplementary Appendix C.4. Define $\widehat{\Xi} = \widehat{\Sigma}^{-1}(\widehat{\Sigma} + \widehat{\Omega})\widehat{\Sigma}^{-1}$.

Theorem 4 *Under Assumptions 1-10, if* $N/T^3 \to 0$, *then* $\widehat{\Xi} \xrightarrow{p} \Sigma_{\theta_0}^{-1}(\Sigma_{\theta_0} + \Omega_{\theta_0})\Sigma_{\theta_0}^{-1}$. *Consequently,* $n^{-1}\Gamma\widehat{\Xi}\Gamma$ *is a consistent estimator of the asymptotic covariance matrix of* $\widehat{\theta}^1$.

4 Extension to TWFE

In some applications, in addition to individual heterogeneity, the dependent variable may also be affected by common period-specific shocks, such as macroeconomic fluctuations, seasonality, or platform-wide demand changes. To accommodate this possibility, we introduce time fixed effects into the baseline genuinely unbalanced model. The individual and time fixed effects are jointly concentrated out from the likelihood, and the asymptotic properties of the resulting TWFE QMLE are then derived. Relative to the OWFE case, the main technical difference is that the incidental-parameter bias must be rederived under the joint projection associated with the two sets of fixed effects.

4.1 Model Specification

For $t = 1, \ldots, T$ and $i \in \mathbb{I}_t$, consider the following extension of model (1):

$$y_{it} = \rho_0 \sum_{j \in \mathbb{I}_t} w_{ij,t} y_{jt} + d_{i,t-1} \left(\lambda_0 \sum_{j \in \mathbb{I}_{t-1}} m_{ij,t} y_{j,t-1} + \nu_0 y_{i,t-1} \right) + d_{it}(1 - d_{i,t-1})\gamma_0 + x_{it}'\beta_0 + \alpha_{i0} + \tau_{t0} + v_{it}, \tag{12}$$

where τ_{t0} denotes the time fixed effect in period t. As in the baseline model, α_{i0} captures unit-specific heterogeneity, while γ_0 represents the common listing effect at a unit's first appearance. The new term τ_{t0} captures shocks common to all units active in period t. To avoid overparameterization, we impose the normalization $\sum_{t=1}^{T} \tau_{t0} = 0$. Let $\tau_0 = (\tau_{20}, \ldots, \tau_{T0})'$. Then, stacking over t, the model can be written as

$$\mathbb{Y} = \rho_0 \mathbb{W}\mathbb{Y} + \mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{J}\tau_0 + \mathbb{V},$$

where $\mathbb{J} \equiv [-\iota_{N_1} \iota_{T-1}'; \text{blkdiag}(\iota_{N_2}, \iota_{N_3}, \cdots, \iota_{N_T})]$, all other objects are defined as in Section 2 for the baseline model.

4.2 Quasi-Maximum Likelihood Estimation

Under the extension with time fixed effects, the corresponding quasi-log-likelihood function is

$$\begin{aligned} L_n(\theta, \alpha, \tau) = &- \frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 + \ln |\mathbb{S}(\rho)| \\ &- \frac{1}{2\sigma^2} (\mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta - \mathbb{D}\alpha - \mathbb{J}\tau)' (\mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta - \mathbb{D}\alpha - \mathbb{J}\tau). \end{aligned}$$

For given ζ, let $(\hat{\alpha}(\zeta)', \hat{\tau}(\zeta)')' = [(\mathbb{D}, \mathbb{J})'(\mathbb{D}, \mathbb{J})]^{-1} (\mathbb{D}, \mathbb{J})' (\mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta)$. Accordingly, define $\bar{\mathbb{P}} \equiv (\mathbb{D}, \mathbb{J}) [(\mathbb{D}, \mathbb{J})'(\mathbb{D}, \mathbb{J})]^{-1} (\mathbb{D}, \mathbb{J})'$, $\bar{\mathbb{Q}} \equiv I_n - \bar{\mathbb{P}}$. Substituting out α and τ yields the concentrated quasi-log-likelihood

$$\bar{L}_n^c(\rho, \delta, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 + \ln |\mathbb{S}(\rho)| - \frac{1}{2\sigma^2} (\mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta)' \bar{\mathbb{Q}} (\mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta).$$

The concentrated quasi-log-likelihood function $\bar{L}_n^c(\rho, \delta, \sigma^2)$ serves as the basis for the subsequent asymptotic analysis. Exactly as in the baseline OWFE case, in practical implementation we further concentrate out δ and σ^2, so that the optimization is ultimately carried out with respect to ρ alone. Specifically, the profile quasi-log-likelihood for ρ is

$$\bar{L}_n^{cc}(\rho) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \hat{\sigma}^2(\rho) + \ln |\mathbb{S}(\rho)| - \frac{n}{2},$$

where $\hat{\delta}(\rho) = \left(\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}\right)^{-1} \mathbb{Z}'\bar{\mathbb{Q}}\mathbb{S}(\rho)\mathbb{Y}$, and $\hat{\sigma}^2(\rho) = \frac{1}{n}\mathbb{Y}'\mathbb{S}(\rho)' \left[\bar{\mathbb{Q}} - \bar{\mathbb{Q}}\mathbb{Z} \left(\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}\right)^{-1} \mathbb{Z}'\bar{\mathbb{Q}}\right] \mathbb{S}(\rho)\mathbb{Y}$.

4.3 Asymptotic properties

The consistency proof proceeds exactly as in the baseline model. To avoid restating conditions that remain unchanged, we continue to maintain Assumptions 2, 3, and 5-7 of the baseline model. Relative to the OWFE case, only Assumptions 1, 4, and 8 need to be modified under the TWFE specification.

Assumption 1′ *The unbalancedness of the model is nonstochastic, which means that d_{it} and D_t are nonstochastic for all i and t. There is a lower bound on the number of observations for all individuals, i.e., $T_i \geqslant c_T T$ for all i, where $c_T > 0$ is a constant that does not vary with N or T. Additionally, $N_0 < c_N N$, where N_0 denotes the number of units observed at $t = 0$ and $0 < c_N < 1$ is a constant that does not vary with N or T. There exists a constant $\bar{c}_N > 0$, not varying with N or T, such that $\inf_{1\leqslant t\leqslant T} \frac{N_t}{N} \geqslant \bar{c}_N$. Moreover, the matrix $\mathbb{J}'\mathbb{Q}\mathbb{J}$ is uniformly nonsingular for N and T.*

Assumption 1′ strengthens Assumption 1 in two aspects that are needed for the TWFE specification. The condition $\inf_{1\leqslant t\leqslant T} \frac{N_t}{N} \geqslant \bar{c}_N$ rules out time periods with too few active units, so that each time fixed effect is estimated from a non-negligible cross-sectional sample. The uniform nonsingularity of $\mathbb{J}'\mathbb{Q}\mathbb{J}$ rules out collinearity between the time fixed effects and the individual fixed effects after projecting out the latter. It ensures that the time fixed effects can be properly concentrated out and that the joint projection matrix is well defined.

The following two assumptions are similar to their OWFE counterparts. We use $\bar{\mathbb{Q}}$ and $\bar{\mathcal{H}}^{(-\gamma)}$ instead to accommodate the additional component involving $\mathbb{J}$.

Assumption 4′ *The elements of x_{it}, α_{i0} and τ_{t0} are nonstochastic and bounded uniformly in i, t, N and T. $\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}$ is nonsingular. Also, $\lim_{n\to\infty} \frac{1}{n} E(\mathbb{Z}^{(-\gamma)\prime}\bar{\mathbb{Q}}\mathbb{Z}^{(-\gamma)})$ exists and is nonsingular.*

Define $\bar{\mathcal{H}}^{(-\gamma)} \equiv \frac{1}{n}\big(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0 + \mathbb{G}_0\mathbb{J}\tau_0,\, \mathbb{Z}^{(-\gamma)}\big)'\bar{\mathbb{Q}}\big(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0 + \mathbb{G}_0\mathbb{J}\tau_0,\, \mathbb{Z}^{(-\gamma)}\big)$ and $\bar{\sigma}_n^2 \equiv \frac{\sigma_0^2}{n}\mathrm{tr}\left[\big(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\big)'\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right]$.

Assumption 8′ *At least one of the following two conditions is satisfied: (i) $\lim_{n\to\infty} E\bar{\mathcal{H}}^{(-\gamma)}$ exists and is nonsingular. (ii) $\lim_{n\to\infty}\left(\frac{1}{n}\ln\left|\sigma_0^2\mathbb{S}_0^{-1}\mathbb{S}_0^{-1\prime}\right| - \frac{1}{n}\ln\left|\bar{\sigma}_n^2\mathbb{S}(\rho)^{-1}\mathbb{S}(\rho)^{-1\prime}\right|\right) \neq 0$ for $\rho \neq \rho_0$.*

Theorem 5 *Under Assumptions 1′, 2, 3, 4′, 5-7 and 8′, $\hat{\theta} \xrightarrow{p} \theta_0$.*

The TWFE score decomposition is detailed in Supplementary Appendix B.5. It is the TWFE counterpart of the decomposition in Section 3.2, with the OWFE projection $\mathbb{Q}$ replaced by the joint projection $\bar{\mathbb{Q}}$. Specifically, Supplementary Appendix B.5 shows that

$$\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^c(\theta_0)}{\partial\theta} = \frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta} - \bar{\Delta}_n,$$

where the first term is a mean-zero score component with limiting covariance $\bar{\Sigma}_{\theta_0} + \bar{\Omega}_{\theta_0}$. Moreover,

$$\bar{\Delta}_n = \sqrt{\frac{N}{c_u T}}\varphi_n(\theta_0) + \sqrt{\frac{T}{c_u N}}\psi_n(\theta_0) + O_p\left(\frac{1}{\sqrt{T}}\right) + O_p\left(\frac{1}{\sqrt{N}}\right),$$

where $c_u = n/(NT)$, $\varphi_n(\theta_0) = O(1)$ is the OWFE bias component, and $\psi_n(\theta_0) = O(1)$ is the additional bias component generated by the time fixed effects. These results, together with the same Taylor expansion argument as in the OWFE case, lead to the following asymptotic distribution.

Assumption 9′ *At least one of the following two conditions is satisfied: (i) $\lim_{n\to\infty} E\bar{\mathcal{H}}^{(-\gamma)}$ exists and is nonsingular. (ii) $\lim_{n\to\infty} \frac{1}{n}\left(\mathrm{tr}(\mathbb{G}_0^2 + \mathbb{G}_0'\mathbb{G}_0) - \frac{2}{n}\left[\mathrm{tr}(\mathbb{G}_0)\right]^2\right) \neq 0$.*

Theorem 6 *Under Assumptions 1′, 2, 3, 4′, 5-7, and 8′-9′,*

$$\sqrt{n}\Gamma^{-1}\left[\hat{\theta}-\left(\theta_0-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0)}{c_uT}-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0)}{c_uN}\right)\right]+O_p\left(\max\left\{\frac{1}{\sqrt{T}},\frac{1}{\sqrt{N}},\sqrt{\frac{N}{T^3}},\sqrt{\frac{T}{N^3}}\right\}\right)$$
$$\xrightarrow{d} N\left(0,\bar{\Sigma}_{\theta_0}^{-1}(\bar{\Sigma}_{\theta_0}+\bar{\Omega}_{\theta_0})\bar{\Sigma}_{\theta_0}^{-1}\right). \quad (13)$$

where $\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0)$ *and* $\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0)$ *are both* $O(1)$.

If error terms v_{it} are normal, $\bar{\Omega}_{\theta_0}=0$ and the asymptotic variance of equation (13) becomes $\bar{\Sigma}_{\theta_0}^{-1}$. Theorem 6 shows that $\hat{\theta}$ exhibits two asymptotic bias components $-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0)}{c_uT}$ and $-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0)}{c_uN}$. It also shows that $\hat{\theta}^{(-\gamma)}$ is $\sqrt{n}$-consistent and $\hat{\gamma}$ is $\sqrt{c_uN}$-consistent.

4.4 Bias Correction

According to (13), $\hat{\theta}$ exhibits the bias $-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0)}{c_uT}-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0)}{c_uN}$, thus a bias-corrected counterpart of $\hat{\theta}$ takes the form

$$\hat{\theta}^1=\hat{\theta}+\frac{\hat{B}_{T,n}}{c_uT}+\frac{\hat{B}_{N,n}}{c_uN},$$

where $\hat{B}_{T,n}=\left[\left(-\frac{1}{n}\frac{\partial^2\bar{L}_n^c(\theta)}{\partial\theta\partial\theta'}\right)^{-1}\varphi_n(\theta)\right]\Big|_{\theta=\hat{\theta}}$, and $\hat{B}_{N,n}=\left[\left(-\frac{1}{n}\frac{\partial^2\bar{L}_n^c(\theta)}{\partial\theta\partial\theta'}\right)^{-1}\psi_n(\theta)\right]\Big|_{\theta=\hat{\theta}}$.

Theorem 7 *Under Assumptions 1′, 2, 3, 4′, 5-7, 8′, 9′ and 10, if* $\frac{N}{T^3}\to 0$ *and* $\frac{T}{N^3}\to 0$, *then*

$$\sqrt{n}\Gamma^{-1}\left(\hat{\theta}^1-\theta_0\right)\xrightarrow{d} N\left(0,\bar{\Sigma}_{\theta_0}^{-1}(\bar{\Sigma}_{\theta_0}+\bar{\Omega}_{\theta_0})\bar{\Sigma}_{\theta_0}^{-1}\right).$$

Hence, the estimator after bias correction is asymptotically normal and centered around θ_0 as long as T grows faster than $N^{\frac{1}{3}}$ and N grows faster than $T^{\frac{1}{3}}$.

4.5 Estimation of the Asymptotic Covariance Matrix

For feasible inference under the TWFE specification, we use the same plug-in principle. The TWFE asymptotic covariance matrix has the sandwich form $\bar{\Sigma}_{\theta_0}^{-1}(\bar{\Sigma}_{\theta_0}+\bar{\Omega}_{\theta_0})\bar{\Sigma}_{\theta_0}^{-1}$, where $\bar{\Sigma}_{\theta_0}$ is the limiting TWFE information matrix and $\bar{\Omega}_{\theta_0}$ is the corresponding non-Gaussian correction term. Let $\hat{\bar{\Sigma}}$ and $\hat{\bar{\Omega}}$ denote the plug-in estimators of the normalized matrices $\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma$ and $\Gamma\bar{\Omega}_{\theta_0,n}\Gamma$, respectively, constructed from the TWFE bias-corrected estimator and the corresponding residual moments. The explicit formulas are reported in Supplementary Appendix C.8. Define $\hat{\bar{\Xi}}=\hat{\bar{\Sigma}}^{-1}(\hat{\bar{\Sigma}}+\hat{\bar{\Omega}})\hat{\bar{\Sigma}}^{-1}$.

Theorem 8 *Under Assumptions 1′, 2, 3, 4′, 5-7, 8′, 9′ and 10, if* $N/T^3\to 0$ *and* $T/N^3\to 0$, *then* $\hat{\bar{\Xi}}\xrightarrow{p}\bar{\Sigma}_{\theta_0}^{-1}(\bar{\Sigma}_{\theta_0}+\bar{\Omega}_{\theta_0})\bar{\Sigma}_{\theta_0}^{-1}$. *Consequently,* $n^{-1}\Gamma\hat{\bar{\Xi}}\Gamma$ *is a consistent estimator of the asymptotic covariance matrix of* $\hat{\theta}^1$.

5 Monte Carlo Simulation

5.1 Data-Generating Process

In our simulation study, we investigate the finite sample performance of the OWFE QMLE, its bias-corrected counterpart, and the feasible inference procedure based on the proposed asymptotic covariance matrix estimator. In the simulation study, we generate an unbalanced panel dataset of size $N\times(T+21)$, where the first 20

periods are used only to generate a stable Y_0 and the remaining $T+1$ periods are used for estimation (set $t=0$ at the 21st period). In order to simulate unbalancedness in the panel, we randomly assign to each node a contiguous observation window within the $T+1$ periods. Specifically, we first draw the observation duration for each observed node: $T_i \sim \text{GE}(\frac{p}{T})+2$, where GE denotes the geometric distribution supported on nonnegative integers and p is the parameter of the geometric distribution. Given the draw T_i, if $T_i \geqslant T+1$, then node i is observed for $T+1$ periods. Otherwise, we place the contiguous observation window of length T_i uniformly at random within the $(T+1)$-period horizon. By tuning the parameter p, we can control the unbalancedness proportion of the panel, which is defined as $\text{UP} \equiv 1-\frac{n}{NT}$, around a target mean level.

We generate the elements of X_t and α_0 independently from standard normal distributions. We simulate the dependent variable using a 20-period pre-sample burn-in. For individuals already observed in period 0, we initialize their outcome at $t=-20$ as $y_{i,-20} \sim N(0,1)$, drawn independently across units. The subsequent observations ($t=-19,\cdots,0$) are then generated iteratively according to equation (2). Given Y_0, we can generate $Y_1,\cdots,Y_T$ iteratively according to equation (2). For the network weight matrix, we first construct an $N\times N$ adjacency matrix A by rook contiguity, and then row-normalize D_tAD_t' and D_tAD_{t-1}', respectively to obtain the weight matrices W_t and M_t.

For the supplementary TWFE simulations reported in Supplementary Appendix D.1, we use the same setup as in the OWFE simulations, except that the outcomes are generated according to the TWFE model in (12). The common time effects are generated from standard normal draws and normalized to satisfy $\sum_{t=1}^{T}\tau_t=0$. During the burn-in stage, independent standard normal common components are generated analogously.

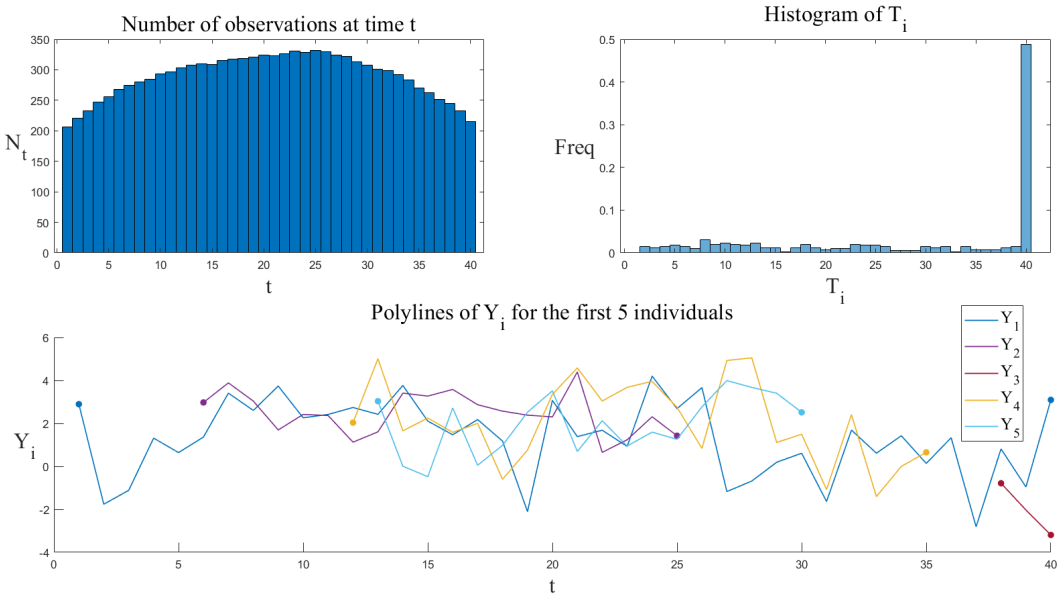


Figure 2: Characteristics of the unbalanced panel we generated

Figure 2 illustrates key features of the simulated unbalanced dynamic network panel data with $N=400$, $T=40$ and $\text{UP}=31.7\%$. It reveals that the number of observed nodes peaks during intermediate stages, with sparser observations in initial and terminal phases. About 50% of nodes are fully observed over the entire $T+1$ periods, while the remaining nodes have shorter observation windows, whose start times are drawn uniformly conditional on their durations.

5.2 Simulation Results

In the main simulation study, we investigate the finite-sample performance of the OWFE QMLE and its bias-corrected counterpart under varying configurations of $N\in\{100,400,1600\}$, $T\in\{10,40\}$, and $\text{UP}\in\{30\%,60\%\}$. For the OWFE simulations, we consider two true parameter settings, $\theta_0=(0.5,0.2,0.1,1,1,1)$

and $\theta_0 = (0.1, 0.3, 0.5, 1, 1, 1)$, where $\theta_0 = (\rho_0, \lambda_0, \nu_0, \gamma_0, \beta_0, \sigma_0^2)$. We also consider three error distributions: normal, centered exponential, and Laplace. Under each setting, the estimation is repeated 1,000 times, and we evaluate the bias, standard deviation (SD), root mean square error (RMSE), and coverage probability (CP) of the estimators.

The OWFE simulation results are presented in Tables 1-2 and Tables D.1–D.6 in Supplementary Appendix D.1. Supplementary Appendix D.1 reports additional TWFE simulation results in Tables D.7–D.12 for $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$ under the same error distributions and UP levels. From the OWFE simulation tables, we obtain the following findings.

(1) The biases of both the original estimator and the bias-corrected estimator decrease significantly as T increases, but show little change as N increases. Moreover, for fixed N and T, a higher UP leads to larger bias. These findings are consistent with the theoretical bias of order $O_p(\frac{1}{c_u T})$.

(2) Regardless of whether bias correction is applied, the standard deviations of most estimators are approximately halved when either N or T is quadrupled, consistent with the theoretical convergence rate of $1/\sqrt{n}$. An exception is $\hat{\gamma}$, whose standard deviation remains essentially unchanged as T increases, aligning with the theoretical convergence rate of $1/\sqrt{c_u N}$. Moreover, the standard deviation increases as the UP rises.

(3) The original estimators exhibit substantial bias, especially for ν and σ^2, while the bias-corrected estimators significantly reduce the bias. The bias-corrected estimators show notable improvements in both bias and RMSE, and their CPs are also closer to the nominal levels of 95% and 90%.

(4) As N increases, there is a downward trend in CPs for both estimators, which is attributable to the asymptotic bias of order $O_p(\sqrt{N/T^3})$.

The supplementary TWFE simulations in Tables D.7–D.12 display a similar pattern. The original TWFE estimator also exhibits noticeable finite-sample bias. After bias correction, the biases and RMSEs are generally reduced, and the CPs move closer to the nominal levels. These results suggest that the proposed bias correction remains effective after introducing time fixed effects.

To examine the correctness of the asymptotic theory, we plot the kernel density curves of $\frac{\hat{\theta}_i - \theta_{0i}}{\mathrm{SE}_i}$, and $\frac{\hat{\theta}_i^1 - \theta_{0i}}{\mathrm{SE}_i}$, where $\hat{\theta}_i$, $\hat{\theta}_i^1$ and θ_{0i} denote the i-th element of $\hat{\theta}$, $\hat{\theta}^1$ and θ_0, respectively, and SE_i denotes the square root of the i-th diagonal element of $\frac{1}{n}\Sigma_{\theta_0,n}^{-1}(\Sigma_{\theta_0,n} + \Omega_{\theta_0,n})\Sigma_{\theta_0,n}^{-1}$. These kernel densities are then compared with the probability density function (PDF) of the standard normal distribution. The results are shown in Figures 3-4 and Figures D.1-D.6 in Supplementary Appendix D.2. As illustrated in the figures, regardless of whether the estimators are bias corrected or not, their distributions become increasingly close to normality as the sample size increases. The bias-corrected estimators exhibit distributions that are closer to a centered normal distribution than the original ones.

Table 1: Results of OWFE estimator with Normal error, UP=30%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0044	0.0327	-0.0753	-0.0348	-0.0098	-0.1594	$N = 100$	BIAS	0.0006	0.0077	-0.0191	-0.0014	-0.0016	-0.0399
	SD	0.0268	0.0367	0.0299	0.1497	0.0427	0.0511		SD	0.0124	0.0178	0.0146	0.1426	0.0191	0.0275
	RMSE	0.0271	0.0492	0.0810	0.1537	0.0438	0.1674		RMSE	0.0124	0.0194	0.0240	0.1426	0.0192	0.0484
	95%CP	0.9340	0.8620	0.2810	0.9470	0.9370	0.1530		95%CP	0.9550	0.9190	0.7070	0.9510	0.9570	0.6860
	90%CP	0.8640	0.7660	0.1920	0.8900	0.8890	0.0930		90%CP	0.8910	0.8740	0.6080	0.8970	0.9060	0.5630
		After bias correction								After bias correction					
	BIAS	**-0.0034**	**0.0061**	**-0.0159**	**-0.0064**	**-0.0005**	**-0.0397**		BIAS	**0.0002**	**0.0001**	**-0.0010**	0.0019	**-0.0007**	**-0.0053**
	SD	**0.0267**	0.0375	0.0311	**0.1495**	**0.0426**	0.0580		SD	0.0124	0.0180	0.0148	**0.1425**	0.0191	0.0284
	RMSE	**0.0269**	**0.0380**	**0.0349**	**0.1497**	**0.0426**	**0.0703**		RMSE	**0.0124**	**0.0180**	**0.0149**	**0.1425**	**0.0191**	**0.0289**
	95%CP	**0.9360**	**0.9380**	**0.9100**	*0.9470*	**0.9410**	**0.8770**		95%CP	*0.9550*	**0.9490**	**0.9430**	0.9530	**0.9560**	**0.9420**
	90%CP	**0.8670**	**0.9000**	**0.8380**	**0.8990**	**0.8920**	**0.7780**		90%CP	0.8890	**0.8950**	**0.8820**	0.8960	**0.9020**	**0.8900**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0021	0.0339	-0.0748	-0.0341	-0.0126	-0.1525	$N = 400$	BIAS	0.0000	0.0082	-0.0194	-0.0027	0.0001	-0.0364
	SD	0.0136	0.0189	0.0146	0.0765	0.0207	0.0259		SD	0.0063	0.0090	0.0070	0.0705	0.0098	0.0142
	RMSE	0.0137	0.0388	0.0762	0.0838	0.0242	0.1546		RMSE	0.0063	0.0122	0.0206	0.0706	0.0098	0.0391
	95%CP	0.9360	0.5630	0.0010	0.9080	0.9070	0.0000		95%CP	0.9540	0.8640	0.2120	0.9590	0.9500	0.2440
	90%CP	0.8730	0.4490	0.0000	0.8450	0.8370	0.0000		90%CP	0.9080	0.7630	0.1390	0.9020	0.8930	0.1700
		After bias correction								After bias correction					
	BIAS	**-0.0012**	**0.0064**	**-0.0152**	**-0.0065**	**-0.0033**	**-0.0319**		BIAS	-0.0004	**0.0006**	**-0.0013**	**0.0004**	0.0010	**-0.0017**
	SD	0.0136	0.0194	0.0151	0.0768	**0.0206**	0.0294		SD	0.0063	0.0091	0.0071	0.0707	**0.0098**	0.0146
	RMSE	**0.0136**	**0.0204**	**0.0214**	**0.0771**	**0.0208**	**0.0434**		RMSE	0.0063	**0.0091**	**0.0072**	0.0707	0.0099	**0.0147**
	95%CP	0.9330	**0.9350**	**0.8290**	**0.9260**	**0.9540**	**0.7590**		95%CP	**0.9500**	**0.9460**	**0.9420**	*0.9590*	0.9450	**0.9310**
	90%CP	**0.8750**	**0.8670**	**0.7260**	**0.8730**	**0.9000**	**0.6690**		90%CP	**0.9020**	**0.9040**	**0.8830**	**0.9010**	**0.9010**	**0.8670**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0020	0.0338	-0.0748	-0.0343	-0.0114	-0.1530	$N = 1600$	BIAS	0.0004	0.0078	-0.0192	-0.0033	-0.0011	-0.0371
	SD	0.0071	0.0090	0.0075	0.0357	0.0102	0.0127		SD	0.0032	0.0047	0.0034	0.0352	0.0048	0.0067
	RMSE	0.0073	0.0350	0.0751	0.0496	0.0153	0.1535		RMSE	0.0032	0.0090	0.0195	0.0354	0.0049	0.0377
	95%CP	0.9300	0.0480	0.0000	0.8550	0.8100	0.0000		95%CP	0.9530	0.6140	0.0010	0.9500	0.9520	0.0010
	90%CP	0.8630	0.0210	0.0000	0.7690	0.6980	0.0000		90%CP	0.9020	0.4960	0.0010	0.8970	0.8920	0.0000
		After bias correction								After bias correction					
	BIAS	**-0.0011**	**0.0062**	**-0.0152**	**-0.0067**	**-0.0022**	**-0.0325**		BIAS	**0.0000**	**0.0002**	**-0.0011**	**-0.0001**	**-0.0002**	**-0.0025**
	SD	0.0071	0.0094	0.0078	0.0359	**0.0102**	0.0144		SD	0.0032	0.0047	0.0034	0.0352	**0.0048**	0.0069
	RMSE	**0.0072**	**0.0112**	**0.0171**	**0.0365**	**0.0104**	**0.0355**		RMSE	**0.0032**	**0.0047**	**0.0036**	**0.0352**	**0.0048**	**0.0073**
	95%CP	**0.9310**	**0.9000**	**0.4840**	**0.9530**	**0.9450**	**0.3380**		95%CP	0.9580	**0.9450**	**0.9450**	0.9490	0.9580	**0.9360**
	90%CP	**0.8720**	**0.8330**	**0.3690**	**0.9040**	**0.9060**	**0.2280**		90%CP	**0.9000**	**0.8900**	**0.8830**	0.8960	**0.9070**	**0.8790**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table 2: Results of OWFE estimator with Normal error, UP=60%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0093	0.0605	-0.1358	-0.0929	-0.0337	-0.2862	$N = 100$	BIAS	0.0004	0.0141	-0.0321	-0.0147	-0.0029	-0.0694
	SD	0.0307	0.0468	0.0440	0.1348	0.0613	0.0613		SD	0.0141	0.0203	0.0185	0.1098	0.0263	0.0352
	RMSE	0.0321	0.0765	0.1428	0.1638	0.0700	0.2927		RMSE	0.0141	0.0247	0.0370	0.1108	0.0264	0.0778
	95%CP	0.9100	0.7650	0.1200	0.8810	0.9000	0.0100		95%CP	0.9470	0.9070	0.6060	0.9570	0.9410	0.5100
	90%CP	0.8370	0.6700	0.0670	0.8260	0.8310	0.0020		90%CP	0.8870	0.8320	0.4840	0.9040	0.8990	0.3840
		After bias correction								After bias correction					
	BIAS	**-0.0062**	**0.0212**	**-0.0481**	**-0.0345**	**-0.0112**	**-0.1083**		BIAS	**0.0000**	**0.0006**	**-0.0031**	**-0.0074**	**-0.0006**	**-0.0105**
	SD	**0.0308**	0.0484	0.0458	**0.1341**	**0.0612**	0.0744		SD	0.0141	0.0206	0.0188	0.1099	**0.0262**	0.0371
	RMSE	**0.0314**	**0.0528**	**0.0664**	**0.1385**	**0.0623**	**0.1314**		RMSE	0.0141	**0.0206**	**0.0191**	**0.1102**	**0.0262**	**0.0386**
	95%CP	**0.9200**	**0.9180**	**0.7850**	**0.9380**	**0.9430**	**0.6750**		95%CP	0.9460	**0.9490**	**0.9500**	0.9590	**0.9460**	**0.9310**
	90%CP	**0.8490**	**0.8530**	**0.6910**	**0.8710**	**0.8740**	**0.5500**		90%CP	**0.8880**	**0.9170**	**0.8890**	0.9090	0.9030	**0.8840**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0061	0.0583	-0.1302	-0.0861	-0.0316	-0.2766	$N = 400$	BIAS	0.0003	0.0145	-0.0317	-0.0069	-0.0026	-0.0660
	SD	0.0153	0.0225	0.0215	0.0645	0.0300	0.0311		SD	0.0068	0.0101	0.0094	0.0573	0.0129	0.0177
	RMSE	0.0165	0.0625	0.1320	0.1076	0.0435	0.2784		RMSE	0.0068	0.0176	0.0331	0.0577	0.0131	0.0683
	95%CP	0.9050	0.2960	0.0000	0.7510	0.8230	0.0000		95%CP	0.9470	0.7290	0.0810	0.9430	0.9450	0.0410
	90%CP	0.8260	0.1980	0.0000	0.6390	0.7240	0.0000		90%CP	0.8990	0.6200	0.0390	0.8840	0.9120	0.0220
		After bias correction								After bias correction					
	BIAS	**-0.0032**	**0.0179**	**-0.0426**	**-0.0282**	**-0.0095**	**-0.0963**		BIAS	**-0.0001**	**0.0009**	**-0.0028**	**0.0004**	**-0.0003**	**-0.0071**
	SD	**0.0152**	0.0232	0.0221	**0.0635**	**0.0299**	0.0378		SD	0.0068	0.0103	0.0096	0.0570	0.0129	0.0187
	RMSE	**0.0156**	**0.0293**	**0.0479**	**0.0695**	**0.0313**	**0.1035**		RMSE	0.0068	**0.0103**	**0.0100**	**0.0570**	**0.0129**	**0.0200**
	95%CP	**0.9230**	**0.8910**	**0.4940**	**0.9320**	**0.9360**	**0.2560**		95%CP	*0.9470*	**0.9470**	**0.9320**	0.9390	**0.9510**	**0.9250**
	90%CP	**0.8560**	**0.8090**	**0.3820**	**0.8830**	**0.8850**	**0.1660**		90%CP	0.8940	**0.9000**	**0.8740**	**0.8980**	**0.9110**	**0.8710**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0055	0.0591	-0.1305	-0.0868	-0.0329	-0.2748	$N = 1600$	BIAS	0.0003	0.0147	-0.0319	-0.0083	-0.0025	-0.0653
	SD	0.0077	0.0118	0.0108	0.0322	0.0152	0.0155		SD	0.0035	0.0050	0.0047	0.0268	0.0066	0.0091
	RMSE	0.0095	0.0602	0.1310	0.0926	0.0363	0.2752		RMSE	0.0035	0.0155	0.0323	0.0281	0.0070	0.0659
	95%CP	0.8560	0.0020	0.0000	0.2400	0.4000	0.0000		95%CP	0.9500	0.1890	0.0000	0.9500	0.9380	0.0000
	90%CP	0.7740	0.0010	0.0000	0.1650	0.2880	0.0000		90%CP	0.8980	0.1120	0.0000	0.8970	0.8770	0.0000
		After bias correction								After bias correction					
	BIAS	**-0.0028**	**0.0185**	**-0.0427**	**-0.0287**	**-0.0109**	**-0.0940**		BIAS	**-0.0002**	**0.0011**	**-0.0031**	**-0.0010**	**-0.0003**	**-0.0064**
	SD	**0.0076**	0.0122	0.0114	**0.0320**	**0.0151**	0.0187		SD	**0.0035**	0.0051	0.0048	0.0269	0.0066	0.0096
	RMSE	**0.0081**	**0.0222**	**0.0442**	**0.0430**	**0.0186**	**0.0959**		RMSE	**0.0035**	**0.0052**	**0.0057**	**0.0269**	**0.0066**	**0.0115**
	95%CP	**0.9020**	**0.6440**	**0.0230**	**0.8600**	**0.8880**	*0.0000*		95%CP	0.9480	**0.9540**	**0.8960**	0.9710	**0.9520**	**0.8850**
	90%CP	**0.8490**	**0.5270**	**0.0100**	**0.7800**	**0.7980**	*0.0000*		90%CP	0.8970	**0.9090**	**0.8180**	0.9070	**0.9080**	**0.7870**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

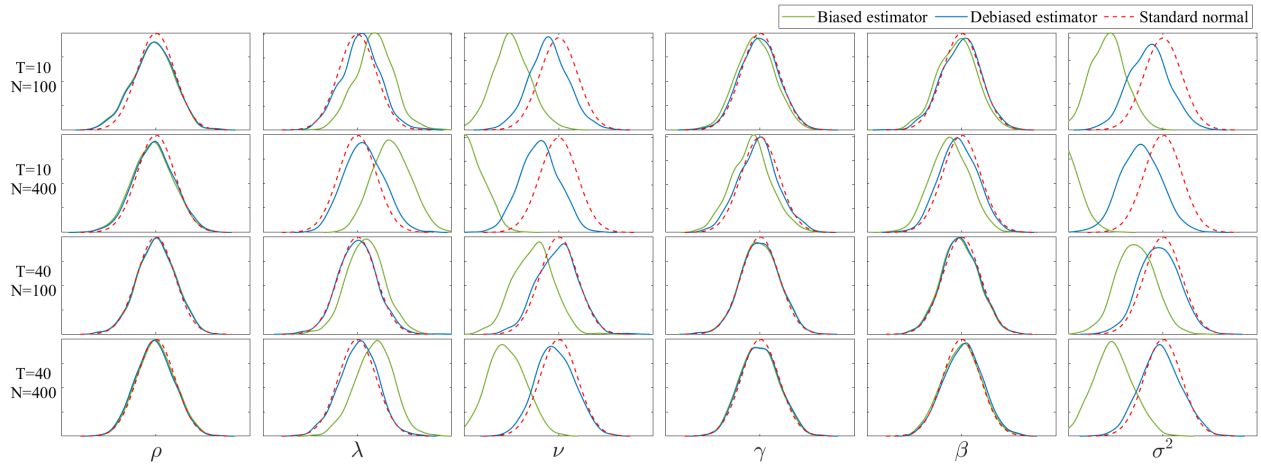


Figure 3: PDF of OWFE standardized estimator with normal error, UP=30%.

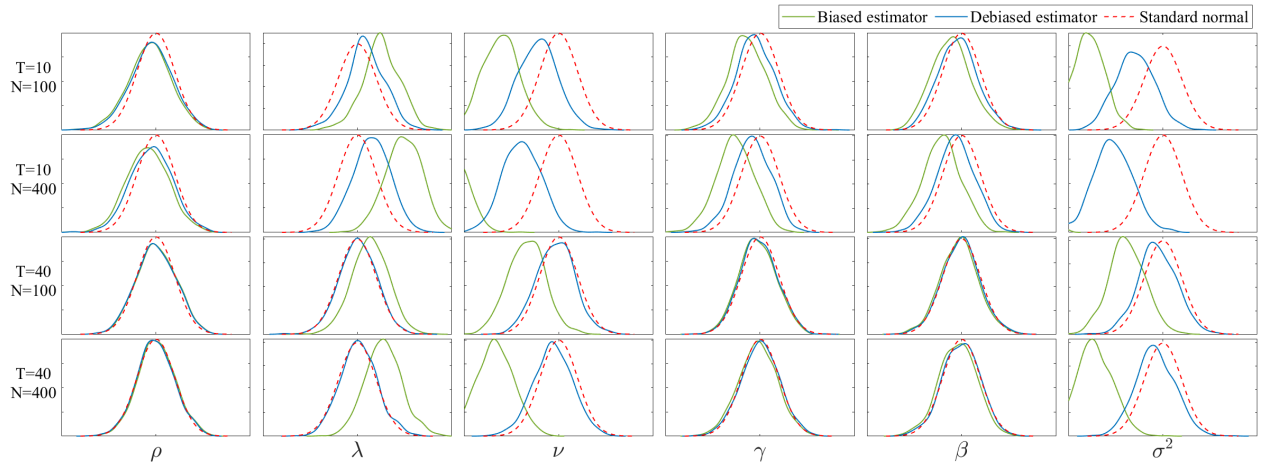


Figure 4: PDF of OWFE standardized estimator with normal error, UP=60%.

6 Real Data Analysis

To assess our model's ability to capture the dynamics of unbalanced spatio-temporal panel data, we take the short-term rental market as the empirical application. In recent years, Airbnb, as a leading global platform for short-term accommodation, has expanded rapidly across cities and regions, and its supply and pricing dynamics have attracted extensive academic attention. More importantly, Airbnb data naturally exhibit an unbalanced panel structure: the set of available listings is not identical across time periods, and many listings are not observed throughout the entire sample period. This provides a representative and complex empirical setting for our study. A large body of literature has already examined the rental pricing of Airbnb properties, such as the studies by Zhang et al. (2017), Gyódi and Nawaro (2021), and Wang and Song (2023), which apply spatial econometric methods to cross-sectional data of Airbnb listings. Jiménez et al. (2022) employ an unbalanced dynamic panel data model to analyze the impact of Airbnb listings on the Spanish tourism markets. Their model does not account for the spatial dependence in the data. Gunter et al. (2020) use a static spatial panel model to explore the substitution or complementarity relationship between Airbnb and traditional accommodations. However, their analysis is restricted to a balanced subsample of listings that are continuously observed over a long period, which accounts for less than 10% of the total sample. This balanced spatial panel model neglects

the spatial spillover effects from short-lived listings and does not account for the temporal dynamics in rental prices. While there have been studies using various models to analyze Airbnb rental prices, none simultaneously account for spatial spillover effects, temporal dynamics (e.g., price lags), and the unbalanced nature of the panel data. This is precisely what our model is designed to achieve.

We collect monthly Airbnb listing data from Inside Airbnb for two regions: New Zealand and New York City[3]. The New Zealand raw data contain 31 monthly snapshots from May 13, 2023 to November 7, 2025, and the New York City raw data contain 18 monthly snapshots from June 4, 2024 to November 1, 2025. Each listing includes information such as rental price, geographic coordinates, availability over the next 30 days, and other listing-level characteristics.

To examine how short-term rental market dynamics vary across areas and spatial scales, we focus on six local markets. In New Zealand, the three samples are ward-level local markets: Waitematā and Gulf Ward, Queenstown-Wakatipu Ward, and Pukehīnau/Lambton Ward. For readability, we refer to them as Auckland, Queenstown, and Wellington, respectively, in the tables. In New York City, the three samples are the neighborhoods of Williamsburg, Midtown, and Hell's Kitchen. Within each local market, the dependent variable is the logarithm of rental price.

The empirical analysis uses the TWFE extension developed in Section 4. The covariates in the empirical specification are the listing's own availability over the next 30 days, denoted by avail30d, and a neighborhood-level availability measure, denoted by avgAvail30d1km.

The unbalanced panel for Auckland, i.e., Waitematā and Gulf Ward, is visualized in Figure 5. This sample contains $N = 5{,}896$ listings, $T = 30$ monthly periods, and $n = 80{,}716$ observations, with an unbalancedness proportion of $\mathrm{UP} = 54.37\%$. The corresponding sample sizes and unbalancedness proportions for the other local markets are reported in Tables 3 and 4. The descriptive statistics of all six datasets are reported in Supplementary Appendix E, Table E.1.

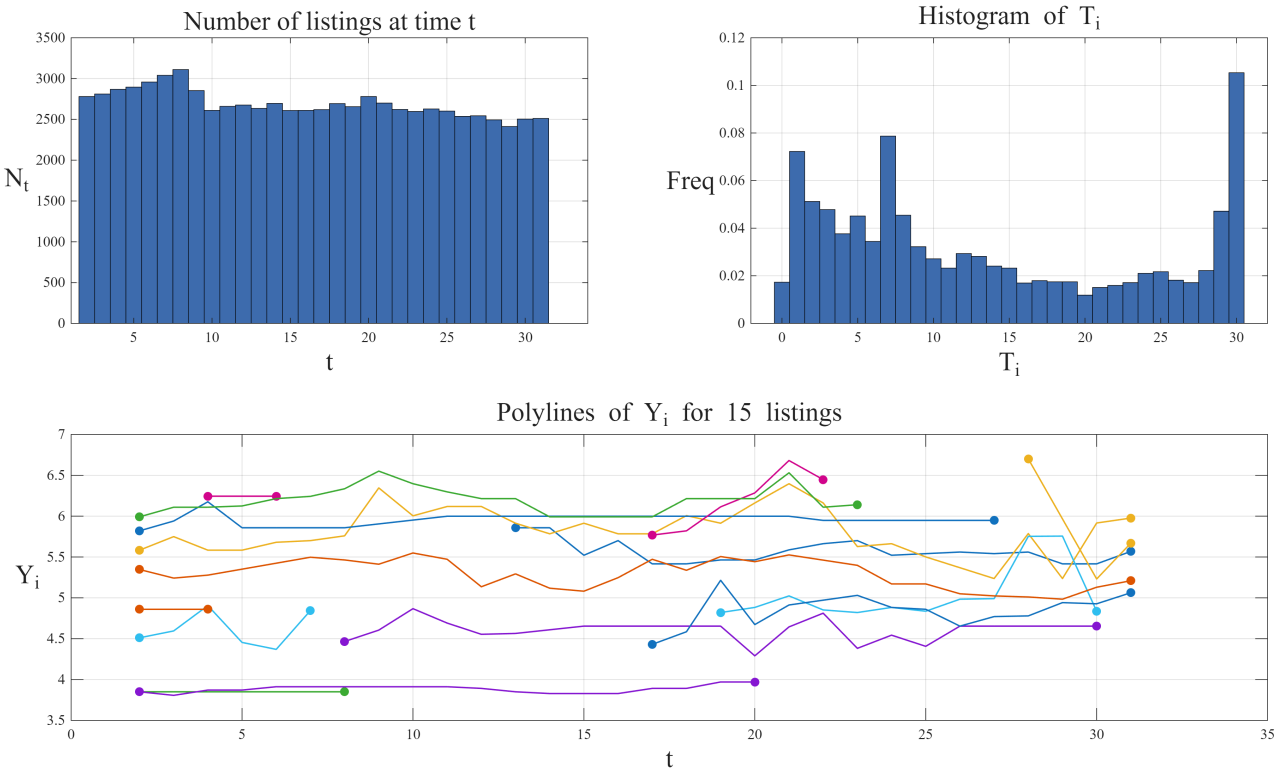


Figure 5: Characteristics of the unbalanced panel of the listing data in Auckland

Figures 6 and 7 display the geographical distribution of log rental prices for listings in Midtown and

[3] https://insideairbnb.com

Queenstown, respectively. The color shading corresponds to the magnitude of log rental price, with darker tones indicating lower values. These figures show that the log rental prices exhibit obvious spatial and temporal correlation.

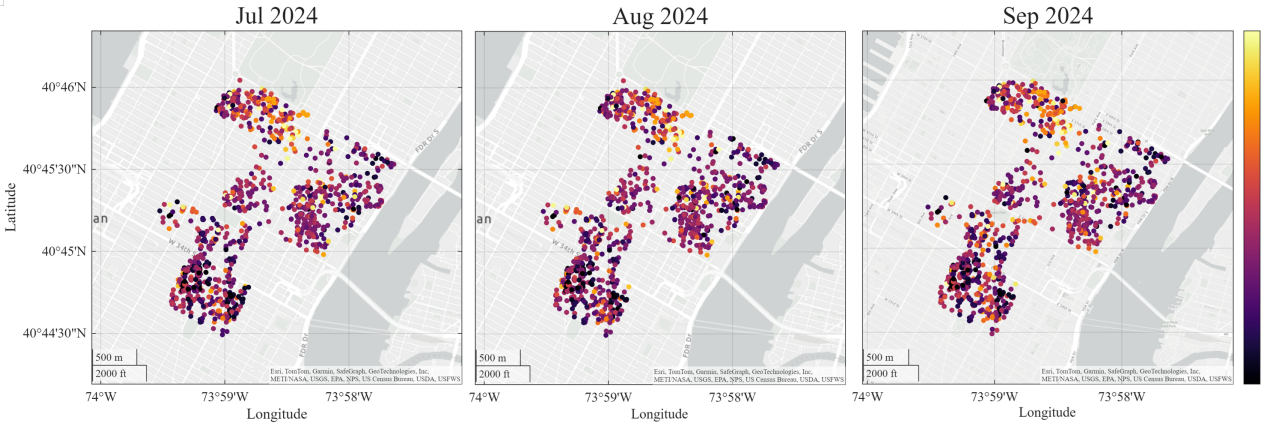


Figure 6: Geographical distribution of log rental prices in Midtown

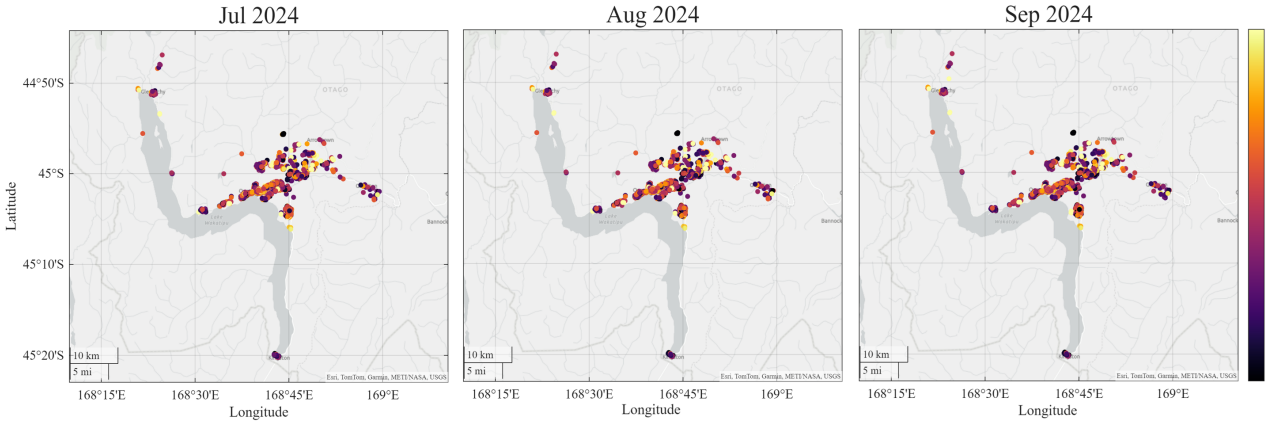


Figure 7: Geographical distribution of log rental prices in Queenstown

Based on the data described above, we estimate the TWFE unbalanced dynamic network panel model using the bias-corrected QMLE. For each local market, W_t and M_t are constructed as row-normalized binary distance-band matrices. Specifically, for a given distance threshold d, two listings are connected if their geographical distance is no larger than d. The contemporaneous matrix W_t is constructed among listings observed in period t, with diagonal elements set to zero. The lagged matrix M_t is constructed between listings observed in period t and listings observed in period $t-1$, with links from a listing to itself in the previous period excluded. The resulting binary adjacency matrices are then row-normalized. Instead of using a common set of distance thresholds for all areas, we choose the thresholds adaptively because the selected local markets differ substantially in spatial scale. Specifically, the thresholds are selected based on the market-specific median nearest-neighbor distance and then rounded to one decimal place. This procedure yields thresholds of 0.4, 0.7, and 1.1 km for the three New Zealand local markets; 0.4, 0.6, and 1.1 km for Williamsburg; 0.1, 0.2, and 0.4 km for Midtown; and 0.2, 0.3, and 0.5 km for Hell's Kitchen. The main tables report the estimates under the largest distance threshold for each local market, while Supplementary Appendix E reports the estimates under all three thresholds.

Tables 3 and 4 compare the bias-corrected TWFE QMLE results across the selected local markets. The results are based on the same class of TWFE dynamic network model, but the distance thresholds are market-specific because of the adaptive construction described above. We obtain the following findings.

(1) The network autoregressive coefficient ρ captures contemporaneous interaction effects among connected listings. It is positive and statistically significant in all six local markets. The estimates are larger in New York City, ranging from 0.4812 to 0.7690, than in New Zealand, where they range from 0.0960 to 0.1612. This difference is consistent with the higher listing density and stronger real-time competitive pressure in New York City. In contrast, listings in the New Zealand local markets are more geographically dispersed, which weakens contemporaneous spatial price transmission.

(2) The lagged network autoregressive effect λ captures the impact of neighboring prices in the previous period on current rental prices. This effect is significantly negative across all six local markets. Moreover, local markets with a larger contemporaneous network autoregressive coefficient ρ generally exhibit a more negative estimate of λ. This pattern suggests an intertemporal dampening mechanism in network spillovers, whereby contemporaneous spatial interactions are partially attenuated by subsequent network effects.

(3) The temporal lagged effect ν is positive and statistically significant in all local markets, indicating rental price persistence. The estimates range from 0.2082 to 0.3982 in New Zealand and from 0.3969 to 0.4638 in New York City.

Table 3: Bias-corrected TWFE QMLE estimates for New Zealand

	Auckland	Queenstown	Wellington
$\rho \quad (W_tY_t)$	0.1612(0.0053)***	0.0960(0.0055)***	0.1591(0.0333)***
$\lambda \quad (M_tY_{t-1})$	-0.1640(0.0053)***	-0.0791(0.0052)***	-0.0735(0.0301)**
$\nu \quad (Y_{t-1})$	0.3982(0.0031)***	0.3006(0.0028)***	0.2082(0.0049)***
avail30d	-0.0006(0.0001)***	-0.0005(0.0001)***	-0.0003(0.0002)**
avgAvail30d1km	-0.0053(0.0004)***	-0.0058(0.0004)***	0.0015(0.0016)
Listing effect	0.2221(0.0059)***	0.2335(0.0064)***	0.1207(0.0274)***
N	5896	6095	1549
T	30	30	30
n	80716	95455	18723
UP	54.37%	47.80%	59.71%

Notes: Estimates are presented with their standard errors in parentheses. Statistical significance is indicated by stars: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.

Table 4: Bias-corrected TWFE QMLE estimates for New York City

	Williamsburg	Midtown	Hell's Kitchen
$\rho \quad (W_tY_t)$	0.7690(0.0310)***	0.4812(0.0238)***	0.5414(0.0267)***
$\lambda \quad (M_tY_{t-1})$	-0.5242(0.0397)***	-0.2740(0.0319)***	-0.4066(0.0328)***
$\nu \quad (Y_{t-1})$	0.4638(0.0064)***	0.4597(0.0068)***	0.3969(0.0063)***
avail30d	0.0006(0.0002)***	-0.0012(0.0003)***	-0.0002(0.0003)
avgAvail30d1km	0.0035(0.0031)	-0.0020(0.0034)	0.0013(0.0093)
Listing effect	-0.0687(0.0438)	0.2528(0.0420)***	-0.0022(0.0380)
N	1566	2390	1943
T	17	17	17
n	14567	23672	16862
UP	45.28%	41.74%	48.95%

Notes: Estimates are presented with their standard errors in parentheses. Statistical significance is indicated by stars: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.

Furthermore, Tables E.2–E.7 in Supplementary Appendix E report detailed estimation results under different network weight matrix specifications for each area. Within each local market, the estimation results are generally similar across different distance-threshold specifications in terms of signs and statistical significance. A common pattern is that the absolute magnitudes of both the network autoregressive coefficient ρ and the lagged network autoregressive effect λ increase as the distance threshold d expands, while the remaining coefficients change only slightly. This pattern is consistent with the role of the distance threshold in defining the network exposure. Larger distance bands include a broader set of local competitors, which strengthens the measured contemporaneous network dependence and the subsequent lagged network adjustment. In contrast, the stability of the non-network coefficients suggests that the estimated effects of listing-level availability, nearby availability, temporal persistence, and listing entry are not driven by the particular distance-band choice.

7 Conclusion

In this paper, we study QMLE for genuinely unbalanced dynamic network panel data models as both N and T go to infinity. We first consider a baseline OWFE specification. For this specification, the QMLE is $\sqrt{NT}$-consistent for all parameters except for the listing effect $\hat{\gamma}$, which achieves $\sqrt{N}$-consistency. When N grows proportionally to T, the QMLE has an asymptotic bias induced by the incidental parameters. The proposed bias-corrected estimator is asymptotically unbiased and normally distributed as long as $N/T^3 \to 0$. We also construct a consistent estimator of the asymptotic covariance matrix, making feasible inference available for the bias-corrected estimator.

We further extend the framework to a TWFE specification. Under this extension, the QMLE remains consistent, and the asymptotic bias contains an additional component generated by the estimation of time fixed effects. We derive the corresponding bias correction and show that the TWFE bias-corrected estimator is asymptotically unbiased and normally distributed when $N/T^3 \to 0$ and $T/N^3 \to 0$. A consistent estimator of the TWFE asymptotic covariance matrix is also provided.

Several extensions remain for future research. First, GMM estimation for this model is worth studying, because it can accommodate heteroskedastic errors and remains applicable when T is finite. Second, building on the current TWFE extension, individual–time interactive fixed effects provide a natural direction for further development. Third, a more flexible mechanism for the listing effect is another promising extension.

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Competing interests

The authors declare none.

Declaration of Generative AI Use

During the preparation of this work the authors used ChatGPT in order to enhance the writing quality. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data Availability Statement

The data that support the findings of this study are available from Inside Airbnb at https://insideairbnb.com. These data were derived from publicly available monthly Airbnb listing snapshots.

Supplementary Material

The supplementary material includes the required lemmas and their proofs, the explicit expressions and properties of the QMLE score function and Hessian matrix, detailed proofs of the theorems, as well as tables and figures for the simulation and empirical results.

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Supplementary Appendix for Quasi-Maximum Likelihood Estimation for a Genuinely Unbalanced Dynamic Network Panel Data Model

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A Some Lemmas

In order to establish the law of large numbers for the quadratic form of $\mathbb{Z}$ and $\mathbb{V}$, we need to study the statistical properties of the quadratic form of two random variables formed by the accumulation of infinite historical errors. Define P_t, Q_t as sequences of $N_t \times N_{t-1}$ matrices, and define their cumulative products as $P_t^{(h)} \equiv P_t P_{t-1} \cdots P_{t-h+1}$ for $h = 1, 2, \cdots$, $P_t^{(0)} = I_{N_t}$ and $Q_t^{(h)} \equiv Q_t Q_{t-1} \cdots Q_{t-h+1}$ for $h = 1, 2, \cdots$, $Q_t^{(0)} = I_{N_t}$ respectively. Then denote two different accumulations of infinite historical errors as

$$L_t = \sum_{h=0}^{\infty} P_{th} V_{t-h}, \quad R_t = \sum_{h=0}^{\infty} Q_{th} V_{t-h},$$

where $P_{th} \equiv B_{1t} P_t^{(h)}$, $Q_{th} \equiv B_{2t} Q_t^{(h)}$, and B_{1t} and B_{2t} are $N_t \times N_t$ matrices. Define their respective lagged versions as

$$L_t^{\ell} = B_{1t}^{\ell} L_{t-1}, \quad R_t^{\ell} = B_{2t}^{\ell} R_{t-1},$$

where B_{1t}^{ℓ} and B_{2t}^{ℓ} are $N_t \times N_{t-1}$ matrices. Denote their respective stacked versions as $\mathbb{L} \equiv (L_1', L_2', \cdots, L_T')'$, $\mathbb{L}^{\ell} \equiv (L_1^{\ell\prime}, L_2^{\ell\prime}, \cdots, L_T^{\ell\prime})'$, and similarly, $\mathbb{R}$ and $\mathbb{R}^{\ell}$ follow the same form. Denote the block diagonal matrices $\mathbb{B}_1 = \text{blkdiag}(B_{11}, B_{12}, \cdots, B_{1T})$ and $\mathbb{B}_2 = \text{blkdiag}(B_{21}, B_{22}, \cdots, B_{2T})$. Denote a nonstochastic n-dimensional vector as $\mathbb{C} = (C_1', C_2', \cdots, C_T')'$. The following assumptions will be used in some of the subsequent lemmas:

Assumption A.1. *$\{B_{1t}\}$, $\{B_{2t}\}$, $\{B_{1t}^{\ell}\}$, $\{B_{2t}^{\ell}\}$, $\{P_t^{(h)}\}$ and $\{Q_t^{(h)}\}$ are nonstochastic matrix sequences such that B_{1t}, B_{2t}, B_{1t}^{ℓ}, B_{2t}^{ℓ}, $\sum_{h=0}^{\infty} \text{abs}(P_t^{(h)} D_{t-h})$, $\sum_{h=t}^{\infty} \text{abs}(D_h' P_h^{(h-t)})$, $\sum_{h=0}^{\infty} \text{abs}(Q_t^{(h)} D_{t-h})$, $\sum_{h=t}^{\infty} \text{abs}(D_h' Q_h^{(h-t)})$ are all UB uniformly for t. Also, $\| \sup_t \text{abs}(D_t' B_{jt} D_t) \|_{\infty} < \infty$ and $\| \sup_t \text{abs}(D_t' B_{jt}^{\ell} D_{t-1}) \|_{\infty} < \infty$, for $j = 1, 2$.*

Assumption A.2. *C_t is an arbitrary nonstochastic N_t-dimensional vector, and elements in C_t are uniformly bounded for t and N_t.*

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Lemma 1. *Under Assumption 2, for $t \geqslant s$,*

$$E(L_t R_s') = \sigma_0^2 \left(\sum_{g=0}^{\infty} P_{t,t-s+g} Q_{sg}' \right),$$

and

$$E(L_t' D_t D_s' R_s) = \sigma_0^2 \mathrm{tr} \left(\sum_{g=0}^{\infty} D_t' P_{t,t-s+g} Q_{sg}' D_s \right).$$

Proof. Since V_t's are independent across t,

$$\begin{aligned} E(L_t R_s') &= E\left[\left(\sum_{h=0}^{t-s-1} P_{th} V_{t-h} + \sum_{g=0}^{\infty} P_{t,t-s+g} V_{s-g} \right) \left(\sum_{g=0}^{\infty} Q_{sg} V_{s-g} \right)' \right] \\ &= E\left[\left(\sum_{g=0}^{\infty} P_{t,t-s+g} V_{s-g} \right) \left(\sum_{g=0}^{\infty} Q_{sg} V_{s-g} \right)' \right] = \sigma_0^2 \left(\sum_{g=0}^{\infty} P_{t,t-s+g} Q_{sg}' \right). \end{aligned}$$

Additionally,

$$E(L_t' D_t D_s' R_s) = E(R_s' D_s D_t' L_t) = \mathrm{tr}\left[D_t' E(L_t R_s') D_s \right] = \sigma_0^2 \mathrm{tr} \left(\sum_{g=0}^{\infty} D_t' P_{t,t-s+g} Q_{sg}' D_s \right)$$

□

Lemma 2. *Under Assumption 2, for any nonstochastic B_1 of size $N_t \times N_s$, B_2 of size $N_g \times N_h$, we have*

$$E\left[(V_t' B_1 V_s)(V_g' B_2 V_h) \right] = \begin{cases} (\mu_4 - 3\sigma_0^4) \sum_{i=1}^{N_t} B_{1,ii} B_{2,ii} & \text{for } t = s = g = h \\ \quad + \sigma_0^4 (\mathrm{tr} B_1 \times \mathrm{tr} B_2 + \mathrm{tr} B_1 B_2 + \mathrm{tr} B_1 B_2') & \\ \sigma_0^4 \mathrm{tr} B_1 \times \mathrm{tr} B_2 & \text{for } t = s \neq g = h \\ \sigma_0^4 \mathrm{tr}(B_1 B_2') & \text{for } t = g \neq s = h \\ \sigma_0^4 \mathrm{tr}(B_1 B_2) & \text{for } t = h \neq s = g \\ 0 & \text{otherwise} \end{cases}.$$

Proof. This lemma is an extension of Lemma 3 in Yu et al. (2008), which assumes that both B_1 and B_2 are square matrices of the same size. Denote the i-th element of V_t as $V_{t,i}$. Then

$$\begin{aligned} E\left[(V_t' B_1 V_s)(V_g' B_2 V_h) \right] &= E\left[\left(\sum_{i=1}^{N_t} \sum_{j=1}^{N_s} V_{t,i} B_{1,ij} V_{s,j} \right) \left(\sum_{k=1}^{N_g} \sum_{l=1}^{N_h} V_{g,k} B_{2,kl} V_{h,l} \right) \right] \\ &= E\left[\sum_{i=1}^{N_t} \sum_{j=1}^{N_s} \sum_{k=1}^{N_g} \sum_{l=1}^{N_h} B_{1,ij} B_{2,kl} V_{t,i} V_{s,j} V_{g,k} V_{h,l} \right] \end{aligned}$$

Since $V_{t,i}$'s are i.i.d across t and i, the equation above will not equal to 0 only when $t = s = g = h$, $t = s \neq g = h$, $t = g \neq s = h$ and $t = h \neq s = g$. For $t = s = g = h$, our conclusion is exactly the same as the corresponding case of Lemma 3 in Yu et al. (2008). Therefore, there is no need for proof. When $t = s \neq g = h$,

$$\begin{aligned} E\left[(V_t' B_1 V_s)(V_g' B_2 V_h) \right] &= E\left[\sum_{i=1}^{N_t} \sum_{j=1}^{N_t} \sum_{k=1}^{N_g} \sum_{l=1}^{N_g} B_{1,ij} B_{2,kl} V_{t,i} V_{t,j} V_{g,k} V_{g,l} \right] \\ &= E\left[\sum_{i=1}^{N_t} \sum_{k=1}^{N_g} B_{1,ii} B_{2,kk} V_{t,i}^2 V_{g,k}^2 \right] \\ &= \sigma_0^4 \mathrm{tr} B_1 \times \mathrm{tr} B_2. \end{aligned}$$

When $t = g \neq s = h$,

$$\begin{aligned}
E\left[(V_t' B_1 V_s)(V_g' B_2 V_h)\right] =& E\left[\sum_{i=1}^{N_t}\sum_{j=1}^{N_s}\sum_{k=1}^{N_t}\sum_{l=1}^{N_s} B_{1,ij}B_{2,kl}V_{t,i}V_{s,j}V_{t,k}V_{s,l}\right] \\
=& E\left[\sum_{i=1}^{N_t}\sum_{j=1}^{N_s} B_{1,ij}B_{2,ij}V_{t,i}^2 V_{s,j}^2\right] \\
=& \sigma_0^4 \mathrm{tr}(B_1 B_2').
\end{aligned}$$

When $t = h \neq s = g$,

$$\begin{aligned}
E\left[(V_t' B_1 V_s)(V_g' B_2 V_h)\right] =& E\left[\sum_{i=1}^{N_t}\sum_{j=1}^{N_s}\sum_{k=1}^{N_s}\sum_{l=1}^{N_t} B_{1,ij}B_{2,kl}V_{t,i}V_{s,j}V_{s,k}V_{t,l}\right] \\
=& E\left[\sum_{i=1}^{N_t}\sum_{j=1}^{N_s} B_{1,ij}B_{2,ji}V_{t,i}^2 V_{s,j}^2\right] \\
=& \sigma_0^4 \mathrm{tr}(B_1 B_2).
\end{aligned}$$

□

Lemma 3. *Under Assumption 2, for $t \geqslant s$,*

$$\begin{aligned}
&\mathrm{Cov}(L_t' R_t, L_s' R_s) \\
=& \sum_{g=0}^{\infty}\left[(\mu_4 - 3\sigma_0^4)\sum_{i=1}^{N_{s-g}}(P_{t,t-s+g}' Q_{t,t-s+g})_{ii}(P_{sg}' Q_{sg})_{ii}\right] + \\
& \sigma_0^4 \mathrm{tr}\left[\left(\sum_{g=0}^{\infty} P_{sg}P_{t,t-s+g}'\right)\left(\sum_{g=0}^{\infty} Q_{t,t-s+g}Q_{sg}'\right) + \left(\sum_{g=0}^{\infty} Q_{sg}P_{t,t-s+g}'\right)\left(\sum_{g=0}^{\infty} Q_{t,t-s+g}P_{sg}'\right)\right].
\end{aligned}$$

Proof. Using Lemma 2, we have

$$\begin{aligned}
E(L_t' R_t L_s' R_s) =& E\left[\left(\sum_{h=0}^{\infty} P_{th}V_{t-h}\right)'\left(\sum_{h=0}^{\infty} Q_{th}V_{t-h}\right)\left(\sum_{g=0}^{\infty} P_{sg}V_{s-g}\right)'\left(\sum_{g=0}^{\infty} Q_{sg}V_{s-g}\right)\right] \\
=& E\left[\left(\sum_{h=0}^{t-s-1} P_{th}V_{t-h} + \sum_{g=0}^{\infty} P_{t,t-s+g}V_{s-g}\right)'\left(\sum_{h=0}^{t-s-1} Q_{th}V_{t-h} + \sum_{g=0}^{\infty} Q_{t,t-s+g}V_{s-g}\right)\right. \\
& \left.\left(\sum_{g=0}^{\infty} P_{sg}V_{s-g}\right)'\left(\sum_{g=0}^{\infty} Q_{sg}V_{s-g}\right)\right] \\
=& E\left[\left(\sum_{h=0}^{t-s-1} P_{th}V_{t-h}\right)'\left(\sum_{h=0}^{t-s-1} Q_{th}V_{t-h}\right)\left(\sum_{g=0}^{\infty} P_{sg}V_{s-g}\right)'\left(\sum_{g=0}^{\infty} Q_{sg}V_{s-g}\right)\right] + \\
& E\left[\left(\sum_{g=0}^{\infty} P_{t,t-s+g}V_{s-g}\right)'\left(\sum_{g=0}^{\infty} Q_{t,t-s+g}V_{s-g}\right)\left(\sum_{g=0}^{\infty} P_{sg}V_{s-g}\right)'\left(\sum_{g=0}^{\infty} Q_{sg}V_{s-g}\right)\right] \\
\equiv& E_1 + E_2,
\end{aligned}$$

where the third equality holds because V_t's are independent across t. Then we have

$$\begin{aligned}
E_1 &= E\left[\left(\sum_{h_1=0}^{t-s-1} P_{th}V_{t-h}\right)'\left(\sum_{h_2=0}^{t-s-1} Q_{th}V_{t-h}\right)\left(\sum_{g_1=0}^{\infty} P_{sg}V_{s-g}\right)'\left(\sum_{g_2=0}^{\infty} Q_{sg}V_{s-g}\right)\right] \\
&= E\left[\left(\sum_{h_1=0}^{t-s-1}\sum_{h_2=0}^{t-s-1} V'_{t-h_1}P'_{th_1}Q_{th_2}V_{t-h_2}\right)\left(\sum_{g_1=0}^{\infty}\sum_{g_2=0}^{\infty} V'_{s-g_1}P'_{sg_1}Q_{sg_2}V_{s-g_2}\right)\right] \\
&= \sigma_0^4\left[\sum_{h=0}^{t-s-1}\mathrm{tr}\left(P'_{th}Q_{th}\right)\right]\left[\sum_{g=0}^{\infty}\mathrm{tr}\left(P'_{sg}Q_{sg}\right)\right],
\end{aligned}$$

where the last equality holds due to the second case in Lemma 2.

$$\begin{aligned}
E_2 &= E\left[\left(\sum_{g_1=0}^{\infty} P_{t,t-s+g}V_{s-g}\right)'\left(\sum_{g_2=0}^{\infty} Q_{t,t-s+g}V_{s-g}\right)\left(\sum_{g_3=0}^{\infty} P_{sg}V_{s-g}\right)'\left(\sum_{g_4=0}^{\infty} Q_{sg}V_{s-g}\right)\right] \\
&= E\left[\sum_{g_1=0}^{\infty}\sum_{g_2=0}^{\infty}\sum_{g_3=0}^{\infty}\sum_{g_4=0}^{\infty}\left(V'_{s-g_1}P'_{t,t-s+g_1}Q_{t,t-s+g_2}V_{s-g_2}\right)\left(V'_{s-g_3}P'_{sg_3}Q_{sg_4}V_{s-g_4}\right)\right] \\
&\equiv E_{21}+E_{22}+E_{23}+E_{24},
\end{aligned}$$

where E_{21}, E_{22}, E_{23} and E_{24} are the four nonzero cases in Lemma 2. Specifically, E_{21} corresponds to the case where $g_1=g_2=g_3=g_4$, so

$$\begin{aligned}
E_{21} = \sum_{g=0}^{\infty}\Bigg[&(\mu_4-3\sigma_0^4)\sum_{i=1}^{N_{s-g}}(P'_{t,t-s+g}Q_{t,t-s+g})_{ii}(P'_{sg}Q_{sg})_{ii} + \sigma_0^4\mathrm{tr}\left(P'_{t,t-s+g}Q_{t,t-s+g}\right)\mathrm{tr}\left(P'_{sg}Q_{sg}\right) \\
&+ \sigma_0^4\mathrm{tr}\left(P'_{t,t-s+g}Q_{t,t-s+g}P'_{sg}Q_{sg}\right) + \sigma_0^4\mathrm{tr}\left(P'_{t,t-s+g}Q_{t,t-s+g}Q'_{sg}P_{sg}\right)\Bigg],
\end{aligned}$$

E_{22} corresponds to the case where $g_1=g_2\neq g_3=g_4$,

$$E_{22} = \sigma_0^4\left[\sum_{g=0}^{\infty}\mathrm{tr}\left(P'_{t,t-s+g}Q_{t,t-s+g}\right)\right]\left[\sum_{g=0}^{\infty}\mathrm{tr}\left(P'_{sg}Q_{sg}\right)\right] - \sigma_0^4\sum_{g=0}^{\infty}\mathrm{tr}\left(P'_{t,t-s+g}Q_{t,t-s+g}\right)\mathrm{tr}\left(P'_{sg}Q_{sg}\right).$$

E_{23} corresponds to the case where $g_1=g_3\neq g_2=g_4$,

$$\begin{aligned}
E_{23} &= \sigma_0^4\left[\sum_{g_1=0}^{\infty}\sum_{g_2=0}^{\infty}\mathrm{tr}\left(P'_{t,t-s+g_1}Q_{t,t-s+g_2}Q'_{sg_2}P_{sg_1}\right)\right] - \sigma_0^4\sum_{g=0}^{\infty}\mathrm{tr}\left(P'_{t,t-s+g}Q_{t,t-s+g}Q'_{sg}P_{sg}\right) \\
&= \sigma_0^4\mathrm{tr}\left[\left(\sum_{g=0}^{\infty}P_{sg}P'_{t,t-s+g}\right)\left(\sum_{g=0}^{\infty}Q_{t,t-s+g}Q'_{sg}\right)\right] - \sigma_0^4\sum_{g=0}^{\infty}\mathrm{tr}\left(P'_{t,t-s+g}Q_{t,t-s+g}Q'_{sg}P_{sg}\right).
\end{aligned}$$

In addition, E_{24} corresponds to the case where $g_1=g_4\neq g_2=g_3$,

$$\begin{aligned}
E_{24} &= \sigma_0^4\left[\sum_{g_1=0}^{\infty}\sum_{g_2=0}^{\infty}\mathrm{tr}(P'_{t,t-s+g_1}Q_{t,t-s+g_2}P'_{sg_2}Q_{sg_1})\right] - \sigma_0^4\sum_{g=0}^{\infty}\mathrm{tr}(P'_{t,t-s+g}Q_{t,t-s+g}P'_{sg}Q_{sg}) \\
&= \sigma_0^4\left[\mathrm{tr}\left(\sum_{g=0}^{\infty}Q_{sg}P'_{t,t-s+g}\right)\left(\sum_{g=0}^{\infty}Q_{t,t-s+g}P'_{sg}\right)\right] - \sigma_0^4\sum_{g=0}^{\infty}\mathrm{tr}(P'_{t,t-s+g}Q_{t,t-s+g}P'_{sg}Q_{sg}).
\end{aligned}$$

Therefore, we have

$$
\begin{aligned}
&\operatorname{Cov}(L_t'R_t, L_s'R_s) = E(L_t'R_tL_s'R_s) - E(L_t'R_t)E(L_s'R_s)\\
=&E_1 + E_{21} + E_{22} + E_{23} + E_{24} - E(L_t'R_t)E(L_s'R_s)\\
=&\sigma_0^4\left[\sum_{h=0}^{t-s-1}\operatorname{tr}\left(P_{th}'Q_{th}\right)\right]\left[\sum_{g=0}^{\infty}\operatorname{tr}\left(P_{sg}'Q_{sg}\right)\right] + \sum_{g=0}^{\infty}\left[(\mu_4 - 3\sigma_0^4)\sum_{i=1}^{N_{s-g}}(P_{t,t-s+g}'Q_{t,t-s+g})_{ii}(P_{sg}'Q_{sg})_{ii}\right] +\\
&\sigma_0^4\left[\sum_{g=0}^{\infty}\operatorname{tr}\left(P_{t,t-s+g}'Q_{t,t-s+g}\right)\right]\left[\sum_{g=0}^{\infty}\operatorname{tr}\left(P_{sg}'Q_{sg}\right)\right] + \sigma_0^4\operatorname{tr}\left[\left(\sum_{g=0}^{\infty}P_{sg}P_{t,t-s+g}'\right)\left(\sum_{g=0}^{\infty}Q_{t,t-s+g}Q_{sg}'\right)\right] +\\
&\sigma_0^4\left[\operatorname{tr}\left(\sum_{g=0}^{\infty}Q_{sg}P_{t,t-s+g}'\right)\left(\sum_{g=0}^{\infty}Q_{t,t-s+g}P_{sg}'\right)\right] - \sigma_0^4\left[\sum_{g=0}^{\infty}\operatorname{tr}\left(P_{tg}'Q_{tg}\right)\right]\left[\sum_{g=0}^{\infty}\operatorname{tr}\left(P_{sg}'Q_{sg}\right)\right]\\
=&\sum_{g=0}^{\infty}\left[(\mu_4 - 3\sigma_0^4)\sum_{i=1}^{N_{s-g}}(P_{t,t-s+g}'Q_{t,t-s+g})_{ii}(P_{sg}'Q_{sg})_{ii}\right] +\\
&\sigma_0^4\operatorname{tr}\left[\left(\sum_{g=0}^{\infty}P_{sg}P_{t,t-s+g}'\right)\left(\sum_{g=0}^{\infty}Q_{t,t-s+g}Q_{sg}'\right) + \left(\sum_{g=0}^{\infty}Q_{sg}P_{t,t-s+g}'\right)\left(\sum_{g=0}^{\infty}Q_{t,t-s+g}P_{sg}'\right)\right],
\end{aligned}
$$

where $E(L_t'R_t)$ and $E(L_s'R_s)$ are calculated according to Lemma 1. □

Lemma 4. *Denote $B_t \equiv B_{1t}B_{2t}$ and $B_t^\ell \equiv B_{1t}^\ell B_{2,t-1}$. Under Assumption A.1, the following facts hold:*

(1) $\|\sup_t \operatorname{abs}(D_t'B_tD_t)\|_\infty < \infty$ *and* $\|\sup_t \operatorname{abs}(D_t'B_t^\ell D_{t-1})\|_\infty < \infty$.

(2) $\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'B_hP_h^{(h-t)}\right)$ *and* $\sum_{h=t}^{\infty}\operatorname{abs}\left(B_t^\ell P_{t-1}^{(h)}D_{t-1-h}\right)$ *are UB uniformly for t.*

(3) $\sum_{h=t}^{\infty}\operatorname{abs}\left(Q_h^{(h-t)\prime}B_{2h}'B_{1h}P_h^{(h-t)}\right)$ *and* $\sum_{h=0}^{\infty}\operatorname{abs}\left(B_t^\ell P_{t-1}^{(h)}P_{t-1}^{(h)\prime}B_t^{\ell\prime}\right)$ *are UB uniformly for t.*

(4) $\sum_{h=0}^{\infty}\operatorname{abs}(P_t^{(h)})\operatorname{abs}(Q_t^{(h)\prime})$ *are UB uniformly for t.*

Proof. **Proof of (1):** According to the definition of $\sup_t \operatorname{abs}(D_t'B_tD_t)$, we have

$$
\begin{aligned}
\|\sup_t \operatorname{abs}(D_t'B_tD_t)\|_\infty &= \sup_{1\leqslant i\leqslant N}\sum_{j=1}^{N}\sup_t\left|(D_t'B_tD_t)_{ij}\right|\\
&\leqslant \sup_{1\leqslant i\leqslant N}\sum_{j=1}^{N}\sup_t\sum_{k=1}^{N}\left|(D_t'B_{1t}D_t)_{ik}\right|\left|(D_t'B_{2t}D_t)_{kj}\right|\\
&\leqslant \sup_{1\leqslant i\leqslant N}\sum_{j=1}^{N}\sum_{k=1}^{N}\left(\sup_t\left|(D_t'B_{1t}D_t)_{ik}\right|\right)\left(\sup_t\left|(D_t'B_{2t}D_t)_{kj}\right|\right)\\
&\leqslant \|\sup_t \operatorname{abs}(D_t'B_{1t}D_t)\|_\infty\|\sup_t \operatorname{abs}(D_t'B_{2t}D_t)\|_\infty\\
&< \infty.
\end{aligned}
$$

Similarly, we can prove $\|\sup_t \operatorname{abs}(D_t'B_t^\ell D_{t-1})\|_\infty < \infty$.

Proof of (2):

Since we assume that B_{1t} and B_{2t}, are UB uniformly for t, B_t is UB because $\|B_t\|_p \leqslant \|B_{1t}\|_p\|B_{2t}\|_p$ holds for both $p = 1$ and $p = \infty$. Moreover, $D_t'B_tD_t$ is also UB uniformly for t, because $D_t'B_tD_t$ only adds some

zero rows and columns to B_t, which means that there exists a constant $c_B > 0$, such that

$$\sup_{N,T}\sup_{t\leqslant T}\max\left(\left\|\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'P_h^{(h-t)}\right)\right\|_1,\left\|\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'P_h^{(h-t)}\right)\right\|_\infty,\left\|D_t'B_tD_t\right\|_1,\left\|D_t'B_tD_t\right\|_\infty\right)\leqslant c_B.$$

Thus, we have

$$\begin{aligned}
\left\|\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'B_hP_h^{(h-t)}\right)\right\|_1 &\leqslant \left\|\sum_{h=t}^{\infty}\operatorname{abs}(D_h'B_hD_h)\operatorname{abs}\left(D_h'P_h^{(h-t)}\right)\right\|_1\\
&= \sup_{1\leqslant j\leqslant N}\sum_{i=1}^{N}\sum_{h=t}^{\infty}\sum_{k=1}^{N}\left|\left(D_h'B_hD_h\right)_{ik}\right|\left|\left(D_h'P_h^{(h-t)}\right)_{kj}\right|\\
&\leqslant \sup_{1\leqslant j\leqslant N}\sum_{h=t}^{\infty}\sum_{k=1}^{N}\left\|D_h'B_hD_h\right\|_1\left|\left(D_h'P_h^{(h-t)}\right)_{kj}\right|\\
&\leqslant c_B\sup_{1\leqslant j\leqslant N}\sum_{k=1}^{N}\sum_{h=t}^{\infty}\left|\left(D_h'P_h^{(h-t)}\right)_{kj}\right|\\
&= c_B\left\|\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'P_h^{(h-t)}\right)\right\|_1\\
&< \infty,
\end{aligned}$$

and

$$\begin{aligned}
\left\|\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'B_hP_h^{(h-t)}\right)\right\|_\infty &\leqslant \left\|\sum_{h=t}^{\infty}\operatorname{abs}(D_h'B_hD_h)\operatorname{abs}\left(D_h'P_h^{(h-t)}\right)\right\|_\infty\\
&= \sup_{1\leqslant i\leqslant N}\sum_{j=1}^{N_t}\sum_{h=t}^{\infty}\sum_{k=1}^{N}\left|\left(D_h'B_hD_h\right)_{ik}\right|\left|\left(D_h'P_h^{(h-t)}\right)_{kj}\right|\\
&\leqslant \sup_{1\leqslant i\leqslant N}\sum_{h=t}^{\infty}\sum_{k=1}^{N}\sup_h\left|\left(D_h'B_hD_h\right)_{ik}\right|\sum_{j=1}^{N_t}\left|\left(D_h'P_h^{(h-t)}\right)_{kj}\right|\\
&\leqslant \sup_{1\leqslant i\leqslant N}\sum_{k=1}^{N}\sup_h\left|\left(D_h'B_hD_h\right)_{ik}\right|\sum_{j=1}^{N_t}\sum_{h=t}^{\infty}\left|\left(D_h'P_h^{(h-t)}\right)_{kj}\right|\\
&\leqslant \|\sup_t\operatorname{abs}(D_t'B_{1t}D_t)\|_\infty\left\|\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'P_h^{(h-t)}\right)\right\|_\infty\\
&< \infty.
\end{aligned}$$

Therefore, $\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'B_hP_h^{(h-t)}\right)$ is UB uniformly for t. Applying the same method, we can prove that $\sum_{h=t}^{\infty}\operatorname{abs}\left(B_t^{\ell}P_{t-1}^{(h)}D_{t-1-h}\right)$ is UB uniformly for t.

Proof of (3):

According to (2), it is obvious that $\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'B_{1h}P_h^{(h-t)}\right)$ and $\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'B_{2h}Q_h^{(h-t)}\right)$ are also

UB uniformly for t. Hence,

$$\begin{aligned}\left\|\sum_{h=t}^{\infty}\operatorname{abs}\left(Q_h^{(h-t)\prime}B_{2h}'B_{1h}P_h^{(h-t)}\right)\right\|_p &\leqslant \left\|\left[\sum_{h=t}^{\infty}\operatorname{abs}\left(Q_h^{(h-t)\prime}B_{2h}'D_h\right)\right]\left[\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'B_{1h}P_h^{(h-t)}\right)\right]\right\|_p \\ &\leqslant \left\|\sum_{h=t}^{\infty}\operatorname{abs}\left(Q_h^{(h-t)\prime}B_{2h}'D_h\right)\right\|_p\left\|\sum_{h=t}^{\infty}\operatorname{abs}\left(D_h'B_{1h}P_h^{(h-t)}\right)\right\|_p \\ &< \infty.\end{aligned}$$

The above inequality holds for both $p=1$ and $p=\infty$. Thus $\sum_{h=t}^{\infty}\operatorname{abs}\left(Q_h^{(h-t)\prime}B_{2h}'B_{1h}P_h^{(h-t)}\right)$ is also UB uniformly for t. Similarly, we can prove that $\sum_{h=0}^{\infty}\operatorname{abs}\left(B_t^{\ell}P_{t-1}^{(h)}P_{t-1}^{(h)\prime}B_t^{\ell\prime}\right)$ is UB uniformly for t.

Proof of (4):

Since $\sum_{h=0}^{\infty}\operatorname{abs}(P_t^{(h)}D_{t-h})$ and $\sum_{h=0}^{\infty}\operatorname{abs}(Q_t^{(h)}D_{t-h})$ are also UB uniformly for t, we similarly have

$$\begin{aligned}\left\|\sum_{h=t}^{\infty}\operatorname{abs}\left(P_t^{(h)}\right)\operatorname{abs}\left(Q_t^{(h)\prime}\right)\right\|_p &\leqslant \left\|\left[\sum_{h=t}^{\infty}\operatorname{abs}\left(P_t^{(h)}D_{t-h}\right)\right]\left[\sum_{h=t}^{\infty}\operatorname{abs}\left(Q_t^{(h)}D_{t-h}\right)'\right]\right\|_p \\ &\leqslant \left\|\sum_{h=0}^{\infty}\operatorname{abs}\left(P_t^{(h)}D_{t-h}\right)\right\|_p\left\|\sum_{h=0}^{\infty}\operatorname{abs}\left(Q_t^{(h)}D_{t-h}\right)'\right\|_p \\ &< \infty.\end{aligned}$$

□

Lemma 5. *Under Assumptions 2, 7 and A.1,*

$$\operatorname{Var}\left(\sum_{t=1}^{T}L_t'R_t\right)=\sum_{t=1}^{T}\sum_{s=1}^{T}\operatorname{Cov}(L_t'R_t,L_s'R_s)=O(NT)$$

Proof.

$$\begin{aligned}\operatorname{Var}\left(\sum_{t=1}^{T}L_t'R_t\right) =& \sum_{t=1}^{T}\operatorname{Cov}(L_t'R_t,L_t'R_t)+2\sum_{t=2}^{T}\sum_{s=1}^{t-1}\operatorname{Cov}(L_t'R_t,L_s'R_s) \\ \leqslant& 2\sum_{t=1}^{T}\sum_{s=1}^{t}\operatorname{Cov}(L_t'R_t,L_s'R_s) \\ =& 2\sigma_0^4\sum_{t=1}^{T}\sum_{s=1}^{t}\operatorname{tr}\left[\left(\sum_{g=0}^{\infty}P_{sg}P_{t,t-s+g}'\right)\left(\sum_{g=0}^{\infty}Q_{t,t-s+g}Q_{sg}'\right)\right]+ \\ & 2\sigma_0^4\sum_{t=1}^{T}\sum_{s=1}^{t}\operatorname{tr}\left[\left(\sum_{g=0}^{\infty}Q_{sg}P_{t,t-s+g}'\right)\left(\sum_{g=0}^{\infty}Q_{t,t-s+g}P_{sg}'\right)\right]+ \\ & 2\sum_{g=0}^{\infty}\left[(\mu_4-3\sigma_0^4)\sum_{t=1}^{T}\sum_{s=1}^{t}\sum_{i=1}^{N_{s-g}}(P_{t,t-s+g}'Q_{t,t-s+g})_{ii}(P_{sg}'Q_{sg})_{ii}\right].\end{aligned} \tag{A.1}$$

The last equality holds from Lemma 3. Denote $\mathcal{A}_{t,PP} \equiv \sum_{g=0}^{\infty} P_t^{(g)} P_t^{(g)\prime}$ and $\mathcal{A}_{t,QQ} \equiv \sum_{g=0}^{\infty} Q_t^{(g)} Q_t^{(g)\prime}$, and they are UB from Lemma 4. We can derive the order of the first term of Equation (A.1) as

$$
\begin{aligned}
&\left| 2\sigma_0^4 \sum_{t=1}^{T} \sum_{s=1}^{t} \operatorname{tr} \left[\left(\sum_{g=0}^{\infty} P_{sg} P'_{t,t-s+g} \right) \left(\sum_{g=0}^{\infty} Q_{t,t-s+g} Q'_{sg} \right) \right] \right| \\
=&\left| 2\sigma_0^4 \sum_{t=1}^{T} \sum_{s=1}^{t} \operatorname{tr} \left[B_{1s} \left(\sum_{g=0}^{\infty} P_s^{(g)} P_s^{(g)\prime} \right) P_t^{(t-s)\prime} B'_{1t} B_{2t} Q_t^{(t-s)} \left(\sum_{g=0}^{\infty} Q_s^{(g)} Q_s^{(g)\prime} \right) B'_{2s} \right] \right| \\
=&\left| 2\sigma_0^4 \sum_{t=1}^{T} \sum_{s=1}^{t} \operatorname{tr} \left[B_{1s} \mathcal{A}_{s,PP} P_t^{(t-s)\prime} B'_{1t} B_{2t} Q_t^{(t-s)} \mathcal{A}_{s,QQ} B'_{2s} \right] \right| \\
=&\left| 2\sigma_0^4 \sum_{s=1}^{T} \sum_{t=s}^{T} \operatorname{tr} \left[P_t^{(t-s)\prime} B'_{1t} B_{2t} Q_t^{(t-s)} \mathcal{A}_{s,QQ} B'_{2s} B_{1s} \mathcal{A}_{s,PP} \right] \right| \\
\leqslant& 2\sigma_0^4 \sum_{s=1}^{T} \operatorname{tr} \left[\sum_{t=s}^{T} \operatorname{abs} \left(P_t^{(t-s)\prime} B'_{1t} B_{2t} Q_t^{(t-s)} \right) \operatorname{abs} \left(\mathcal{A}_{s,QQ} B'_{2s} B_{1s} \mathcal{A}_{s,PP} \right) \right] \\
\leqslant& 2\sigma_0^4 \sum_{s=1}^{T} \operatorname{tr} \left[\sum_{t=s}^{\infty} \operatorname{abs} \left(P_t^{(t-s)\prime} B'_{1t} B_{2t} Q_t^{(t-s)} \right) \operatorname{abs} \left(\mathcal{A}_{s,QQ} B'_{2s} B_{1s} \mathcal{A}_{s,PP} \right) \right] \\
=& O(NT).
\end{aligned}
$$

The first equality holds because $P_{th} = B_{1t} P_t^{(h)}$ and $P_t^{(t-s+g)} = P_t^{(t-s)} P_s^{(g)}$. By (3) in Lemma 4, we know that $\sum_{t=s}^{\infty} \operatorname{abs} \left(P_t^{(t-s)\prime} B'_{1t} B_{2t} Q_t^{(t-s)} \right)$ is UB, then the term inside the trace operator of the second-to-last equation is an $N_s \times N_s$ UB matrix, so the last equality holds. Since P_{th} and Q_{th} are arbitrary matrices, under the same assumptions, we can similarly derive the order of the second term in equation (A.1) as

$$
2\sigma_0^4 \sum_{t=1}^{T} \sum_{s=1}^{t} \operatorname{tr} \left[\left(\sum_{g=0}^{\infty} Q_{sg} P'_{t,t-s+g} \right) \left(\sum_{g=0}^{\infty} Q_{t,t-s+g} P'_{sg} \right) \right] = O(NT).
$$

Since $\sum_{g=0}^{\infty} \operatorname{abs}(P'_{sg} Q_{sg})$ is UB by Lemma 4, there exists a finite constant c_{PQ} such that

$$
\sup_N \sup_s \sup_{i=1,\cdots,N_s} \sum_{g=0}^{\infty} \left| (P'_{sg} Q_{sg})_{ii} \right| \leqslant c_{PQ},
$$

therefore we have

$$
\begin{aligned}
&\left| \sum_{g=0}^{\infty} \sum_{t=1}^{T} \sum_{s=1}^{t} \sum_{i=1}^{N_{s-g}} (P'_{t,t-s+g} Q_{t,t-s+g})_{ii} (P'_{sg} Q_{sg})_{ii} \right| \\
\leqslant& c_{PQ} \sum_{t=1}^{T} \sum_{s=1}^{t} \sum_{g=0}^{\infty} \sum_{i=1}^{N_{s-g}} \left| (P'_{t,t-s+g} Q_{t,t-s+g})_{ii} \right| \\
\leqslant& c_{PQ} \sum_{t=1}^{T} \sum_{s=1}^{t} \sum_{g=0}^{\infty} \operatorname{tr} \left[\operatorname{abs} \left(P_s^{(g)\prime} P_t^{(t-s)\prime} B'_{1t} B_{2t} Q_t^{(t-s)} Q_s^{(g)} \right) \right] \\
\leqslant& c_{PQ} \sum_{s=1}^{T} \sum_{t=s}^{T} \operatorname{tr} \left[\sum_{g=0}^{\infty} \operatorname{abs} \left(Q_s^{(g)} \right) \operatorname{abs} \left(P_s^{(g)\prime} \right) \operatorname{abs} \left(P_t^{(t-s)\prime} B'_{1t} B_{2t} Q_t^{(t-s)} \right) \right] \\
\leqslant& c_{PQ} \sum_{s=1}^{T} \operatorname{tr} \left[\sum_{g=0}^{\infty} \operatorname{abs} \left(Q_s^{(g)} \right) \operatorname{abs} \left(P_s^{(g)\prime} \right) \sum_{t=s}^{\infty} \operatorname{abs} \left(P_t^{(t-s)\prime} B'_{1t} B_{2t} Q_t^{(t-s)} \right) \right] \\
=& O(NT).
\end{aligned}
$$

The last term in equation (A.1) is also $O(NT)$, so $\text{Var}\left(\sum_{t=1}^{T} L_t' R_t\right) = O(NT)$. □

Lemma 6. $\mathbb{L}'\mathbb{P}\mathbb{R} = \mathbb{L}'\mathbb{D}(\mathbb{D}'\mathbb{D})^{-1}\mathbb{D}'\mathbb{R} = (\sum_{h=0}^{\infty} \ddot{P}_h V_{T-h})'(\sum_{h=0}^{\infty} \ddot{Q}_h V_{T-h})$, *where*

$$\ddot{P}_h = \begin{cases} \mathcal{T}^{-\frac{1}{2}} \sum_{t=T-h}^{T} D_t' P_{t,t+h-T} & \text{for } h < T \\ \mathcal{T}^{-\frac{1}{2}} \sum_{t=1}^{T} D_t' P_{t,t+h-T} & \text{for } h \geqslant T \end{cases},$$

$\ddot{Q}_h$ *has the same pattern, and* $\mathcal{T} \equiv \text{diag}(T_1, T_2, \cdots, T_N)$.

Proof. According to the definitions of $\mathbb{L}$, $\mathbb{R}$ and $\mathbb{D}$, we have

$$\mathbb{D}'\mathbb{D} = \sum_{t=1}^{T} D_t' D_t = \sum_{t=1}^{T} \text{diag}(d_{1t}, d_{2t}, \cdots, d_{N,t}) = \text{diag}(T_1, T_2, \cdots, T_N) = \mathcal{T}.$$

Then

$$\mathbb{L}'\mathbb{D}(\mathbb{D}'\mathbb{D})^{-1}\mathbb{D}'\mathbb{R} = \left(\mathcal{T}^{-\frac{1}{2}} \sum_{t=1}^{T} D_t' L_t\right)' \left(\mathcal{T}^{-\frac{1}{2}} \sum_{t=1}^{T} D_t' R_t\right).$$

Since

$$\begin{aligned} L_1 =& P_{10}V_1 + P_{11}V_0 + P_{12}V_{-1} + \cdots, \\ L_2 =& P_{20}V_2 + P_{21}V_1 + P_{22}V_0 + P_{23}V_{-1} + \cdots, \\ & \vdots \\ L_T =& P_{T0}V_T \cdots + P_{T,T-1}V_1 + P_{T,T}V_0 + P_{T,T+1}V_{-1} + \cdots, \end{aligned}$$

we have

$$\begin{aligned} \sum_{t=1}^{T} D_t' L_t =& D_T' P_{T0} V_T + (D_{T-1}' P_{T-1,0} + D_T' P_{T1}) V_{T-1} + \cdots \\ & + (D_1' P_{11} + D_2' P_{22} + \cdots + D_T' P_{TT}) V_0 + (D_1' P_{12} + D_2' P_{23} + \cdots + D_T' P_{T,T+1}) V_{-1} + \cdots. \end{aligned}$$

Therefore

$$\mathcal{T}^{-\frac{1}{2}} \sum_{t=1}^{T} D_t' L_t = \mathcal{T}^{-\frac{1}{2}} \sum_{t=1}^{T} D_t' \sum_{h=0}^{\infty} P_{th} V_{t-h} = \sum_{h=0}^{\infty} \ddot{P}_h V_{T-h},$$

where

$$\ddot{P}_h = \begin{cases} \mathcal{T}^{-\frac{1}{2}} \sum_{t=T-h}^{T} D_t' P_{t,t+h-T} & \text{for } h < T \\ \mathcal{T}^{-\frac{1}{2}} \sum_{t=1}^{T} D_t' P_{t,t+h-T} & \text{for } h \geqslant T \end{cases}.$$

Similarly, $\ddot{Q}_h$ has the same pattern. □

Lemma 7. *Under Assumptions 1, 2, 7 and A.1,*

$$\frac{1}{NT}\mathbb{L}'\mathbb{R} - \frac{1}{NT} E\left(\mathbb{L}'\mathbb{R}\right) = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.2}$$

$$\frac{1}{NT}\mathbb{L}'\mathbb{P}\mathbb{R} - \frac{1}{NT} E\left(\mathbb{L}'\mathbb{P}\mathbb{R}\right) = O_p\left(\frac{1}{\sqrt{NT^2}}\right), \tag{A.3}$$

$$\frac{1}{NT}\mathbb{L}'\mathbb{Q}\mathbb{R} - \frac{1}{NT} E\left(\mathbb{L}'\mathbb{Q}\mathbb{R}\right) = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.4}$$

where $\frac{1}{NT} E\left(\mathbb{L}'\mathbb{R}\right) = O(1)$ *and* $\frac{1}{NT} E\left(\mathbb{L}'\mathbb{P}\mathbb{R}\right) = O(\frac{1}{T})$.

Proof. **Proof of equation (A.2)**

According to Lemmas 1 and 4, we have

$$\frac{1}{NT}E\left(\mathbb{L}'\mathbb{R}\right)=\frac{1}{NT}E\left(\sum_{t=1}^{T}L_t'R_t\right)=\frac{1}{NT}\sigma_0^2\sum_{t=1}^{T}\mathrm{tr}\left(\sum_{h=0}^{\infty}P_{th}Q_{th}'\right)=O(1)$$

and according to Lemma 5, $\mathrm{Var}(\mathbb{L}'\mathbb{R})=\mathrm{Var}(\sum_{t=1}^{T}L_t'R_t)=O(NT)$. Using Chebyshev's inequality, we can prove equation (A.2).

Proof of equation (A.3)

According to Lemmas 6 and 3, we have

$$\begin{aligned}\mathrm{Var}\left(\mathbb{L}'\mathbb{P}\mathbb{R}\right)=&\mathrm{Cov}\left[\left(\sum_{h=0}^{\infty}\ddot{P}_hV_{T-h}\right)'\left(\sum_{h=0}^{\infty}\ddot{Q}_hV_{T-h}\right),\left(\sum_{h=0}^{\infty}\ddot{P}_hV_{T-h}\right)'\left(\sum_{h=0}^{\infty}\ddot{Q}_hV_{T-h}\right)\right]\\=&\sigma_0^4\mathrm{tr}\left[\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right)\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right)+\left(\sum_{g=0}^{\infty}\ddot{P}_g\ddot{P}_g'\right)\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{Q}_g'\right)\right]+\\&\sum_{g=0}^{\infty}\left[(\mu_4-3\sigma_0^4)\sum_{i=1}^{N_{T-g}}\left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\right],\end{aligned}\tag{A.5}$$

where $\ddot{P}_g$ and $\ddot{Q}_g$ are defined as in Lemma 6. $\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'=\sum_{g=0}^{T-1}\ddot{Q}_g\ddot{P}_g'+\sum_{g=T}^{\infty}\ddot{Q}_g\ddot{P}_g'$, $p=1$ and $p=\infty$,

$$\begin{aligned}\left\|\sum_{g=0}^{T-1}\ddot{Q}_g\ddot{P}_g'\right\|_p=&\left\|\mathcal{T}^{-\frac{1}{2}}\sum_{g=0}^{T-1}\left(\sum_{t=T-g}^{T}D_t'Q_{t,t+g-T}\right)\left(\sum_{t=T-g}^{T}D_t'P_{t,t+g-T}\right)'\mathcal{T}^{-\frac{1}{2}}\right\|_p\\\leqslant&\|\mathcal{T}^{-\frac{1}{2}}\|_p^2\sum_{g=0}^{T-1}\left\|\sum_{t=T-g}^{T}D_t'Q_{t,t+g-T}\right\|_p\left\|\left(\sum_{t=T-g}^{T}D_t'P_{t,t+g-T}\right)'\right\|_p\\\leqslant&\frac{1}{c_TT}\sum_{g=0}^{T-1}\left\|\sum_{t=T-g}^{T}D_t'B_{2t}Q_t^{(t+g-T)}\right\|_p\left\|\left(\sum_{t=T-g}^{T}D_t'B_{1t}P_t^{(t+g-T)}\right)'\right\|_p\\\leqslant&\frac{1}{c_TT}\sum_{g=0}^{T-1}\left\|\sum_{t=T-g}^{\infty}\mathrm{abs}\left(D_t'B_{2t}Q_t^{(t+g-T)}\right)\right\|_p\left\|\sum_{t=T-g}^{\infty}\mathrm{abs}\left(D_t'B_{1t}P_t^{(t+g-T)}\right)'\right\|_p\\=&O(1),\end{aligned}$$

where the second inequality holds because $T_i\geqslant c_TT$ in Assumption 1 and the last equality follows from Lemma 4. Likewise, for $p=1$ and $p=\infty$,

$$\begin{aligned}\left\|\sum_{g=T}^{\infty}\ddot{Q}_g\ddot{P}_g'\right\|_p=&\left\|\mathcal{T}^{-\frac{1}{2}}\sum_{g=T}^{\infty}\left(\sum_{t=1}^{T}D_t'Q_{t,t+g-T}\right)\left(\sum_{t=1}^{T}D_t'P_{t,t+g-T}\right)'\mathcal{T}^{-\frac{1}{2}}\right\|_p\\=&\left\|\mathcal{T}^{-\frac{1}{2}}\sum_{g=T}^{\infty}\left(\sum_{t=1}^{T}D_t'B_{2t}Q_t^{(t)}\right)Q_0^{(g-T)}P_0^{(g-T)\prime}\left(\sum_{t=1}^{T}D_t'B_{1t}P_t^{(t)}\right)'\mathcal{T}^{-\frac{1}{2}}\right\|_p\\\leqslant&\|\mathcal{T}^{-\frac{1}{2}}\|_p^2\left\|\sum_{t=1}^{T}D_t'B_{2t}Q_t^{(t)}\right\|_p\left\|\sum_{g=T}^{\infty}Q_0^{(g-T)}P_0^{(g-T)\prime}\right\|_p\left\|\left(\sum_{t=1}^{T}D_t'B_{1t}P_t^{(t)}\right)'\right\|_p\\=&O\left(\frac{1}{T}\right),\end{aligned}$$

where the second equality holds because $P_{t,t+g-T} = B_{1t}P_t^{(t)}P_0^{(g-T)}$ and $Q_{t,t+g-T} = B_{2t}Q_t^{(t)}Q_0^{(g-T)}$. The last quality holds because of Assumption 1 and Lemma 4. Therefore,

$$\left\|\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right\|_p \leqslant \left\|\sum_{g=0}^{T-1}\ddot{Q}_g\ddot{P}_g'\right\|_p + \left\|\sum_{g=T}^{\infty}\ddot{Q}_g\ddot{P}_g'\right\|_p = O(1) + O\left(\frac{1}{T}\right) = O(1).$$

Since P_{th} and Q_{th} are arbitrary matrices, under the same assumptions, similarly we have $\left\|\sum_{g=0}^{\infty}\ddot{P}_g\ddot{P}_g'\right\|_p = O(1)$ and $\left\|\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{Q}_g'\right\|_p = O(1)$ for $p = 1$ and $p = \infty$. Then

$$\begin{aligned}
&\left\|\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right)\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right) + \left(\sum_{g=0}^{\infty}\ddot{P}_g\ddot{P}_g'\right)\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{Q}_g'\right)\right\|_p \\
\leqslant &\left\|\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right\|_p^2 + \left\|\sum_{g=0}^{\infty}\ddot{P}_g\ddot{P}_g'\right\|_p\left\|\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{Q}_g'\right\|_p \\
= &O(1),
\end{aligned}$$

for $p = 1$ and $p = \infty$. So the order of the first term in equation (A.5) can be derived as

$$\sigma_0^4\mathrm{tr}\left[\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right)\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right) + \left(\sum_{g=0}^{\infty}\ddot{P}_g\ddot{P}_g'\right)\left(\sum_{g=0}^{\infty}\ddot{Q}_g'\ddot{Q}_g'\right)\right] = O(N).$$

Furthermore,

$$\begin{aligned}
\left|\sum_{g=0}^{\infty}\sum_{i=1}^{N_{T-g}}\left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\right| &\leqslant \sup_{g,i}\left|\left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\right|\left(\sum_{i=1}^{N_{T-g}}\left|\sum_{g=0}^{\infty}\left(D_g'\ddot{P}_g'\ddot{Q}_gD_g\right)_{ii}\right|\right) \\
&\leqslant \sup_g\left\|D_g'\ddot{P}_g'\ddot{Q}_gD_g\right\|_1\mathrm{tr}\left[\mathrm{abs}\left(\sum_{g=0}^{\infty}D_g'\ddot{P}_g'\ddot{Q}_gD_g\right)\right] \\
&= O(N),
\end{aligned}$$

where the last equality holds because both $\|\sum_{g=0}^{\infty}D_g'\ddot{P}_g'\ddot{Q}_gD_g\|_1$ and $\|\sum_{g=0}^{\infty}D_g'\ddot{P}_g'\ddot{Q}_gD_g\|_\infty$ are $O(1)$, so the sup in the expression is $O(1)$, and the trace is $O(N)$. Thus, the second term in equation A.5 is also $O(N)$. Therefore, $\mathrm{Var}(\mathbb{L}'\mathbb{P}\mathbb{R}) = O(N)$, then

$$\frac{1}{NT}\mathbb{L}'\mathbb{P}\mathbb{R} - \frac{1}{NT}E\left(\mathbb{L}'\mathbb{P}\mathbb{R}\right) = O_p\left(\frac{1}{\sqrt{NT^2}}\right)$$

according to Chebyshev's inequality. Also, according to Lemma 1,

$$\frac{1}{NT}E\left(\mathbb{L}'\mathbb{P}\mathbb{R}\right) = \frac{1}{NT}\sigma_0^2\mathrm{tr}\left(\sum_{h=0}^{\infty}\ddot{P}_h\ddot{Q}_h'\right) = O\left(\frac{1}{T}\right).$$

Proof of equation (A.4)
Since $\mathbb{Q} = I_n - \mathbb{P}$, combining equations (A.2) and (A.3), the results follows.

□

Lemma 8. *Under Assumptions 2, 7, A.1 and A.2, we have*

$$\frac{1}{NT}\mathbb{C}'\mathbb{L} = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.6}$$

$$\frac{1}{NT}\mathbb{C}'\mathbb{Q}\mathbb{L} = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.7}$$

$$\frac{1}{NT}\mathbb{C}'\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.8}$$

$$\frac{1}{NT}\mathbb{C}'\mathbb{Q}\mathbb{B}_1\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.9}$$

where $E(\mathbb{L}) = 0$ *and* $E(\mathbb{V}) = 0$.

Proof. **Proof of equations (A.6) and (A.7)**

Since $E(L_t) = 0$ and C_t, B_t, D_t are nonstochastic, $E(\mathbb{L}) = E(\mathbb{C}'\mathbb{L}) = E(\mathbb{C}'\mathbb{Q}\mathbb{L}) = 0$. Additionally,

$$\begin{aligned}\operatorname{Var}\left(\mathbb{C}'\mathbb{L}\right) &= \operatorname{Var}\left(\sum_{t=1}^{T} C_t'L_t\right) = \sum_{t=1}^{T}\sum_{s=1}^{T}\operatorname{Cov}\left(C_t'L_t, C_s'L_s\right) \leqslant \sum_{t=1}^{T}\sum_{s=1}^{T}\left|C_t'E\left(L_tL_s'\right)C_s\right| \\ &\leqslant 2\sum_{t=1}^{T}\sum_{s=1}^{t}\left|C_t'E\left(L_tL_s'\right)C_s\right| = 2\sigma_0^2\sum_{t=1}^{T}\sum_{s=1}^{t}\left|C_t'\left(\sum_{g=0}^{\infty}P_{t,t-s+g}P_{sg}'\right)C_s\right|,\end{aligned}$$

where the last equality follows from Lemma 1. According to Assumption A.1, $\{B_{1t}\}$ is UB uniformly for t so there exists a constant $c_B > 0$ such that

$$\sup_{N,T}\sup_{t\leqslant T}\max\left(\|B_t\|_1, \|B_t\|_\infty\right) \leqslant c_B.$$

Moreover, by Assumption A.2, elements in $\mathbb{C}$ are uniformly bounded, and the same holds for $\mathbb{Q}\mathbb{C}$ as $\mathbb{Q}$ is the projection matrix. Thus, there exists a constant $c_C > 0$ such that

$$\sup_{N}\sup_{T}\sup_{1\leqslant i\leqslant n}\max\left(|(\mathbb{C})_i|, |(\mathbb{Q}\mathbb{C})_i|\right) \leqslant c_C.$$

Hence, for any t and $1 \leqslant i \leqslant N_t$, we have

$$\left|\left(B_{1t}'C_t\right)_i\right| = \left|\sum_{j=1}^{N_t}\left(B_{1t}'\right)_{ij}\left(C_t\right)_j\right| \leqslant c_Bc_C,$$

therefore,

$$\begin{aligned}\left|C_t'\left(\sum_{g=0}^{\infty}P_{t,t-s+g}P_{sg}'\right)C_s\right| &= \left|C_t'B_{1t}P_t^{(t-s)}\sum_{g=0}^{\infty}\left(P_s^{(g)}P_s^{(g)\prime}\right)B_{1s}'C_s\right| \\ &\leqslant \sum_{i=1}^{N_t}\sum_{j=1}^{N_s}\left|\left(B_{1t}'C_t\right)_i\left(P_t^{(t-s)}\sum_{g=0}^{\infty}P_s^{(g)}P_s^{(g)\prime}\right)_{ij}\left(B_{1s}'C_s\right)_j\right| \\ &\leqslant c_B^2c_C^2\sum_{i=1}^{N_t}\sum_{j=1}^{N_s}\left|\left(P_t^{(t-s)}\sum_{g=0}^{\infty}P_s^{(g)}P_s^{(g)\prime}\right)_{ij}\right|,\end{aligned}$$

where the first equation holds by $P_{th} = B_{1t}P_t^{(h)}$ and $P_t^{(t-s+g)} = P_t^{(t-s)}P_s^{(g)}$. So

$$\begin{aligned}
\mathrm{Var}\left(\mathbb{C}'\mathbb{L}\right) &\leqslant 2c_B^2c_C^2\sigma_0^2\sum_{t=1}^{T}\sum_{s=1}^{t}\sum_{i=1}^{N_t}\sum_{j=1}^{N_s}\left|\left(P_t^{(t-s)}\sum_{g=0}^{\infty}P_s^{(g)}P_s^{(g)\prime}\right)_{ij}\right| \\
&= 2c_B^2c_C^2\sigma_0^2\sum_{s=1}^{T}\sum_{t=s}^{T}\sum_{i=1}^{N}\sum_{j=1}^{N_s}\left|\left(D_t'P_t^{(t-s)}\sum_{g=0}^{\infty}P_s^{(g)}P_s^{(g)\prime}\right)_{ij}\right| \\
&\leqslant 2c_B^2c_C^2\sigma_0^2\sum_{s=1}^{T}\sum_{i=1}^{N_t}\sum_{j=1}^{N_s}\left|\left[\sum_{t=s}^{\infty}\mathrm{abs}\left(D_t'P_t^{(t-s)}\right)\sum_{g=0}^{\infty}\mathrm{abs}\left(P_s^{(g)}P_s^{(g)\prime}\right)\right]_{ij}\right| \\
&= O(NT),
\end{aligned}$$

where the last equality holds because $\sum_{t=s}^{\infty}\mathrm{abs}\left(D_t'P_t^{(t-s)}\right)\sum_{g=0}^{\infty}\mathrm{abs}\left(P_s^{(g)}P_s^{(g)\prime}\right)$ is UB by Assumption A.1 and Lemma 4. Then we can obtain equation (A.6) by Chebyshev's inequality. By the same reasoning, we can obtain the same order for $\mathrm{Var}\left(\mathbb{C}'\mathbb{Q}\mathbb{L}\right)$ by replacing $\mathbb{C}$ with $\mathbb{Q}\mathbb{C}$ from the previous analysis. Therefore, equation (A.7) follows.

Proof of equations (A.8) and (A.9)

Since $E(V_t) = 0$, $E(\mathbb{C}'\mathbb{V}) = E(\mathbb{C}'\mathbb{Q}\mathbb{B}_1\mathbb{V}) = 0$. Because the error terms v_{it} are i.i.d. across i and t, we have

$$\mathrm{Var}(\mathbb{C}'\mathbb{V}) = \mathrm{Var}(\sum_{t=1}^{T}C_t'V_t) = \sum_{t=1}^{T}\mathrm{Var}(C_t'V_t) = \sigma_0^2\sum_{t=1}^{T}C_t'C_t = O(NT),$$

where the last equality holds because elements in C_t are uniformly bounded. Then we can obtain equation (A.8) by Chebyshev's inequality. Since $\{B_{1t}\}$ is UB uniformly for t, elements in $\mathbb{B}_1'\mathbb{Q}\mathbb{C}$ are also uniformly bounded, we therefore have $\mathrm{Var}\left(\mathbb{C}'\mathbb{Q}\mathbb{B}_1\mathbb{V}\right) = O(NT)$ by the same reasoning. Therefore, equation (A.9) follows. □

Lemma 9. *Under Assumptions 1, 2, 7, A.1, we have*

$$\frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{L} = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.10}$$

$$\frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{Q}\mathbb{L} = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.11}$$

and

$$\frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{B}_1\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.12}$$

$$\frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{Q}\mathbb{B}_1\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right). \tag{A.13}$$

Proof. **Proof of equation (A.10) and (A.11)**

As $\mathbb{Z}^{(\gamma)} = ((D_1F_1\iota_N)', \cdots, (D_TF_T\iota_N)')'$, we have

$$
\begin{aligned}
&\operatorname{Var}\left(\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{L}\right) = \operatorname{Var}\left[\sum_{t=1}^{T}(D_tF_t\iota_N)'B_{2t}'L_t\right] \\
\leqslant & 2\sum_{t=1}^{T}\sum_{s=1}^{t}\left|\operatorname{Cov}\left[(D_tF_t\iota_N)'B_{2t}'L_t, (D_sF_s\iota_N)'B_{2s}'L_s\right]\right| \\
\leqslant & 2\iota_N'\sum_{t=1}^{T}\sum_{s=1}^{t}\operatorname{abs}\left[F_t'D_t'B_{2t}'E\left(L_tL_s'\right)B_{2s}D_sF_s\right]\iota_N \\
= & 2\sigma_0^2\iota_N'\sum_{s=1}^{T}\sum_{t=s}^{T}\operatorname{abs}\left[F_t'D_t'B_{2t}'B_{1t}\left(\sum_{g=0}^{\infty}P_t^{(t-s+g)}P_s^{(g)\prime}\right)B_{1s}'B_{2s}D_sF_s\right]\iota_N \\
\leqslant & 2\sigma_0^2\iota_N'\sum_{s=1}^{T}\sum_{t=s}^{T}\operatorname{abs}\left(F_t'D_t'B_{2t}'B_{1t}P_t^{(t-s)}\right)\operatorname{abs}\left[\left(\sum_{g=0}^{\infty}P_s^{(g)}P_s^{(g)\prime}\right)B_{1s}'B_{2s}D_s\right]F_s\iota_N \\
\leqslant & 2\sigma_0^2\sum_{s=1}^{T}\left\{\iota_N'\left[\sum_{t=s}^{\infty}\operatorname{abs}\left(D_t'B_{2t}'B_{1t}P_t^{(t-s)}\right)\right]\operatorname{abs}\left[\left(\sum_{g=0}^{\infty}P_s^{(g)}P_s^{(g)\prime}\right)B_{1s}'B_{2s}D_s\right]\right\}F_s\iota_N \\
= & O(N),
\end{aligned}
$$

where the second equality follows from Lemma 1, and the third inequality is due to $P_t^{(t-s+g)} = P_t^{(t-s)}P_s^{(g)}$. The final inequality holds because F_t is a diagonal matrix with diagonal entries equal to either 0 or 1. Furthermore, the last equality holds because $\sum_{t=s}^{\infty}\operatorname{abs}\left(D_t'B_{2t}'B_{1t}P_t^{(t-s)}\right)$ and $\operatorname{abs}\left(\sum_{g=0}^{\infty}P_s^{(g)}P_s^{(g)\prime}\right)$ are UB by Lemma 4, then each element of the vector inside the braces is $O(1)$. Therefore, the entire expression is $O(N)$ because $F_s\iota_N$ is a sparse binary vector with fewer than N nonzero entries in total over $s = 1, \cdots, T$. Due to $E(\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{L}) = 0$, by Chebyshev's inequality we have $\frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{L} = O_p\left(\frac{1}{\sqrt{NT}}\right)$. Additionally,

$$
\begin{aligned}
&\operatorname{Var}(\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{P}\mathbb{L}) \\
= & \operatorname{Var}\left[\left(\sum_{t=1}^{T}D_t'B_{2t}D_tF_t\iota_N\right)'\mathcal{T}^{-1}\left(\sum_{t=1}^{T}D_t'B_{1t}\sum_{h=0}^{\infty}P_t^{(h)}V_{t-h}\right)\right] \\
= & \operatorname{Var}\left[\left(\mathcal{T}^{-\frac{1}{2}}\sum_{t=1}^{T}D_t'B_{2t}D_tF_t\iota_N\right)'\left(\sum_{h=0}^{\infty}\ddot{P}_hV_{T-h}\right)\right] \\
= & \sigma_0^2\left(\mathcal{T}^{-\frac{1}{2}}\sum_{t=1}^{T}D_t'B_{2t}D_tF_t\iota_N\right)'\left(\sum_{h=0}^{\infty}\ddot{P}_h\ddot{P}_h'\right)\left(\mathcal{T}^{-\frac{1}{2}}\sum_{t=1}^{T}D_t'B_{2t}D_tF_t\iota_N\right) \\
\leqslant & \frac{\sigma_0^2}{c_TT}\left(\sup_t\operatorname{abs}(D_t'B_{2t}D_t)\sum_{t=1}^{T}F_t\iota_N\right)'\operatorname{abs}\left(\sum_{h=0}^{\infty}\ddot{P}_h\ddot{P}_h'\right)\left(\sup_t\operatorname{abs}(D_t'B_{2t}D_t)\sum_{t=1}^{T}F_t\iota_N\right) \\
\leqslant & \sigma_0^2\|\sup_t\operatorname{abs}(D_t'B_{2t}D_t)\|_\infty^2\left[\iota_N'\operatorname{abs}\left(\sum_{h=0}^{\infty}\ddot{P}_h\ddot{P}_h'\right)\iota_N\right] \\
= & O\left(\frac{N}{T}\right),
\end{aligned}
$$

where $\ddot{P}_h$ is defined in Lemma 6. The first inequality holds because $T_i \geqslant c_TT$ by Assumption 1. The second inequality above holds due to the sparsity of $F_t\iota_N$, specifically, for each i, $(F_t)_{ii} = d_{it}(1 - d_{i,t-1})$ equals 1

at most once over $t = 1, \cdots, T$ (indicating the first appearance of the i-th individual) and is zero for other t. Therefore, for $i = 1, \cdots, N$, we have

$$\left[\sup_t \operatorname{abs}(D_t' B_{2t} D_t) \sum_{t=1}^{T} F_t \iota_N\right]_i \leqslant \left[\sup_t \operatorname{abs}(D_t' B_{2t} D_t)\iota_N\right]_i \leqslant \|\sup_t \operatorname{abs}(D_t' B_{2t} D_t)\|_\infty.$$

In addition, the last equality holds because, similar to the corresponding term in equation (A.5) in the proof of Lemma 7, $\operatorname{abs}\left(\sum_{h=0}^{\infty} \ddot{P}_h \ddot{P}_h'\right)$ is UB, then the value inside the brackets is $O(N)$. Hence, we have $\frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{P}\mathbb{L} = O_p\left(\frac{1}{\sqrt{NT^2}}\right)$ by Chebyshev's inequality. Combining with equation (A.10), we can prove (A.11).

Proof of equations (A.12) and (A.13)

Since $\mathbb{Z}^{(\gamma)} = ((D_1 F_1 \iota_N)', \cdots, (D_T F_T \iota_N)')'$ is a sparse binary vector with at most N nonzero elements and $\mathbb{B}_2'\mathbb{B}_1\mathbb{B}_1'\mathbb{B}_2$ is UB, $\operatorname{Var}(\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{B}_1\mathbb{V}) = \sigma_0^2\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{B}_1\mathbb{B}_1'\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(N)$. Then $\frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{B}_1\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right)$ by Chebyshev's inequality. Moreover, since D_t is an $N_t \times N$ matrix obtained by removing the corresponding rows of unobserved nodes during period t from I_N, we have

$$\begin{aligned}\|\mathbb{P}\|_1 = \|\mathbb{P}\|_\infty &= \|\operatorname{abs}\left(\mathbb{D}\mathcal{T}^{-1}\mathbb{D}'\right)\iota_n\|_\infty = \|(D_1', \cdots, D_T')'\mathcal{T}^{-1}(D_1', \cdots, D_T')\iota_n\|_\infty \\ &= \|(D_1', \cdots, D_T')'\mathcal{T}^{-1}(T_1, \cdots, T_N)'\|_\infty = (D_1', \cdots, D_T')'\iota_N = \|\iota_n\|_\infty = 1,\end{aligned}$$

where the third equality holds because all entries of the matrices $\{D_t\}$ and $\mathcal{T}^{-1}$ are all nonnegative. The above means that $\mathbb{P}$ is also UB. Thus, $\operatorname{Var}(\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{P}\mathbb{B}_1\mathbb{V}) = \sigma_0^2\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{P}\mathbb{B}_1\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(N)$ because $\mathbb{B}_2'\mathbb{P}\mathbb{B}_1\mathbb{B}_1'\mathbb{P}\mathbb{B}_2$ is UB. By Chebyshev's inequality, we also have $\frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_2'\mathbb{P}\mathbb{B}_1\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right)$, then equation (A.13) follows.

□

Lemma 10. *Lemmas 7-9 still hold by replacing $\mathbb{L}$ and $\mathbb{R}$ with $\mathbb{L}^\ell$ and $\mathbb{R}^\ell$ respectively.*

Proof. By definition, we have $L_t^\ell = \sum_{h=0}^{\infty} B_{1t}^\ell B_{1,t-1} P_{t-1}^{(h)} V_{t-h-1}$ and $R_t^\ell = \sum_{h=0}^{\infty} B_{2t}^\ell B_{2,t-1} Q_{t-1}^{(h)} V_{t-h-1}$, where $\|\sup_t \operatorname{abs}(D_t' B_{1t}^\ell B_{1,t-1} D_{t-1})\|_\infty < \infty$ and $\|\sup_t \operatorname{abs}(D_t' B_{2t}^\ell B_{2,t-1} D_{t-1})\|_\infty < \infty$ according to Lemma 4. Moreover, since v_{it}'s are i.i.d. with respect to i and t, $\mathbb{L}^\ell$ and $\mathbb{R}^\ell$ also satisfy the conditions of Lemmas 7-9, leading to the same result. □

Lemma 11. *Under Assumptions 1, 2, 7 and A.1,*

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\mathbb{B}_2\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.14}$$

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{B}_2\mathbb{V}\right) = O_p\left(\frac{1}{\sqrt{NT^2}}\right), \tag{A.15}$$

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{B}_2\mathbb{V}\right) = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.16}$$

where

$$E\left(\mathbb{L}^{\ell\prime}\mathbb{B}_2\mathbb{V}\right) = 0, \tag{A.17}$$

and

$$E\left(\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{B}_2\mathbb{V}\right) = \sigma_0^2 \operatorname{tr}\left[\mathcal{T}^{-1}\sum_{h=1}^{T-1}\sum_{t=T-h}^{T-1} D_{t+1}' B_{t+1}^\ell P_t^{(t+h-T)} B_{2,T-h}' D_{T-h}\right] = O(N). \tag{A.18}$$

Proof. **Proof of equation (A.14)**

For simplicity, let $B_t^\ell \equiv B_{1t}^\ell B_{1,t-1}$. Since v_{it} are i.i.d across i and t,

$$E(\mathbb{L}^{\ell\prime}\mathbb{B}_2\mathbb{V}) = E\left(\sum_{t=1}^{T} L_t^{\ell\prime} B_{2t} V_t\right) = \sum_{t=1}^{T}\sum_{h=0}^{\infty} E\left(V_{t-h-1}' P_{t-1}^{(h)\prime} B_t^{\ell\prime} B_{2t} V_t\right) = 0.$$

Then

$$\begin{aligned}
\mathrm{Var}(\mathbb{L}^{\ell\prime}\mathbb{B}_2\mathbb{V}) &= \mathrm{Var}\left(\sum_{t=1}^{T} L_t^{\ell\prime} B_{2t} V_t\right) = \sum_{t=1}^{T}\sum_{s=1}^{T} E\left(L_t^{\ell\prime} B_{2t} V_t L_s^{\ell\prime} B_{2s} V_s\right) \\
&= \sum_{t=1}^{T}\sum_{s=1}^{T}\sum_{h=0}^{\infty}\sum_{g=0}^{\infty} E\left(V_{t-h-1}' P_{t-1}^{(h)\prime} B_t^{\ell\prime} B_{2t} V_t V_{s-g-1}' P_{s-1}^{(g)\prime} B_s^{\ell\prime} B_{2s} V_s\right) \\
&= \sum_{t=1}^{T}\sum_{h=0}^{\infty} E\left(V_{t-h-1}' P_{t-1}^{(h)\prime} B_t^{\ell\prime} B_{2t} V_t V_{t-h-1}' P_{t-1}^{(h)\prime} B_t^{\ell\prime} B_{2t} V_t\right) \\
&= \sigma_0^4 \sum_{t=1}^{T}\sum_{h=0}^{\infty} \mathrm{tr}\left(P_{t-1}^{(h)\prime} B_t^{\ell\prime} B_{2t} B_{2t}' B_t^{\ell} P_{t-1}^{(h)}\right) \\
&= \sigma_0^4 \sum_{t=1}^{T} \mathrm{tr}\left(B_t^{\ell\prime} B_{2t} B_{2t}' B_t^{\ell} \sum_{h=0}^{\infty} P_{t-1}^{(h)} P_{t-1}^{(h)\prime}\right) \\
&= O(NT),
\end{aligned}$$

where the fourth equality holds because the expectation is nonzero only when $t = s$ and $h = g$ according to Lemma 2, and the last equality holds because $B_t^{\ell\prime} B_{2t} B_{2t}' B_t^{\ell}$ and $\sum_{h=0}^{\infty} P_{t-1}^{(h)} P_{t-1}^{(h)\prime}$ are UB uniformly for t by Assumption A.1 and Lemma 4. Therefore, we can prove equation (A.14) by Chebyshev's inequality.

Proof of equation (A.15)

Similar to Lemma 6, we have

$$\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{B}_2\mathbb{V} = \left(\sum_{t=1}^{T} D_t L_t^{\ell}\right)' \mathcal{T}^{-1} \left(\sum_{t=1}^{T} D_t' B_{2t} V_t\right) \equiv \left(\sum_{h=0}^{\infty} \ddot{P}_h V_{T-h}\right)' \left(\sum_{h=0}^{\infty} \ddot{Q}_h V_{T-h}\right),$$

where

$$\ddot{P}_h \equiv \begin{cases} 0 & \text{for } h = 0 \\ \mathcal{T}^{-\frac{1}{2}} \sum_{t=T-h}^{T-1} D_{t+1}' B_{t+1}^{\ell} P_t^{(t+h-T)} & \text{for } 0 < h < T \\ \mathcal{T}^{-\frac{1}{2}} \sum_{t=0}^{T-1} D_{t+1}' B_{t+1}^{\ell} P_t^{(t+h-T)} & \text{for } h \geqslant T \end{cases} \text{ and } \ddot{Q}_h \equiv \begin{cases} \mathcal{T}^{-\frac{1}{2}} D_{T-h}' B_{2,T-h} & \text{for } h < T \\ 0 & \text{for } h \geqslant T \end{cases}.$$

Therefore, by Lemma 1,

$$\begin{aligned}
&\left|E\left(\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{B}_2\mathbb{V}\right) = \sigma_0^2 \mathrm{tr}\left(\sum_{h=0}^{\infty} \ddot{P}_h \ddot{Q}_h'\right)\right| \\
=&\left|\sigma_0^2 \mathrm{tr}\left[\sum_{h=1}^{T-1} \mathcal{T}^{-\frac{1}{2}} \left(\sum_{t=T-h}^{T-1} D_{t+1}' B_{t+1}^{\ell} P_t^{(t+h-T)}\right) B_{2,T-h}' D_{T-h} \mathcal{T}^{-\frac{1}{2}}\right]\right| \\
\leqslant& \sigma_0^2 \mathrm{tr}\left[\mathcal{T}^{-1} \sum_{h=1}^{T-1} \mathrm{abs}\left(\sum_{t=T-h}^{T-1} D_{t+1}' B_{t+1}^{\ell} P_t^{(t+h-T)}\right) \mathrm{abs}\left(B_{2,T-h}' D_{T-h}\right)\right] \\
\leqslant& \frac{\sigma_0^2}{c_T T} \sum_{h=1}^{T-1} \mathrm{tr}\left[\mathrm{abs}\left(\sum_{t=T-h}^{T-1} D_{t+1}' B_{t+1}^{\ell} P_t^{(t+h-T)}\right) \mathrm{abs}\left(B_{2,T-h}' D_{T-h}\right)\right] \\
=& O(N),
\end{aligned}$$

where the second inequality holds because $T_i > c_T T$ by Assumption 1 and the last equality holds because the matrix under the trace is UB uniformly for h according to Lemma 4. By Lemma 3, we have

$$\begin{aligned}\operatorname{Var}\left(\mathbb{L}^{\ell\prime}\mathbb{PB}_2\mathbb{V}\right) =& \operatorname{Cov}\left[\left(\sum_{h=0}^{\infty}\ddot{P}_h V_{T-h}\right)'\left(\sum_{h=0}^{\infty}\ddot{Q}_h V_{T-h}\right),\left(\sum_{h=0}^{\infty}\ddot{P}_h V_{T-h}\right)'\left(\sum_{h=0}^{\infty}\ddot{Q}_h V_{T-h}\right)\right] \\ =& \sigma_0^4 \operatorname{tr}\left[\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right)\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right)+\left(\sum_{g=0}^{\infty}\ddot{P}_g\ddot{P}_g'\right)\left(\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{Q}_g'\right)\right]+ \\ & \sum_{g=0}^{\infty}\left[(\mu_4-3\sigma_0^4)\sum_{i=1}^{N_{T-g}}\left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\right], \end{aligned} \tag{A.19}$$

where

$$\begin{aligned}\left\|\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{P}_g'\right\|_p =& \left\|\sum_{h=1}^{T-1}\mathcal{T}^{-\frac{1}{2}}D_{T-h}'B_{2,T-h}\left(\sum_{t=T-h}^{T-1}D_{t+1}'B_{t+1}^{\ell}P_t^{(t+h-T)}\right)'\mathcal{T}^{-\frac{1}{2}}\right\|_p \\ \leqslant& \frac{1}{c_T T}\sum_{h=1}^{T-1}\left\|D_{T-h}'B_{2,T-h}\left(\sum_{t=T-h}^{\infty}D_{t+1}'B_{t+1}^{\ell}P_t^{(t+h-T)}\right)'\right\|_p \\ =& O(1)\end{aligned}$$

holds for $p=1$ and $p=\infty$. Since the $\ddot{P}_h$ here can be regarded as a lagged version of that defined in Lemma 6, and $\|\sup_t \operatorname{abs}(B_t^{\ell})\|_{\infty} < \infty$ by Lemma 4, similar to the derivation of the upper bound for the corresponding term in equation (A.5), we have

$$\left\|\sum_{g=0}^{\infty}\ddot{P}_g\ddot{P}_g'\right\|_p = O(1)$$

for $p=1$ and $p=\infty$. Additionally,

$$\begin{aligned}\left\|\sum_{g=0}^{\infty}\ddot{Q}_g\ddot{Q}_g'\right\|_p &= \left\|\sum_{g=0}^{T-1}\mathcal{T}^{-\frac{1}{2}}D_{T-g}'B_{2,T-g}B_{2,T-g}'D_{T-g}\mathcal{T}^{-\frac{1}{2}}\right\|_p \\ &\leqslant \frac{1}{c_T T}\sum_{g=0}^{T-1}\left\|D_{T-g}'B_{2,T-g}B_{2,T-g}'D_{T-g}\right\|_p \\ &= \frac{1}{c_T T}\sum_{g=0}^{T-1}\left\|B_{2,T-g}B_{2,T-g}'\right\|_p \\ &= O(1),\end{aligned}$$

where the first inequality holds by $T_i \geqslant c_T T$ in Assumption 1, the second one holds because $D_{T-g}'B_{2,T-g}B_{2,T-g}'D_{T-g}$ is merely an augmented version of $B_{2,T-g}B_{2,T-g}'$ with additional all-zero rows and columns, and the last equality holds because $B_{2,T-g}B_{2,T-g}'$ is UB uniformly for g. Therefore, the matrix within the trace in the first term of equation (A.19) is UB, which means that the first term is $O(N)$. For the second term in equation (A.19),

we have

$$\begin{aligned}\left|\sum_{g=0}^{\infty}\sum_{i=1}^{N_{T-g}}\left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\right| &\leqslant \sup_{g,i}\left|\left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\right|\left(\sum_{i=1}^{N_{T-g}}\left|\sum_{g=0}^{\infty}\left(D_g'\ddot{P}_g'\ddot{Q}_gD_g\right)_{ii}\right|\right)\\ &\leqslant \sup_g\left\|D_g'\ddot{P}_g'\ddot{Q}_gD_g\right\|_1\operatorname{tr}\left[\operatorname{abs}\left(\sum_{g=0}^{\infty}D_g'\ddot{P}_g'\ddot{Q}_gD_g\right)\right]\\ &= O(N),\end{aligned}$$

where the last equality holds because $\sum_{g=0}^{\infty}\ddot{P}_g'\ddot{Q}_g$ is UB, so $\sup_g\left\|D_g'\ddot{P}_g'\ddot{Q}_gD_g\right\|_1$ is $O(1)$, and the trace above is $O(N)$. So the second term in equation (A.19) is also $O(N)$, which means that the whole term is $O(N)$. Therefore, we can get equation (A.15) by Chebyshev's inequality.

Proof of equation (A.16)

Since $\mathbb{Q}=I_n-\mathbb{P}$, combining equations (A.14) and (A.15), the results follows. □

Lemma 12. *Under Assumptions 1, 2, 7 and A.1,*

$$\frac{1}{NT}\mathbb{V}'\mathbb{B}_1'\mathbb{B}_2\mathbb{V}-\frac{1}{NT}E\left(\mathbb{V}\mathbb{B}_1'\mathbb{B}_2\mathbb{V}\right)=O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.20}$$

$$\frac{1}{NT}\mathbb{V}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}-\frac{1}{NT}E\left(\mathbb{V}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}\right)=O_p\left(\frac{1}{\sqrt{NT^2}}\right), \tag{A.21}$$

$$\frac{1}{NT}\mathbb{V}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}-\frac{1}{NT}E\left(\mathbb{V}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}\right)=O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.22}$$

where $\frac{1}{NT}E(\mathbb{V}'\mathbb{B}_1'\mathbb{B}_2\mathbb{V})=O(1)$ *and* $\frac{1}{NT}E(\mathbb{V}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V})=O(\frac{1}{T})$.

Proof. **Proof of equation (A.20)**

$$\begin{aligned}\frac{1}{NT}E\left(\mathbb{V}'\mathbb{B}_1'\mathbb{B}_2\mathbb{V}\right)&=\frac{1}{NT}E\left(\sum_{t=1}^{T}V_t'B_{1t}'B_{2t}V_t\right)=\frac{1}{NT}E\left[\sum_{t=1}^{T}\operatorname{tr}(B_{2t}V_tV_t'B_{1t}')\right]\\ &=\frac{1}{NT}\sigma_0^2\sum_{t=1}^{T}\operatorname{tr}(B_{1t}'B_{2t})=O(1),\end{aligned}$$

where the last equality is because $\{B_{1t}\}$ and $\{B_{2t}\}$ are UB uniform for t. Moreover by Lemma 2,

$$\begin{aligned}&\operatorname{Var}\left(\mathbb{V}'\mathbb{B}_1'\mathbb{B}_2\mathbb{V}\right)\\ =&\sum_{t=1}^{T}\sum_{s=1}^{T}\operatorname{Cov}(V_t'B_{1t}'B_{2t}V_t,V_s'B_{1s}'B_{2s}V_s)\\ =&\sum_{t=1}^{T}\sum_{s=1}^{T}\left[E(V_t'B_{1t}'B_{2t}V_tV_s'B_{1s}'B_{2s}V_s)-E(V_t'B_{1t}'B_{2t}V_t)E(V_s'B_{1s}'B_{2s}V_s)\right]\\ =&\sum_{t=1}^{T}\left[(\mu_4-3\sigma_0^4)\sum_{i=1}^{N_t}(B_{1t}'B_{2t})_{ii}^2+\sigma_0^4\left(\operatorname{tr}(B_{1t}'B_{2t})^2+\operatorname{tr}(B_{1t}'B_{2t}B_{1t}'B_{2t})+\operatorname{tr}(B_{1t}'B_{2t}B_{2t}'B_{1t})\right)\right]+\\ &\sum_{t=1}^{T}\sum_{s=1}^{T}\sigma_0^4\operatorname{tr}(B_{1t}'B_{2t})\operatorname{tr}(B_{1s}'B_{2s})-\sum_{t=1}^{T}\sigma_0^4\operatorname{tr}(B_{1t}'B_{2t})^2-\sigma_0^4\left[\sum_{t=1}^{T}\operatorname{tr}(B_{1t}'B_{2t})\right]^2\\ =&\sum_{t=1}^{T}\left[(\mu_4-3\sigma_0^4)\sum_{i=1}^{N_t}(B_{1t}'B_{2t})_{ii}^2+\sigma_0^4\left(\operatorname{tr}(B_{1t}'B_{2t}B_{1t}'B_{2t})+\operatorname{tr}(B_{1t}'B_{2t}B_{2t}'B_{1t})\right)\right]\\ =&O(NT).\end{aligned}$$

Then we can get equation (A.20) by Chebyshev's inequality.

Proof of equation (A.21)

Similar to Lemma 6, we have

$$\mathbb{V}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} = \left(\sum_{t=1}^{T} D_t'B_{1t}V_t\right)' \mathcal{T}^{-1} \left(\sum_{t=1}^{T} D_t'B_{2t}V_t\right) \equiv \left(\sum_{h=0}^{\infty} \ddot{P}_h V_{T-h}\right)' \left(\sum_{h=0}^{\infty} \ddot{Q}_h V_{T-h}\right),$$

where

$$\ddot{P}_h \equiv \begin{cases} \mathcal{T}^{-\frac{1}{2}} D_{T-h}' B_{1,T-h} & \text{for } h < T \\ 0 & \text{for } h \geqslant T \end{cases}$$

and $\ddot{Q}_h$ has the same pattern. Then according to Lemma 1,

$$\begin{aligned}
&\left|\frac{1}{NT} E\left(\mathbb{V}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}\right)\right| = \frac{1}{NT}\left|E\left[\left(\sum_{h=0}^{\infty} \ddot{P}_h V_{T-h}\right)' \left(\sum_{h=0}^{\infty} \ddot{Q}_h V_{T-h}\right)\right]\right| \\
=&\frac{1}{NT}\sigma_0^2 \left|\mathrm{tr}\left(\sum_{h=0}^{\infty} \ddot{P}_h \ddot{Q}_h'\right)\right| \leqslant \frac{1}{NT}\sigma_0^2 \sum_{t=1}^{T} \left|\mathrm{tr}\left(\mathcal{T}^{-\frac{1}{2}} D_t' B_{1t} B_{1t}' D_t \mathcal{T}^{-\frac{1}{2}}\right)\right| \\
\leqslant&\frac{1}{c_T NT^2}\sigma_0^2 \sum_{t=1}^{T} \mathrm{tr}\left(\mathrm{abs}(B_{1t}'B_{2t})\right) = O\left(\frac{1}{T}\right),
\end{aligned}$$

The second equality holds because V_t's are independent across t. The second inequality holds due to Assumption 1 with $T_i \geqslant c_T T$. The last equality holds because $\{B_{1t}\}$ and $\{B_{2t}\}$ are UB uniform for t. Moreover according to Lemma 3,

$$\begin{aligned}
&\mathrm{Var}\left(\mathbb{V}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}\right) \\
=&\mathrm{Cov}\left[\left(\sum_{h=0}^{\infty} \ddot{P}_h V_{T-h}\right)' \left(\sum_{h=0}^{\infty} \ddot{Q}_h V_{T-h}\right), \left(\sum_{h=0}^{\infty} \ddot{P}_h V_{T-h}\right)' \left(\sum_{h=0}^{\infty} \ddot{Q}_h V_{T-h}\right)\right] \\
=&\sigma_0^4 \mathrm{tr}\left[\left(\sum_{g=0}^{\infty} \ddot{Q}_g \ddot{P}_g'\right)\left(\sum_{g=0}^{\infty} \ddot{Q}_g \ddot{P}_g'\right) + \left(\sum_{g=0}^{\infty} \ddot{P}_g \ddot{P}_g'\right)\left(\sum_{g=0}^{\infty} \ddot{Q}_g' \ddot{Q}_g'\right)\right] + \\
&\sum_{g=0}^{\infty}\left[(\mu_4 - 3\sigma_0^4) \sum_{i=1}^{N_{T-g}} \left(\ddot{P}_g'\ddot{Q}_g\right)_{ii} \left(\ddot{P}_g'\ddot{Q}_g\right)_{ii}\right] \\
=&\sigma_0^4 \sum_{g=1}^{T}\sum_{h=1}^{T} \mathrm{tr}\left[\left(\mathcal{T}^{-\frac{1}{2}} D_g' B_{2g} B_{1g}' D_g \mathcal{T}^{-\frac{1}{2}}\right)\left(\mathcal{T}^{-\frac{1}{2}} D_h' B_{2h} B_{1h}' D_h \mathcal{T}^{-\frac{1}{2}}\right)\right] + \\
&\sigma_0^4 \sum_{g=1}^{T}\sum_{h=1}^{T} \mathrm{tr}\left[\left(\mathcal{T}^{-\frac{1}{2}} D_g' B_{1g} B_{1g}' D_g \mathcal{T}^{-\frac{1}{2}}\right)\left(\mathcal{T}^{-\frac{1}{2}} D_h' B_{2h} B_{2h}' D_h \mathcal{T}^{-\frac{1}{2}}\right)\right] + \\
&\sum_{g=1}^{T}\left[(\mu_4 - 3\sigma_0^4) \sum_{i=1}^{N_g} \left(B_{1g}' D_g \mathcal{T}^{-1} D_g' B_{2g}\right)_{ii} \left(B_{1g}' D_g \mathcal{T}^{-1} D_g' B_{2g}\right)_{ii}\right].
\end{aligned} \tag{A.23}$$

Since for any g and h,

$$\begin{aligned}
&\left|\mathrm{tr}\left[\left(\mathcal{T}^{-\frac{1}{2}}D_g'B_{2g}B_{1g}'D_g\mathcal{T}^{-\frac{1}{2}}\right)\left(\mathcal{T}^{-\frac{1}{2}}D_h'B_{2h}B_{1h}'D_h\mathcal{T}^{-\frac{1}{2}}\right)\right]\right| \\
=&\left|\mathrm{tr}\left[D_h\mathcal{T}^{-1}D_g'B_{2g}B_{1g}'D_g\mathcal{T}^{-1}D_h'B_{2h}B_{1h}'\right]\right| \\
\leqslant&\mathrm{tr}\left[\mathrm{abs}\left(D_h\mathcal{T}^{-1}D_g'\right)\mathrm{abs}\left(B_{2g}B_{1g}'\right)\mathrm{abs}\left(D_g\mathcal{T}^{-1}D_h'\right)\mathrm{abs}\left(B_{2h}B_{1h}'\right)\right] \\
\leqslant&\frac{1}{c_T^2T^2}\mathrm{tr}\left[D_hD_g'\mathrm{abs}\left(B_{2g}B_{1g}'\right)D_gD_h'\mathrm{abs}\left(B_{2h}B_{1h}'\right)\right] \\
=&O\left(\frac{N}{T^2}\right),
\end{aligned}$$

where the second inequality holds due to Assumption 1 with $T_i \geqslant c_T T$, and the last equality holds because $\{B_{1t}\}$ and $\{B_{2t}\}$ are UB uniform for t. Similarly, we have

$$\mathrm{tr}\left[\left(\mathcal{T}^{-\frac{1}{2}}D_g'B_{1g}B_{1g}'D_g\mathcal{T}^{-\frac{1}{2}}\right)\left(\mathcal{T}^{-\frac{1}{2}}D_h'B_{2h}B_{2h}'D_h\mathcal{T}^{-\frac{1}{2}}\right)\right] = O\left(\frac{N}{T^2}\right).$$

Therefore, the first two terms in equation (A.23) is $O(N)$. Then, by the same argument, we have

$$\begin{aligned}
\sup_{g,i}\left|\left(B_{1g}'D_g\mathcal{T}^{-1}D_g'B_{2g}\right)_{ii}\right| &\leqslant \sup_{g,i}\sum_{k=1}^{N}\mathrm{abs}(B_{1g}'D_g)_{ik}T_k^{-1}\mathrm{abs}(D_g'B_{2g})_{ki} \\
&\leqslant \frac{1}{c_T T}\sup_{g,i}\sum_{k=1}^{N}\mathrm{abs}(B_{1g}'D_g)_{ik}\mathrm{abs}(D_g'B_{2g})_{ki} = O\left(\frac{1}{T}\right).
\end{aligned}$$

Thus,

$$\begin{aligned}
&\left|\sum_{g=1}^{T}\sum_{i=1}^{N_g}\left(B_{1g}'D_g\mathcal{T}^{-1}D_g'B_{2g}\right)_{ii}\left(B_{1g}'D_g\mathcal{T}^{-1}D_g'B_{2g}\right)_{ii}\right| \\
\leqslant&\sup_{i,g}\left|\left(B_{1g}'D_g\mathcal{T}^{-1}D_g'B_{2g}\right)_{ii}\right|\sum_{g=1}^{T}\mathrm{tr}\left[\mathrm{abs}\left(B_{1g}'D_g\mathcal{T}^{-1}D_g'B_{2g}\right)\right] \\
=&O\left(\frac{N}{T}\right),
\end{aligned}$$

that is, the last term in equation (A.23) is $O\left(\frac{N}{T}\right)$. Therefore, $\mathrm{Var}(\mathbb{V}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}) = O(N) + O\left(\frac{N}{T}\right) = O(N)$. Then by Chebyshev's inequality, equation (A.21) follows.

Proof of equation (A.22)

Since $\mathbb{Q} = I_n - \mathbb{P}$, combining equations (A.20) and (A.21), the results follows. □

Lemma 13. *Under Assumptions 1-7 and A.1, we have*

$$\frac{1}{NT}\mathbb{V}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E[\mathbb{V}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}] = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.24}$$

$$\frac{1}{NT}\mathbb{Z}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E[\mathbb{Z}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}] = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.25}$$

$$\frac{1}{NT}\mathbb{Z}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z} - \frac{1}{NT}E[\mathbb{Z}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}] = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.26}$$

where $\frac{1}{NT}E[\mathbb{V}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}] = O(1)$, $\frac{1}{NT}E[\mathbb{Z}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}] = O(\frac{1}{T})$ *and* $\frac{1}{NT}E[\mathbb{Z}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}] = O(1)$.

Proof. **Proof of equation (A.24)**

This result is just equation (A.22) in Lemma 12.

Proof of equation (A.25)

Recall that $Z_t = (D_t(I_N - F_t)D_t'M_tY_{t-1}, D_tD_{t-1}'Y_{t-1}, D_tF_t\iota_N, X_t)$ and $\mathbb{Z} = (Z_1', Z_2', \cdots, Z_T')'$. Denote $\mathbb{Z}_1$, $\mathbb{Z}_2$ and $\mathbb{Z}^{(\gamma)}$ as the first three columns of $\mathbb{Z}$, and let $\mathbb{Z}_4$ denote the remaining k columns. Since

$$\mathbb{Z}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} = \begin{pmatrix} \mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} \\ \mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} \\ \mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} \\ \mathbb{Z}_4'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} \end{pmatrix},$$

we only need to derive the quantitative law of large numbers for each element of the above vectors. According to equation (3),

$$Y_t = \sum_{h=0}^{\infty} S_t^{-1}A_t^{(h)}[D_{t-h}(\gamma_0 F_{t-h}\iota_N + \alpha_0) + X_{t-h}\beta_0 + V_{t-h}] = \mathcal{E}_t + \mathcal{X}_t\beta_0 + \mathcal{V}_t,$$

where $\mathcal{E}_t = \sum_{h=0}^{\infty} S_t^{-1}A_t^{(h)}D_{t-h}(\gamma_0 F_{t-h}\iota_N + \alpha_0)$, $\mathcal{X}_t = \sum_{h=0}^{\infty} S_t^{-1}A_t^{(h)}X_{t-h}$ and $\mathcal{V}_t = \sum_{h=0}^{\infty} S_t^{-1}A_t^{(h)}V_{t-h}$. Therefore, $\mathbb{B}_1\mathbb{Z}_1$ is the stack version of vector $B_{1t}D_t(I_N - F_t)D_t'M_t(\mathcal{E}_{t-1} + \mathcal{X}_{t-1}\beta_0 + \mathcal{V}_{t-1})$, which can be viewed as $\mathbb{C} + \mathbb{L}^{\ell}$, by specifying $C_t = B_{1t}D_t(I_N - F_t)D_t'M_t(\mathcal{E}_{t-1} + \mathcal{X}_{t-1}\beta_0)$ and $L_t^{\ell} = B_{1t}D_t(I_N - F_t)D_t'M_t\mathcal{V}_{t-1}$. Then we have

$$\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} \equiv \mathbb{C}'\mathbb{Q}\mathbb{B}_2\mathbb{V} + \mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{B}_2\mathbb{V},$$

where $\frac{1}{NT}\mathbb{C}'\mathbb{Q}\mathbb{B}_2\mathbb{V} = O_p(\frac{1}{\sqrt{NT}})$ by Lemma 8, $\frac{1}{NT}\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{B}_2\mathbb{V}\right) = O_p(\frac{1}{\sqrt{NT}})$, and $\frac{1}{NT}E\left[\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{B}_2\mathbb{V}\right] = O(\frac{1}{T})$ by Lemma 11. Therefore,

$$\frac{1}{NT}\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\left[\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}\right] = O_p\left(\frac{1}{\sqrt{NT}}\right)$$

and $\frac{1}{NT}E\left(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}\right) = O(\frac{1}{T})$. Similarly, $\mathbb{B}_1\mathbb{Z}_2$ can also be viewed as $\mathbb{C} + \mathbb{L}^{\ell}$ by specifying $C_t = B_{1t}D_tD_{t-1}(\mathcal{E}_{t-1} + \mathcal{X}_{t-1}\beta_0)$ and $L_t^{\ell} = B_{1t}D_tD_{t-1}\mathcal{V}_{t-1}$. Hence,

$$\frac{1}{NT}\mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\left[\mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}\right] = O_p\left(\frac{1}{\sqrt{NT}}\right)$$

and $\frac{1}{NT}E\left(\mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}\right) = O(\frac{1}{T})$ also hold. Each column of $\mathbb{B}_1\mathbb{Z}^{(\gamma)}$ and $\mathbb{B}_1\mathbb{Z}_4$ can be viewed as $\mathbb{C}$ because they are both vectors whose elements are uniformly bounded. Then $\mathbb{Z}_3$ and $\mathbb{Z}_4$ satisfy the same conclusion because $\frac{1}{NT}\mathbb{C}'\mathbb{Q}\mathbb{B}_2\mathbb{V} = O_p(\frac{1}{\sqrt{NT}})$ by Lemma 8. Thus, equation (A.25) holds and $\frac{1}{NT}E\left(\mathbb{Z}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V}\right) = O(\frac{1}{T})$.

Proof of equation (A.26)

We have

$$\mathbb{Z}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z} = \begin{pmatrix} \mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1 & \mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_2 & \mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} & \mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_4 \\ \mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1 & \mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_2 & \mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} & \mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_4 \\ \mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1 & \mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_2 & \mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} & \mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_4 \\ \mathbb{Z}_4'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1 & \mathbb{Z}_4'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_2 & \mathbb{Z}_4'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} & \mathbb{Z}_4'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_4 \end{pmatrix}.$$

Firstly, we can decompose

$$\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1 \equiv (\mathbb{C}_1 + \mathbb{L}^{\ell})'\mathbb{Q}(\mathbb{C}_2 + \mathbb{R}^{\ell}) = \mathbb{C}_1'\mathbb{Q}\mathbb{C}_2 + \mathbb{C}_1'\mathbb{Q}\mathbb{R}^{\ell} + \mathbb{C}_2'\mathbb{Q}\mathbb{L}^{\ell} + \mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{R}^{\ell},$$

where $\frac{1}{NT}\mathbb{C}'_1\mathbb{Q}\mathbb{R}^{\ell} = O_p\left(\frac{1}{\sqrt{NT}}\right)$ and $\frac{1}{NT}\mathbb{C}'_2\mathbb{Q}\mathbb{L}^{\ell} = O_p\left(\frac{1}{\sqrt{NT}}\right)$ by Lemmas 8 and 10, and $\frac{1}{NT}\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{R}^{\ell} - \frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{R}^{\ell}\right) = O_p\left(\frac{1}{\sqrt{NT}}\right)$, $\frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{R}^{\ell}\right) = O(1)$ by Lemmas 7 and 10. We have $\frac{1}{NT}\mathbb{C}'_1\mathbb{Q}\mathbb{C}_2 = O(1)$ because elements in $\mathbb{C}_1$ and $\mathbb{C}_2$ are uniformly bounded and $\mathbb{Q}$ is a projection matrix. Hence,

$$\frac{1}{NT}\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1 - \frac{1}{NT}E\left[\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1\right] = O_p\left(\frac{1}{\sqrt{NT}}\right)$$

and $\frac{1}{NT}E\left(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1\right) = O(1)$. The same conclusion holds for $\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_2$, $\mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1$ and $\mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_2$ as they can undergo similar decomposition.

Secondly, we have $\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} \equiv (\mathbb{C}_1 + \mathbb{L}^{\ell})'\mathbb{Q}\mathbb{C}_2 = \mathbb{C}_2'\mathbb{Q}\mathbb{C}_1 + \mathbb{C}_2'\mathbb{Q}\mathbb{L}^{\ell}$. Similarly, we have

$$\frac{1}{NT}\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} - \frac{1}{NT}E\left[\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)}\right] = O_p\left(\frac{1}{\sqrt{NT}}\right)$$

and $\frac{1}{NT}E\left(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)}\right) = O(1)$ by Lemmas 8 and 10. The same conclusion holds for $\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1$, $\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_4$, $\mathbb{Z}_4'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_1$, $\mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)}$, $\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_2$, $\mathbb{Z}_2'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_4$ and $\mathbb{Z}_4'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_2$.

Lastly, The scalar $\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)}$, as well as each element of the vector $\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_4$, $\mathbb{Z}_4\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)}$, and the matrix $\mathbb{Z}_4'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}_4$, can be regarded as $\mathbb{C}_1'\mathbb{Q}\mathbb{C}_2$ and $\frac{1}{NT}\mathbb{C}_1'\mathbb{Q}\mathbb{C}_2 = O(1)$. Therefore, equation (A.26) follows and $\frac{1}{NT}E[\mathbb{Z}'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}] = O(1)$.

□

Lemma 14. *Under Assumptions 1-7 and A.1, we have*

$$\frac{1}{NT}\mathbb{V}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E[\mathbb{V}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}] = O_p\left(\frac{1}{\sqrt{NT^2}}\right), \tag{A.27}$$

$$\frac{1}{NT}\mathbb{Z}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E[\mathbb{Z}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}] = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.28}$$

$$\frac{1}{NT}\mathbb{Z}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z} - \frac{1}{NT}E[\mathbb{Z}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}] = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.29}$$

where $\frac{1}{NT}E[\mathbb{V}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}] = O(\frac{1}{T})$, $\frac{1}{NT}E[\mathbb{Z}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}] = O(\frac{1}{T})$ *and* $\frac{1}{NT}E[\mathbb{Z}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}] = O(1)$.

Proof. **Proof of equation (A.27)**

This result is just equation (A.21) in Lemma 12.

Proof of equation (A.28)

Similar to the proof of Lemma 13, we have

$$\mathbb{Z}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} = \begin{pmatrix} \mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} \\ \mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} \\ \mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} \\ \mathbb{Z}_4'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} \end{pmatrix},$$

where $\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} \equiv \mathbb{C}'\mathbb{V} + \mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{B}_2\mathbb{V}$. We have $\frac{1}{NT}\mathbb{C}'\mathbb{V} = O_p(\frac{1}{\sqrt{NT}})$ by Lemmas 8 and 10. According to Lemma 11, we have $\frac{1}{NT}\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{B}_2\mathbb{V}\right) = O_p(\frac{1}{\sqrt{NT^2}})$ and $\frac{1}{NT}E\left[\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{B}_2\mathbb{V}\right] = O(\frac{1}{T})$. Therefore,

$$\frac{1}{NT}\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\left[\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}\right] = O_p\left(\frac{1}{\sqrt{NT}}\right)$$

and $\frac{1}{NT}E\left(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}\right) = O(\frac{1}{T})$. Similarly,

$$\frac{1}{NT}\mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\left[\mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}\right] = O_p\left(\frac{1}{\sqrt{NT}}\right)$$

holds and $\frac{1}{NT}E\left(\mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}\right) = O(\frac{1}{T})$.

$\mathbb{B}_2'\mathbb{P}\mathbb{B}_1\mathbb{Z}^{(\gamma)}$ and each column of $\mathbb{B}_2'\mathbb{P}\mathbb{B}_1\mathbb{Z}_4$ can be viewed as $\mathbb{C}$ because elements of all these vectors are uniformly bounded. Then $\mathbb{Z}_3$ and $\mathbb{Z}_4$ satisfy the same conclusion because $\frac{1}{NT}\mathbb{C}'\mathbb{V} = O_p(\frac{1}{\sqrt{NT}})$ by Lemma 8. Thus, equation (A.28) holds and $\frac{1}{NT}E\left(\mathbb{Z}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V}\right) = O(\frac{1}{T})$.

Proof of equation (A.29)

We have

$$\mathbb{Z}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z} = \begin{pmatrix} \mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1 & \mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_2 & \mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} & \mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_4 \\ \mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1 & \mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_2 & \mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} & \mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_4 \\ \mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1 & \mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_2 & \mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} & \mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_4 \\ \mathbb{Z}_4'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1 & \mathbb{Z}_4'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_2 & \mathbb{Z}_4'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} & \mathbb{Z}_4'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_4 \end{pmatrix}.$$

Similar to the proof of Lemma 13, we can first decompose $\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1 \equiv (\mathbb{C}_1 + \mathbb{L}^\ell)'\mathbb{P}(\mathbb{C}_2 + \mathbb{R}^\ell) = \mathbb{C}_1'\mathbb{P}\mathbb{C}_2 + \mathbb{C}_1'\mathbb{P}\mathbb{R}^\ell + \mathbb{C}_2'\mathbb{P}\mathbb{L}^\ell + \mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{R}^\ell$, where $\frac{1}{NT}\mathbb{C}'_1\mathbb{P}\mathbb{R}^\ell = O_p\left(\frac{1}{\sqrt{NT}}\right)$ and $\frac{1}{NT}\mathbb{C}'_2\mathbb{P}\mathbb{L}^\ell = O_p\left(\frac{1}{\sqrt{NT}}\right)$ by Lemmas 8 and 10, and $\frac{1}{NT}\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{R}^\ell - \frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{R}^\ell\right) = O_p\left(\frac{1}{\sqrt{NT^2}}\right)$, $\frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{R}^\ell\right) = O(\frac{1}{T})$ by Lemmas 7 and 10. Moreover, $\frac{1}{NT}\left|\mathbb{C}_1'\mathbb{P}\mathbb{C}_2\right| = O(1)$ because elements in $\mathbb{C}$ are uniformly bounded and $\mathbb{P}$ is a projection matrix. Hence,

$$\frac{1}{NT}\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1 - \frac{1}{NT}E\left[\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1\right] = O_p\left(\frac{1}{\sqrt{NT}}\right)$$

and $\frac{1}{NT}E\left(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1\right) = O(1)$. The same conclusion holds for $\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_2$, $\mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1$ and $\mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_2$ as they can undergo similar decomposition.

Secondly, we have $\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} \equiv (\mathbb{C}_1 + \mathbb{L}^\ell)'\mathbb{P}\mathbb{C}_2 = \mathbb{C}_1'\mathbb{P}\mathbb{C}_2 + \mathbb{C}_2'\mathbb{P}\mathbb{L}^\ell$. Similarly, we have

$$\frac{1}{NT}\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} - \frac{1}{NT}E\left[\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)}\right] = O_p\left(\frac{1}{\sqrt{NT}}\right)$$

and $\frac{1}{NT}E\left(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)}\right) = O(1)$ by Lemmas 8 and 10, which also holds for $\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1$, $\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_4$, $\mathbb{Z}_4'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_1$, $\mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)}$, $\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_2$, $\mathbb{Z}_2'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_4$ and $\mathbb{Z}_4'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_2$.

Lastly, the scalar $\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)}$, as well as each element of the vector $\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_4$, $\mathbb{Z}_4\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)}$ and the matrix $\mathbb{Z}_4'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}_4$ is of the form $\mathbb{C}_1'\mathbb{P}\mathbb{C}_2$ and $\frac{1}{NT}\mathbb{C}_1'\mathbb{P}\mathbb{C}_2 = O(1)$. Therefore, equation (A.29) follows and $\frac{1}{NT}E[\mathbb{Z}'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}] = O(1)$. □

Lemma 15. *The results in Lemmas 13 and 14 still hold when $\mathbb{Z}$ is replaced by $\mathbb{Z}\widetilde{\Gamma}$, where $\widetilde{\Gamma} = \text{blkdiag}(I_2, \sqrt{T}, I_k)$. Furthermore,*

$$\sqrt{T}\mathbb{Z}^{(-\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O_p\left(N\sqrt{T}\right) \qquad T\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(NT).$$

Proof. For equations (A.25) and (A.28) with $\mathbb{Z}$ replaced by $\mathbb{Z}\widetilde{\Gamma}$, we only need to further show that

$$\frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right) \qquad \text{and} \qquad \frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right),$$

which have been proved in Lemma 9.

For equations (A.26) and (A.29) with $\mathbb{Z}$ replaced by $\mathbb{Z}\widetilde{\Gamma}$, since $\mathbb{Z}_1$ and $\mathbb{Z}_2$ take the same form and $\mathbb{Z}^{(\gamma)}$ and $\mathbb{Z}_4$ are nonstochastic, we need to further show that

$$\frac{1}{N}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(1), \qquad \frac{1}{N}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(1),$$
$$\frac{\sqrt{T}}{NT}\mathbb{Z}_4'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O\left(\frac{1}{\sqrt{T}}\right), \qquad \frac{\sqrt{T}}{NT}\mathbb{Z}_4'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O\left(\frac{1}{\sqrt{T}}\right),$$

and

$$\frac{\sqrt{T}}{NT}\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} - \frac{\sqrt{T}}{NT}E(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)}) = O_p\left(\frac{1}{\sqrt{NT}}\right),$$
$$\frac{\sqrt{T}}{NT}\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} - \frac{\sqrt{T}}{NT}E(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)}) = O_p\left(\frac{1}{\sqrt{NT}}\right).$$

Firstly, we can prove $\frac{1}{N}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(1)$ and $\frac{1}{N}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(1)$ because $\mathbb{B}_1'\mathbb{P}\mathbb{B}_2$ and $\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2$ are UB and $\mathbb{Z}^{(\gamma)}$ is a sparse binary vector with at most N elements.

Secondly, since elements in $\mathbb{Z}_4$ are uniformly bounded, $\|\mathbb{Z}_4\|_\infty = O(1)$, $\|\mathbb{B}_2'\mathbb{Q}\mathbb{B}_1\mathbb{Z}_4\|_\infty \leqslant \|\mathbb{B}_2'\mathbb{Q}\mathbb{B}_1\|_\infty\|\mathbb{Z}_4\|_\infty = O(1)$, which means that elements in $\mathbb{B}_2'\mathbb{Q}\mathbb{B}_1\mathbb{Z}_4$ are also uniformly bounded. Therefore, $\frac{\sqrt{T}}{NT}\mathbb{Z}_4'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O\left(\frac{1}{\sqrt{T}}\right)$ by sparsity of $\mathbb{Z}^{(\gamma)}$ and we can similarly prove $\frac{\sqrt{T}}{NT}\mathbb{Z}_4'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O\left(\frac{1}{\sqrt{T}}\right)$.

Lastly, $\mathbb{B}_1\mathbb{Z}_1$ can be viewed as $\mathbb{L}^\ell + \mathbb{C}$, where $\frac{\sqrt{T}}{NT}\mathbb{C}\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O\left(\frac{1}{\sqrt{T}}\right)$ and $\frac{\sqrt{T}}{NT}\mathbb{C}\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O\left(\frac{1}{\sqrt{T}}\right)$. Moreover, we have $\frac{\sqrt{T}}{NT}\mathbb{L}^{\ell\prime}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O_p\left(\frac{1}{\sqrt{NT}}\right)$ and $\frac{\sqrt{T}}{NT}\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O_p\left(\frac{1}{\sqrt{NT}}\right)$ by Lemma 9. Taken together, we can prove

$$\frac{\sqrt{T}}{NT}\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} - \frac{\sqrt{T}}{NT}E(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)}) = O_p\left(\frac{1}{\sqrt{NT}}\right),$$
$$\frac{\sqrt{T}}{NT}\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)} - \frac{\sqrt{T}}{NT}E(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)}) = O_p\left(\frac{1}{\sqrt{NT}}\right),$$

and $\frac{\sqrt{T}}{NT}E(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)}) = O\left(\frac{1}{\sqrt{T}}\right)$, $\frac{\sqrt{T}}{NT}E(\mathbb{Z}_1'\mathbb{B}_1'\mathbb{P}\mathbb{B}_2\mathbb{Z}^{(\gamma)}) = O\left(\frac{1}{\sqrt{T}}\right)$.

The above proof also indicates that $\sqrt{T}\mathbb{Z}^{(-\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O_p\left(N\sqrt{T}\right)$ and $T\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\mathbb{Q}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(NT)$. □

Based on the definition of $\widetilde{\mathbb{Z}}^*$ in Section 3.2, the following lemma will be used in Appendix B.2 to show that the expectation of the Hessian matrix and the corresponding part of the variance of the score function are asymptotically equivalent up to an $O(1/T)$ remainder.

Lemma 16. *Under Assumptions 1-7 and A.1, we have*

$$\frac{1}{NT}E\left(\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma}\right) - \frac{1}{NT}E\left(\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{*\prime}\widetilde{\mathbb{Z}}^*\widetilde{\Gamma}\right) = O\left(\frac{1}{T}\right) \tag{A.30}$$
$$\frac{1}{NT}E\left(\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{G}_0\mathbb{Z}\right) - \frac{1}{NT}E\left(\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{*\prime}\widetilde{\mathbb{G}_0\mathbb{Z}}^*\right) = O\left(\frac{1}{T}\right) \tag{A.31}$$
$$\frac{1}{NT}E\left(\mathbb{Z}'\mathbb{G}_0'\mathbb{Q}\mathbb{G}_0\mathbb{Z}\right) - \frac{1}{NT}E\left(\widetilde{\mathbb{G}_0\mathbb{Z}}^{*\prime}\widetilde{\mathbb{G}_0\mathbb{Z}}^*\right) = O\left(\frac{1}{T}\right) \tag{A.32}$$

where $\frac{1}{NT}E\left(\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{*\prime}\widetilde{\mathbb{Z}}^*\widetilde{\Gamma}\right)$, $\frac{1}{NT}E\left(\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{*\prime}\widetilde{\mathbb{G}_0\mathbb{Z}}^*\right)$ *and* $\frac{1}{NT}E\left(\widetilde{\mathbb{G}_0\mathbb{Z}}^{*\prime}\widetilde{\mathbb{G}_0\mathbb{Z}}^*\right)$ *are all* $O(1)$.

Proof. Recall that $\widetilde{\mathbb{Z}}^* = \mathbb{Q}\mathbb{Z} + \mathbb{P}\mathbb{Z}^\circ$,

$$\frac{1}{NT}E\left(\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma}\right) - \frac{1}{NT}E\left(\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{*\prime}\widetilde{\mathbb{Z}}^*\widetilde{\Gamma}\right) = \frac{1}{NT}E\left(\widetilde{\Gamma}\mathbb{Z}^{\circ\prime}\mathbb{P}\mathbb{Z}^\circ\widetilde{\Gamma}\right) = \frac{1}{NT}E\left(\mathbb{Z}^{\circ\prime}\mathbb{P}\mathbb{Z}^\circ\right),$$

where the last equality holds by $\mathbb{Z}^\circ = (Z_1^{\circ\prime}, \cdots, Z_T^{\circ\prime})'$ and $Z_t^\circ = \left(D_t(I_N - F_t)D_t'M_t\mathcal{V}_{t-1}, D_tD_{t-1}'\mathcal{V}_{t-1}, \mathbf{0}_{N_t\times(k+1)}\right)$. Since $\mathbb{Z}^\circ$ takes the form of $(\mathbb{L}^\ell, \mathbb{R}^\ell, \mathbf{0}_{n\times(k+1)})$, we have $\frac{1}{NT}E\left(\mathbb{Z}^{\circ\prime}\mathbb{P}\mathbb{Z}^\circ\right) = O\left(\frac{1}{T}\right)$ by Lemma 7. Additionally, we have $\frac{1}{NT}E\left(\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma}\right) = O(1)$ by Lemmas 13 and 15. Thus, equation (A.30) holds and $\frac{1}{NT}E\left(\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{*\prime}\widetilde{\mathbb{Z}}^*\widetilde{\Gamma}\right) = O(1)$.

Equations (A.31) and (A.32) can be proved analogously by the aforementioned method. □

Lemma 17. *Under Assumptions 2 and A.1,* $E\left[\left(L_t^\ell\right)_i^4\right] = O(1)$ *uniformly for* i, t, N.

Proof. Denote e_i as an N_t dimensional vector where the i-th component equals 1 and all other components are 0. For simplicity, let $B_t^\ell \equiv B_{1t}^\ell B_{1,t-1}$. Thus, $(L_t^\ell)_i = \sum_{h=0}^\infty e_i' B_t^\ell P_{t-1}^{(h)} V_{t-1-h}$. Then using Lemma 2, we have

$$
\begin{aligned}
&E\left[\left(L_t^\ell\right)_i^4\right] \\
=&E\left[\left(\sum_{h_1=0}^\infty e_i' B_t^\ell P_{t-1}^{(h_1)} V_{t-1-h_1}\right)'\left(\sum_{h_2=0}^\infty e_i' B_t^\ell P_{t-1}^{(h_2)} V_{t-1-h_2}\right)\left(\sum_{h_3=0}^\infty e_i' B_t^\ell P_{t-1}^{(h_3)} V_{t-1-h_3}\right)'\right. \\
&\left.\left(\sum_{h_4=0}^\infty e_i' B_t^\ell P_{t-1}^{(h_4)} V_{t-1-h_4}\right)\right] \\
=&\sum_{h_1,h_2,h_3,h_4=0}^\infty E\left[\left(V_{t-1-h_1}' P_{t-1}^{(h_1)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h_2)} V_{t-1-h_2}\right)\left(V_{t-1-h_3}' P_{t-1}^{(h_3)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h_4)} V_{t-1-h_4}\right)\right] \\
=&\sum_{h=0}^\infty(\mu_4-3\sigma_0^4)\sum_{j=1}^{N_{t-1-h}}\left[\left(P_{t-1}^{(h)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h)}\right)_{jj}\right]^2 + \\
&\sum_{h_1=0}^\infty\sum_{h_2=0}^\infty \sigma_0^4 \mathrm{tr}\left(P_{t-1}^{(h_1)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h_1)}\right)\mathrm{tr}\left(P_{t-1}^{(h_2)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h_2)}\right) + \\
&\sum_{h_1=0}^\infty\sum_{h_2=0}^\infty \sigma_0^4 \mathrm{tr}\left[P_{t-1}^{(h_1)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h_2)}\left(P_{t-1}^{(h_1)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h_2)}\right)'\right] + \\
&\sum_{h_1=0}^\infty\sum_{h_2=0}^\infty \sigma_0^4 \mathrm{tr}\left[P_{t-1}^{(h_1)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h_2)}\left(P_{t-1}^{(h_2)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h_1)}\right)'\right] \\
=&\sum_{h=0}^\infty(\mu_4-3\sigma_0^4)\sum_{j=1}^{N_{t-1-h}}\left[\left(P_{t-1}^{(h)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h)}\right)_{jj}\right]^2 + \\
&\sigma_0^4\sum_{h_1=0}^\infty\sum_{h_2=0}^\infty\left[\left(e_i' B_t^\ell P_{t-1}^{(h_1)} P_{t-1}^{(h_1)\prime} B_t^{\ell\prime} e_i\right)\left(e_i' B_t^\ell P_{t-1}^{(h_2)} P_{t-1}^{(h_2)\prime} B_t^{\ell\prime} e_i\right) + \right. \\
&\left.\left(e_i' B_t^\ell P_{t-1}^{(h_1)} P_{t-1}^{(h_1)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h_2)} P_{t-1}^{(h_2)\prime} B_t^{\ell\prime} e_i\right) + \left(e_i' B_t^\ell P_{t-1}^{(h_2)} P_{t-1}^{(h_1)\prime} B_t^{\ell\prime} e_i e_i' B_t^\ell P_{t-1}^{(h_2)} P_{t-1}^{(h_1)\prime} B_t^{\ell\prime} e_i\right)\right] \\
=&\sum_{h=0}^\infty(\mu_4-3\sigma_0^4)\sum_{j=1}^{N_{t-1-h}}\left[\left(B_t^\ell P_{t-1}^{(h)}\right)_{ij}\right]^4 + 2\sigma_0^4\left[\sum_{h=0}^\infty\left(B_t^\ell P_{t-1}^{(h)} P_{t-1}^{(h)\prime} B_t^{\ell\prime}\right)_{ii}\right]^2 + \\
&\sigma_0^4\sum_{h_1=0}^\infty\sum_{h_2=0}^\infty\left[\left(B_t^\ell P_{t-1}^{(h_2)} P_{t-1}^{(h_1)\prime} B_t^{\ell\prime}\right)_{ii}\right]^2.
\end{aligned}
$$

$$
\begin{aligned}
\sum_{h=0}^\infty(\mu_4-3\sigma_0^4)\sum_{j=1}^{N_{t-1-h}}\left[\left(B_t^\ell P_{t-1}^{(h)}\right)_{ij}\right]^4 &= \sum_{h=0}^\infty(\mu_4-3\sigma_0^4)\sum_{j=1}^{N}\left[\left(B_t^\ell P_{t-1}^{(h)} D_{t-1-h}\right)_{ij}\right]^4 \\
&\leqslant (\mu_4-3\sigma_0^4)\left[\sum_{j=1}^N\sum_{h=0}^\infty \mathrm{abs}\left(B_t^\ell P_{t-1}^{(h)} D_{t-1-h}\right)_{ij}\right]^4 \\
&= O(1),
\end{aligned}
$$

where the last equality holds because matrix $\sum_{h=0}^\infty \mathrm{abs}\left(B_t^\ell P_{t-1}^{(h)} D_{t-1-h}\right)$ is UB uniformly for t according to

Lemma 4. Moreover,

$$\left[\sum_{h=0}^{\infty}\left(B_t^{\ell}P_{t-1}^{(h)}P_{t-1}^{(h)\prime}B_t^{\ell\prime}\right)_{ii}\right]^2 \leqslant \left\{\left[\sum_{h=0}^{\infty}\text{abs}\left(B_t^{\ell}P_{t-1}^{(h)}P_{t-1}^{(h)\prime}B_t^{\ell\prime}\right)\right]_{ii}\right\}^2 = O(1),$$

where the equality holds because $\sum_{h=0}^{\infty}\text{abs}\left(B_t^{\ell}P_{t-1}^{(h)}P_{t-1}^{(h)\prime}B_t^{\ell\prime}\right)$ is UB according to Lemma 4. Similarly,

$$\sum_{h_1=0}^{\infty}\sum_{h_2=0}^{\infty}\left[\left(B_t^{\ell}P_{t-1}^{(h_2)}P_{t-1}^{(h_1)\prime}B_t^{\ell\prime}\right)_{ii}\right]^2 \leqslant \left\{\left[\sum_{h_1=0}^{\infty}\sum_{h_2=0}^{\infty}\text{abs}\left(B_t^{\ell}P_{t-1}^{(h_2)}D_{t-1-h}D_{t-1-h}'P_{t-1}^{(h_1)\prime}B_t^{\ell\prime}\right)\right]_{ii}\right\}^2$$
$$\leqslant \left[\left(\sum_{h_2=0}^{\infty}\text{abs}\left(B_t^{\ell}P_{t-1}^{(h_2)}D_{t-1-h}\right)\right)\left(\sum_{h_1=0}^{\infty}\text{abs}\left(B_t^{\ell}P_{t-1}^{(h_1)}D_{t-1-h}\right)\right)\right]_{ii}^2 = O(1),$$

where the equality holds because $\text{abs}\left(B_t^{\ell}P_{t-1}^{(h)}D_{t-1-h}\right)$ is UB uniformly for t according to Lemma 4. Therefore, $E\left[\left(L_t^{\ell}\right)_i^4\right] = O(1)$. □

To establish the central limit theorem for $\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}$, let

$$\begin{aligned}\mathcal{Q}_n &\equiv \sum_{t=1}^{T}\left[V_t'B_{3t}V_t + V_t'L_t^{\ell} + V_t'\widetilde{C}_t - \sigma_0^2\text{tr}(B_{3t})\right]\\ &= \sum_{t=1}^{T}\left[V_t'B_{3t}V_t + V_t'B_{1t}^{\ell}L_{t-1} + V_t'\left(C_t + a\sqrt{T}D_tF_t\iota_N\right) - \sigma_0^2\text{tr}(B_{3t})\right] \equiv \sum_{t=1}^{T}\sum_{i=1}^{N_t}\mathfrak{q}_{n,ti},\end{aligned} \tag{A.33}$$

where B_{3t} is an arbitrary symmetric[1] UB matrix uniformly for t. The N_t-dimensional vector $\widetilde{C}_t \equiv C_t + a\sqrt{T}D_tF_t\iota_N$, where C_t is an arbitrary vector satisfying Assumption A.2, and a is a constant. Furthermore, B_{1t}^{ℓ} is an arbitrary matrix satisfying Assumption A.1. $\mathfrak{q}_{n,ti} \equiv 2\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j}V_{t,i} + [(L_t^{\ell})_i + (\widetilde{C}_t)_i]V_{t,i} + (B_{3t})_{ii}(V_{t,i}^2 - \sigma_0^2)$, and $V_{t,i}$ is the i-th element in V_t. Linear combinations of terms satisfying equation (A.33) can still be written in the same form and do not violate the original assumptions, because

$$\begin{aligned}&w^{(1)}\mathcal{Q}_n^{(1)} + w^{(2)}\mathcal{Q}_n^{(2)}\\ &= \sum_{t=1}^{T}\left[V_t'\left(w^{(1)}B_{3t}^{(1)} + w^{(2)}B_{3t}^{(2)}\right)V_t + V_t'\left(w^{(1)}B_{1t}^{\ell(1)} + w^{(2)}B_{1t}^{\ell(2)}\right)L_{t-1} +\right.\\ &\left. V_t'\left(w^{(1)}C_t^{(1)} + w^{(2)}C_t^{(2)} + (w^{(1)}a^{(1)} + w^{(2)}a^{(2)})\sqrt{T}D_tF_t\iota_N\right) - \sigma_0^2\text{tr}\left(w^{(1)}B_{3t}^{(1)} + w^{(2)}B_{3t}^{(2)}\right)\right].\end{aligned}$$

[1]The symmetry of B_3 can be imposed without loss of generality because $V_t'B_{3t}V_t = V_t'[(B_{3t} + B_{3t}')/2]V_t$.

Then we have $E(\mathcal{Q}_n) = 0$ and the variance is

$$
\begin{aligned}
\sigma^2_{\mathcal{Q}_n} =& E\left[\left(\sum_{t=1}^{T} V_t' B_{3t} V_t + V_t' L_t^{\ell} + V_t' \widetilde{C}_t - \sigma_0^2 \mathrm{tr}(B_{3t})\right)^2\right] \\
=& E\Bigg[\sum_{t=1}^{T}\sum_{s=1}^{T}\Big(V_t' B_{3t} V_t V_s' B_{3s} V_s + 2V_t' B_{3t} V_t V_s'(L_s^{\ell} + \widetilde{C}_s) - 2V_t' B_{3t} V_t \sigma_0^2 \mathrm{tr}(B_{3s}) + \\
& V_t'(L_t^{\ell} + \widetilde{C}_t) V_s'(L_s^{\ell} + \widetilde{C}_s) - 2V_t'(L_t^{\ell} + \widetilde{C}_t)\sigma_0^2 \mathrm{tr}(B_{3s}) + \sigma_0^4 \mathrm{tr}(B_{3t})\mathrm{tr}(B_{3s})\Big)\Bigg] \\
=& \sum_{t=1}^{T}\left[(\mu_4 - 3\sigma_0^4)\sum_{i=1}^{N_t}(B_{3t})_{ii}^2 + 2\sigma_0^4 \mathrm{tr}(B_{3t}^2)\right] + 2\mu_3 \sum_{t=1}^{T}\sum_{i=1}^{N_t}(B_{3t})_{ii}(\widetilde{C}_t)_i + \\
& \sigma_0^4 \sum_{t=1}^{T}\sum_{h=0}^{\infty}\mathrm{tr}(B_{1t}^{\ell} P_{t-1,h} P_{t-1,h}' B_{1t}^{\ell\prime}) + \sigma_0^2 \sum_{t=1}^{T}\widetilde{C}_t'\widetilde{C}_t \\
=& \sum_{t=1}^{T}\left[(\mu_4 - 3\sigma_0^4)\sum_{i=1}^{N_t}(B_{3t})_{ii}^2 + 2\sigma_0^4 \mathrm{tr}(B_{3t}^2)\right] + 2\mu_3 \sum_{t=1}^{T}\sum_{i=1}^{N_t}(B_{3t})_{ii}\left[(C_t)_i + a\sqrt{T}(D_t F_t \iota_N)_i\right] + \\
& + \sigma_0^4 \sum_{t=1}^{T}\mathrm{tr}\left[B_{1t}^{\ell}\sum_{h=0}^{\infty}\left(P_{t-1,h} P_{t-1,h}'\right) B_{1t}^{\ell\prime}\right] + \sigma_0^2 \sum_{t=1}^{T}\left(C_t + a\sqrt{T} D_t F_t \iota_N\right)'\left(C_t + a\sqrt{T} D_t F_t \iota_N\right) \\
=& O(NT),
\end{aligned}
$$

where the last equality holds because B_{3t} is UB uniformly for t, elements in C_t are uniformly bounded, $F_t \iota_N$ is a sparse vector with fewer than N nonzero entries in total over $t = 1, \cdots, T$, and the matrix inside the trace operation is UB by Lemma 4.

Lemma 18. *Under Assumptions 2, 7, A.1 and A.2, if the sequence* $\frac{1}{n}\sigma^2_{\mathcal{Q}_n}$ *is bounded away from zero, then* $\frac{\mathcal{Q}_n}{\sigma_{\mathcal{Q}_n}} \xrightarrow{d} N(0,1)$.

Proof. The proof of central limit theorem follows the idea in Yu et al. (2008). Denote the σ-field

$$
\mathcal{F}_{n,ti} = \sigma\left(\left(\bigcup_{s=1}^{\infty}\{V_{t-s,j} \mid j = 1, 2, \cdots, N_{t-s}\}\right) \cup \{V_{t,j} \mid j = 1, 2, \cdots, i\}\right),
$$
$$
\text{for } t = 1, 2 \cdots, T, \text{ and } i = 1, 2, \cdots, N_t,
$$

which is generated by the i-th node's error observed at time t and all its historical errors, where "historical" includes both errors from all time periods preceding t and those within period t that are ordered before the i-th node. Define $\mathcal{F}_{n,t0} = \mathcal{F}_{n,t-1,N_{t-1}}$. To merge the two indices t and i into a single index, define a global index $j = \sum_{s=1}^{t-1} N_s + i$, where j ranges from 1 to n. The σ-field $\mathcal{F}_{n,ti}$ is then simplified as $\mathcal{F}_{n,j}$, and $\mathfrak{q}_{n,ti}$ is simplified as $\mathfrak{q}_{n,j}$. Thus, $E(\mathfrak{q}_{n,j}|\mathcal{F}_{n,j-1}) = 0$, which forms a martingale difference array. To apply the martingale CLT to $\sum_{j=1}^{n}\mathfrak{q}_{n,j} = \sum_{t=1}^{T}\sum_{i=1}^{N_t}\mathfrak{q}_{n,ti}$, the sufficient conditions are (i) $\frac{1}{\sigma^{2+\eta}_{\mathcal{Q}_n}}\sum_{t=1}^{T}\sum_{i=1}^{N_t}|\mathfrak{q}_{n,ti}|^{2+\eta} \to 0$ and (ii) $\frac{1}{\sigma^2_{\mathcal{Q}_n}}\sum_{t=1}^{T}\sum_{i=1}^{N_t}E(\mathfrak{q}^2_{n,ti}|\mathcal{F}_{n,t,i-1}) \xrightarrow{p} 1$.

Proof of (i)

For any $p > 0$, $q > 0$ with $\frac{1}{p} + \frac{1}{q} = 1$, we have

$$
\begin{aligned}
&|\mathfrak{q}_{n,ti}| \\
\leqslant& \left(\left|(L_t^\ell)_i\right| + \left|(\widetilde{C}_t)_i\right|\right)|V_{t,i}| + |(B_{3t})_{ii}|^{\frac{1}{p}}|(B_{3t})_{ii}|^{\frac{1}{q}}\left|V_{t,i}^2 - \sigma_0^2\right| + \sum_{j=1}^{i-1}|(B_{3t})_{ij}|^{\frac{1}{p}}\left(2|(B_{3t})_{ij}|^{\frac{1}{q}}|V_{t,j}||V_{t,i}|\right) \\
\leqslant& \left[\left(\left|(L_t^\ell)_i\right| + \left|(\widetilde{C}_t)_i\right|\right)^p + \sum_{j=1}^{i}|(B_{3t})_{ij}|\right]^{\frac{1}{p}}\left[|V_{t,i}|^q + |(B_{3t})_{ii}|\left|V_{t,i}^2 - \sigma_0^2\right|^q + \sum_{j=1}^{i-1}2^q|(B_{3t})_{ij}||V_{t,j}|^q|V_{t,i}|^q\right]^{\frac{1}{q}}
\end{aligned}
$$

where the second inequality is derived by applying Hölder's inequality. Thus,

$$
\begin{aligned}
E\left[|\mathfrak{q}_{n,ti}|^q\right] \leqslant& E\left[\left(\left(\left|(L_t^\ell)_i\right| + \left|(\widetilde{C}_t)_i\right|\right)^p + \sum_{j=1}^{i}|(B_{3t})_{ij}|\right)^{\frac{q}{p}}\right] \times \\
&\left[E\left(|V_{t,i}|^q\right) + |(B_{3t})_{ii}|\,E\left(\left|V_{t,i}^2 - \sigma_0^2\right|^q\right) + 2^q E\left(|V_{t,i}|^q\right)\sum_{j=1}^{i-1}|(B_{3t})_{ij}|E\left(|V_{t,j}|^q\right)\right],
\end{aligned}
$$

by the independence between V_t and L_t^ℓ. Let $q = 2 + \eta$ with a small η defined in Assumption 2, then $p = \frac{2+\eta}{1+\eta}$. There exists $c_1 > 0$ such that

$$
\sup_{t,i}\left[E|V_{t,i}|^q + |(B_{3t})_{ii}|E\left|V_{t,i}^2 - \sigma_0^2\right|^q + 2^q E|V_{t,i}|^q\sum_{j=1}^{i-1}|(B_{3t})_{ij}|E|V_{t,j}|^q\right] < c_1,
$$

due to $E|v_{it}|^{4+2\eta} < \infty$ by Assumption 2 and B_{3t} is UB uniformly for t. Therefore,

$$
\begin{aligned}
&\sum_{t=1}^{T}\sum_{i=1}^{N_t}E|\mathfrak{q}_{n,ti}|^{2+\eta} \\
\leqslant& c_1\sum_{t=1}^{T}\sum_{i=1}^{N_t}E\left[\left(\left(\left|(L_t^\ell)_i\right| + \left|(\widetilde{C}_t)_i\right|\right)^p + \sum_{j=1}^{i}|(B_{3t})_{ij}|\right)^{\frac{q}{p}}\right] \\
\leqslant& 2^\eta c_1\sum_{t=1}^{T}\sum_{i=1}^{N_t}E\left[\left(\left|(L_t^\ell)_i\right| + |(C_t)_i| + \left|a\sqrt{T}(D_tF_t\iota_N)_i\right|\right)^{2+\eta}\right] + \sum_{t=1}^{T}\sum_{i=1}^{N_t}2^\eta c_1\left(\sum_{j=1}^{i}|(B_{3t})_{ij}|\right)^{1+\eta} \\
\leqslant& 2^{1+2\eta}c_1\sum_{t=1}^{T}\sum_{i=1}^{N_t}\left[E\left(\left|(L_t^\ell)_i\right|^{2+\eta}\right) + |(C_t)_i|^{2+\eta} + T^{1+\frac{\eta}{2}}\left|a(D_tF_t\iota_N)_i\right|^{2+\eta} + \left(\sum_{j=1}^{n}|(B_{3t})_{ij}|\right)^{1+\eta}\right] \\
=& O\left(NT^{1+\frac{\eta}{2}}\right),
\end{aligned}
$$

where the last two inequalities are derived by applying the C_r-inequality. By Lemma 17, which establishes $E\left(\left|(L_t^\ell)_i\right|^4\right) = O(1)$ uniformly for t and i, combined with the boundedness of C_t, the sparsity of $F_t\iota_N$, and B_{3t} being UB uniformly for t, the last equality follows. Since the sequence $\frac{1}{n}\sigma_{\mathcal{Q}_n}^2$ is bounded away from zero, then $\sigma_{\mathcal{Q}_n}^{2+\eta} = O\left((NT)^{1+\frac{\eta}{2}}\right)$ and it is nonnegligible at this order, we have $\frac{1}{\sigma_{\mathcal{Q}_n}^{2+\eta}}\sum_{j=1}^{n}E|\mathfrak{q}_{n,j}|^{2+\eta} = O\left(N^{-\frac{\eta}{2}}\right)$, which proves condition (i).

Proof of (ii)

Since

$$E\left(\mathfrak{q}_{n,ti}^2|\mathcal{F}_{n,t,i-1}\right) = \left[2\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j} + (L_t^\ell)_i + (\widetilde{C}_t)_i\right]^2 E(V_{t,i}^2) + (B_{3t})_{ii}^2 E\left[(V_{t,i}^2-\sigma_0^2)^2\right] +$$
$$2\left[2\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j} + (L_t^\ell)_i + (\widetilde{C}_t)_i\right](B_{3t})_{ii}E\left[V_{t,i}(V_{t,i}^2-\sigma_0^2)\right]$$
$$=\sigma_0^2\left[2\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j} + (L_t^\ell)_i + (\widetilde{C}_t)_i\right]^2 + (\mu_4-\sigma_0^4)(B_{3t})_{ii}^2 +$$
$$2\mu_3(B_{3t})_{ii}\left[2\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j} + (L_t^\ell)_i + (\widetilde{C}_t)_i\right].$$

Then

$$\sum_{t=1}^{T}\sum_{i=1}^{N_t} E\left(\mathfrak{q}_{n,ti}^2|\mathcal{F}_{n,t,i-1}\right)$$
$$= \sum_{t=1}^{T}\sum_{i=1}^{N_t}\sigma_0^2\left[2\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j} + (L_t^\ell)_i + (\widetilde{C}_t)_i\right]^2 + \sum_{t=1}^{T}\sum_{i=1}^{N_t}(\mu_4-\sigma_0^4)(B_{3t})_{ii}^2$$
$$+ \sum_{t=1}^{T}\sum_{i=1}^{N_t} 2\mu_3(B_{3t})_{ii}\left[2\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j} + (L_t^\ell)_i + (\widetilde{C}_t)_i\right]$$
$$= \sum_{t=1}^{T}\sum_{i=1}^{N_t} 4\sigma_0^2\left[\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j}\right]^2 + \sum_{t=1}^{T}\sum_{i=1}^{N_t}\sigma_0^2(L_t^\ell)_i^2 + \sum_{t=1}^{T}\sum_{i=1}^{N_t}\sigma_0^2(\widetilde{C}_t)_i^2$$
$$+ \sum_{t=1}^{T}\sum_{i=1}^{N_t}\left[4\sigma_0^2(C_t)_i + 4\mu_3(B_{3t})_{ii}\right]\left[\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j}\right] + \sum_{t=1}^{T}\sum_{i=1}^{N_t}[2\sigma_0^2(C_t)_i + 2\mu_3(B_{3t})_{ii}](L_t^\ell)_i$$
$$+ \sum_{t=1}^{T}\sum_{i=1}^{N_t} 4a\sqrt{T}(D_tF_t\iota_N)_i\left[\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j}\right] + \sum_{t=1}^{T}\sum_{i=1}^{N_t} 2a\sqrt{T}(D_tF_t\iota_N)_i(L_t^\ell)_i$$
$$+ \sum_{t=1}^{T}\sum_{i=1}^{N_t} 4\sigma_0^2\left[\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j}\right](L_t^\ell)_i + \sum_{t=1}^{T}(\mu_4-\sigma_0^4)\sum_{i=1}^{N_t}(B_{3t})_{ii}^2 + 2\mu_3\sum_{t=1}^{T}\sum_{i=1}^{N_t}(B_{3t})_{ii}(\widetilde{C}_t)_i.$$

Denote $\mathbb{B}_3 \equiv \text{blkdiag}(B_{31}, B_{32}, \cdots, B_{3T})$, then

$$\sum_{t=1}^{T}\sum_{i=1}^{N_t}\left[\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j}\right]^2 = \sum_{t=1}^{T}\sum_{i=1}^{N_t} V_t'B_{3t}^{-\prime}B_{3t}^{-}V_t = \mathbb{V}'\mathbb{B}_3^{-\prime}\mathbb{B}_3^{-}\mathbb{V},$$

thus by equation (A.20) from Lemma 12,

$$\frac{1}{NT}\mathbb{V}'\mathbb{B}_3^{-\prime}\mathbb{B}_3^{-}\mathbb{V} - \frac{1}{NT}E(\mathbb{V}'\mathbb{B}_3^{-\prime}\mathbb{B}_3^{-}\mathbb{V}) = O_p\left(\frac{1}{\sqrt{NT}}\right),$$

where $E(\mathbb{V}'\mathbb{B}_3^{-\prime}\mathbb{B}_3^{-}\mathbb{V}) = \sigma_0^2\sum_{t=1}^{T}\text{tr}(B_{3t}^{-\prime}B_{3t}^{-})$. Additionally, $\sum_{t=1}^{T}\sum_{i=1}^{N_t}\sigma_0^2(L_t^\ell)_i^2 = \sigma_0^2\sum_{t=1}^{T}L_t^{\ell\prime}L_t^\ell = \sigma_0^2\mathbb{L}^{\ell\prime}\mathbb{L}^\ell$, according to equation (A.2) and Lemma 10,

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\mathbb{L}^\ell - \frac{1}{NT}E(\mathbb{L}^{\ell\prime}\mathbb{L}^\ell) = O_P\left(\frac{1}{\sqrt{NT}}\right),$$

where $E(\mathbb{L}^{\ell\prime}\mathbb{L}^{\ell}) = \sigma_0^2 \sum_{t=1}^{T} \text{tr}(\sum_{h=0}^{\infty} B_{1t}^{\ell} P_{t-1,h} P_{t-1,h}' B_{1t}^{\ell\prime})$. Denote C_{2t} as an N_t dimensional vector such that elements in C_{2t} are uniformly bounded and $(C_{2t})_i = 4\sigma_0^2 (C_t)_i + 4\mu_3 (B_{3t})_{ii}$, then

$$\sum_{t=1}^{T}\sum_{i=1}^{N_t}\left[4\sigma_0^2(C_t)_i + 4\mu_3(B_{3t})_{ii}\right]\left[\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j}\right] = \sum_{t=1}^{T} C_{2t}' B_{3t}^{-} V_t = O_p\left(\sqrt{NT}\right)$$

by equation (A.8) in Lemma 8. Similarly,

$$\sum_{t=1}^{T}\sum_{i=1}^{N_t}[2\sigma_0^2(C_t)_i + 2\mu_3(B_{3t})_{ii}](L_t^{\ell})_i = \sum_{t=1}^{T} C_{2t}' L_t^{\ell} = O_p\left(\sqrt{NT}\right)$$

by equation (A.6) from Lemma 8. Moreover

$$\sum_{t=1}^{T}\sum_{i=1}^{N_t} a\sqrt{T}(D_t F_t \iota_N)_i \left[\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j}\right] = \sum_{t=1}^{T} a\sqrt{T}(D_t F_t \iota_N)' B_{3t}^{-} V_t = a\sqrt{T}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_3^{-}\mathbb{V}$$

and

$$\sum_{t=1}^{T}\sum_{i=1}^{N_t} a\sqrt{T}(D_t F_t \iota_N)_i (L_t^{\ell})_i = a\sqrt{T}\mathbb{Z}^{(\gamma)\prime}\mathbb{L}^{\ell},$$

which are both $O_p(\sqrt{NT})$ by equations (A.12) and (A.10) by Lemma 9. Furthermore, we have

$$\sum_{t=1}^{T}\sum_{i=1}^{N_t}\left[\sum_{j=1}^{i-1}(B_{3t})_{ij}V_{t,j}\right](L_t^{\ell})_i = \sum_{t=1}^{T} L_t^{\ell\prime} B_{3t}^{-} V_t = \mathbb{L}^{\ell\prime}\mathbb{B}_3^{-}\mathbb{V} = O_p\left(\sqrt{NT}\right)$$

by equation (A.14) in Lemma 11. Therefore,

$$\begin{aligned}
&\sum_{t=1}^{T}\sum_{i=1}^{N_t} E\left(\mathfrak{q}_{n,ti}^2 | \mathcal{F}_{n,t,i-1}\right) \\
&= \sum_{t=1}^{T} 4\sigma_0^4 \text{tr}\left(B_{3t}^{-\prime} B_{3t}^{-}\right) + \sigma_0^4 \sum_{t=1}^{T} \text{tr}\left(\sum_{h=0}^{\infty} B_{1t}^{\ell} P_{t-1,h} P_{t-1,h}' B_{1t}^{\ell\prime}\right) + \sum_{t=1}^{T}\sum_{i=1}^{N_t}\sigma_0^2(\widetilde{C}_t)_i^2 \\
&\quad + \sum_{t=1}^{T}(\mu_4 - \sigma_0^4)\sum_{i=1}^{N_t}(B_{3t})_{ii}^2 + 2\mu_3 \sum_{t=1}^{T}\sum_{i=1}^{N_t}(B_{3t})_{ii}(\widetilde{C}_t)_i + O_p(\sqrt{NT}).
\end{aligned}$$

Since B_{3t} is symmetric, which implies that $\text{tr}(B_{3t}^2) = 2tr(B_{3t}^{-\prime} B_{3t}^{-}) + \sum_{i=1}^{N_t}(B_{3t})_{ii}^2$. So

$$\frac{1}{NT} E\left(\mathfrak{q}_{n,ti}^2 | \mathcal{F}_{n,t,i-1}\right) = \frac{1}{NT}\sigma_{\mathcal{Q}_n}^2 + O_p\left(\frac{1}{\sqrt{NT}}\right),$$

which proves condition (ii). □

Additional lemmas for the TWFE case

Recall that $\mathbb{L}^{\ell}$ and $\mathbb{R}^{\ell}$ are defined above. We can write $\mathbb{L}^{\ell} = \mathbb{B}_1^{\ell}\mathcal{P}_{\infty}^{\ell}\mathbb{V}_{\infty}$ and $\mathbb{R}^{\ell} = \mathbb{B}_2^{\ell}\mathcal{Q}_{\infty}^{\ell}\mathbb{V}_{\infty}$, where $\mathbb{B}_1^{\ell} = \text{blkdiag}(B_{11}^{\ell}, \ldots, B_{1T}^{\ell}), \mathbb{B}_2^{\ell} = \text{blkdiag}(B_{21}^{\ell}, \ldots, B_{2T}^{\ell})$ are $n \times (\sum_{t=0}^{T-1} N_t)$ matrices, $\mathbb{V}_{\infty} \equiv (V_T', V_{T-1}', \ldots, V_1', V_0', V_{-1}', \ldots)'$ is infinity dimensional vector,

$$\mathcal{P}_{\infty}^{\ell} = \begin{pmatrix}
\mathbf{0}_{N_0 \times N_T} & \mathbf{0}_{N_0 \times N_{T-1}} & \cdots & \mathbf{0}_{N_0 \times N_1} & B_{10}P_0^{(0)} & B_{10}P_0^{(1)} & \cdots \\
\mathbf{0}_{N_1 \times N_T} & \mathbf{0}_{N_1 \times N_{T-1}} & \cdots & B_{11}P_1^{(0)} & B_{11}P_1^{(1)} & B_{11}P_1^{(2)} & \cdots \\
\vdots & \vdots & \iddots & \vdots & \vdots & \vdots & \ddots \\
\mathbf{0}_{N_{T-1} \times N_T} & B_{1,T-1}P_{T-1}^{(0)} & \cdots & B_{1,T-1}P_{T-1}^{(T-2)} & B_{1,T-1}P_{T-1}^{(T-1)} & B_{1,T-1}P_{T-1}^{(T)} & \cdots
\end{pmatrix},$$

is $(\sum_{t=0}^{T-1} N_t) \times \infty$ matrix, and $\mathcal{Q}_{\infty}^{\ell}$ is defined analogously.

Lemma 19. *Under Assumption A.1, both $\mathcal{P}_\infty^\ell$ and $\mathcal{Q}_\infty^\ell$ are UB. Consequently, $\mathbb{B}_1^\ell \mathcal{P}_\infty^\ell$ and $\mathbb{B}_2^\ell \mathcal{Q}_\infty^\ell$ are also UB.*

Proof. We only prove the claim for $\mathcal{P}_\infty^\ell$, since the proof for $\mathcal{Q}_\infty^\ell$ is identical. By Assumption A.1 and Lemma 4,

$$\sup_{N,T} \|\mathcal{P}_\infty^\ell\|_\infty \leqslant \sup_{N,T} \sup_{0\leqslant t\leqslant T-1} \left\| \sum_{h=0}^{\infty} \operatorname{abs}\Big(B_{1t} P_t^{(h)} D_{t-h}\Big) \right\|_\infty < \infty,$$

and

$$\sup_{N,T} \|\mathcal{P}_\infty^\ell\|_1 \leqslant \sup_{N,T} \sup_{t\leqslant T-1} \left\| \sum_{h=t}^{\infty} \operatorname{abs}\Big(D_h' B_{1h} P_h^{(h-t)}\Big) \right\|_1 < \infty.$$

Hence $\mathcal{P}_\infty^\ell$ is UB. Since $\mathbb{B}_1^\ell$ is UB by Assumption A.1, $\mathbb{B}_1^\ell \mathcal{P}_\infty^\ell$ is also UB. The proof for $\mathbb{B}_2^\ell \mathcal{Q}_\infty^\ell$ is the same. □

Lemma 20. *Suppose Assumption 2 holds. Let Π_n be an $n \times n$ nonstochastic matrix that is UB, and let $\mathbb{A}_{1,\infty}$ and $\mathbb{A}_{2,\infty}$ be two nonstochastic $n \times \infty$ matrices that are UB. Define $\mathbb{M}_n \equiv \mathbb{A}_{1,\infty}' \Pi_n \mathbb{A}_{2,\infty}$. Then*

$$E\big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\big] = \sigma_0^2 \operatorname{tr}(\mathbb{M}_n), \tag{A.34}$$

and

$$\operatorname{Var}\big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\big] = (\mu_4 - 3\sigma_0^4) \sum_{j=1}^{\infty} (\mathbb{M}_n)_{jj}^2 + \sigma_0^4 \operatorname{tr}(\mathbb{M}_n \mathbb{M}_n') + \sigma_0^4 \operatorname{tr}(\mathbb{M}_n^2).$$

Proof. We have

$$(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty) = \mathbb{V}_\infty' \mathbb{M}_n \mathbb{V}_\infty = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} (\mathbb{M}_n)_{jk} (\mathbb{V}_\infty)_j (\mathbb{V}_\infty)_k.$$

Since Π_n, $\mathbb{A}_{1,\infty}$ and $\mathbb{A}_{2,\infty}$ are UB, the $n \times n$ matrix $\Pi_n \mathbb{A}_{2,\infty} \mathbb{A}_{1,\infty}'$ is UB, and hence $\operatorname{tr}\big(\Pi_n \mathbb{A}_{2,\infty} \mathbb{A}_{1,\infty}'\big) = O(n)$. Similarly, the $n \times n$ matrices $\Pi_n \mathbb{A}_{2,\infty} \mathbb{A}_{2,\infty}' \Pi_n' \mathbb{A}_{1,\infty} \mathbb{A}_{1,\infty}'$ and $\Pi_n \mathbb{A}_{2,\infty} \mathbb{A}_{1,\infty}' \Pi_n \mathbb{A}_{2,\infty} \mathbb{A}_{1,\infty}'$ are also UB, and thus their traces are $O(n)$. Therefore, we have

$$\sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \big|(\mathbb{M}_n)_{jk}\big|^2 = \operatorname{tr}(\mathbb{M}_n \mathbb{M}_n') = \operatorname{tr}\big(\Pi_n \mathbb{A}_{2,\infty} \mathbb{A}_{2,\infty}' \Pi_n' \mathbb{A}_{1,\infty} \mathbb{A}_{1,\infty}'\big) < \infty.$$

where the cyclic permutation of factors is legitimate because the matrix $\Pi_n \mathbb{A}_{2,\infty} \mathbb{A}_{2,\infty}' \Pi_n' \mathbb{A}_{1,\infty} \mathbb{A}_{1,\infty}'$ is $n \times n$ and hence finite-dimensional. Additionally, $\sum_{j=1}^{\infty} \big|(\mathbb{M}_n)_{jj}\big|^2 < \infty$, and

$$\sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \big|(\mathbb{M}_n)_{jk}(\mathbb{M}_n)_{kj}\big| \leqslant \frac{1}{2} \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \big[(\mathbb{M}_n)_{jk}^2 + (\mathbb{M}_n)_{kj}^2\big] = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} (\mathbb{M}_n)_{jk}^2 = \operatorname{tr}(\mathbb{M}_n \mathbb{M}_n') < \infty.$$

Hence the relevant series are absolutely summable. In particular, Tonelli's theorem is used in the following two steps. First, for the expectation,

$$E\left[\sum_{j=1}^{\infty}\sum_{k=1}^{\infty} (\mathbb{M}_n)_{jk} (\mathbb{V}_\infty)_j (\mathbb{V}_\infty)_k\right] = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} (\mathbb{M}_n)_{jk} E\big[(\mathbb{V}_\infty)_j (\mathbb{V}_\infty)_k\big].$$

Since the elements of $\mathbb{V}_\infty$ are i.i.d. with mean zero and variance σ_0^2, only the terms with $j = k$ survive, so

$$E\big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\big] = \sigma_0^2 \sum_{j=1}^{\infty} (\mathbb{M}_n)_{jj} = \sigma_0^2 \operatorname{tr}(\mathbb{M}_n).$$

Second, for the second moment, Tonelli's theorem justifies the expansion

$$E\left[\left(\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}(\mathbb{M}_n)_{jk}(\mathbb{V}_\infty)_j(\mathbb{V}_\infty)_k\right)^2\right] = \sum_{j,k,l,m\geqslant 1}(\mathbb{M}_n)_{jk}(\mathbb{M}_n)_{lm}E[(\mathbb{V}_\infty)_j(\mathbb{V}_\infty)_k(\mathbb{V}_\infty)_l(\mathbb{V}_\infty)_m].$$

The same case-by-case classification as in the proof of Lemma 2 applies. The only nonzero contributions are

$$j=k=l=m, \qquad j=k\neq l=m, \qquad j=l\neq k=m, \qquad j=m\neq k=l.$$

Therefore,

$$\mathrm{Var}\big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\big] = (\mu_4-3\sigma_0^4)\sum_{j=1}^{\infty}(\mathbb{M}_n)_{jj}^2 + \sigma_0^4\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}(\mathbb{M}_n)_{jk}^2 + \sigma_0^4\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}(\mathbb{M}_n)_{jk}(\mathbb{M}_n)_{kj}.$$

Finally,

$$\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}(\mathbb{M}_n)_{jk}^2 = \mathrm{tr}(\mathbb{M}_n\mathbb{M}_n'), \qquad \sum_{j=1}^{\infty}\sum_{k=1}^{\infty}(\mathbb{M}_n)_{jk}(\mathbb{M}_n)_{kj} = \mathrm{tr}(\mathbb{M}_n^2),$$

which gives the desired result. □

Lemma 21. *Suppose Assumptions 1′, 2, 7 and A.1 hold. Let Π_n be either $\bar{\mathbb{P}}$ or $\bar{\mathbb{Q}}$, and let $\mathbb{A}_{1,\infty}$ and $\mathbb{A}_{2,\infty}$ be two nonstochastic $n\times\infty$ matrices that are UB. Then*

$$E\Big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\Big] = O\big(\mathrm{rank}(\Pi_n)\big).$$

Moreover,

$$\mathrm{Var}\Big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\Big] = O\big(\mathrm{rank}(\Pi_n)\big).$$

Consequently,

$$\frac{1}{NT}(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\bar{\mathbb{Q}}(\mathbb{A}_{2,\infty}\mathbb{V}_\infty) - \frac{1}{NT}E\Big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\bar{\mathbb{Q}}(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\Big] = O_p\left(\frac{1}{\sqrt{NT}}\right),$$

$$\frac{1}{NT}(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\bar{\mathbb{P}}(\mathbb{A}_{2,\infty}\mathbb{V}_\infty) - \frac{1}{NT}E\Big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\bar{\mathbb{P}}(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\Big] = O_p\left(\frac{1}{T\sqrt{N}}+\frac{1}{N\sqrt{T}}\right),$$

where $\frac{1}{NT}E\Big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\bar{\mathbb{Q}}(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\Big] = O(1)$ *and* $\frac{1}{NT}E\Big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\bar{\mathbb{P}}(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\Big] = O\big(\frac{1}{T}\big)+O\big(\frac{1}{N}\big)$.

Proof. By Lemma B.3 of Meng and Yang (2026b), both $\bar{\mathbb{P}}$ and $\bar{\mathbb{Q}}$ are UB. Hence Π_n is UB in either case. The expectation formula follows directly from Lemma 20:

$$E\Big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\Big] = \sigma_0^2\,\mathrm{tr}\Big(\Pi_n\mathbb{A}_{2,\infty}\mathbb{A}_{1,\infty}'\Big).$$

Moreover,

$$\begin{aligned}\big|\mathrm{tr}\big(\Pi_n\mathbb{A}_{2,\infty}\mathbb{A}_{1,\infty}'\big)\big| &\leqslant \mathrm{rank}\big(\Pi_n\mathbb{A}_{2,\infty}\mathbb{A}_{1,\infty}'\big)\,\big\|\Pi_n\mathbb{A}_{2,\infty}\mathbb{A}_{1,\infty}'\big\|_2 \\ &\leqslant \mathrm{rank}(\Pi_n)\,\big\|\Pi_n\mathbb{A}_{2,\infty}\mathbb{A}_{1,\infty}'\big\|_1^{1/2}\big\|\Pi_n\mathbb{A}_{2,\infty}\mathbb{A}_{1,\infty}'\big\|_\infty^{1/2} = O\big(\mathrm{rank}(\Pi_n)\big), \qquad \text{(A.35)}\end{aligned}$$

Therefore,

$$E\Big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\Big] = O\big(\mathrm{rank}(\Pi_n)\big).$$

For the variance, let $\mathbb{M}_n \equiv \mathbb{A}'_{1,\infty}\Pi_n\mathbb{A}_{2,\infty}$. By Lemma 20,

$$\mathrm{Var}\Big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\Big] = (\mu_4 - 3\sigma_0^4)\sum_{j=1}^{\infty}(\mathbb{M}_n)_{jj}^2 + \sigma_0^4\,\mathrm{tr}(\mathbb{M}_n\mathbb{M}_n') + \sigma_0^4\,\mathrm{tr}(\mathbb{M}_n^2).$$

As in the proof of Lemma 20,

$$\mathrm{tr}(\mathbb{M}_n\mathbb{M}_n') = \mathrm{tr}\big(\Pi_n\mathbb{A}_{2,\infty}\mathbb{A}'_{2,\infty}\Pi'_n\mathbb{A}_{1,\infty}\mathbb{A}'_{1,\infty}\big)\,, \qquad \mathrm{tr}(\mathbb{M}_n^2) = \mathrm{tr}\big(\Pi_n\mathbb{A}_{2,\infty}\mathbb{A}'_{1,\infty}\Pi_n\mathbb{A}_{2,\infty}\mathbb{A}'_{1,\infty}\big)\,.$$

Moreover, $\sum_{j=1}^{\infty}(\mathbb{M}_n)_{jj}^2 \leqslant \mathrm{tr}(\mathbb{M}_n\mathbb{M}_n')$. Hence, by the same argument as for the equation (A.35),

$$\big|\mathrm{tr}(\mathbb{M}_n\mathbb{M}_n')\big| = O\big(\mathrm{rank}(\Pi_n)\big), \qquad \big|\mathrm{tr}(\mathbb{M}_n^2)\big| = O\big(\mathrm{rank}(\Pi_n)\big).$$

Therefore,

$$\mathrm{Var}\Big[(\mathbb{A}_{1,\infty}\mathbb{V}_\infty)'\Pi_n(\mathbb{A}_{2,\infty}\mathbb{V}_\infty)\Big] = O\big(\mathrm{rank}(\Pi_n)\big).$$

Since $\bar{\mathbb{P}}$ is the orthogonal projection matrix onto the column space of the two-way fixed effects design matrix $(\mathbb{D},\mathbb{J})$, we have $\mathrm{rank}(\bar{\mathbb{P}}) = N + T - 1$, where the subtraction of one degree of freedom is due to the normalization on the time fixed effects. Moreover, $\mathrm{rank}(\bar{\mathbb{Q}}) = n - N - T + 1 = O(NT)$, so the stated expectation and stochastic orders follow. The stochastic orders follow from Chebyshev's inequality. □

Lemma 22. *Under the Assumptions 1′, 2, 7 and A.1, the following results hold:*

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\bar{\mathbb{Q}}\mathbb{R}^{\ell} - \frac{1}{NT}E\big(\mathbb{L}^{\ell\prime}\bar{\mathbb{Q}}\mathbb{R}^{\ell}\big) = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.36}$$

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\bar{\mathbb{P}}\mathbb{R}^{\ell} - \frac{1}{NT}E\big(\mathbb{L}^{\ell\prime}\bar{\mathbb{P}}\mathbb{R}^{\ell}\big) = O_p\left(\frac{1}{T\sqrt{N}} + \frac{1}{N\sqrt{T}}\right), \tag{A.37}$$

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\big(\mathbb{L}^{\ell\prime}\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{V}\big) = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.38}$$

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\bar{\mathbb{P}}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\big(\mathbb{L}^{\ell\prime}\bar{\mathbb{P}}\mathbb{B}_2\mathbb{V}\big) = O_p\left(\frac{1}{T\sqrt{N}} + \frac{1}{N\sqrt{T}}\right). \tag{A.39}$$

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathbb{R}^{\ell} - \frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathbb{R}^{\ell}\right) = O_p\left(\frac{1}{N\sqrt{T}}\right), \tag{A.40}$$

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathbb{B}_2\mathbb{V}\right) = O_p\left(\frac{1}{N\sqrt{T}}\right). \tag{A.41}$$

The expectations satisfy:

$$\frac{1}{NT}E\big(\mathbb{L}^{\ell\prime}\bar{\mathbb{Q}}\mathbb{R}^{\ell}\big) = O(1), \qquad \frac{1}{NT}E\big(\mathbb{L}^{\ell\prime}\bar{\mathbb{P}}\mathbb{R}^{\ell}\big) = O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right), \tag{A.42}$$

$$\frac{1}{NT}E\big(\mathbb{L}^{\ell\prime}\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{V}\big) = O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right). \tag{A.43}$$

$$\frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathbb{R}^{\ell}\right) = O\left(\frac{1}{N}\right), \qquad \frac{1}{NT}E\left(\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathbb{B}_2\mathbb{V}\right) = O\left(\frac{1}{N}\right). \tag{A.44}$$

Specifically,

$$\begin{aligned}\frac{1}{NT}E\big(\mathbb{L}^{\ell\prime}\bar{\mathbb{P}}\mathbb{B}_2\mathbb{V}\big) &= \frac{\sigma_0^2}{NT}\mathrm{tr}\left[\mathcal{T}^{-1}\sum_{h=1}^{T-1}\sum_{t=T-h}^{T-1}D'_{t+1}B_{t+1}^{\ell}P_t^{(t+h-T)}B'_{2,T-h}D_{T-h}\right]\\ &\quad + \frac{\sigma_0^2}{NT}\mathrm{tr}\left(\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathcal{B}_{2\infty}\mathcal{P}_{\infty}^{\ell\prime}\mathbb{B}_1^{\ell\prime}\right) = O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right),\end{aligned} \tag{A.45}$$

where $B_{t+1}^{\ell} \equiv B_{1,t+1}^{\ell}B_{1,t}$.

Proof. We have $\mathbb{L}^{\ell} = \mathbb{B}_1^{\ell}\mathcal{P}_{\infty}^{\ell}\mathbb{V}_{\infty}$ and $\mathbb{R}^{\ell} = \mathbb{B}_2^{\ell}\mathcal{Q}_{\infty}^{\ell}\mathbb{V}_{\infty}$. Moreover, we can write $\mathbb{B}_2\mathbb{V} = \mathcal{B}_{2\infty}\mathbb{V}_{\infty}$, where $\mathcal{B}_{2\infty}$ is an $n \times \infty$ matrix defined as

$$\mathcal{B}_{2\infty} = \begin{pmatrix} \mathbf{0}_{N_1\times N_T} & \dots & \mathbf{0}_{N_1\times N_2} & B_{21} & \mathbf{0}_{N_1\times N_0} & \dots \\ \mathbf{0}_{N_2\times N_T} & \dots & B_{22} & \mathbf{0}_{N_2\times N_1} & \mathbf{0}_{N_2\times N_0} & \dots \\ \vdots & \iddots & \vdots & \vdots & \vdots & \ddots \\ B_{2T} & \dots & \mathbf{0}_{N_T\times N_2} & \mathbf{0}_{N_T\times N_1} & \mathbf{0}_{N_T\times N_0} & \dots \end{pmatrix}.$$

Since the matrices $\mathbb{B}_1^{\ell}\mathcal{P}_{\infty}^{\ell}$, $\mathbb{B}_2^{\ell}\mathcal{Q}_{\infty}^{\ell}$, and $\mathcal{B}_{2\infty}$ are all UB according to Assumption A.1 and Lemma 19, Lemma 21 can be directly applied to establish (A.36)–(A.39), and (A.42)–(A.43).

To derive the explicit expression (A.45), we notice that $\bar{\mathbb{P}}$ is the projection matrix onto the column space of $[\mathbb{D}, \mathbb{J}]$. We can naturally decompose this into the projection onto $\mathbb{D}$ plus the projection onto the part of $\mathbb{J}$ that is orthogonal to $\mathbb{D}$. Therefore, we have $\bar{\mathbb{P}} = \mathbb{P} + \mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}$. Then the expectation splits into two terms:

$$E\big(\mathbb{L}^{\ell\prime}\bar{\mathbb{P}}\mathbb{B}_2\mathbb{V}\big) = E\big(\mathbb{L}^{\ell\prime}\mathbb{P}\mathbb{B}_2\mathbb{V}\big) + E\big(\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathbb{B}_2\mathbb{V}\big).$$

By Lemma 11, the first term yields the double-sum trace formula exactly as in (A.18). For the second term, applying equation (A.34), we have:

$$\begin{aligned} E\big(\mathbb{L}^{\ell\prime}\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathbb{B}_2\mathbb{V}\big) &= E\big(\mathbb{V}'_{\infty}\mathcal{P}_{\infty}^{\ell\prime}\mathbb{B}_1^{\ell\prime}\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathcal{B}_{2\infty}\mathbb{V}_{\infty}\big) \\ &= \sigma_0^2\,\mathrm{tr}\big(\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathcal{B}_{2\infty}\mathcal{P}_{\infty}^{\ell\prime}\mathbb{B}_1^{\ell\prime}\big). \end{aligned}$$

Combining these two parts directly yields (A.45).

The same argument also applies to the residualized time-effect projection $\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}$. Since $\mathrm{rank}(\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}) = T-1$, Lemma 21 gives (A.40)–(A.41), and their expectations satisfy (A.44). □

Lemma 23. *Under Assumptions 1′, 2, 3, 4′, 5–7 and A.1, we have*

$$\frac{1}{NT}\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\big(\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{V}\big) = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.46}$$

$$\frac{1}{NT}\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{P}}\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\big(\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{P}}\mathbb{B}_2\mathbb{V}\big) = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.47}$$

$$\frac{1}{NT}\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{Z} - \frac{1}{NT}E\big(\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{Z}\big) = O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{A.48}$$

$$\frac{1}{NT}\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{P}}\mathbb{B}_2\mathbb{Z} - \frac{1}{NT}E\big(\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{P}}\mathbb{B}_2\mathbb{Z}\big) = O_p\left(\frac{1}{\sqrt{NT}}\right). \tag{A.49}$$

Moreover,

$$\frac{1}{NT}E\big(\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{V}\big) = O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right), \qquad \frac{1}{NT}E\big(\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{P}}\mathbb{B}_2\mathbb{V}\big) = O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right), \tag{A.50}$$

$$\frac{1}{NT}E\big(\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{Z}\big) = O(1), \qquad \frac{1}{NT}E\big(\mathbb{Z}'\mathbb{B}_1'\bar{\mathbb{P}}\mathbb{B}_2\mathbb{Z}\big) = O(1). \tag{A.51}$$

The results in equations (A.46)–(A.49) remain valid when $\mathbb{Z}$ is replaced by $\mathbb{Z}\widetilde{\Gamma}$, where $\widetilde{\Gamma} = \mathrm{blkdiag}(I_2, \sqrt{T}, I_k)$. Furthermore,

$$\sqrt{T}\,\mathbb{Z}^{(-\gamma)\prime}\mathbb{B}_1'\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O_p(N\sqrt{T}), \qquad T\,\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(NT). \tag{A.52}$$

Proof. Write $\mathbb{Z} = (\mathbb{Z}_1, \mathbb{Z}_2, \mathbb{Z}^{(\gamma)}, \mathbb{Z}_4)$, where $\mathbb{Z}^{(\gamma)}$ is the listing regressor and $\mathbb{Z}_4$ collects the remaining k columns. As in the one-way case, $\mathbb{B}_1\mathbb{Z}_1, \mathbb{B}_1\mathbb{Z}_2, \mathbb{B}_2\mathbb{Z}_1$, and $\mathbb{B}_2\mathbb{Z}_2$ are all of the form $\mathbb{C} + \mathbb{L}^\ell$ or $\mathbb{C} + \mathbb{R}^\ell$, while $\mathbb{Z}^{(\gamma)}$ and the columns of $\mathbb{Z}_4$ are of the form $\mathbb{C}$.

We first consider the $\mathbb{Z}$-$\mathbb{V}$ terms. Let $\Pi_n \in \{\bar{\mathbb{Q}}, \bar{\mathbb{P}}\}$. Every element of $\mathbb{Z}'\mathbb{B}_1'\Pi_n\mathbb{B}_2\mathbb{V}$ can be reduced to terms of the form

$$\mathbb{C}'\Pi_n\mathbb{B}_2\mathbb{V}, \qquad \mathbb{L}^{\ell\prime}\Pi_n\mathbb{B}_2\mathbb{V}.$$

Since Lemma B.3 of Meng and Yang (2026) implies that $\bar{\mathbb{P}}$ and $\bar{\mathbb{Q}}$ are UB, elements in $\mathbb{B}_2'\Pi_n\mathbb{C}$ are still uniformly bounded. Therefore, by equation (A.8) in Lemma 8,

$$\frac{1}{NT}\mathbb{C}'\Pi_n\mathbb{B}_2\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right).$$

For the second component, Lemma 22 gives

$$\frac{1}{NT}\mathbb{L}^{\ell\prime}\Pi_n\mathbb{B}_2\mathbb{V} - \frac{1}{NT}E\big(\mathbb{L}^{\ell\prime}\Pi_n\mathbb{B}_2\mathbb{V}\big) = O_p\left(\frac{1}{\sqrt{NT}}\right),$$

with

$$\frac{1}{NT}E\big(\mathbb{L}^{\ell\prime}\Pi_n\mathbb{B}_2\mathbb{V}\big) = O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right).$$

Combining these bounds proves equations (A.46), (A.47), and (A.50).

Next consider the $\mathbb{Z}$-$\mathbb{Z}$ terms. Expanding block by block, every element of $\mathbb{Z}'\mathbb{B}_1'\Pi_n\mathbb{B}_2\mathbb{Z}$ is a sum of terms of the following types:

$$\mathbb{C}_1'\Pi_n\mathbb{C}_2, \qquad \mathbb{C}'\Pi_n\mathbb{L}^\ell, \qquad \mathbb{L}^{\ell\prime}\Pi_n\mathbb{R}^\ell.$$

The first type is deterministic, and $\mathbb{C}_1'\Pi_n\mathbb{C}_2 = O(NT)$, because Π_n is UB and $\mathbb{C}_1, \mathbb{C}_2$ have uniformly bounded elements.

For the second type, $\Pi_n\mathbb{C}$ is deterministic with uniformly bounded elements. Applying equation (A.6), together with Lemma 10, gives

$$\frac{1}{NT}\mathbb{C}'\Pi_n\mathbb{L}^\ell = O_p\left(\frac{1}{\sqrt{NT}}\right).$$

The expectation is zero.

The third type is controlled by equations (A.36)–(A.37) and the expectation bounds in (A.42). Therefore, equations (A.48) and (A.49) hold, and their expectations satisfy equation (A.51).

It remains to verify the statements involving $\widetilde{\Gamma}$. The matrix $\widetilde{\Gamma}$ only multiplies the listing column by $\sqrt{T}$. Since $\Pi_n \in \{\bar{\mathbb{Q}}, \bar{\mathbb{P}}\}$ is still UB, the same sparse-vector calculations as in Lemma 15 give

$$\frac{1}{N}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\Pi_n\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(1), \qquad \frac{\sqrt{T}}{NT}\mathbb{C}'\Pi_n\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O\left(\frac{1}{\sqrt{T}}\right),$$

and

$$\frac{\sqrt{T}}{NT}\mathbb{L}^{\ell\prime}\Pi_n\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O_p\left(\frac{1}{\sqrt{NT}}\right), \qquad \frac{\sqrt{T}}{NT}\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\Pi_n\mathbb{B}_2\mathbb{V} = O_p\left(\frac{1}{\sqrt{NT}}\right).$$

Together with the bounds already established for the non-listing columns, these equations imply that (A.46)–(A.49) remain valid after replacing $\mathbb{Z}$ by $\mathbb{Z}\widetilde{\Gamma}$. They also imply

$$\sqrt{T}\,\mathbb{Z}^{(-\gamma)\prime}\mathbb{B}_1'\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O_p\left(N\sqrt{T}\right), \qquad T\,\mathbb{Z}^{(\gamma)\prime}\mathbb{B}_1'\bar{\mathbb{Q}}\mathbb{B}_2\mathbb{Z}^{(\gamma)} = O(NT),$$

which is equation (A.52). □

Lemma 24. *Under Assumptions 1′, 2, 3, 4′, 5–7 and A.1, we have*

$$\frac{1}{NT}E\big(\widetilde{\Gamma}\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}\widetilde{\Gamma}\big)-\frac{1}{NT}E\big(\widetilde{\Gamma}\bar{\bar{\mathbb{Z}}}^{*\prime}\bar{\bar{\mathbb{Z}}}^{*}\widetilde{\Gamma}\big)=O\left(\frac{1}{T}\right)+O\left(\frac{1}{N}\right), \tag{A.53}$$

$$\frac{1}{NT}E\big(\widetilde{\Gamma}\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{G}_0\mathbb{Z}\big)-\frac{1}{NT}E\big(\widetilde{\Gamma}\bar{\bar{\mathbb{Z}}}^{*\prime}\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^{*}\big)=O\left(\frac{1}{T}\right)+O\left(\frac{1}{N}\right), \tag{A.54}$$

$$\frac{1}{NT}E\big(\mathbb{Z}'\mathbb{G}_0'\bar{\mathbb{Q}}\mathbb{G}_0\mathbb{Z}\big)-\frac{1}{NT}E\big(\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^{*\prime}\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^{*}\big)=O\left(\frac{1}{T}\right)+O\left(\frac{1}{N}\right), \tag{A.55}$$

where $\frac{1}{NT}E\big(\widetilde{\Gamma}\bar{\bar{\mathbb{Z}}}^{*\prime}\bar{\bar{\mathbb{Z}}}^{*}\widetilde{\Gamma}\big)=O(1)$, $\frac{1}{NT}E\big(\widetilde{\Gamma}\bar{\bar{\mathbb{Z}}}^{*\prime}\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^{*}\big)=O(1)$, *and* $\frac{1}{NT}E\big(\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^{*\prime}\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^{*}\big)=O(1)$.

Proof. Recall that $\bar{\bar{\mathbb{Z}}}^{\circ}=\bar{\mathbb{P}}\mathbb{Z}^{\circ}$, $\bar{\bar{\mathbb{Z}}}^{*}=\bar{\mathbb{Q}}\mathbb{Z}+\bar{\bar{\mathbb{Z}}}^{\circ}$, and similarly $\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^{\circ}=\bar{\mathbb{P}}\mathbb{G}_0\mathbb{Z}^{\circ}$, $\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^{*}=\bar{\mathbb{Q}}\mathbb{G}_0\mathbb{Z}+\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^{\circ}$. Therefore,

$$\frac{1}{NT}E\big(\widetilde{\Gamma}\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}\widetilde{\Gamma}\big)-\frac{1}{NT}E\big(\widetilde{\Gamma}\bar{\bar{\mathbb{Z}}}^{*\prime}\bar{\bar{\mathbb{Z}}}^{*}\widetilde{\Gamma}\big)=\frac{1}{NT}E\big(\widetilde{\Gamma}\mathbb{Z}^{\circ\prime}\bar{\mathbb{P}}\mathbb{Z}^{\circ}\widetilde{\Gamma}\big),$$

where $\mathbb{Z}^{\circ}$ has the same structure as in the one-way case, namely $\mathbb{Z}^{\circ}=\big(\mathbb{L}^{\ell},\mathbb{R}^{\ell},0_{n\times(k+1)}\big)$, up to relabeling of the coefficient matrices. Hence, Lemma 22 implies that the right-hand side is of order $O(\frac{1}{T})+O(\frac{1}{N})$, which proves (A.53). Since $\frac{1}{NT}E\big(\widetilde{\Gamma}\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}\widetilde{\Gamma}\big)$ is $O(1)$ by Lemma 23, therefore $\frac{1}{NT}E\big(\widetilde{\Gamma}\bar{\bar{\mathbb{Z}}}^{*\prime}\bar{\bar{\mathbb{Z}}}^{*}\widetilde{\Gamma}\big)=O(1)$.

Equations (A.54) and (A.55) are proved in the same way after replacing $\mathbb{Z}^{\circ}$ by $\mathbb{G}_0\mathbb{Z}^{\circ}$. Under Assumptions 3 and 5, $\mathbb{G}_0=\mathbb{W}\mathbb{S}_0^{-1}$ is UB, so Lemmas 22 and 23 continue to apply. □

B QMLE

B.1 The first and second order conditions of QMLE

For the concentrated likelihood function (7), the first order derivatives are

$$\frac{\partial L_n^c(\theta)}{\partial\theta}=\begin{pmatrix}\frac{\partial L_n^c(\theta)}{\partial\rho}\\ \frac{\partial L_n^c(\theta)}{\partial\delta}\\ \frac{\partial L_n^c(\theta)}{\partial\sigma^2}\end{pmatrix}=\begin{pmatrix}\frac{1}{\sigma^2}(\mathbb{W}\mathbb{Y})'\widetilde{\mathbb{V}}(\zeta)-\mathrm{tr}\left(\mathbb{S}(\rho)^{-1}\mathbb{W}\right)\\ \frac{1}{\sigma^2}\mathbb{Z}'\widetilde{\mathbb{V}}(\zeta)\\ -\frac{n}{2\sigma^2}+\frac{1}{2\sigma^4}\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta)\end{pmatrix},$$

where we denote $\widetilde{\mathbb{V}}(\zeta)=\mathbb{Q}(\mathbb{S}(\rho)\mathbb{Y}-\mathbb{Z}\delta)$ for simplicity. The second order derivatives are

$$\frac{\partial^2 L_n^c(\theta)}{\partial\theta\partial\theta'}=-\begin{pmatrix}\frac{1}{\sigma^2}(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y}+\mathrm{tr}\left(\mathbb{S}(\rho)^{-1}\mathbb{W}\mathbb{S}(\rho)^{-1}\mathbb{W}\right) & * & *\\ \frac{1}{\sigma^2}\mathbb{Z}'\mathbb{Q}\mathbb{W}\mathbb{Y} & \frac{1}{\sigma^2}\mathbb{Z}'\mathbb{Q}\mathbb{Z} & *\\ \frac{1}{\sigma^4}\widetilde{\mathbb{V}}(\zeta)'\mathbb{Q}\mathbb{W}\mathbb{Y} & \frac{1}{\sigma^4}\widetilde{\mathbb{V}}(\zeta)'\mathbb{Q}\mathbb{Z} & -\frac{n}{2\sigma^4}+\frac{1}{\sigma^6}\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta)\end{pmatrix}.$$

At θ_0, we have

$$\frac{\partial L_n^c(\theta_0)}{\partial\theta}=\begin{pmatrix}\frac{1}{\sigma_0^2}(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{V}-\mathrm{tr}(\mathbb{G}_0)\\ \frac{1}{\sigma_0^2}\mathbb{Z}'\mathbb{Q}\mathbb{V}\\ -\frac{n}{2\sigma_0^2}+\frac{1}{2\sigma_0^4}\mathbb{V}'\mathbb{Q}\mathbb{V}\end{pmatrix}=\begin{pmatrix}\frac{1}{\sigma_0^2}(\mathbb{G}_0\mathbb{D}\alpha_0+\mathbb{G}_0\mathbb{V})'\mathbb{Q}\mathbb{V}+\frac{1}{\sigma_0^2}(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{V}-\mathrm{tr}(\mathbb{G}_0)\\ \frac{1}{\sigma_0^2}\mathbb{Z}'\mathbb{Q}\mathbb{V}\\ -\frac{n}{2\sigma_0^2}+\frac{1}{2\sigma_0^4}\mathbb{V}'\mathbb{Q}\mathbb{V}\end{pmatrix},$$

and the information matrix is $\Sigma_{\theta_0,n} \equiv E\left(-\frac{1}{n}\frac{\partial^2 L_n^c(\theta_0)}{\partial\theta\partial\theta'}\right) = \Sigma_{\theta_0,n}^{(1)} - \Sigma_{\theta_0,n}^{(2)}$, where $\Sigma_{\theta_0,n}^{(1)} =$

$$\begin{pmatrix} \frac{1}{n\sigma_0^2}E\left[(\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{G}_0\mathbb{D}\alpha_0)\right]+\frac{1}{n}\mathrm{tr}\left(\mathbb{G}_0^2+\mathbb{G}_0'\mathbb{G}_0\right) & * & * \\ \frac{1}{n\sigma_0^2}E\left[\mathbb{Z}'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{G}_0\mathbb{D}\alpha_0)\right] & \frac{1}{n\sigma_0^2}E\left(\mathbb{Z}'\mathbb{Q}\mathbb{Z}\right) & * \\ \frac{1}{n\sigma_0^2}\mathrm{tr}\left(\mathbb{G}_0\right) & \mathbf{0}_{1\times(k+3)} & \frac{1}{2\sigma_0^4} \end{pmatrix},$$

$\Sigma_{\theta_0,n}^{(2)} =$

$$\begin{pmatrix} \frac{2}{n\sigma_0^2}E\left[(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{P}\mathbb{G}_0\mathbb{V}\right]+\frac{1}{n}\mathrm{tr}\left(\mathbb{G}_0'\mathbb{P}\mathbb{G}_0\right) & * & * \\ \frac{1}{n\sigma_0^2}E\left[\mathbb{Z}'\mathbb{P}\mathbb{G}_0\mathbb{V}\right] & \mathbf{0}_{(k+3)\times(k+3)} & * \\ \frac{1}{n\sigma_0^4}E\left(\mathbb{V}'\mathbb{P}\mathbb{G}_0\mathbb{Z}\delta_0\right)+\frac{1}{n\sigma_0^2}\mathrm{tr}\left(\mathbb{P}\mathbb{G}_0\right) & \frac{1}{n\sigma_0^4}E\left(\mathbb{V}'\mathbb{P}\mathbb{Z}\right) & \frac{N}{n\sigma_0^4} \end{pmatrix}.$$

By Lemma 14, all expectation terms in $\Sigma_{\theta_0,n}^{(2)}$ are $O(\frac{1}{T})$. Since $\mathbb{G}_0 = \mathrm{blkdiag}(W_1S_1^{-1},\cdots,W_TS_T^{-1})$,

$$\begin{aligned}\frac{1}{n}\left|\mathrm{tr}\left(\mathbb{P}\mathbb{G}_0\right)\right| &= \frac{1}{n}\left|\mathrm{tr}\left(\mathbb{D}\mathcal{T}^{-1}\mathbb{D}'\mathbb{G}_0\right)\right| = \frac{1}{n}\left|\mathrm{tr}\left(\mathcal{T}^{-1}\mathbb{D}'\mathbb{G}_0\mathbb{D}\right)\right| \leqslant \frac{1}{c_TnT}\left|\mathrm{tr}\left(\mathbb{D}'\mathbb{G}_0\mathbb{D}\right)\right| \\ &\leqslant \frac{1}{c_TnT}\sum_{t=1}^{T}\left|\mathrm{tr}\left(D_t'W_tS_t^{-1}D_t\right)\right| = O\left(\frac{1}{T}\right),\end{aligned}$$

where the first inequality holds because $T_i > c_TT$ by Assumption 1 and the last equality holds because $D_t'W_tS_t^{-1}D_t$ is an N-dimensional UB matrix. Similarly, we can prove $\frac{1}{n}\mathrm{tr}\left(\mathbb{G}_0'\mathbb{P}\mathbb{G}_0\right) = O\left(\frac{1}{T}\right)$. Thus, $\Sigma_{\theta_0,n}^{(2)} = O(\frac{1}{T})$, which means that $\Sigma_{\theta_0,n} = \Sigma_{\theta_0,n}^{(1)} + O(\frac{1}{T})$. We also have $\Gamma\Sigma_{\theta_0,n}\Gamma = \Gamma\Sigma_{\theta_0,n}^{(1)}\Gamma + O(\frac{1}{T})$ by Lemmas 13-15.

B.2 The variance of the gradient

Denote

$$E\left(\frac{1}{\sqrt{n}}\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}\cdot\frac{1}{\sqrt{n}}\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta'}\right) \equiv \begin{pmatrix} E_{\rho\rho} & * & * \\ E_{\delta\rho} & E_{\delta\delta'} & * \\ E_{\sigma^2\rho} & E_{\sigma^2\delta'} & E_{\sigma^2\sigma^2} \end{pmatrix},$$

where

$$
\begin{aligned}
E_{\rho\rho} =& \frac{1}{n}E\left[\left(\frac{1}{\sigma_0^2}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V}+\frac{1}{\sigma_0^2}\mathbb{V}'\mathbb{G}_0'\mathbb{V}+\frac{1}{\sigma_0^2}(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)'\mathbb{V}-\mathrm{tr}(\mathbb{G}_0)\right)^2\right]\\
=& \frac{1}{n}E\left[\frac{1}{\sigma_0^4}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V}\mathbb{V}'\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0+\frac{1}{\sigma_0^4}\mathbb{V}'\mathbb{G}_0'\mathbb{V}\mathbb{V}'\mathbb{G}_0'\mathbb{V}+\frac{1}{\sigma_0^4}(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)'\mathbb{V}\mathbb{V}'\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0+\mathrm{tr}(\mathbb{G}_0)^2+\right.\\
&\frac{2}{\sigma_0^4}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V}\mathbb{V}'\mathbb{G}_0'\mathbb{V}+\frac{2}{\sigma_0^4}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V}\mathbb{V}'\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0-\frac{2}{\sigma_0^2}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V}\mathrm{tr}(\mathbb{G}_0)+\\
&\left.\frac{2}{\sigma_0^4}\mathbb{V}'\mathbb{G}_0'\mathbb{V}\mathbb{V}'\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0-\frac{2}{\sigma_0^2}\mathbb{V}'\mathbb{G}_0'\mathbb{V}\mathrm{tr}(\mathbb{G}_0)-\frac{2}{\sigma_0^2}(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)'\mathbb{V}\mathrm{tr}(\mathbb{G}_0)\right]\\
=& \frac{1}{n\sigma_0^2}E\left[\left(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0+\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right)'\left(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0+\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right)\right]+\frac{\mu_4-3\sigma_0^4}{n\sigma_0^4}\sum_{i=1}^n\mathbb{G}_{0,ii}^2\\
&+\frac{1}{n}\mathrm{tr}(\mathbb{G}_0^2+\mathbb{G}_0\mathbb{G}_0')+\frac{2\mu_3}{n\sigma_0^4}\sum_{i=1}^n(\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0)_i\mathbb{G}_{0,ii}+\frac{2\mu_3}{n\sigma_0^4}\sum_{i=1}^nE(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)_i\mathbb{G}_{0,ii},
\end{aligned}
$$

$$
\begin{aligned}
E_{\delta\rho} =& \frac{1}{n}E\left[\frac{1}{\sigma_0^2}\widetilde{\mathbb{Z}}^{*\prime}\mathbb{V}\left(\frac{1}{\sigma_0^2}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V}+\frac{1}{\sigma_0^2}\mathbb{V}'\mathbb{G}_0'\mathbb{V}+\frac{1}{\sigma_0^2}(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)'\mathbb{V}-\mathrm{tr}(\mathbb{G}_0)\right)\right]\\
=& \frac{1}{n}E\left[\frac{1}{\sigma_0^4}\widetilde{\mathbb{Z}}^{*\prime}\mathbb{V}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V}+\frac{1}{\sigma_0^4}\widetilde{\mathbb{Z}}^{*\prime}\mathbb{V}\mathbb{V}'\mathbb{G}_0'\mathbb{V}+\frac{1}{\sigma_0^4}\widetilde{\mathbb{Z}}^{*\prime}\mathbb{V}(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)'\mathbb{V}-\frac{1}{\sigma_0^2}\widetilde{\mathbb{Z}}^{*\prime}\mathbb{V}\mathrm{tr}(\mathbb{G}_0)\right]\\
=& \frac{1}{n\sigma_0^2}E\left(\widetilde{\mathbb{Z}}^{*\prime}\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right)+\frac{\mu_3}{n\sigma_0^4}\sum_{i=1}^nE(\widetilde{\mathbb{Z}}^{*\prime})_{\cdot,i}\mathbb{G}_{0,ii}+\frac{1}{n\sigma_0^2}E\left(\widetilde{\mathbb{Z}}^{*\prime}\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0\right),
\end{aligned}
$$

$$
\begin{aligned}
E_{\sigma^2\rho} =& \frac{1}{n}E\left[\left(-\frac{n}{2\sigma_0^2}+\frac{1}{2\sigma_0^4}\mathbb{V}'\mathbb{V}\right)\left(\frac{1}{\sigma_0^2}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V}+\frac{1}{\sigma_0^2}\mathbb{V}'\mathbb{G}_0'\mathbb{V}+\frac{1}{\sigma_0^2}(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)'\mathbb{V}-\mathrm{tr}(\mathbb{G}_0)\right)\right]\\
=& \frac{1}{n}E\left[-\frac{n}{2\sigma_0^4}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V}-\frac{n}{2\sigma_0^4}\mathbb{V}'\mathbb{G}_0'\mathbb{V}-\frac{n}{2\sigma_0^4}(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)'\mathbb{V}+\frac{n}{2\sigma_0^2}\mathrm{tr}(\mathbb{G}_0)+\right.\\
&\left.\frac{1}{2\sigma_0^6}\mathbb{V}'\mathbb{V}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V}+\frac{1}{2\sigma_0^6}\mathbb{V}'\mathbb{V}\mathbb{V}'\mathbb{G}_0'\mathbb{V}+\frac{1}{2\sigma_0^6}\mathbb{V}'\mathbb{V}(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)'\mathbb{V}-\frac{1}{2\sigma_0^4}\mathbb{V}'\mathbb{V}\mathrm{tr}(\mathbb{G}_0)\right]\\
=& \frac{\mu_4-\sigma_0^4}{2n\sigma_0^6}\mathrm{tr}(\mathbb{G}_0)+\frac{\mu_3}{2n\sigma_0^6}\sum_{i=1}^n(\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0)_i+\frac{\mu_3}{2n\sigma_0^6}\sum_{i=1}^nE(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)_i,
\end{aligned}
$$

$$
\begin{aligned}
E_{\delta\delta'} =& \frac{1}{n}E\left(\frac{1}{\sigma_0^4}\widetilde{\mathbb{Z}}^{*\prime}\mathbb{V}\mathbb{V}'\widetilde{\mathbb{Z}}^*\right)=\frac{1}{n}E\left(\frac{1}{\sigma_0^4}\sum_{t=1}^T\sum_{s=1}^T\widetilde{Z}_t^{*\prime}V_tV_s'\widetilde{Z}_s^*\right)\\
=& \frac{1}{n}E\left(\frac{1}{\sigma_0^4}\sum_{t=1}^T\widetilde{Z}_t^{*\prime}V_tV_t'\widetilde{Z}_t^*\right)=\frac{1}{n\sigma_0^2}E\left(\widetilde{\mathbb{Z}}^{*\prime}\widetilde{\mathbb{Z}}^*\right),
\end{aligned}
$$

where the third equality holds because the fact that $\widetilde{Z}_t^*$ is independent with V_s when $t\leqslant s$.

$$
E_{\sigma^2\delta'} =\frac{1}{n}E\left[\left(-\frac{n}{2\sigma_0^2}+\frac{1}{2\sigma_0^4}\mathbb{V}'\mathbb{V}\right)\mathbb{V}'\widetilde{\mathbb{Z}}^*\right]=\frac{\mu_3}{2n\sigma_0^6}\sum_{i=1}^nE(\widetilde{\mathbb{Z}}^*)_{i,\cdot},
$$

$$
E_{\sigma^2\sigma^2} =\frac{1}{n}E\left[\left(-\frac{n}{2\sigma_0^2}+\frac{1}{2\sigma_0^4}\mathbb{V}'\mathbb{V}\right)^2\right]=\frac{\mu_4-\sigma_0^4}{4\sigma_0^8}.
$$

Therefore,

$$E\left(\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}\cdot\frac{1}{\sqrt{n}}\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta'}\Gamma\right)=$$

$$\begin{pmatrix}\left\{\begin{array}{c}\frac{1}{n\sigma_0^2}E\left[\left(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0+\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right)'\left(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0+\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right)\right]\\+\frac{1}{n}\mathrm{tr}(\mathbb{G}_0^2+\mathbb{G}_0\mathbb{G}_0')\end{array}\right\} & * & *\\ \frac{1}{n\sigma_0^2}E\left[\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{*\prime}\left(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0+\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right)\right] & \frac{1}{n\sigma_0^2}\widetilde{\Gamma}E\left(\widetilde{\mathbb{Z}}^{*\prime}\widetilde{\mathbb{Z}}^*\right)\widetilde{\Gamma} & *\\ \frac{1}{n\sigma_0^2}\mathrm{tr}(\mathbb{G}_0) & \mathbf{0}_{1\times(k+3)} & \frac{1}{2\sigma_0^4}\end{pmatrix}+$$

$$\begin{pmatrix}\frac{\mu_4-3\sigma_0^4}{n\sigma_0^4}\sum_{i=1}^n\mathbb{G}_{0,ii}^2+\frac{2\mu_3}{n\sigma_0^4}\sum_{i=1}^n\left[E(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)+\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right]_i\mathbb{G}_{0,ii} & * & *\\ \frac{\mu_3}{n\sigma_0^4}\sum_{i=1}^n E(\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{*\prime})_{\cdot,i}\mathbb{G}_{0,ii} & \mathbf{0}_{(k+3)\times(k+3)} & *\\ \frac{\mu_4-3\sigma_0^4}{2n\sigma_0^6}\mathrm{tr}(\mathbb{G}_0)+\frac{\mu_3}{2n\sigma_0^6}\sum_{i=1}^n\left[E(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)+\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right]_i & \frac{\mu_3}{2n\sigma_0^6}\sum_{i=1}^n E(\widetilde{\mathbb{Z}}^*\widetilde{\Gamma})_{i,\cdot} & \frac{\mu_4-3\sigma_0^4}{4\sigma_0^8}\end{pmatrix}.$$

The first matrix above is equal to $\Gamma\Sigma_{\theta_0,n}\Gamma+O\left(\frac{1}{T}\right)$ by Lemma 16. Denote the second matrix as $\Gamma\Omega_{\theta_0,n}\Gamma$, Since $\widetilde{\mathbb{G}_0\mathbb{Z}}^*=\mathbb{Q}\mathbb{G}_0\mathbb{Z}+\widetilde{\mathbb{G}_0\mathbb{Z}}^\circ$ and $\widetilde{\mathbb{Z}}^*=\mathbb{Q}\mathbb{Z}+\widetilde{\mathbb{Z}}^\circ$, where $E(\widetilde{\mathbb{G}_0\mathbb{Z}}^\circ)=E(\widetilde{\mathbb{Z}}^\circ)=0$, it follows that $\sum_{i=1}^n E(\widetilde{\mathbb{G}_0\mathbb{Z}}^*)_{i,\cdot}=\sum_{i=1}^n E(\widetilde{\mathbb{Z}}^*\widetilde{\Gamma})_{i,\cdot}=0$ and $\sum_{i=1}^n(\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0)_i=0$ because $\iota_n'\mathbb{Q}=0$. Therefore, we have

$$\Gamma\Omega_{\theta_0,n}\Gamma=\begin{pmatrix}\frac{\mu_4-3\sigma_0^4}{n\sigma_0^4}\sum_{i=1}^n\mathbb{G}_{0,ii}^2+\frac{2\mu_3}{n\sigma_0^4}\sum_{i=1}^n\left[E(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)+\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right]_i\mathbb{G}_{0,ii} & * & *\\ \frac{\mu_3}{n\sigma_0^4}\sum_{i=1}^n E(\widetilde{\Gamma}\widetilde{\mathbb{Z}}^*)_{\cdot,i}\mathbb{G}_{0,ii} & \mathbf{0}_{(k+3)\times(k+3)} & *\\ \frac{\mu_4-3\sigma_0^4}{2n\sigma_0^6}\mathrm{tr}(\mathbb{G}_0) & \mathbf{0}_{1\times(k+3)} & \frac{\mu_4-3\sigma_0^4}{4\sigma_0^8}\end{pmatrix}.$$

When v_{it} are normally distributed, $\Gamma\Omega_{\theta_0,n}\Gamma=0$ because $\mu_3=0$ and $\mu_4-3\sigma^4=0$ for normal distribution. Hence,

$$E\left(\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}\cdot\frac{1}{\sqrt{n}}\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta'}\Gamma\right)=\Gamma\Sigma_{\theta_0,n}\Gamma+\Gamma\Omega_{\theta_0,n}\Gamma+O\left(\frac{1}{T}\right).$$

B.3 Some properties of the Hessian matrix

For Hessian matrix $\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\theta)}{\partial\theta\partial\theta'}\Gamma$, we have the following results:

(1). Hessian matrix is Lipschitz continuous at θ_0, which means

$$-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\theta)}{\partial\theta\partial\theta'}\Gamma-\left(-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma\right)=\|\Gamma^{-1}(\theta-\theta_0)\|_2\cdot O_p(1),\tag{B.1}$$

(2). Hessian matrix is consistent at θ_0, which means

$$-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma-\Gamma\Sigma_{\theta_0,n}\Gamma=O_p\left(\frac{1}{\sqrt{NT}}\right),\tag{B.2}$$

(3). The information matrix $\Sigma_{\theta_0} \equiv \lim_{n\to\infty} \Gamma\Sigma_{\theta_0,n}\Gamma$ is nonsingular.

Proof. **Proof of (1)**

Since

$$-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\theta)}{\partial\theta\partial\theta'}\Gamma - \left(-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma\right) = -\frac{1}{n}\times$$

$$\begin{pmatrix} \left(\frac{1}{\sigma^2}-\frac{1}{\sigma_0^2}\right)(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y} + \mathrm{tr}\left[\mathbb{G}(\rho)^2-\mathbb{G}_0^2\right] & * & * \\ \left(\frac{1}{\sigma^2}-\frac{1}{\sigma_0^2}\right)\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{W}\mathbb{Y} & \left(\frac{1}{\sigma^2}-\frac{1}{\sigma_0^2}\right)\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma} & * \\ \left[\frac{1}{\sigma^4}\widetilde{\mathbb{V}}(\zeta)-\frac{1}{\sigma_0^4}\mathbb{V}\right]'\mathbb{Q}\mathbb{W}\mathbb{Y} & \left[\frac{1}{\sigma^4}\widetilde{\mathbb{V}}(\zeta)-\frac{1}{\sigma_0^4}\mathbb{V}\right]'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma} & \begin{bmatrix} -\frac{n}{2\sigma_0^4}+\frac{1}{\sigma^6}\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta) \\ +\frac{n}{2\sigma^4}-\frac{1}{\sigma_0^6}\mathbb{V}'\mathbb{Q}\mathbb{V}\end{bmatrix} \end{pmatrix}$$

where $(\frac{1}{n\sigma^2}-\frac{1}{n\sigma_0^2})(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y} = \frac{1}{n\bar{\sigma}^4}(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y}(\sigma_0^2-\sigma^2)$, $\bar{\sigma}^2$ lies between σ^2 and σ_0^2. Since $\mathbb{W}\mathbb{Y} = \mathbb{G}_0(\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{V})$, the term $(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y}$ can be decomposed as a sum of quadratic forms involving $\mathbb{Z}$ and $\mathbb{V}$, as shown in equation (C.2). According to Lemma 13, it follows that $\frac{1}{n\bar{\sigma}^4}(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y} = O_p(1)$. Thus, $\|(\frac{1}{n\sigma^2}-\frac{1}{n\sigma_0^2})(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y}\|_2 \leqslant \|(\sigma_0^2-\sigma^2)\|_2\cdot O_p(1)$. Similarly, we can prove $\|(\frac{1}{n\sigma^2}-\frac{1}{n\sigma_0^2})\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{W}\mathbb{Y}\|_2 \leqslant \|(\sigma_0^2-\sigma^2)\|_2\cdot O_p(1)$ and $\|(\frac{1}{n\sigma^2}-\frac{1}{n\sigma_0^2})\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma}\|_2 \leqslant \|(\sigma_0^2-\sigma^2)\|_2\cdot O_p(1)$ by Lemmas 13 and 15. Additionally, $\frac{1}{n}\mathrm{tr}[\mathbb{G}(\rho)^2-\mathbb{G}_0^2] = \frac{2}{n}\mathrm{tr}[\mathbb{G}(\bar{\rho})^3](\rho-\rho_0)$, where $\bar{\rho}$ lies between ρ and ρ_0. Note that

$$\begin{aligned}\sup_{N,T}\|\mathbb{G}(\bar{\rho})\|_\infty &= \sup_{N,T}\|\mathbb{W}\mathbb{S}^{-1}(\bar{\rho})\|_\infty \leqslant \sup_{N,T}\|\mathbb{W}\|_\infty \sup_{N,T}\|\mathbb{S}^{-1}(\bar{\rho})\|_\infty \\ &= \left(\sup_{N,T}\sup_t\|W_t\|_\infty\right)\left(\sup_{N,T}\sup_t\|S_t(\bar{\rho})^{-1}\|_\infty\right) < \infty, \end{aligned} \tag{B.3}$$

where the second equality holds because $\mathbb{W}$ and $\mathbb{S}^{-1}(\bar{\rho})$ are block diagonal matrix and the last inequality holds by equation (9) and Assumption 5. Thus, $\sup_{N,T}\|\mathbb{G}(\bar{\rho})^3\|_\infty \leqslant \sup_{N,T}\|\mathbb{G}(\bar{\rho})\|_\infty^3 \leqslant \infty$, which implies that $\frac{2}{n}\mathrm{tr}[\mathbb{G}(\bar{\rho})^3] = O(1)$. So $\|\frac{1}{n}\mathrm{tr}[\mathbb{G}(\rho)^2-\mathbb{G}_0^2]\|_2 \leqslant \|\rho-\rho_0\|_2\cdot O(1)$. Moreover,

$$\begin{aligned}&\frac{1}{n}\left[\frac{1}{\sigma^4}\widetilde{\mathbb{V}}(\zeta)-\frac{1}{\sigma_0^4}\mathbb{V}\right]'\mathbb{Q}\mathbb{W}\mathbb{Y} \\ =&\frac{1}{n}\left[\frac{1}{\sigma^4}(\rho_0-\rho)\mathbb{W}\mathbb{Y}+\frac{1}{\sigma^4}\mathbb{Z}(\delta_0-\delta)+\left(\frac{1}{\sigma^4}-\frac{1}{\sigma_0^4}\right)\mathbb{V}\right]'\mathbb{Q}\mathbb{W}\mathbb{Y} \\ =&(\rho_0-\rho)\frac{1}{n\sigma^4}(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y}+[\widetilde{\Gamma}^{-1}(\delta_0-\delta)]'\frac{1}{n\sigma^4}\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{W}\mathbb{Y}-(\sigma^2-\sigma_0^2)\frac{2}{n\bar{\sigma}^6}\mathbb{V}'\mathbb{Q}\mathbb{W}\mathbb{Y}.\end{aligned}$$

Since all the coefficients of $\Gamma^{-1}(\theta-\theta_0)$ in the above expression are $O_p(1)$ by Lemmas 13 and 15, we therefore have $\|\frac{1}{n}[\frac{1}{\sigma^4}\widetilde{\mathbb{V}}(\zeta)-\frac{1}{\sigma_0^4}\mathbb{V}]'\mathbb{Q}\mathbb{W}\mathbb{Y}\|_2 \leqslant \|\Gamma^{-1}(\theta-\theta_0)\|_2\cdot O_p(1)$ by Cauchy-Schwarz inequality. Therefore, we can similarly prove that the remaining elements of the above matrix are $\|\Gamma^{-1}(\theta-\theta_0)\|_2\cdot O_p(1)$.

Proof of (2)

We have

$$-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma-\Gamma\Sigma_{\theta_0,n}\Gamma=-\frac{1}{n}\times$$
$$\begin{pmatrix}\frac{1}{\sigma_0^2}\left[(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y}-E\left((\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y}\right)\right] & * & * \\ \frac{1}{\sigma_0^2}\left[\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{W}\mathbb{Y}-E\left(\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{W}\mathbb{Y}\right)\right] & \frac{1}{\sigma_0^2}\left[\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma}-E\left(\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma}\right)\right] & * \\ \frac{1}{\sigma_0^4}\left[\mathbb{V}'\mathbb{Q}\mathbb{W}\mathbb{Y}-E\left(\mathbb{V}'\mathbb{Q}\mathbb{W}\mathbb{Y}\right)\right] & \frac{1}{\sigma_0^4}\left[\mathbb{V}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma}-E\left(\mathbb{V}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma}\right)\right] & \frac{1}{\sigma_0^6}\left[\mathbb{V}'\mathbb{Q}\mathbb{V}-E\left(\mathbb{V}'\mathbb{Q}\mathbb{V}\right)\right]\end{pmatrix}.$$

Since $\mathbb{W}\mathbb{Y}=\mathbb{G}_0(\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{V})$, which is a combination of $\mathbb{Z}$ and $\mathbb{V}$, it follows that elements in the matrix above are $O_p(\frac{1}{\sqrt{NT}})$ by Lemmas 13 and 15.

Proof of (3)

The information matrix is nonsingular if and only if for any vector $\boldsymbol{a}$, $\Sigma_{\theta_0}\boldsymbol{a}=0$ implies $\boldsymbol{a}=0$. Let $\boldsymbol{a}=(a_1,\boldsymbol{a}_2',a_3)'$, where a_1 and a_3 are scalars and $\boldsymbol{a}_2$ is $(k+3)$-dimensional vector. As

$$\begin{aligned}\Sigma_{\theta_0}=&\lim_{n\to\infty}\Gamma\Sigma_{\theta_0,n}\Gamma=\lim_{n\to\infty}\Gamma\Sigma_{\theta_0,n}^{(1)}\Gamma\\ \equiv&\frac{1}{\sigma_0^2}\times\begin{pmatrix}\mathcal{H}_{\rho\rho}+\lim_{n\to\infty}\frac{\sigma_0^2}{n}\mathrm{tr}\left(\mathbb{G}_0^2+\mathbb{G}_0'\mathbb{G}_0\right) & * & * \\ \mathcal{H}_{\rho\delta} & \mathcal{H}_{\delta\delta} & * \\ \lim_{n\to\infty}\frac{1}{n}\mathrm{tr}\left(\mathbb{G}_0\right) & \mathbf{0}_{1\times(k+3)} & \frac{1}{2\sigma_0^2}\end{pmatrix},\end{aligned}$$

where the second equality holds because $\Gamma\Sigma_{\theta_0,n}\Gamma=\Gamma\left(\Sigma_{\theta_0,n}^{(1)}-\Sigma_{\theta_0,n}^{(2)}\right)\Gamma=\Gamma\Sigma_{\theta_0,n}^{(1)}\Gamma+O(\frac{1}{T})$, as derived in B.1. Additionally, $\mathcal{H}_{\rho\rho}\equiv\lim_{n\to\infty}\frac{1}{n}E\left[(\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{G}_0\mathbb{D}\alpha_0)\right]$, $\mathcal{H}_{\rho\delta}\equiv\lim_{n\to\infty}\frac{1}{n}E[\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{G}_0\mathbb{D}\alpha_0)]$ and $\mathcal{H}_{\delta\delta}\equiv\lim_{n\to\infty}\frac{1}{n}E(\widetilde{\Gamma}\mathbb{Z}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma})$. $\Sigma_{\theta_0}\boldsymbol{a}=0$ means

$$\begin{cases}\frac{1}{\sigma_0^2}\left[\mathcal{H}_{\rho\rho}+\lim_{n\to\infty}\frac{\sigma_0^2}{n}\mathrm{tr}\left(\mathbb{G}_0^2+\mathbb{G}_0'\mathbb{G}_0\right)\right]a_1+\frac{1}{\sigma_0^2}\mathcal{H}_{\rho\delta}'\boldsymbol{a}_2+\frac{1}{\sigma_0^2}\lim_{n\to\infty}\frac{1}{n}\mathrm{tr}\left(\mathbb{G}_0\right)a_3=0\\ \frac{1}{\sigma_0^2}\mathcal{H}_{\rho\delta}a_1+\frac{1}{\sigma_0^2}\mathcal{H}_{\delta\delta}\boldsymbol{a}_2=0\\ \frac{1}{\sigma_0^2}\lim_{n\to\infty}\frac{1}{n}\mathrm{tr}\left(\mathbb{G}_0\right)a_1+\frac{1}{2\sigma_0^4}a_3=0\end{cases}.$$

According to the last two equations, we have $a_3=-2\sigma_0^2\lim_{n\to\infty}\frac{1}{n}\mathrm{tr}(\mathbb{G}_0)a_1$ and $\boldsymbol{a}_2=-\mathcal{H}_{\delta\delta}^{-1}\mathcal{H}_{\rho\delta}a_1$. Substituting them into the first equation gives

$$\left[\mathcal{H}_{\rho\rho}-\mathcal{H}_{\rho\delta}'\mathcal{H}_{\delta\delta}^{-1}\mathcal{H}_{\rho\delta}+\sigma_0^2\lim_{n\to\infty}\frac{1}{n}\left(\mathrm{tr}(\mathbb{G}_0^2+\mathbb{G}_0'\mathbb{G}_0)-\frac{2}{n}\left(\mathrm{tr}(\mathbb{G}_0)\right)^2\right)\right]a_1=0. \tag{B.4}$$

Note that

$$
\begin{aligned}
&\begin{pmatrix} \mathcal{H}_{\rho\rho} & \mathcal{H}'_{\rho\delta} \\ \mathcal{H}_{\rho\delta} & \mathcal{H}_{\delta\delta} \end{pmatrix} \\
&= \lim_{n\to\infty} \frac{1}{n} E\left[(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0, \widetilde{\Gamma}\mathbb{Z})'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0, \widetilde{\Gamma}\mathbb{Z}) \right] \\
&= \lim_{n\to\infty} \frac{1}{n} \\
&\quad E\begin{pmatrix}
(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0) & * & * & * & * \\
\mathbb{Z}_1'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0) & \mathbb{Z}_1'\mathbb{Q}\mathbb{Z}_1 & * & * & * \\
\mathbb{Z}_2'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0) & \mathbb{Z}_2'\mathbb{Q}\mathbb{Z}_1 & \mathbb{Z}_2'\mathbb{Q}\mathbb{Z}_2 & * & * \\
\sqrt{T}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0) & \sqrt{T}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}_1 & \sqrt{T}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}_2 & T\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)} & * \\
\mathbb{Z}_4'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0) & \mathbb{Z}_4'\mathbb{Q}\mathbb{Z}_1 & \mathbb{Z}_4'\mathbb{Q}\mathbb{Z}_2 & \sqrt{T}\mathbb{Z}_4'\mathbb{Q}\mathbb{Z}^{(\gamma)} & \mathbb{Z}_4'\mathbb{Q}\mathbb{Z}_4
\end{pmatrix},
\end{aligned}
\tag{B.5}
$$

where $\mathbb{Z}_1$, $\mathbb{Z}_2$ and $\mathbb{Z}^{(\gamma)}$ are the first three columns of $\mathbb{Z}$, and $\mathbb{Z}_4$ denotes the remaining k columns of $\mathbb{Z}$. Then $\mathcal{H}_{\rho\rho} - \mathcal{H}'_{\rho\delta}\mathcal{H}_{\delta\delta}^{-1}\mathcal{H}_{\rho\delta} \geqslant 0$ because $(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0, \widetilde{\Gamma}\mathbb{Z})'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0, \widetilde{\Gamma}\mathbb{Z})$ is positive semi-definite. Moreover, note that

$$
\operatorname{tr}(\mathbb{G}_0^2 + \mathbb{G}_0'\mathbb{G}_0) - \frac{2}{n}\operatorname{tr}(\mathbb{G}_0)^2 = \frac{1}{2}\operatorname{tr}\left[\left(\mathbb{G}_0 + \mathbb{G}_0' - \frac{2}{n}\operatorname{tr}(\mathbb{G}_0) I_n \right) \left(\mathbb{G}_0 + \mathbb{G}_0' - \frac{2}{n}\operatorname{tr}(\mathbb{G}_0) I_n \right)' \right] \geqslant 0.
$$

Under the above conditions, we will use Assumption 9 (i) and Assumption 9 (ii) separately to show that either $\mathcal{H}_{\rho\rho} - \mathcal{H}'_{\rho\delta}\mathcal{H}_{\delta\delta}^{-1}\mathcal{H}_{\rho\delta}$ or $\lim_{n\to\infty} \frac{1}{n}\left(\operatorname{tr}(\mathbb{G}_0^2 + \mathbb{G}_0'\mathbb{G}_0) - \frac{2}{n}(\operatorname{tr}(\mathbb{G}_0))^2\right)$ is greater than 0, which then implies from equation (B.4) that $a_1 = 0$.

1. (Using Assumption 9 (i)) Because $\mathbb{Z}^{(\gamma)}$ is sparse binary vector with at most N nonzero elements, elements in $E(\mathbb{Z})$ and α_0 are uniformly bounded, and $\mathbb{G}_0$ is UB, we have $\sqrt{T}E[\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0)] = O(N\sqrt{T})$, $\sqrt{T}E[\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}_1] = O(N\sqrt{T})$, $\sqrt{T}E[\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}_2] = O(N\sqrt{T})$, $\sqrt{T}\mathbb{Z}_4'\mathbb{Q}\mathbb{Z}^{(\gamma)} = O(N\sqrt{T})$ and $T\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)} = O(NT)$. Moreover,

$$
\begin{aligned}
\frac{T}{n}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)} &= \frac{T}{n}\sum_{t=1}^{T} \iota_N' F_t' D_t' D_t F_t \iota_N - \frac{T}{n}\left(\sum_{t=1}^{T} D_t' D_t F_t \iota_N \right)' \mathcal{T}^{-1} \left(\sum_{t=1}^{T} D_t' D_t F_t \iota_N \right) \\
&= \frac{T}{n}\sum_{t=1}^{T} \iota_N' F_t \iota_N - \frac{T}{n}\operatorname{tr}\left[\mathcal{T}^{-1} \left(\sum_{t=1}^{T} F_t \iota_N \right) \left(\sum_{t=1}^{T} F_t \iota_N \right)' \right] \\
&\geqslant \frac{T(1-c_N)N}{n} - \frac{T}{n}\sum_{i=1}^{N} \frac{1}{T_i} \\
&\geqslant \frac{1-c_N}{c_u} - \frac{1}{c_u c_T T},
\end{aligned}
$$

where the second equality holds because $D_t'D_t = \operatorname{diag}(d_{1t}, \cdots, d_{Nt})$ and $F_t = \operatorname{diag}(d_{1t}(1-d_{1,t-1}), \cdots, d_{Nt}(1-d_{N,t-1}))$. The first inequality follows from Assumption 1, which states that the listing effect

appears at least $(1-c_N)N$ times and at most N times over the entire panel. Thus $\sum_{t=1}^{T}\iota_N' F_t \iota_N \geqslant (1-c_N)N$, and elements in vector $\sum_{t=1}^{T} F_t\iota_N$ is bounded above by 1. The second inequality holds by $T_i \geqslant c_T T$ in Assumption 1. Therefore, $\lim_{n\to\infty}\frac{T}{n}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)}$ exists and bounded away from zero.

Then according to Assumption 9 (i), the determinant of the matrix (B.5) is,

$$\lim_{n\to\infty}\frac{1}{n}T\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)}\left|\frac{1}{n}E\left[\mathcal{H}^{(-\gamma)}\right]\right| > 0.$$

Then the matrix (B.5) is positive definite, and therefore $\mathcal{H}_{\rho\rho} - \mathcal{H}_{\rho\delta}'\mathcal{H}_{\delta\delta}^{-1}\mathcal{H}_{\rho\delta} > 0$.

2. (Using Assumption 9 (ii)) It is trivial.

Once we have $a_1 = 0$, we then have $\boldsymbol{a}_2 = \mathbf{0}_{(k+3)\times 1}$ and $a_3 = 0$, which means that the information matrix is nonsingular. □

B.4 Asymptotic Representation of Δ_n

Given

$$\Delta_n = \begin{pmatrix} \frac{1}{\sqrt{n}\sigma_0^2}\mathbb{V}'\mathbb{G}_0'\mathbb{P}\mathbb{V} + \frac{1}{\sqrt{n}\sigma_0^2}(\widetilde{\mathbb{G}_0\mathbb{Z}}^{\circ}\delta_0)'\mathbb{V} \\ \frac{1}{\sqrt{n}\sigma_0^2}\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{\circ\prime}\mathbb{V} \\ \frac{1}{2\sqrt{n}\sigma_0^4}\mathbb{V}'\mathbb{P}\mathbb{V} \end{pmatrix},$$

where

$$\begin{aligned}\frac{1}{\sqrt{n}\sigma_0^2}\mathbb{V}'\mathbb{G}_0'\mathbb{P}\mathbb{V} &= \frac{1}{\sqrt{n}}\mathrm{tr}(\mathbb{G}_0'\mathbb{P}) + O_p\left(\frac{1}{\sqrt{T}}\right) = \sqrt{\frac{N}{c_u T}}\frac{1}{N}\mathrm{tr}(\mathbb{G}_0'\mathbb{P}) + O_p\left(\frac{1}{\sqrt{T}}\right), \\ \frac{1}{2\sqrt{n}\sigma_0^4}\mathbb{V}'\mathbb{P}\mathbb{V} &= \frac{1}{2\sqrt{n}\sigma_0^2}\mathrm{tr}(\mathbb{P}) + O_p\left(\frac{1}{\sqrt{T}}\right) = \sqrt{\frac{N}{c_u T}}\frac{1}{2\sigma_0^2} + O_p\left(\frac{1}{\sqrt{T}}\right)\end{aligned}$$

by equation (A.21) in Lemma 12. Additionally, we have

$$\begin{aligned}\frac{1}{\sqrt{n}\sigma_0^2}(\widetilde{\mathbb{G}_0\mathbb{Z}}^{\circ}\delta_0)'\mathbb{V} &= \sqrt{\frac{N}{c_u T}}\frac{1}{N}E\left[(\widetilde{\mathbb{G}_0\mathbb{Z}}^{\circ}\delta_0)'\mathbb{V}\right] + O_p\left(\frac{1}{\sqrt{T}}\right), \\ \frac{1}{\sqrt{n}\sigma_0^2}\left(\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{\circ\prime}\mathbb{V}\right) &= \sqrt{\frac{N}{c_u T}}\frac{1}{N}E\left(\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{\circ\prime}\mathbb{V}\right) + O_p\left(\frac{1}{\sqrt{T}}\right)\end{aligned}$$

by equation (A.15) in Lemma 11. Since $\widetilde{\mathbb{Z}}^{\circ} = \mathbb{P}\mathbb{Z}^{\circ}$, and $Z_t^{\circ} = (D_t(I_N - F_t)D_t'M_t\mathcal{V}_t, D_tD_{t-1}'\mathcal{V}_t, \mathbf{0}_{N_t\times(k+1)})$, thus

$$E[\widetilde{\Gamma}\widetilde{\mathbb{Z}}^{\circ\prime}\mathbb{V}] = \begin{pmatrix} \frac{1}{N}\mathrm{tr}\left(\mathcal{T}^{-1}\sum_{h=1}^{T-1}\sum_{t=T-h}^{T-1} D_{t+1}'D_{t+1}(I_N - F_{t+1})D_{t+1}'M_{t+1}S_t(\rho)^{-1}A_t^{(t+h-T)}(\theta)D_{T-h}\right) \\ \frac{1}{N}\mathrm{tr}\left(\mathcal{T}^{-1}\sum_{h=1}^{T-1}\sum_{t=T-h}^{T-1} D_{t+1}'D_{t+1}D_t'S_t(\rho)^{-1}A_t^{(t+h-T)}(\theta)D_{T-h}\right) \\ \mathbf{0}_{(k+1)\times 1} \end{pmatrix}$$

by equations (A.17) and (A.18) in Lemma 11. Similarly, we have

$$E\left[(\widetilde{\mathbb{G}_0\mathbb{Z}}^{\circ}\delta_0)'\mathbb{V}\right] = \operatorname{tr}\left(\mathcal{T}^{-1}\sum_{h=1}^{T-1}\sum_{t=T-h}^{T-1}\lambda D'_{t+1}G_{t+1}D_{t+1}(I_N - F_{t+1})D'_{t+1}M_{t+1}S_t^{-1}A_t^{(t+h-T)}D_{T-h}\right) + \operatorname{tr}\left(\mathcal{T}^{-1}\sum_{h=1}^{T-1}\sum_{t=T-h}^{T-1}\nu D'_{t+1}G_{t+1}D_{t+1}D'_tS_t^{-1}A_t^{(t+h-T)}D_{T-h}\right),$$

where $G_t \equiv W_tS_t^{-1}$. Thus, $\Delta_n = \sqrt{\frac{N}{c_uT}}\varphi_n(\theta_0) + O_p\left(\frac{1}{\sqrt{T}}\right)$, $\varphi_n(\theta_0) = O(1)$ and

$$\varphi_n(\theta) = \begin{pmatrix} \left[\begin{array}{l} \frac{1}{N}\operatorname{tr}\left(\mathcal{T}^{-1}\sum_{h=1}^{T-1}\sum_{t=T-h}^{T-1}\lambda D'_{t+1}G_{t+1}(\rho)D_{t+1}(I_N - F_{t+1})D'_{t+1}M_{t+1}S_t(\rho)^{-1}A_t^{(t+h-T)}(\theta)D_{T-h}\right) \\ + \frac{1}{N}\operatorname{tr}\left(\mathcal{T}^{-1}\sum_{h=1}^{T-1}\sum_{t=T-h}^{T-1}\nu D'_{t+1}G_{t+1}(\rho)D_{t+1}D'_tS_t(\rho)^{-1}A_t^{(t+h-T)}(\theta)D_{T-h}\right) + \frac{1}{N}\operatorname{tr}(\mathbb{G}(\rho)'\mathbb{P}) \end{array}\right] \\ \frac{1}{N}\operatorname{tr}\left(\mathcal{T}^{-1}\sum_{h=1}^{T-1}\sum_{t=T-h}^{T-1}D'_{t+1}D_{t+1}(I_N - F_{t+1})D'_{t+1}M_{t+1}S_t(\rho)^{-1}A_t^{(t+h-T)}(\theta)D_{T-h}\right) \\ \frac{1}{N}\operatorname{tr}\left(\mathcal{T}^{-1}\sum_{h=1}^{T-1}\sum_{t=T-h}^{T-1}D'_{t+1}D_{t+1}D'_tS_t(\rho)^{-1}A_t^{(t+h-T)}(\theta)D_{T-h}\right) \\ \mathbf{0}_{(k+1)\times 1} \\ \frac{1}{2\sigma^2} \end{pmatrix}, \tag{B.6}$$

where $G_t(\rho) = W_tS_t(\rho)^{-1}$, $A_t^{(h)}(\theta) = A_t(\theta)A_{t-1}(\theta)\cdots A_{t-h+1}(\theta)$ for $h = 1, 2, \cdots$, $A_t^{(0)}(\theta) = I_{N_t}$, and $A_t(\theta) = D_t[\lambda_0(I_N - F_t)D'_tM_t + \nu_0D'_{t-1}]S_{t-1}(\rho)^{-1}$.

B.5 Score, variance, and Hessian formulas for the TWFE QMLE

This subsection provides the TWFE counterparts of the formulas in Sections B.1–B.4. Recall that $\bar{\mathbb{P}}$ is the projection matrix onto the space spanned by the individual and time effects, and $\bar{\mathbb{Q}} = I_n - \bar{\mathbb{P}}$. For $\zeta = (\rho, \delta')'$, denote

$$\bar{\mathbb{V}}(\zeta) = \bar{\mathbb{Q}}\left(\mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta\right).$$

Then the TWFE concentrated log-likelihood function is

$$\bar{L}_n^c(\theta) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 + \log|\mathbb{S}(\rho)| - \frac{1}{2\sigma^2}\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta).$$

The first and second order conditions

For simplicity, denote $\mathbb{G}(\rho) = \mathbb{W}\mathbb{S}^{-1}(\rho)$. The first order derivatives are

$$\frac{\partial\bar{L}_n^c(\theta)}{\partial\theta} = \begin{pmatrix} \frac{1}{\sigma^2}(\mathbb{W}\mathbb{Y})'\bar{\mathbb{V}}(\zeta) - \operatorname{tr}\{\mathbb{G}(\rho)\} \\ \frac{1}{\sigma^2}\mathbb{Z}'\bar{\mathbb{V}}(\zeta) \\ -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta) \end{pmatrix}.$$

The second order derivatives are

$$\frac{\partial^2 \bar{L}_n^c(\theta)}{\partial\theta\partial\theta'} = -\begin{pmatrix} \frac{1}{\sigma^2}(\mathbb{W}\mathbb{Y})'\bar{\mathbb{Q}}\mathbb{W}\mathbb{Y} + \text{tr}\{\mathbb{G}^2(\rho)\} & * & * \\ \frac{1}{\sigma^2}\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{W}\mathbb{Y} & \frac{1}{\sigma^2}\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z} & * \\ \frac{1}{\sigma^4}\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{Q}}\mathbb{W}\mathbb{Y} & \frac{1}{\sigma^4}\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{Q}}\mathbb{Z} & -\frac{n}{2\sigma^4} + \frac{1}{\sigma^6}\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta) \end{pmatrix}.$$

Moreover, the information matrix is $\bar{\Sigma}_{\theta_0,n} \equiv E\left(-\frac{1}{n}\frac{\partial^2 \bar{L}_n^c(\theta_0)}{\partial\theta\partial\theta'}\right) = \bar{\Sigma}^{(1)}_{\theta_0,n} - \bar{\Sigma}^{(2)}_{\theta_0,n}$, where $\bar{\Sigma}^{(1)}_{\theta_0,n} =$

$$\begin{pmatrix} \left[\begin{array}{r} \frac{1}{n\sigma_0^2}E\left[\{\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0 + \mathbb{G}_0\mathbb{J}\tau_0\}'\bar{\mathbb{Q}}\{\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0 + \mathbb{G}_0\mathbb{J}\tau_0\}\right] \\ + \frac{1}{n}\text{tr}\left(\mathbb{G}_0^2 + \mathbb{G}_0'\mathbb{G}_0\right) \end{array}\right] & * & * \\ \frac{1}{n\sigma_0^2}E\left[\mathbb{Z}'\bar{\mathbb{Q}}\{\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0 + \mathbb{G}_0\mathbb{J}\tau_0\}\right] & \frac{1}{n\sigma_0^2}E\left(\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}\right) & * \\ \frac{1}{n\sigma_0^2}\text{tr}\left(\mathbb{G}_0\right) & \mathbf{0}_{1\times(k+3)} & \frac{1}{2\sigma_0^4} \end{pmatrix},$$

and $\bar{\Sigma}^{(2)}_{\theta_0,n} =$

$$\begin{pmatrix} \frac{2}{n\sigma_0^2}E\left[(\mathbb{G}_0\mathbb{Z}\delta_0)'\bar{\mathbb{P}}\mathbb{G}_0\mathbb{V}\right] + \frac{1}{n}\text{tr}\left(\mathbb{G}_0'\bar{\mathbb{P}}\mathbb{G}_0\right) & * & * \\ \frac{1}{n\sigma_0^2}E\left[\mathbb{Z}'\bar{\mathbb{P}}\mathbb{G}_0\mathbb{V}\right] & \mathbf{0}_{(k+3)\times(k+3)} & * \\ \frac{1}{n\sigma_0^4}E\left(\mathbb{V}'\bar{\mathbb{P}}\mathbb{G}_0\mathbb{Z}\delta_0\right) + \frac{1}{n\sigma_0^2}\text{tr}\left(\bar{\mathbb{P}}\mathbb{G}_0\right) & \frac{1}{n\sigma_0^4}E\left(\mathbb{V}'\bar{\mathbb{P}}\mathbb{Z}\right) & \frac{N+T-1}{n\sigma_0^4} \end{pmatrix}.$$

By Lemma 23, all expectation terms in $\bar{\Sigma}^{(2)}_{\theta_0,n}$ are $O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right)$. It remains to control the trace terms. Since $\bar{\mathbb{P}}$ is an orthogonal projection matrix with rank $N+T-1$, and $\mathbb{G}_0$, $\mathbb{G}_0'\mathbb{G}_0$ are UB under Assumptions 3 and 5, we have

$$\frac{1}{n}\left|\text{tr}\left(\bar{\mathbb{P}}\mathbb{G}_0\right)\right| \leqslant \frac{1}{n}\text{rank}\left(\bar{\mathbb{P}}\right)\|\mathbb{G}_0\|_2 = O\left(\frac{N+T}{NT}\right) = O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right).$$

Similarly,

$$\frac{1}{n}\left|\text{tr}\left(\mathbb{G}_0'\bar{\mathbb{P}}\mathbb{G}_0\right)\right| \leqslant \frac{1}{n}\text{rank}\left(\bar{\mathbb{P}}\right)\|\mathbb{G}_0\|_2^2 = O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right).$$

Thus, $\bar{\Sigma}^{(2)}_{\theta_0,n} = O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right)$, which means that $\bar{\Sigma}_{\theta_0,n} = \bar{\Sigma}^{(1)}_{\theta_0,n} + O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right)$. We also have $\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma = \Gamma\bar{\Sigma}^{(1)}_{\theta_0,n}\Gamma + O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right)$ by Lemma 23.

TWFE score decomposition

Let $\bar{\mathbb{P}} = I_n - \bar{\mathbb{Q}}$. Define $\bar{\tilde{\mathbb{Z}}}^\circ = \bar{\mathbb{P}}\mathbb{Z}^\circ$, $\bar{\tilde{\mathbb{Z}}}^* = \bar{\mathbb{Q}}\mathbb{Z} + \bar{\tilde{\mathbb{Z}}}^\circ$, $\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^\circ = \bar{\mathbb{P}}\mathbb{G}_0\mathbb{Z}^\circ$, and $\overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^* = \bar{\mathbb{Q}}\mathbb{G}_0\mathbb{Z} + \overline{\widetilde{\mathbb{G}_0\mathbb{Z}}}^\circ$. At θ_0, the scaled TWFE score can be decomposed as

$$\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^c(\theta_0)}{\partial\theta} = \frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta} - \bar{\Delta}_n,$$

where

$$\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta} = \begin{pmatrix} \frac{1}{\sqrt{n}\sigma_0^2}\left[\left\{\bar{\mathbb{Q}}\mathbb{G}_0(\mathbb{D}\alpha_0+\mathbb{J}\tau_0)+\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^*\delta_0\right\}'\mathbb{V}+\mathbb{V}'\mathbb{G}_0\mathbb{V}\right]-\frac{1}{\sqrt{n}}\operatorname{tr}(\mathbb{G}_0) \\ \frac{1}{\sqrt{n}\sigma_0^2}\tilde{\Gamma}'\bar{\tilde{\mathbb{Z}}}^{*\prime}\mathbb{V} \\ -\frac{\sqrt{n}}{2\sigma_0^2}+\frac{1}{2\sqrt{n}\sigma_0^4}\mathbb{V}'\mathbb{V} \end{pmatrix},$$

and

$$\bar{\Delta}_n = \begin{pmatrix} \frac{1}{\sqrt{n}\sigma_0^2}\mathbb{V}'\mathbb{G}_0'\bar{\mathbb{P}}\mathbb{V}+\frac{1}{\sqrt{n}\sigma_0^2}\left(\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^\circ\delta_0\right)'\mathbb{V} \\ \frac{1}{\sqrt{n}\sigma_0^2}\tilde{\Gamma}'\bar{\tilde{\mathbb{Z}}}^{\circ\prime}\mathbb{V} \\ \frac{1}{2\sqrt{n}\sigma_0^4}\mathbb{V}'\bar{\mathbb{P}}\mathbb{V} \end{pmatrix}.$$

The first component is a mean-zero score component, while $\bar{\Delta}_n$ collects the nonzero mean terms generated by concentrating out the individual and time fixed effects.

Variance of the mean-zero score

The mean-zero score can be written as

$$\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta} = \begin{pmatrix} \frac{1}{\sqrt{n}\sigma_0^2}\left[\bar{\mathbb{Q}}\mathbb{G}_0\left(\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)+\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^*\delta_0\right]'\mathbb{V}+\frac{1}{\sqrt{n}\sigma_0^2}\mathbb{V}'\mathbb{G}_0\mathbb{V}-\frac{1}{\sqrt{n}}\operatorname{tr}(\mathbb{G}_0) \\ \frac{1}{\sqrt{n}\sigma_0^2}\tilde{\Gamma}\bar{\tilde{\mathbb{Z}}}^{*\prime}\mathbb{V} \\ -\frac{\sqrt{n}}{2\sigma_0^2}+\frac{1}{2\sqrt{n}\sigma_0^4}\mathbb{V}'\mathbb{V} \end{pmatrix}.$$

By the same calculation as in Section B.2 and Lemma 24, with $\mathbb{Q}$ replaced by $\bar{\mathbb{Q}}$ and $\mathbb{D}\alpha_0$ replaced by $\mathbb{D}\alpha_0+\mathbb{J}\tau_0$, we have

$$\begin{aligned} &E\left[\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta}\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta'}\Gamma\frac{1}{\sqrt{n}}\right] \\ &= \begin{pmatrix} \bar{\Sigma}_{\rho\rho,n} & \bar{\Sigma}_{\rho\delta,n} & \bar{\Sigma}_{\rho\sigma,n} \\ * & \bar{\Sigma}_{\delta\delta,n} & 0 \\ * & * & \frac{1}{2\sigma_0^4} \end{pmatrix} + \begin{pmatrix} \bar{\Omega}_{\rho\rho,n} & \bar{\Omega}_{\rho\delta,n} & \bar{\Omega}_{\rho\sigma,n} \\ * & \mathbf{0}_{(k+3)\times(k+3)} & \bar{\Omega}_{\delta\sigma,n} \\ * & * & \frac{\mu_4-3\sigma_0^4}{4\sigma_0^8} \end{pmatrix} + O\left(\frac{1}{T}\right)+O\left(\frac{1}{N}\right), \end{aligned} \tag{B.7}$$

where

$$\begin{aligned}
\bar{\Sigma}_{\rho\rho,n} =& \frac{1}{n\sigma_0^2} E\left[\left\{\bar{\mathbb{Q}}\mathbb{G}_0\left(\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)+\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^*\delta_0\right\}'\left\{\bar{\mathbb{Q}}\mathbb{G}_0\left(\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)+\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^*\delta_0\right\}\right] \\
&+\frac{1}{n}\operatorname{tr}\left(\mathbb{G}_0^2+\mathbb{G}_0'\mathbb{G}_0\right), \\
\bar{\Sigma}_{\rho\delta,n} =& \frac{1}{n\sigma_0^2} E\left[\left\{\bar{\mathbb{Q}}\mathbb{G}_0\left(\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)+\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^*\delta_0\right\}'\bar{\tilde{\mathbb{Z}}}^*\right]\tilde{\Gamma}, \\
\bar{\Sigma}_{\delta\delta,n} =& \frac{1}{n\sigma_0^2}\tilde{\Gamma}E\left(\bar{\tilde{\mathbb{Z}}}^{*\prime}\bar{\tilde{\mathbb{Z}}}^*\right)\tilde{\Gamma}, \\
\bar{\Sigma}_{\sigma\rho,n} =& \frac{1}{n\sigma_0^2}\operatorname{tr}(\mathbb{G}_0), \\
\bar{\Omega}_{\rho\rho,n} =& \frac{\mu_4-3\sigma_0^4}{n\sigma_0^4}\sum_{i=1}^n(\mathbb{G}_0)_{ii}^2+\frac{2\mu_3}{n\sigma_0^4}\sum_{i=1}^n E\left[\left\{\bar{\mathbb{Q}}\mathbb{G}_0\left(\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)+\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^*\delta_0\right\}_i\right](\mathbb{G}_0)_{ii}, \\
\bar{\Omega}_{\rho\delta,n} =& \frac{\mu_3}{n\sigma_0^4}\sum_{i=1}^n(\mathbb{G}_0)_{ii}E\left(\bar{\tilde{\mathbb{Z}}}^*\tilde{\Gamma}\right)_{i,\cdot}, \\
\bar{\Omega}_{\rho\sigma,n} =& \frac{\mu_4-3\sigma_0^4}{2n\sigma_0^6}\operatorname{tr}(\mathbb{G}_0)+\frac{\mu_3}{2n\sigma_0^6}\sum_{i=1}^n E\left[\left\{\bar{\mathbb{Q}}\mathbb{G}_0\left(\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)+\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^*\delta_0\right\}_i\right], \\
\bar{\Omega}_{\delta\sigma,n} =& \frac{\mu_3}{2n\sigma_0^6}\sum_{i=1}^n E\left[\left(\bar{\tilde{\mathbb{Z}}}^*\tilde{\Gamma}\right)_{i,\cdot}'\right].
\end{aligned}$$

Since $\bar{\tilde{\mathbb{Z}}}^* = \bar{\mathbb{Q}}\mathbb{Z}+\bar{\tilde{\mathbb{Z}}}^\circ$, and $\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^* = \bar{\mathbb{Q}}\mathbb{G}_0\mathbb{Z}+\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^\circ$, where $E\left(\bar{\tilde{\mathbb{Z}}}^\circ\right)=0$ and $E\left(\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^\circ\right)=0$, and since $\iota_n'\bar{\mathbb{Q}}=0$, we have

$$\sum_{i=1}^n E\left[\left\{\bar{\mathbb{Q}}\mathbb{G}_0\left(\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)+\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^*\delta_0\right\}_i\right]=\iota_n'\bar{\mathbb{Q}}\mathbb{G}_0\left(\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)+\iota_n'\bar{\mathbb{Q}}\mathbb{G}_0E(\mathbb{Z})\delta_0=0,$$

and

$$\sum_{i=1}^n E\left[\left(\bar{\tilde{\mathbb{Z}}}^*\tilde{\Gamma}\right)_{i,\cdot}'\right]=\left\{\iota_n'\bar{\mathbb{Q}}E(\mathbb{Z})\tilde{\Gamma}\right\}'=\mathbf{0}_{(k+3)\times 1}.$$

Therefore,

$$\bar{\Omega}_{\rho\sigma,n}=\frac{\mu_4-3\sigma_0^4}{2n\sigma_0^6}\operatorname{tr}(\mathbb{G}_0),\qquad \bar{\Omega}_{\delta\sigma,n}=\mathbf{0}_{(k+3)\times 1}.$$

If v_{it}'s are normally distributed, then $\mu_3=0$ and $\mu_4-3\sigma_0^4=0$. Hence $\Gamma\bar{\Omega}_{\theta_0,n}\Gamma=0$.

Some properties of the Hessian matrix

For the Hessian matrix $\frac{1}{n}\Gamma\frac{\partial^2\bar{L}_n^c(\theta)}{\partial\theta\partial\theta'}\Gamma$, we have the following results.

(1). The Hessian matrix is Lipschitz continuous at θ_0, that is,

$$-\frac{1}{n}\Gamma\frac{\partial^2\bar{L}_n^c(\theta)}{\partial\theta\partial\theta'}\Gamma-\left(-\frac{1}{n}\Gamma\frac{\partial^2\bar{L}_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma\right)=\left\|\Gamma^{-1}(\theta-\theta_0)\right\|_2 O_p(1). \tag{B.8}$$

(2). The Hessian matrix is consistent at θ_0, that is,

$$-\frac{1}{n}\Gamma\frac{\partial^2\bar{L}_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma-\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma=O_p\left(\frac{1}{\sqrt{NT}}\right). \tag{B.9}$$

(3). The information matrix $\bar{\Sigma}_{\theta_0}=\lim_{n\to\infty}\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma$ is nonsingular.

Proof. For (1), the proof is the same as that of the Lipschitz continuity of the Hessian matrix in Section B.3. The only difference is that $\mathbb{Q}$ is replaced by $\bar{\mathbb{Q}}$. Since $\bar{\mathbb{Q}}$ is also an orthogonal projection matrix, the TWFE linear–quadratic form bounds in Lemma 23 give the same orders. Therefore,

$$-\frac{1}{n}\Gamma\frac{\partial^2\bar{L}_n^c(\theta)}{\partial\theta\partial\theta'}\Gamma-\left(-\frac{1}{n}\Gamma\frac{\partial^2\bar{L}_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma\right)=\left\|\Gamma^{-1}(\theta-\theta_0)\right\|_2 O_p(1).$$

For (2), at θ_0, we have

$$-\frac{1}{n}\Gamma\frac{\partial^2\bar{L}_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma=\begin{pmatrix}\frac{1}{n\sigma_0^2}(\mathbb{W}\mathbb{Y})'\bar{\mathbb{Q}}\mathbb{W}\mathbb{Y}+\frac{1}{n}\operatorname{tr}(\mathbb{G}_0^2) & \frac{1}{n\sigma_0^2}(\mathbb{W}\mathbb{Y})'\bar{\mathbb{Q}}\mathbb{Z}\tilde{\Gamma} & \frac{1}{n\sigma_0^4}(\mathbb{W}\mathbb{Y})'\bar{\mathbb{Q}}\mathbb{V}\\ * & \frac{1}{n\sigma_0^2}\tilde{\Gamma}\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}\tilde{\Gamma} & \frac{1}{n\sigma_0^4}\tilde{\Gamma}\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{V}\\ * & * & -\frac{1}{2\sigma_0^4}+\frac{1}{n\sigma_0^6}\mathbb{V}'\bar{\mathbb{Q}}\mathbb{V}\end{pmatrix}.$$

Moreover, $\mathbb{W}\mathbb{Y}=\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{G}_0\left(\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)+\mathbb{G}_0\mathbb{V}$. Hence, after substituting this expression into the above Hessian matrix, its stochastic blocks are finite linear combinations of TWFE linear–quadratic forms involving $\mathbb{Z}$ and $\mathbb{V}$, together with nonstochastic terms. By Lemma 23, we have

$$-\frac{1}{n}\Gamma\frac{\partial^2\bar{L}_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma-\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma=O_p\left(\frac{1}{\sqrt{NT}}\right).$$

This proves (2).

For (3), the proof is the same as the proof of nonsingularity in Section B.3. Specifically, after replacing the OWFE information matrix by $\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma$, the Schur complement argument gives the same two alternatives as in Assumption 9′. Thus, under Assumption 9′, $\bar{\Sigma}_{\theta_0}$ is nonsingular. This completes the proof of (1)–(3). □

TWFE bias representation

Using the decomposition of $\bar{\Delta}_n$ above and the identity $\bar{\mathbb{P}}=\mathbb{P}+\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}$, we have

$$\bar{\tilde{\mathbb{Z}}}^{\circ}=\tilde{\mathbb{Z}}^{\circ}+\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathbb{Z}^{\circ},$$

and

$$\widetilde{\overline{\mathbb{G}_0\mathbb{Z}}}^{\circ}=\widetilde{\mathbb{G}_0\mathbb{Z}}^{\circ}+\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}\mathbb{G}_0\mathbb{Z}^{\circ}.$$

Therefore, $\bar{\Delta}_n$ can be decomposed into the OWFE component generated by $\mathbb{P}$ and the time-effect component generated by $\mathbb{Q}\mathbb{J}(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}\mathbb{J}'\mathbb{Q}$. The leading deterministic part of the OWFE component is summarized by $\varphi_n(\theta)$, whose explicit expression is given in equation (B.6).

For the time-effect component, let K_t be the $N_t\times(T-1)$ matrix such that $(K_1',K_2',\cdots,K_T')'=\mathbb{Q}\mathbb{J}$. Then according to equations (A.40) and (A.41), the additional time-effect bias component is

$$\psi_n(\theta)\equiv\begin{pmatrix}\frac{1}{T}\sum_{t=1}^{T}\operatorname{tr}\left[(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}K_t'G_t(\rho)K_t\right]+\frac{1}{T}\sum_{t=1}^{T-1}\sum_{h=T-t}^{T-1}\operatorname{tr}\left[(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}K_{t+1}'\mathcal{M}_{\rho,t,h}(\theta)K_{T-h}\right]\\ \frac{1}{T}\sum_{t=1}^{T-1}\sum_{h=T-t}^{T-1}\operatorname{tr}\left[(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}K_{t+1}'\mathcal{M}_{\lambda,t,h}(\theta)K_{T-h}\right]\\ \frac{1}{T}\sum_{t=1}^{T-1}\sum_{h=T-t}^{T-1}\operatorname{tr}\left[(\mathbb{J}'\mathbb{Q}\mathbb{J})^{-1}K_{t+1}'\mathcal{M}_{\nu,t,h}(\theta)K_{T-h}\right]\\ \mathbf{0}_{k+1}\\ \frac{T-1}{2T\sigma^2}\end{pmatrix},\tag{B.10}$$

where

$$\begin{aligned}\mathcal{M}_{\lambda,t,h}(\theta) =& D_{t+1}(I_N - F_{t+1})D_{t+1}'M_{t+1}S_t^{-1}(\rho)A_t^{(t+h-T)}(\theta),\\ \mathcal{M}_{\nu,t,h}(\theta) =& D_{t+1}D_t'S_t^{-1}(\rho)A_t^{(t+h-T)}(\theta),\\ \mathcal{M}_{\rho,t,h}(\theta) =& G_{t+1}(\rho)\left[\lambda\mathcal{M}_{\lambda,t,h}(\theta) + \nu\mathcal{M}_{\nu,t,h}(\theta)\right].\end{aligned}$$

We have $\psi_n(\theta_0) = O(1)$ by equation (A.44). Thus, by (A.39) and (A.45),

$$\bar{\Delta}_n = \sqrt{\frac{N}{c_u T}}\varphi_n(\theta_0) + \sqrt{\frac{T}{c_u N}}\psi_n(\theta_0) + O_p\left(\frac{1}{\sqrt{T}}\right) + O_p\left(\frac{1}{\sqrt{N}}\right), \tag{B.11}$$

where $c_u = n/(NT)$, $\varphi_n(\theta_0) = O(1)$, and $\psi_n(\theta_0) = O(1)$.

Furthermore, by the nonsingularity of $\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma$ established above and the same argument as in Section B.4, $\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0) = O_p(1)$, and $\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0) = O_p(1)$.

C Proof of Theorems

C.1 Proof of Theorem 1

Since only $O(N)$ observations contain information about γ, this parameter exhibits a slower convergence rate compared to others. Denote $\theta^{(-\gamma)}$ as the vector obtained by removing γ from θ, with $\hat{\theta}^{(-\gamma)}$ and $\theta_0^{(-\gamma)}$ defined analogously. We will first prove that $\hat{\theta}^{(-\gamma)} \xrightarrow{p} \theta_0^{(-\gamma)}$, then the consistency of $\hat{\gamma}$ follows directly from its explicit expression in terms of $\hat{\theta}^{(-\gamma)}$.

Recall that

$$Q_n(\theta) = E[L_n^c(\theta)] = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\sigma^2 + \ln|\mathbb{S}(\rho)| - \frac{1}{2\sigma^2}E\left[\widetilde{\mathbb{V}}'(\zeta)\widetilde{\mathbb{V}}(\zeta)\right].$$

The consistency of $\hat{\theta}^{(-\gamma)}$ comes from uniform convergence and identifiable uniqueness, we first show the uniform convergence of $\frac{1}{n}(L_n^c(\theta) - Q_n(\theta))$, and then show the identifiable uniqueness by proving the equicontinuity of $Q_n(\theta)$ and the global identified condition. Our proof follows the idea in Yu et al. (2008).

Proof of uniform convergence

The uniform convergence is specifically defined as $\sup_{\theta\in\Theta}|\frac{1}{n}(L_n^c(\theta) - Q_n(\theta))| \xrightarrow{p} 0$. Denote the true value of each parameter as $\theta_0 = (\rho_0, \delta_0', \sigma_0^2)$, $\zeta_0 = (\rho_0, \delta_0')$. As

$$\frac{1}{n}(L_n^c(\theta) - Q_n(\theta)) = \frac{1}{2\sigma^2}\left(\frac{1}{n}\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta) - \frac{1}{n}E\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta)\right),$$

we will first show $\sup_{\theta\in\Theta}\frac{1}{n}\left|\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta) - E\left[\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta)\right]\right| \xrightarrow{p} 0$. Since $\widetilde{\mathbb{V}}(\zeta) = \mathbb{Q}(\mathbb{Y} - \rho\mathbb{W}\mathbb{Y} - \mathbb{Z}\delta)$, substituting $\mathbb{Y}$ for $\mathbb{Y} = \rho_0\mathbb{W}\mathbb{Y} + \mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{V}$, we have $\widetilde{\mathbb{V}}(\zeta) = \mathbb{Q}[\mathbb{V} - (\rho - \rho_0)\mathbb{W}\mathbb{Y} - \mathbb{Z}(\delta - \delta_0)]$. Therefore

$$\begin{aligned}\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta) =& \mathbb{V}'\mathbb{Q}\mathbb{V} + (\rho-\rho_0)^2(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y} + (\delta-\delta_0)'\mathbb{Z}'\mathbb{Q}\mathbb{Z}(\delta-\delta_0) -\\ & 2(\rho-\rho_0)\mathbb{V}'\mathbb{Q}\mathbb{W}\mathbb{Y} - 2\mathbb{V}'\mathbb{Q}\mathbb{Z}(\delta-\delta_0) + 2(\rho-\rho_0)(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{Z}(\delta-\delta_0).\end{aligned} \tag{C.1}$$

Since $\mathbb{W}\mathbb{Y} = \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{V})$, $\mathbb{V}'\mathbb{Q}\mathbb{W}\mathbb{Y} = \mathbb{V}'\mathbb{Q}\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{V}'\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0 + \mathbb{V}'\mathbb{Q}\mathbb{G}_0\mathbb{V}$, $(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{Z} = (\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{Z} + (\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{Z} + (\mathbb{G}_0\mathbb{V})'\mathbb{Q}\mathbb{Z}$, and

$$\begin{aligned}(\mathbb{W}\mathbb{Y})'\mathbb{Q}\mathbb{W}\mathbb{Y} =& (\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{G}_0\mathbb{Z}\delta_0 + (\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0 + (\mathbb{G}_0\mathbb{V})'\mathbb{Q}\mathbb{G}_0\mathbb{V} +\\ & 2(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0 + 2(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{G}_0\mathbb{V} + 2(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{G}_0\mathbb{V}.\end{aligned} \tag{C.2}$$

Since $\mathbb{G}_0 = \mathbb{W}\mathbb{S}_0^{-1} = \text{blkdiag}(W_1S_1^{-1}, W_2S_2^{-1}, \cdots, W_TS_T^{-1})$, where $W_tS_t^{-1}$ is UB uniformly for t and $\|\sup_t \text{abs}(W_tS_t^{-1})\|_\infty < \infty$ is ensured by Assumption 5 and Lemma 4, it follows from Lemma 13 that

$$\frac{1}{n}\mathbb{V}'\mathbb{Q}\mathbb{V} - \frac{1}{n}E[\mathbb{V}'\mathbb{Q}\mathbb{V}] \xrightarrow{p} 0, \qquad \frac{1}{n}\mathbb{V}'\mathbb{Q}\mathbb{G}_0\mathbb{V} - \frac{1}{n}E[\mathbb{V}'\mathbb{Q}\mathbb{G}_0\mathbb{V}] \xrightarrow{p} 0,$$
$$\frac{1}{n}(\mathbb{G}_0\mathbb{V})'\mathbb{Q}\mathbb{G}_0\mathbb{V} - \frac{1}{n}E[(\mathbb{G}_0\mathbb{V})'\mathbb{Q}\mathbb{G}_0\mathbb{V}] \xrightarrow{p} 0, \qquad \frac{1}{n}\mathbb{Z}'\mathbb{Q}\mathbb{Z} - \frac{1}{n}E[\mathbb{Z}'\mathbb{Q}\mathbb{Z}] \xrightarrow{p} 0,$$
$$\frac{1}{n}(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{Z} - \frac{1}{n}E[(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{Z}] \xrightarrow{p} 0, \qquad \frac{1}{n}(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{G}_0\mathbb{Z}\delta_0 - \frac{1}{n}E[(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{G}_0\mathbb{Z}\delta_0] \xrightarrow{p} 0,$$
$$\frac{1}{n}\mathbb{V}'\mathbb{Q}\mathbb{Z} - \frac{1}{n}E[\mathbb{V}'\mathbb{Q}\mathbb{Z}] \xrightarrow{p} 0, \qquad \frac{1}{n}\mathbb{V}'\mathbb{Q}\mathbb{G}_0\mathbb{Z}\delta_0 - \frac{1}{n}E[\mathbb{V}'\mathbb{Q}\mathbb{G}_0\mathbb{Z}\delta_0] \xrightarrow{p} 0,$$
$$\frac{1}{n}(\mathbb{G}_0\mathbb{V})'\mathbb{Q}\mathbb{Z} - \frac{1}{n}E[(\mathbb{G}_0\mathbb{V})'\mathbb{Q}\mathbb{Z}] \xrightarrow{p} 0, \qquad \frac{1}{n}(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{G}_0\mathbb{V} - \frac{1}{n}E[(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{G}_0\mathbb{V}] \xrightarrow{p} 0.$$

Additionally, according to Lemma 8,

$$\frac{1}{n}\mathbb{V}'\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0 \xrightarrow{p} 0, \qquad \frac{1}{n}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{G}_0\mathbb{V} \xrightarrow{p} 0,$$
$$\frac{1}{n}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{Z} - \frac{1}{n}E[(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{Z}] \xrightarrow{p} 0, \qquad \frac{1}{n}(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0 - \frac{1}{n}E[(\mathbb{G}_0\mathbb{Z}\delta_0)'\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0] \xrightarrow{p} 0.$$

Therefore, all polynomial coefficients in $\frac{1}{n}\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta)$ will converge to their expected values. Since ρ and δ are bounded in Θ, it follows that $\sup_{\theta\in\Theta}\frac{1}{n}|\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta) - E[\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta)]| \xrightarrow{p} 0$. Since σ^2 is bounded away from zero, we therefore have $\sup_{\theta\in\Theta}\frac{1}{n}|L_n^c(\theta) - Q_n(\theta)| \xrightarrow{p} 0$.

Proof of equicontinuity

Next we prove that $\frac{1}{n}Q_n(\theta)$ is uniformly equicontinuous for $\theta \in \Theta$, which is specifically defined as for any $\epsilon > 0$, there exists $\eta > 0$ and $n_0 \in \mathbb{N}$, such that whenever $d(\theta, \bar{\theta}) < \eta$ and $n > n_0$, we have

$$\left|\frac{1}{n}\left(Q_n(\theta) - Q_n(\bar{\theta})\right)\right| < \epsilon.$$

Since

$$\frac{1}{n}Q_n(\theta) = -\frac{1}{2}\ln 2\pi - \frac{1}{2}\ln\sigma^2 + \frac{1}{n}\ln|\mathbb{S}(\rho)| - \frac{1}{2n\sigma^2}E\left[\widetilde{\mathbb{V}}'(\zeta)\widetilde{\mathbb{V}}(\zeta)\right],$$

where $E[\widetilde{\mathbb{V}}'(\zeta)\widetilde{\mathbb{V}}(\zeta)] = (\mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta)'\mathbb{Q}(\mathbb{S}(\rho)\mathbb{Y} - \mathbb{Z}\delta)$, substitute $\mathbb{Y}$ for $\mathbb{Y} = \mathbb{S}_0^{-1}(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{V})$, then

$$\begin{aligned}
E[\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta)] =& E\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{V}) - \mathbb{Z}\delta\right)'\mathbb{Q}\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{V}) - \mathbb{Z}\delta\right)\right] \\
=& E\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0) - \mathbb{Z}\delta\right)'\mathbb{Q}\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0) - \mathbb{Z}\delta\right)\right] \\
&+ E\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\mathbb{V}\right)'\mathbb{Q}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\mathbb{V}\right] + 2E\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0) - \mathbb{Z}\delta\right)'\mathbb{Q}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\mathbb{V}\right],
\end{aligned} \tag{C.3}$$

where $E\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\mathbb{V}\right)'\mathbb{Q}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\mathbb{V}\right] = \sigma_0^2\text{tr}\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)'\mathbb{Q}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right]$, and

$$E\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0) - \mathbb{Z}\delta\right)'\mathbb{Q}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\mathbb{V}\right] = O(N)$$

by Lemma 13.

Denote $\sigma_n^2 \equiv \frac{\sigma_0^2}{n}\text{tr}[(\mathbb{S}(\rho)\mathbb{S}_0^{-1})'\mathbb{S}(\rho)\mathbb{S}_0^{-1}]$. Since

$$\frac{\sigma_0^2}{n}\text{tr}\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)'\mathbb{P}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right] = \frac{\sigma_0^2}{n}\text{tr}\left[\mathbb{D}\mathcal{T}^{-1}\mathbb{D}'\mathbb{S}(\rho)\mathbb{S}_0^{-1}\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)'\right]$$
$$\leqslant \frac{\sigma_0^2}{c_T nT}\text{tr}\left[\text{abs}\left(\mathbb{D}'\mathbb{S}(\rho)\mathbb{S}_0^{-1}\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)'\mathbb{D}\right)\right] = O\left(\frac{1}{T}\right),$$

where the inequality holds because $T_i > c_T T$ by Assumption 1 and the matrix inside the trace operation is UB. Thus, $\frac{\sigma_0^2}{n}\text{tr}\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)'\mathbb{Q}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right] = \sigma_n^2 + O\left(\frac{1}{T}\right)$. Further substitute $\mathbb{S}(\rho)\mathbb{S}_0^{-1}$ for $\mathbb{S}(\rho)\mathbb{S}_0^{-1} = I_n + (\rho_0 - \rho)\mathbb{G}_0$, then we have

$$\frac{1}{n}E[\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta)] = \frac{1}{n}E\left\{[(\rho_0-\rho)\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0) + \mathbb{Z}(\delta_0 - \delta)]'\mathbb{Q}\left[(\rho_0-\rho)\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0) + \mathbb{Z}(\delta_0 - \delta)\right]\right\}$$
$$+ \sigma_n^2 + o(1), \tag{C.4}$$

where the first term is equal to $(\rho-\rho_0, \delta'-\delta_0')E[\mathcal{H}](\rho-\rho_0, \delta'-\delta_0')'$, and $\mathcal{H} \equiv \frac{1}{n}(\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{G}_0\mathbb{D}\alpha_0, \mathbb{Z})'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{G}_0\mathbb{D}\alpha_0, \mathbb{Z})$. Therefore, the uniform equicontinuity of $\frac{1}{n}Q_n(\theta)$ is sufficient to show the following is uniformly equicontinuous: (i) $-\frac{1}{2}\ln\sigma^2$; (ii) $\frac{1}{n}\ln|\mathbb{S}(\rho)|$; (iii) $(\rho-\rho_0, \delta'-\delta_0')E[\mathcal{H}](\rho-\rho_0, \delta'-\delta_0')'$; (iv) σ_n^2.

Since σ^2 is bounded away from zero, (i) is obviously uniformly equicontinuous. For (ii), $\frac{1}{n}(\ln|\mathbb{S}(\rho_1)| - \ln|\mathbb{S}(\rho_2)|) = (\rho_1-\rho_2)\frac{1}{n}\text{tr}(\mathbb{W}\mathbb{S}^{-1}(\bar{\rho}))$, where $\bar{\rho}$ lies between ρ_1 and ρ_2. According to equation (B.3), $\sup_{N,T}\|\mathbb{W}\mathbb{S}^{-1}(\bar{\rho})\|_\infty < \infty$, so $\frac{1}{n}\text{tr}(\mathbb{W}\mathbb{S}^{-1}(\bar{\rho}))$ is bounded, which means that (ii) is uniformly equicontinuous. (iii) is uniformly equicontinuous because the parameter space (ρ, δ) is bounded and $E[\mathcal{H}]$ is $O(1)$. For (iv),

$$\frac{\sigma_0^2}{n}\{\text{tr}[(\mathbb{S}(\rho_1)\mathbb{S}_0^{-1})'\mathbb{Q}\mathbb{S}(\rho_1)\mathbb{S}_0^{-1}] - \text{tr}[(\mathbb{S}(\rho_2)\mathbb{S}_0^{-1})'\mathbb{Q}\mathbb{S}(\rho_2)\mathbb{S}_0^{-1}]\}$$
$$= \frac{\sigma_0^2}{n}\{(\rho_0-\rho_1)^2\text{tr}(\mathbb{G}_0'\mathbb{Q}\mathbb{G}_0) + 2(\rho_0-\rho_1)\text{tr}(\mathbb{G}_0\mathbb{Q}) - (\rho_0-\rho_2)^2\text{tr}(\mathbb{G}_0'\mathbb{Q}\mathbb{G}_0) - 2(\rho_0-\rho_2)\text{tr}(\mathbb{G}_0\mathbb{Q})\}$$
$$= 2(\rho_2-\rho_1)\sigma_0^2\frac{\text{tr}(\mathbb{G}_0\mathbb{Q})}{n} + (2\rho_0-\rho_1-\rho_2)(\rho_2-\rho_1)\sigma_0^2\frac{\text{tr}(\mathbb{G}_0'\mathbb{Q}\mathbb{G}_0)}{n}.$$

Since $\mathbb{G}_0$ and $\mathbb{Q}$ are UB, similarly (iv) is uniformly equicontinuous.

Proof of global identification condition

Then we prove that θ_0, except γ_0, is identified, which is specifically defined as $\forall\theta^{(-\gamma)} \neq \theta_0^{(-\gamma)}$,

$$\limsup_{n\to\infty}\frac{1}{n}[Q_n(\theta) - Q_n(\theta_0)] < 0.$$

Note that

$$\frac{1}{n}\left(Q_n(\theta) - Q_n(\theta_0)\right) = -\frac{1}{2}\left(\ln\sigma^2 - \ln\sigma_0^2\right) + \frac{1}{n}\left(\ln|\mathbb{S}(\rho)| - \ln|\mathbb{S}_0|\right) - \frac{1}{2n\sigma^2}E\left[\widetilde{\mathbb{V}}(\zeta)'\widetilde{\mathbb{V}}(\zeta)\right] + \frac{c_u T - 1}{2c_u T}, \tag{C.5}$$

where $c_u \equiv \frac{n}{NT}$. According to equation (C.4), we can decompose equation (C.5) into

$$\frac{1}{n}\left(Q_n(\theta) - Q_n(\theta_0)\right) = L_1(\rho, \sigma^2) - \frac{1}{2\sigma^2}L_2(\rho, \delta) + o(1),$$

where $L_1(\rho, \sigma^2) \equiv -\frac{1}{2}\left(\ln\sigma^2 - \ln\sigma_0^2\right) + \frac{1}{n}\left(\ln|\mathbb{S}(\rho)| - \ln|\mathbb{S}_0|\right) - \frac{1}{2\sigma^2}(\sigma_n^2 - \sigma^2)$ and $L_2(\rho, \delta) \equiv \frac{1}{n}E\{[(\rho_0 - \rho)\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0) + \mathbb{Z}(\delta_0 - \delta)]'\mathbb{Q}[(\rho_0 - \rho)\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0) + \mathbb{Z}(\delta_0 - \delta)]\}$.

For a spatial autoregressive model $\mathbb{Y} = \rho_0\mathbb{W}\mathbb{Y} + \mathbb{V}$ with $v_{it} \sim \text{N}(0, \sigma_0^2)$, the information inequality of this model ensures that

$$-\frac{1}{2}\left(\ln\sigma^2 - \ln\sigma_0^2\right) + \frac{1}{n}\left(\ln|\mathbb{S}(\rho)| - \ln|\mathbb{S}_0|\right) - \frac{1}{2\sigma^2}(\sigma_n^2 - \sigma^2) \leqslant 0.$$

Note that left-hand side of the above equation is precisely $L_1(\rho, \sigma^2)$.

Denote $\delta^{(-\gamma)}$ as the vector obtained by removing γ from δ, with $\hat{\delta}^{(-\gamma)}$ and $\delta_0^{(-\gamma)}$ defined analogously. By decomposing $\mathbb{Z}\delta$ into $\mathbb{Z}^{(-\gamma)}\delta^{(-\gamma)} + \mathbb{Z}^{(\gamma)}\gamma$, $L_2(\rho, \delta)$ can be viewed as a quadratic function of ρ, $\delta^{(-\gamma)}$ and γ, i.e.,

$$L_2(\rho, \delta^{(-\gamma)}, \gamma) = \left(\rho - \rho_0, \delta^{(-\gamma)\prime} - \delta_0^{(-\gamma)\prime}\right) E\left[\mathcal{H}^{(-\gamma)}\right] \left(\rho - \rho_0, \delta^{(-\gamma)\prime} - \delta_0^{(-\gamma)\prime}\right)' + \\ + \frac{2}{n}\left[(\rho_0 - \rho)\mathbb{G}_0(E\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0) + E\mathbb{Z}^{(-\gamma)}(\delta^{(-\gamma)} - \delta_0^{(-\gamma)})\right]' \mathbb{Q}\mathbb{Z}^{(\gamma)}\gamma + \frac{1}{n}\gamma'\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)}\gamma,$$

where $\mathcal{H}^{(-\gamma)} = \frac{1}{n}'(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0, \mathbb{Z}^{(-\gamma)})'\mathbb{Q}(\mathbb{G}_0\mathbb{Z}\delta_0 + \mathbb{G}_0\mathbb{D}\alpha_0, \mathbb{Z}^{(-\gamma)})$, and $\mathbb{Z}^{(\gamma)} = ((D_1F_1\iota_N)', \cdots, (D_TF_T\iota_N)')'$ is a sparse binary vector with at most N nonzero elements, as we can only observe at most N listed nodes. Therefore,

$$\frac{2}{n}\left[(\rho_0 - \rho)\mathbb{G}_0(E\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0) + E\mathbb{Z}^{(-\gamma)}(\delta^{(-\gamma)} - \delta_0^{(-\gamma)})\right]' \mathbb{Q}\mathbb{Z}^{(\gamma)}\gamma + \frac{1}{n}\gamma'\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)}\gamma = o(1),$$

because elements in $E\mathbb{Z}$ and $E\mathbb{Z}^{(-\gamma)}$ are uniformly bounded and $\mathbb{G}_0$ is UB. Thus

$$\lim_{n\to\infty} L_2(\rho, \delta^{(-\gamma)}, \gamma) = \left(\rho - \rho_0, \delta^{(-\gamma)\prime} - \delta_0^{(-\gamma)\prime}\right) E\left[\mathcal{H}^{(-\gamma)}\right] \left(\rho - \rho_0, \delta^{(-\gamma)\prime} - \delta_0^{(-\gamma)\prime}\right)' \geqslant 0,$$

because $\mathcal{H}^{(-\gamma)}$ is a quadratic form. Below we will show that $(\rho, \delta^{(-\gamma)})$ is globally identified by using either Assumption 8 (i) or Assumption 8 (ii).

1. (Using Assumption 8 (i)) According to Assumption 8 (i), we have $\lim_{n\to\infty} L_2(\rho, \delta^{(-\gamma)}, \gamma) > 0$ whenever $(\rho, \delta^{(-\gamma)}) \neq (\rho_0, \delta_0^{(-\gamma)})$. Combined with $L_1(\rho, \sigma^2) \leqslant 0$, it implies that $(\rho, \delta^{(-\gamma)})$ is globally identified.

2. (Using Assumption 8 (ii)) Since σ_n^2 is uniquely maximizer of $L_1(\rho, \sigma^2)$ over σ^2, the information inequality of the SAR model imples that $L_1(\rho, \sigma^2) \leqslant L_1(\rho, \sigma_n^2) \leqslant 0$. Moreover, $L_1(\rho, \sigma_n^2) = 0$ if and only if the pdf $f(\mathbb{Y}|\rho, \sigma_n^2) = f(\mathbb{Y}|\rho_0, \sigma_0^2)$, where

$$f(\mathbb{Y}|\rho, \sigma_n^2) = \frac{1}{(2\pi)^{\frac{n}{2}}\left|\sigma_n^2\mathbb{S}(\rho)^{-1}\mathbb{S}(\rho)^{-1\prime}\right|^{\frac{1}{2}}} \exp\left[-\frac{1}{2\sigma_n^2}\mathbb{Y}'\mathbb{S}(\rho)'\mathbb{S}(\rho)\mathbb{Y}\right].$$

   If Assumption 8 (ii) holds, $f(\mathbb{Y}|\rho, \sigma_n^2) \neq f(\mathbb{Y}|\rho_0, \sigma_0^2)$ for $\rho \neq \rho_0$ as $n \to \infty$. Then $\limsup_{n\to\infty} L_1(\rho, \sigma^2) < 0$ whenever $\rho \neq \rho_0$. Combined with $\lim_{n\to\infty} L_2(\rho, \delta^{(-\gamma)}, \gamma) \geqslant 0$, ρ_0 is globally identified. Then $\delta^{(-\gamma)}$ will also be identified because,

$$\lim_{n\to\infty} L_2(\rho_0, \delta^{(-\gamma)}, \gamma) = \left(\delta^{(-\gamma)} - \delta_0^{(-\gamma)}\right)' \left[\lim_{n\to\infty} \frac{1}{n} E\left(\mathbb{Z}^{(-\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(-\gamma)}\right)\right] \left(\delta^{(-\gamma)} - \delta_0^{(-\gamma)}\right) > 0$$

   whenever $\delta^{(-\gamma)} \neq \delta_0^{(-\gamma)}$ by Assumption 4.

Once $(\rho, \delta^{(-\gamma)})$ is globally identified, σ^2 is also globally identified because $\limsup_{n\to\infty} L_1(\rho_0, \sigma^2) = \frac{1}{2}(\ln\frac{\sigma_0^2}{\sigma^2} - \frac{\sigma_0^2}{\sigma^2} + 1) \leqslant 0$ and the equality holds if and only if $\sigma^2 = \sigma_0^2$, because $\ln x - x + 1 \leqslant 0$ and the equality holds only when $x = 1$. Therefore, $\limsup_{n\to\infty} \frac{1}{n}\left(Q_n(\theta) - Q_n(\theta_0)\right) < 0$ for any $\theta^{(-\gamma)} \neq \theta_0^{(-\gamma)}$, which means $\theta^{(-\gamma)}$ is globally identified.

Since Θ is compact and $Q_n(\theta)$ is uniformly equicontinuous, combined with uniformly convergence and global identification of $\theta^{(-\gamma)}$, we can prove the consistency of $\hat{\theta}^{(-\gamma)}$. Given $\hat{\theta}^{(-\gamma)}$, $L_n^c(\theta)$ is maximized at

$$\begin{aligned}\hat{\gamma} &= \left(\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)}\right)^{-1}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\left(\mathbb{S}(\hat{\rho})\mathbb{Y} - \mathbb{Z}^{(-\gamma)}\hat{\delta}^{(-\gamma)}\right) \\ &= \left(\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)}\right)^{-1}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\left(\mathbb{S}(\hat{\rho})\mathbb{S}_0^{-1}\left(\mathbb{Z}^{(-\gamma)}\delta_0^{(-\gamma)} + \mathbb{Z}^{(\gamma)}\gamma_0 + \mathbb{D}\alpha_0 + \mathbb{V}\right) - \mathbb{Z}^{(-\gamma)}\hat{\delta}^{(-\gamma)}\right).\end{aligned}$$

Since $\hat{\theta}^{(-\gamma)} \xrightarrow{p} \theta_0^{(-\gamma)}$, $\hat{\gamma} = \gamma_0 + (\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)})^{-1}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{V} + o_p(1)$. Additionally, $(\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)})^{-1}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{V} = o_p(1)$ because $E[(\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)})^{-1}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{V}] = 0$ and

$$\begin{aligned}\operatorname{Var}\left[\left(\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)}\right)^{-1}\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{V}\right] &= \sigma_0^2\left(\mathbb{Z}^{(\gamma)\prime}\mathbb{Q}\mathbb{Z}^{(\gamma)}\right)^{-1} = \sigma_0^2\left(\mathbb{Z}^{(\gamma)\prime}\mathbb{Z}^{(\gamma)} - \mathbb{Z}^{(\gamma)\prime}\mathbb{D}\mathcal{T}^{-1}\mathbb{D}'\mathbb{Z}^{(\gamma)}\right)^{-1} \\ &\leqslant \sigma_0^2\left[(1-c_N)N - \frac{N}{c_T T}\right]^{-1} = O\left(\frac{1}{N}\right),\end{aligned}$$

where the inequality holds because $\mathbb{Z}^{(\gamma)}$ is a binary vector with at least $(1-c_N)N$ nonzero elements and $T_i \geqslant c_T T$ by Assumption 1. Therefore, $\hat{\gamma} \xrightarrow{p} \gamma_0$, so $\hat{\theta} \xrightarrow{p} \theta_0$.

C.2 Proof of Theorem 2

According to the Taylor expansion $\frac{\partial L_n^c(\hat{\theta})}{\partial\theta} = \frac{\partial L_n^c(\theta_0)}{\partial\theta} + \frac{\partial^2 L_n^c(\bar{\theta})}{\partial\theta\partial\theta'}(\hat{\theta}-\theta_0)$, where $\frac{\partial L_n^c(\hat{\theta})}{\partial\theta} = 0$ and $\bar{\theta}$ lies between $\hat{\theta}$ and θ_0, we further have

$$\sqrt{n}\Gamma^{-1}(\hat{\theta}-\theta_0) = \left(-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\bar{\theta})}{\partial\theta\partial\theta'}\Gamma\right)^{-1}\left(\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta} - \Delta_n\right), \tag{C.6}$$

where $\Delta_n = \sqrt{\frac{N}{c_u T}}\varphi_n(\theta_0) + O_p\left(\frac{1}{\sqrt{T}}\right)$, and

$$\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta} = \begin{pmatrix} \frac{1}{\sqrt{n}\sigma_0^2}(\mathbb{G}_0\mathbb{D}\alpha_0)'\mathbb{Q}\mathbb{V} + \frac{1}{\sqrt{n}\sigma_0^2}\mathbb{V}'\mathbb{G}_0'\mathbb{V} + \frac{1}{\sqrt{n}\sigma_0^2}(\widetilde{\mathbb{G}_0\mathbb{Z}}^*\delta_0)'\mathbb{V} - \frac{1}{\sqrt{n}}\operatorname{tr}(\mathbb{G}_0) \\ \frac{1}{\sqrt{n}\sigma_0^2}\tilde{\Gamma}\tilde{\mathbb{Z}}^{*\prime}\mathbb{V} \\ -\frac{n}{2\sqrt{n}\sigma_0^2} + \frac{1}{2\sqrt{n}\sigma_0^4}\mathbb{V}'\mathbb{V} \end{pmatrix}.$$

Since $\tilde{\mathbb{Z}}^* = \mathbb{Q}\mathbb{Z} + \tilde{\mathbb{Z}}^\circ = \mathbb{Z} - \mathbb{P}(\mathbb{Z}-\mathbb{Z}^\circ)$ and $\widetilde{\mathbb{G}_0\mathbb{Z}}^* = \mathbb{Q}\mathbb{G}_0\mathbb{Z} + \widetilde{\mathbb{G}_0\mathbb{Z}}^\circ = \mathbb{G}_0\mathbb{Z} - \mathbb{P}\mathbb{G}_0(\mathbb{Z}-\mathbb{Z}^\circ)$, where

$$Z_t - Z_t^\circ = \left(D_t(I_N - F_t)D_t'M_t(\mathcal{X}_{t-1}\beta_0 + \mathcal{E}_{t-1}), D_tD_{t-1}'(\mathcal{X}_{t-1}\beta_0 + \mathcal{E}_{t-1}), D_tF_t\iota_N, X_t\right)$$

can be viewed as a constant matrix, elements in $\frac{1}{\sqrt{n}}\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}$ follow the form of equation (A.33). Moreover, since linear combinations of different $\mathcal{Q}_n$ still preserve the form of equation (A.33), and by Lemma 18 and Cramér-Wold Theorem, we have

$$\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta} \xrightarrow{d} N(0, \Sigma_{\theta_0} + \Omega_{\theta_0}),$$

where $\Sigma_{\theta_0} + \Omega_{\theta_0}$ is the asymptotic variance of $\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}$ according to equation (10). Moreover,

$$-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\bar{\theta})}{\partial\theta\partial\theta'}\Gamma = \left[-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\bar{\theta})}{\partial\theta\partial\theta'}\Gamma - \left(-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma\right)\right] + \left(-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma - \Gamma\Sigma_{\theta_0,n}\Gamma\right) + \Gamma\Sigma_{\theta_0,n}\Gamma,$$

where the first term is $\|\Gamma^{-1}(\bar{\theta}-\theta_0)\|_2 \cdot O_p(1)$ and the second term is $O_p(\frac{1}{\sqrt{NT}})$ by equations (B.1) and (B.2), respectively. By the consistency of $\hat{\theta}$, $-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\bar{\theta})}{\partial\theta\partial\theta'}\Gamma = \Gamma\Sigma_{\theta_0,n}\Gamma + o_p(1)$. Moreover, $\Gamma\Sigma_{\theta_0,n}\Gamma$ is nonsingular, which has been proved in Appendix B.3. So $\sqrt{n}\Gamma^{-1}(\hat{\theta}-\theta_0) = O_p(1)\cdot[O_p(1) + O(\sqrt{N/T})]$ according to equation (C.6), which means that

$$\Gamma^{-1}(\hat{\theta}-\theta_0) = O_p\left(\max\left\{\frac{1}{\sqrt{NT}}, \frac{1}{T}\right\}\right). \tag{C.7}$$

Hence,

$$\begin{aligned}\sqrt{n}\Gamma^{-1}(\hat{\theta}-\theta_0) &= \left[\Gamma\Sigma_{\theta_0,n}\Gamma + O_p\left(\max\left\{\frac{1}{\sqrt{NT}},\frac{1}{T}\right\}\right)\right]^{-1}\left(\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}-\Delta_n\right)\\ &= \left[(\Gamma\Sigma_{\theta_0,n}\Gamma)^{-1} + O_p\left(\max\left\{\frac{1}{\sqrt{NT}},\frac{1}{T}\right\}\right)\right]\left(\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}-\Delta_n\right),\end{aligned}$$

where the second equality holds by the fact that for any matrix sequences A_n and B_n such that $\lim_{n\to\infty} A_n$ exists and nonsingular, and B_n is $O_p(n^{-\eta})$ for some $\eta > 0$, we have

$$(A_n+B_n)^{-1} = (I+A_n^{-1}B_n)^{-1}A_n^{-1} = A_n^{-1} - A_n^{-1}B_nA_n^{-1} + A_n^{-1}B_nA_n^{-1}B_nA_n^{-1} - \cdots = A_n^{-1} + O_p(n^{-\eta}). \tag{C.8}$$

Therefore, we have

$$\sqrt{n}\Gamma^{-1}(\hat{\theta}-\theta_0) + \left[(\Gamma\Sigma_{\theta_0,n}\Gamma)^{-1} + O_p\left(\max\left\{\frac{1}{\sqrt{NT}},\frac{1}{T}\right\}\right)\right]\Delta_n = \left[(\Gamma\Sigma_{\theta_0,n}\Gamma)^{-1}+o_p(1)\right]\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}.$$

As $\lim_{n\to\infty}\Gamma\Sigma_{\theta_0,n}\Gamma = \Sigma_{\theta_0}$, and $\Delta_n = \sqrt{\frac{N}{c_uT}}\varphi_n(\theta_0) + O_p\left(\frac{1}{\sqrt{T}}\right)$, combined with the limiting distribution of $\frac{1}{\sqrt{n}}\Gamma\frac{\partial L_n^{c*}(\theta_0)}{\partial\theta}$, we finally have

$$\sqrt{n}\Gamma^{-1}(\hat{\theta}-\theta_0)+\sqrt{\frac{N}{c_uT}}(\Gamma\Sigma_{\theta_0,n}\Gamma)^{-1}\varphi_n(\theta_0)+O_p\left(\max\left\{\frac{1}{\sqrt{T}},\sqrt{\frac{N}{T^3}}\right\}\right)\xrightarrow{d} N\left(0,\Sigma_{\theta_0}^{-1}(\Sigma_{\theta_0}+\Omega_{\theta_0})\Sigma_{\theta_0}^{-1}\right),$$

which means that

$$\sqrt{n}\Gamma^{-1}\left[\hat{\theta}-\left(\theta_0-\frac{\Sigma_{\theta_0,n}^{-1}\varphi_n(\theta_0)}{c_uT}\right)\right]+O_p\left(\max\left\{\frac{1}{\sqrt{T}},\sqrt{\frac{N}{T^3}}\right\}\right)\xrightarrow{d} N\left(0,\Sigma_{\theta_0}^{-1}(\Sigma_{\theta_0}+\Omega_{\theta_0})\Sigma_{\theta_0}^{-1}\right),$$

where one of the multipliers Γ can be omitted because the γ-component of the vector $\varphi_n(\theta_0)$ is zero.

C.3 Proof of Theorem 3

Since $\hat{\theta}^1 = \hat{\theta} + \frac{\hat{B}_n}{c_uT}$, where $\hat{B}_n = \left[\left(-\frac{1}{n}\frac{\partial^2 L_n^c(\theta)}{\partial\theta\partial\theta'}\right)^{-1}\varphi_n(\theta)\right]\Big|_{\theta=\hat{\theta}}$. According to Theorem 2, if $\frac{N}{T^3}\to 0$, we need to prove

$$\sqrt{\frac{N}{c_uT}}\Gamma^{-1}\left[\left(-\frac{1}{n}\frac{\partial^2 L_n^c(\hat{\theta})}{\partial\theta\partial\theta'}\right)^{-1}\varphi_n(\hat{\theta})-\Sigma_{\theta_0,n}^{-1}\varphi_n(\theta_0)\right]\xrightarrow{p} 0. \tag{C.9}$$

Based on equations (B.1) and (B.2), we have

$$\begin{aligned}-\Gamma\frac{1}{n}\frac{\partial^2 L_n^c(\hat{\theta})}{\partial\theta\partial\theta'}\Gamma - \Gamma\Sigma_{\theta_0,n}\Gamma &= -\Gamma\frac{1}{n}\frac{\partial^2 L_n^c(\hat{\theta})}{\partial\theta\partial\theta'}\Gamma - \left(-\Gamma\frac{1}{n}\frac{\partial^2 L_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma\right) + \left(-\Gamma\frac{1}{n}\frac{\partial^2 L_n^c(\theta_0)}{\partial\theta\partial\theta'}\Gamma\right) - \Gamma\Sigma_{\theta_0,n}\Gamma\\ &= \|\Gamma^{-1}(\hat{\theta}-\theta_0)\|_2\cdot O_p(1) + O_p\left(\frac{1}{\sqrt{NT}}\right)\\ &= O_p\left(\max\left\{\frac{1}{\sqrt{NT}},\frac{1}{T}\right\}\right),\end{aligned}$$

where the last equality holds by equation (C.7). Then using equation (C.8), we have

$$\left(-\Gamma\frac{1}{n}\frac{\partial^2 L_n^c(\hat{\theta})}{\partial\theta\partial\theta'}\Gamma\right)^{-1} = (\Gamma\Sigma_{\theta_0,n}\Gamma)^{-1} + O_p\left(\max\left\{\frac{1}{\sqrt{NT}},\frac{1}{T}\right\}\right).$$

Recall that $\Gamma^{-1}\varphi_n(\theta) = \varphi_n(\theta)$, thus

$$
\begin{aligned}
&\sqrt{\frac{N}{c_u T}}\Gamma^{-1}\left[\left(-\frac{1}{n}\frac{\partial^2 L_n^c(\hat{\theta})}{\partial\theta\partial\theta'}\right)^{-1}\varphi_n(\hat{\theta}) - \Sigma_{\theta_0,n}^{-1}\varphi_n(\theta_0)\right] \\
=&\sqrt{\frac{N}{c_u T}}\left[\left(-\frac{1}{n}\Gamma\frac{\partial^2 L_n^c(\hat{\theta})}{\partial\theta\partial\theta'}\Gamma\right)^{-1}\varphi_n(\hat{\theta}) - (\Gamma\Sigma_{\theta_0,n}\Gamma)^{-1}\varphi_n(\theta_0)\right] \\
=&\sqrt{\frac{N}{c_u T}}(\Gamma\Sigma_{\theta_0,n}\Gamma)^{-1}\left[\varphi_n(\hat{\theta}) - \varphi_n(\theta_0)\right] + \sqrt{\frac{N}{c_u T}}\varphi_n(\hat{\theta}) \times O_p\left(\max\left\{\frac{1}{\sqrt{NT}}, \frac{1}{T}\right\}\right) \\
=&O_p\left(\sqrt{\frac{N}{T}}\right)\cdot\left[\frac{\partial\varphi_n(\bar{\theta})}{\partial\theta'}\Gamma\right]\left[\Gamma^{-1}(\hat{\theta}-\theta_0)\right] + \varphi_n(\hat{\theta}) \times O_p\left(\max\left\{\frac{1}{T}, \sqrt{\frac{N}{T^3}}\right\}\right),
\end{aligned}
$$

where $\bar{\theta}$ lies between $\hat{\theta}$ and θ_0. Since $\Gamma^{-1}(\hat{\theta}-\theta_0) = O_p\left(\max\left\{\frac{1}{\sqrt{NT}}, \frac{1}{T}\right\}\right)$, in order to prove equation (C.9), it suffices to show that $\frac{\partial\varphi_n(\bar{\theta})}{\partial\theta'}\Gamma$ is $O_p(1)$. By exchanging the order of summation in equation (B.6) and adjusting the summation indices, we obtain

$$
\varphi_n(\theta) = \begin{pmatrix}
\left[\begin{array}{l}
\frac{1}{N}\mathrm{tr}\left(\mathcal{T}^{-1}\sum_{t=1}^{T-1}\lambda D_{t+1}' G_{t+1}(\rho) D_{t+1}(I_N - F_{t+1}) D_{t+1}' M_{t+1} S_t(\rho)^{-1}\sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\right) \\
+\frac{1}{N}\mathrm{tr}\left(\mathcal{T}^{-1}\sum_{t=1}^{T-1}\nu D_{t+1}' G_{t+1}(\rho) D_{t+1} D_t' S_t(\rho)^{-1}\sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\right) + \frac{1}{N}\mathrm{tr}(\mathbb{G}(\rho)'\mathbb{P})
\end{array}\right] \\
\frac{1}{N}\mathrm{tr}\left(\mathcal{T}^{-1}\sum_{t=T-h}^{T-1} D_{t+1}' D_{t+1}(I_N - F_{t+1}) D_{t+1}' M_{t+1} S_t(\rho)^{-1}\sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\right) \\
\frac{1}{N}\mathrm{tr}\left(\mathcal{T}^{-1}\sum_{t=T-h}^{T-1} D_{t+1}' D_{t+1} D_t' S_t(\rho)^{-1}\sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\right) \\
\mathbf{0}_{(k+1)\times 1} \\
\frac{1}{2\sigma^2}
\end{pmatrix}.
$$

Firstly, we will prove that elements in $\frac{\partial \varphi_n(\bar{\theta})}{\partial \rho}$ are $O_p(1)$. As

$$
\begin{aligned}
&\Big|\frac{\partial}{\partial \rho}\frac{1}{N}\mathrm{tr}\Big(\mathcal{T}^{-1}\sum_{t=1}^{T-1}\lambda D_{t+1}' G_{t+1}(\rho) D_{t+1} \\
&\qquad (I_N - F_{t+1}) D_{t+1}' M_{t+1} S_t(\rho)^{-1} \sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\Big)\Big| \\
\leqslant &\frac{1}{N}\Big|\mathrm{tr}\Big[\mathcal{T}^{-1}\sum_{t=1}^{T-1}\lambda D_{t+1}' W_{t+1}\Big(\frac{\partial}{\partial \rho} S_{t+1}^{-1}(\rho)\Big) D_{t+1} \\
&\qquad (I_N - F_{t+1}) D_{t+1}' M_{t+1} S_t(\rho)^{-1} \sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\Big]\Big| \\
&+ \frac{1}{N}\Big|\mathrm{tr}\Big[\mathcal{T}^{-1}\sum_{t=1}^{T-1}\lambda D_{t+1}' G_{t+1}(\rho) D_{t+1} \\
&\qquad (I_N - F_{t+1}) D_{t+1}' M_{t+1} \Big(\frac{\partial}{\partial \rho} S_t^{-1}(\rho)\Big) \sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\Big]\Big| \\
&+ \frac{1}{N}\Big|\mathrm{tr}\Big[\mathcal{T}^{-1}\sum_{t=1}^{T-1}\lambda D_{t+1}' G_{t+1}(\rho) D_{t+1} \\
&\qquad (I_N - F_{t+1}) D_{t+1}' M_{t+1} S_t^{-1}(\rho) \Big(\frac{\partial}{\partial \rho} \sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\Big)\Big]\Big| \\
\leqslant &\frac{1}{c_T NT}\sum_{t=1}^{T-1}\mathrm{tr}\Big\{\mathrm{abs}\Big[\lambda D_{t+1}' W_{t+1}\Big(\frac{\partial}{\partial \rho} S_{t+1}^{-1}(\rho)\Big) D_{t+1} \\
&\qquad (I_N - F_{t+1}) D_{t+1}' M_{t+1} S_t(\rho)^{-1}\Big] \\
&\qquad \mathrm{abs}\Big[\sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\Big]\Big\} \\
&+ \frac{1}{c_T NT}\sum_{t=1}^{T-1}\mathrm{tr}\Big\{\mathrm{abs}\Big[\lambda D_{t+1}' G_{t+1}(\rho) D_{t+1} \\
&\qquad (I_N - F_{t+1}) D_{t+1}' M_{t+1} \Big(\frac{\partial}{\partial \rho} S_t^{-1}(\rho)\Big)\Big] \\
&\qquad \mathrm{abs}\Big[\sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\Big]\Big\} \\
&+ \frac{1}{c_T NT}\sum_{t=1}^{T-1}\mathrm{tr}\Big\{\mathrm{abs}\Big[\lambda D_{t+1}' G_{t+1}(\rho) D_{t+1} \\
&\qquad (I_N - F_{t+1}) D_{t+1}' M_{t+1} S_t^{-1}(\rho)\Big] \\
&\qquad \mathrm{abs}\Big[\frac{\partial}{\partial \rho}\sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}\Big]\Big\}
\end{aligned}
$$

where the last inequality holds because $T_i \geqslant c_T T$ by Assumption 1. The last equality holds because W_t and M_t are UB uniformly for t by Assumption 5. Additionally, $S_t(\rho)^{-1}$, $G_t(\rho)$, $\frac{\partial}{\partial \rho} S_t^{-1}(\rho) = S_t^{-1}(\rho) W_t S_t^{-1}(\rho)$, $\sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}$ and $\frac{\partial}{\partial \rho}\sum_{h=0}^{T-t} A_t^{(h)}(\theta) D_{t-h}$ are all UB uniformly in a neighborhood of ρ_0 for t by Assump-

tions 5 and 10. Therefore, all the matrices inside the trace operations are UB uniformly for t, which means that the derivative above is $O(1)$ in a neighborhood of ρ_0. Since $\hat{\theta} \xrightarrow{p} \theta_0$, and $\bar{\theta}$ lies between θ_0 and $\hat{\theta}$, $\bar{\theta}$ lies in a neighborhood of θ_0 with probability approaching one. Thus, the derivative above is $O_p(1)$ when evaluated at $\bar{\theta}$. Using the same approach, we can similarly show that all elements in $\frac{\partial \varphi_n(\bar{\theta})}{\partial \rho}$, as well as $\frac{\partial \varphi_n(\bar{\theta})}{\partial \lambda}$ and $\frac{\partial \varphi_n(\bar{\theta})}{\partial \nu}$ are $O_p(1)$. Since $\sqrt{T}\frac{\partial \varphi_n(\bar{\theta})}{\partial \gamma} = \mathbf{0}_{(k+5)\times 1}$, $\frac{\partial \varphi_n(\bar{\theta})}{\partial \beta'} = \mathbf{0}_{(k+5)\times k}$ and $\frac{\partial \varphi_n(\bar{\theta})}{\partial \sigma^2} = (\mathbf{0}_{1\times(k+4)}, -\frac{1}{2\bar{\sigma}^4})'$, it follows that elements in $\frac{\partial \varphi_n(\bar{\theta})}{\partial \theta'}\Gamma$ are $O_p(1)$, which completes the proof.

C.4 Proof of Theorem 4

Let $\widehat{\theta}^1 = (\widehat{\rho}^1, \widehat{\delta}^{1\prime}, \widehat{\sigma}^{2,1})'$ be the bias-corrected estimator. Define $\widehat{\alpha}^1 = (\mathbb{D}'\mathbb{D})^{-1}\mathbb{D}'\{\mathbb{S}(\widehat{\rho}^1)\mathbb{Y} - \mathbb{Z}\widehat{\delta}^1\}$, and $\widehat{\mathbb{V}} = \mathbb{Q}\{\mathbb{S}(\widehat{\rho}^1)\mathbb{Y} - \mathbb{Z}\widehat{\delta}^1\}$. Let $\check{\sigma}^2 = n^{-1}\widehat{\mathbb{V}}'\widehat{\mathbb{V}}$, $\check{\mu}_3 = n^{-1}\sum_{r=1}^n \widehat{\mathbb{V}}_r^3$, $\check{\mu}_4 = n^{-1}\sum_{r=1}^n \widehat{\mathbb{V}}_r^4$, and $\check{c} = (\check{\mu}_4 - 3\check{\sigma}^4)/\check{\sigma}^4$, where $\check{\sigma}^4 = (\check{\sigma}^2)^2$ and $\check{\sigma}^6 = (\check{\sigma}^2)^3$. Let $\widehat{\mathbb{G}} = \mathbb{W}\mathbb{S}(\widehat{\rho}^1)^{-1}$, $\widehat{g} = (\widehat{\mathbb{G}}_{11}, \ldots, \widehat{\mathbb{G}}_{nn})'$. Define

$$\widehat{\Sigma} = \begin{pmatrix} \frac{1}{n\check{\sigma}^2}\{\widehat{\mathbb{G}}(\mathbb{Z}\widehat{\delta}^1 + \mathbb{D}\widehat{\alpha}^1)\}'\mathbb{Q}\{\widehat{\mathbb{G}}(\mathbb{Z}\widehat{\delta}^1 + \mathbb{D}\widehat{\alpha}^1)\} + \frac{1}{n}\operatorname{tr}\{\widehat{\mathbb{G}}(\widehat{\mathbb{G}} + \widehat{\mathbb{G}}')\} & * & * \\ \frac{1}{n\check{\sigma}^2}\widetilde{\Gamma}'\mathbb{Z}'\mathbb{Q}\widehat{\mathbb{G}}(\mathbb{Z}\widehat{\delta}^1 + \mathbb{D}\widehat{\alpha}^1) & \frac{1}{n\check{\sigma}^2}\widetilde{\Gamma}'\mathbb{Z}'\mathbb{Q}\mathbb{Z}\widetilde{\Gamma} & * \\ \frac{1}{n\check{\sigma}^2}\operatorname{tr}(\widehat{\mathbb{G}}) & \mathbf{0}_{1\times(k+3)} & \frac{1}{2\check{\sigma}^4} \end{pmatrix}.$$

Next, define

$$\widehat{\Omega} = \begin{pmatrix} \frac{\check{c}}{n}\widehat{g}'\widehat{g} + \frac{2\check{\mu}_3}{n\check{\sigma}^4}\left[\widehat{\mathbb{G}}\mathbb{Z}\widehat{\delta}^1 + \mathbb{Q}\widehat{\mathbb{G}}\mathbb{D}\widehat{\alpha}^1\right]'\widehat{g} & * & * \\ \frac{\check{\mu}_3}{n\check{\sigma}^4}\widetilde{\Gamma}'\mathbb{Z}'\widehat{g} & \mathbf{0}_{(k+3)\times(k+3)} & * \\ \frac{\check{c}}{2n\check{\sigma}^2}\operatorname{tr}(\widehat{\mathbb{G}}) & \mathbf{0}_{1\times(k+3)} & \frac{\check{c}}{4\check{\sigma}^4} \end{pmatrix}.$$

Set

$$\widehat{\Xi} = \widehat{\Sigma}^{-1}(\widehat{\Sigma} + \widehat{\Omega})\widehat{\Sigma}^{-1}.$$

Proof. We first show the consistency of the residual moment estimators. Since

$$\widehat{\mathbb{V}} - \mathbb{Q}\mathbb{V} = \mathbb{Q}\left[-(\widehat{\rho}^1 - \rho_0)\mathbb{W}\mathbb{Y} - \mathbb{Z}(\widehat{\delta}^1 - \delta_0)\right],$$

we need to control the average moments of the right-hand side. For any $n \times n$ UB matrix B and any vector a, Hölder's inequality implies that, for $q = 2, 3, 4$,

$$\begin{aligned} \sum_{r=1}^n |(Ba)_r|^q = \sum_{r=1}^n \left|\sum_{s=1}^n B_{rs}a_s\right|^q &\leqslant \sum_{r=1}^n \left(\sum_{s=1}^n |B_{rs}||a_s|\right)^q \leqslant \sum_{r=1}^n \left(\sum_{s=1}^n |B_{rs}|\right)^{q-1} \sum_{s=1}^n |B_{rs}||a_s|^q \\ &\leqslant \|B\|_\infty^{q-1} \sum_{s=1}^n |a_s|^q \left(\sum_{r=1}^n |B_{rs}|\right) \leqslant \|B\|_\infty^{q-1}\|B\|_1 \sum_{s=1}^n |a_s|^q. \end{aligned} \tag{C.10}$$

Thus,

$$\begin{aligned}
&\frac{1}{n}\sum_{r=1}^{n}\left|\widehat{\mathbb{V}}_r-(\mathbb{Q}\mathbb{V})_r\right|^q\\
\leqslant&\|\mathbb{Q}\|_\infty^{q-1}\|\mathbb{Q}\|_1\frac{1}{n}\sum_{r=1}^{n}\left|\left[-(\widehat{\rho}^1-\rho_0)\mathbb{W}\mathbb{Y}-\mathbb{Z}(\widehat{\delta}^1-\delta_0)\right]_r\right|^q\\
\leqslant&\|\mathbb{Q}\|_\infty^{q-1}\|\mathbb{Q}\|_1(k+4)^{q-1}\left[|\widehat{\rho}^1-\rho_0|^q\frac{1}{n}\sum_{r=1}^{n}|(\mathbb{W}\mathbb{Y})_r|^q+\sum_{j=1}^{k+3}|\widehat{\delta}_j^1-\delta_{0j}|^q\frac{1}{n}\sum_{r=1}^{n}|\mathbb{Z}_{rj}|^q\right],
\end{aligned}\tag{C.11}$$

where the second inequality holds because $|x_1+\cdots+x_{k+4}|^q\leqslant(k+4)^{q-1}\sum_{j=1}^{k+4}|x_j|^q$.

By the moving-average representation derived in equation (3), $Y_t=\mathcal{E}_t+\mathcal{X}_t\beta_0+\mathcal{V}_t$. The deterministic component $\mathcal{E}_t+\mathcal{X}_t\beta_0$ has uniformly bounded elements by Assumptions 4–6. Moreover, $\mathcal{V}_t$ is an infinite historical-error accumulation of the same form as that considered in Lemma 17. Hence, by the same argument as in Lemma 17, $\sup_{i,t,N}E|\mathcal{V}_{t,i}|^4=O(1)$. Therefore, applying Markov's inequality gives $\frac{1}{n}\sum_{r=1}^{n}|\mathbb{Y}_r|^4=O_p(1)$. Since $\mathbb{W}$ is UB, the equation (C.10) gives

$$\frac{1}{n}\sum_{r=1}^{n}|(\mathbb{W}\mathbb{Y})_r|^q=O_p(1).$$

The cases $q=2,3$ follow from the fourth-moment bound. Similarly, for each j, we obtain

$$\frac{1}{n}\sum_{r=1}^{n}|\mathbb{Z}_{rj}|^q=O_p(1).$$

Together with $\widehat{\rho}^1-\rho_0=o_p(1)$ and $\widehat{\delta}_j^1-\delta_{0j}=o_p(1)$ for each j, equation (C.11) gives

$$\frac{1}{n}\sum_{r=1}^{n}\left|\widehat{\mathbb{V}}_r-(\mathbb{Q}\mathbb{V})_r\right|^q=o_p(1),\qquad q=2,3,4.$$

Since $\widehat{\mathbb{V}}=\mathbb{Q}\mathbb{V}+\widehat{\mathbb{V}}-\mathbb{Q}\mathbb{V}$, the binomial expansion gives

$$\begin{aligned}
\frac{1}{n}\sum_{r=1}^{n}\left|\widehat{\mathbb{V}}_r^q-(\mathbb{Q}\mathbb{V})_r^q\right|&\leqslant\frac{1}{n}\sum_{l=1}^{q}\binom{q}{l}\sum_{r=1}^{n}|(\mathbb{Q}\mathbb{V})_r|^{q-l}\left|\widehat{\mathbb{V}}_r-(\mathbb{Q}\mathbb{V})_r\right|^l\\
&\leqslant\sum_{l=1}^{q}\binom{q}{l}\left[\frac{1}{n}\sum_{r=1}^{n}|(\mathbb{Q}\mathbb{V})_r|^q\right]^{(q-l)/q}\left[\frac{1}{n}\sum_{r=1}^{n}\left|\widehat{\mathbb{V}}_r-(\mathbb{Q}\mathbb{V})_r\right|^q\right]^{l/q},
\end{aligned}\tag{C.12}$$

where the second inequality follows from Hölder's inequality. Since $\mathbb{Q}$ is UB, equation (C.10) gives

$$\frac{1}{n}\sum_{r=1}^{n}|(\mathbb{Q}\mathbb{V})_r|^q\leqslant\|\mathbb{Q}\|_\infty^{q-1}\|\mathbb{Q}\|_1\frac{1}{n}\sum_{r=1}^{n}|\mathbb{V}_r|^q=O_p(1),\qquad q=2,3,4.$$

Combining this with $\frac{1}{n}\sum_{r=1}^{n}\left|\widehat{\mathbb{V}}_r-(\mathbb{Q}\mathbb{V})_r\right|^q=o_p(1)$, we obtain

$$\frac{1}{n}\sum_{r=1}^{n}\widehat{\mathbb{V}}_r^q-\frac{1}{n}\sum_{r=1}^{n}(\mathbb{Q}\mathbb{V})_r^q=o_p(1),\qquad q=2,3,4.$$

Next, we show that the sample moments of $\mathbb{Q}\mathbb{V}$ and $\mathbb{V}$ are asymptotically equivalent. Since $\mathbb{Q}\mathbb{V}=\mathbb{V}-\mathbb{P}\mathbb{V}$, it suffices to control the average moments of $\mathbb{P}\mathbb{V}$. Each element of $\mathbb{P}\mathbb{V}$ is an individual-specific average of the

errors over the observed periods. For observation r corresponding to individual i, $(\mathbb{PV})_r = \frac{1}{T_i}\sum_{t:d_{it}=1} v_{it}$. Since the errors are independent with mean zero, variance σ_0^2, and fourth moment μ_4, we have

$$E|(\mathbb{PV})_r|^2 = E\left(\frac{1}{T_i}\sum_{t:d_{it}=1} v_{it}\right)^2 = \frac{1}{T_i^2}\sum_{t:d_{it}=1} E(v_{it}^2) = \frac{\sigma_0^2}{T_i} = O(T^{-1})$$

Similarly,

$$\begin{aligned} E|(\mathbb{PV})_r|^4 &= E\left(\frac{1}{T_i}\sum_{t:d_{it}=1} v_{it}\right)^4 = \frac{1}{T_i^4}\left[\sum_{t:d_{it}=1} E(v_{it}^4) + 6\sum_{\substack{s<t \\ d_{is}=d_{it}=1}} E(v_{is}^2)E(v_{it}^2)\right] \\ &= \frac{1}{T_i^4}\left[T_i\mu_4 + 3T_i(T_i-1)\sigma_0^4\right] = O(T^{-2}). \end{aligned}$$

Moreover, by Hölder's inequality,

$$E|(\mathbb{PV})_r|^3 \leqslant \left(E|(\mathbb{PV})_r|^4\right)^{3/4} = O(T^{-3/2}).$$

It follows that, for $q = 2, 3, 4$, $E|(\mathbb{PV})_r|^q = O(T^{-q/2})$. Therefore,

$$\frac{1}{n}\sum_{r=1}^n |(\mathbb{PV})_r|^q = O_p(T^{-q/2}) = o_p(1),$$

by Markov's inequality. Moreover, by Assumption 2, $\frac{1}{n}\sum_{r=1}^n |\mathbb{V}_r|^q = O_p(1)$. Thus, similar to equation (C.12), we have

$$\begin{aligned} \frac{1}{n}\sum_{r=1}^n |(\mathbb{QV})_r^q - \mathbb{V}_r^q| &\leqslant \frac{1}{n}\sum_{l=1}^q \binom{q}{l}\sum_{r=1}^n |\mathbb{V}_r|^{q-l}|(\mathbb{PV})_r|^l \\ &\leqslant \sum_{l=1}^q \binom{q}{l}\left[\frac{1}{n}\sum_{r=1}^n |\mathbb{V}_r|^q\right]^{(q-l)/q}\left[\frac{1}{n}\sum_{r=1}^n |(\mathbb{PV})_r|^q\right]^{l/q} = o_p(1). \end{aligned}$$

Consequently, $\frac{1}{n}\sum_{r=1}^n (\mathbb{QV})_r^q - \frac{1}{n}\sum_{r=1}^n \mathbb{V}_r^q = o_p(1)$, for $q = 2, 3, 4$, which implies

$$\frac{1}{n}\sum_{r=1}^n \widehat{\mathbb{V}}_r^q - \frac{1}{n}\sum_{r=1}^n \mathbb{V}_r^q = o_p(1), \qquad q = 2, 3, 4.$$

By the law of large numbers,

$$\frac{1}{n}\sum_{r=1}^n \mathbb{V}_r^2 \xrightarrow{p} \sigma_0^2, \qquad \frac{1}{n}\sum_{r=1}^n \mathbb{V}_r^3 \xrightarrow{p} \mu_3, \qquad \frac{1}{n}\sum_{r=1}^n \mathbb{V}_r^4 \xrightarrow{p} \mu_4.$$

Therefore,

$$\check{\sigma}^2 \xrightarrow{p} \sigma_0^2, \qquad \check{\mu}_3 \xrightarrow{p} \mu_3, \qquad \check{\mu}_4 \xrightarrow{p} \mu_4, \qquad \check{c} \xrightarrow{p} \frac{\mu_4 - 3\sigma_0^4}{\sigma_0^4}.$$

We next show that $\widehat{\Sigma}$ is consistent. By Theorem 3, we have $\widehat{\theta}^1 \xrightarrow{p} \theta_0$. In particular, $\widehat{\rho}^1 - \rho_0 = o_p(1)$. Since $\mathbb{S}(\widehat{\rho}^1) = \mathbb{S}(\rho_0) - (\widehat{\rho}^1 - \rho_0)\mathbb{W}$, applying (C.8) gives, for $p = 1, \infty$,

$$\left\|\mathbb{S}(\widehat{\rho}^1)^{-1} - \mathbb{S}(\rho_0)^{-1}\right\|_p = O_p(|\widehat{\rho}^1 - \rho_0|) = o_p(1).$$

Since $\mathbb{W}$ is UB, for $p = 1, \infty$, we have

$$\left\|\widehat{\mathbb{G}} - \mathbb{G}_0\right\|_p = \left\|\mathbb{W}\left[\mathbb{S}(\widehat{\rho}^{\,1})^{-1} - \mathbb{S}(\rho_0)^{-1}\right]\right\|_p \leqslant \|\mathbb{W}\|_p \left\|\mathbb{S}(\widehat{\rho}^{\,1})^{-1} - \mathbb{S}(\rho_0)^{-1}\right\|_p = o_p(1).$$

Hence, $\|\widehat{\mathbb{G}} - \mathbb{G}_0\|_1 = o_p(1)$ and $\|\widehat{\mathbb{G}} - \mathbb{G}_0\|_\infty = o_p(1)$.

From the definition of $\widehat{\alpha}^{\,1}$, we have

$$\mathbb{D}\widehat{\alpha}^{\,1} - \mathbb{D}\alpha_0 = \mathbb{P}\left[\mathbb{S}(\widehat{\rho}^{\,1})\mathbb{Y} - \mathbb{Z}\widehat{\delta}^{\,1}\right] - \mathbb{D}\alpha_0 = \mathbb{P}\mathbb{V} - \mathbb{P}\left[(\widehat{\rho}^{\,1} - \rho_0)\mathbb{W}\mathbb{Y} + \mathbb{Z}(\widehat{\delta}^{\,1} - \delta_0)\right].$$

By Lemma 12, we have $\frac{1}{n}(\mathbb{P}\mathbb{V})'\mathbb{P}\mathbb{V} = o_p(1)$. Moreover, applying the same argument as in (C.11) gives $\frac{1}{n}\left\|\mathbb{P}\left[(\widehat{\rho}^{\,1} - \rho_0)\mathbb{W}\mathbb{Y} + \mathbb{Z}(\widehat{\delta}^{\,1} - \delta_0)\right]\right\|^2 = o_p(1)$. Combining the two bounds above, we obtain

$$\frac{1}{n}\|\mathbb{D}\widehat{\alpha}^{\,1} - \mathbb{D}\alpha_0\|^2 = o_p(1).$$

Since $\widehat{\mathbb{G}}$ and $\mathbb{G}_0$ are UB, $\|\widehat{\mathbb{G}} - \mathbb{G}_0\|_1 = o_p(1)$ and $\|\widehat{\mathbb{G}} - \mathbb{G}_0\|_\infty = o_p(1)$, we have

$$\frac{1}{n}\left\|\widehat{\mathbb{G}}\mathbb{Z}\widehat{\delta}^{\,1} - \mathbb{G}_0\mathbb{Z}\delta_0\right\|^2 = o_p(1), \qquad \frac{1}{n}\left\|\widehat{\mathbb{G}}\mathbb{D}\widehat{\alpha}^{\,1} - \mathbb{G}_0\mathbb{D}\alpha_0\right\|^2 = o_p(1). \tag{C.13}$$

Consequently,

$$\frac{1}{n}\left\|\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right\|^2 = o_p(1).$$

Moreover, $\frac{1}{n}\|\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\|^2 = O_p(1)$. Therefore,

$$\begin{aligned}
&\frac{1}{n}\left[\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right)\right]'\mathbb{Q}\left[\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right)\right] - \frac{1}{n}\left[\mathbb{G}_0\left(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0\right)\right]'\mathbb{Q}\left[\mathbb{G}_0\left(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0\right)\right] \\
=&\frac{1}{n}\left[\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right]'\mathbb{Q}\left[\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right] \\
&+ \frac{2}{n}\left[\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right]'\mathbb{Q}\left[\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right].
\end{aligned}$$

For the first term above, we have

$$\begin{aligned}
&\left|\frac{1}{n}\left[\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right]'\mathbb{Q}\left[\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right]\right| \\
\leqslant&\|\mathbb{Q}\|_\infty^{1/2}\|\mathbb{Q}\|_1^{1/2}\frac{1}{n}\left\|\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right\|^2 = o_p(1).
\end{aligned}$$

For the second term, by the Cauchy–Schwarz inequality,

$$\begin{aligned}
&\left|\frac{2}{n}\left[\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right]'\mathbb{Q}\left[\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right]\right| \\
\leqslant&2\left[\frac{1}{n}\|\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\|^2\right]^{1/2}\left[\frac{1}{n}\left\|\mathbb{Q}\left\{\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right\}\right\|^2\right]^{1/2} \\
\leqslant&2\|\mathbb{Q}\|_\infty^{1/2}\|\mathbb{Q}\|_1^{1/2}\left[\frac{1}{n}\|\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\|^2\right]^{1/2}\left[\frac{1}{n}\left\|\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0)\right\|^2\right]^{1/2} = o_p(1).
\end{aligned}$$

Hence,

$$\frac{1}{n}\left[\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right)\right]'\mathbb{Q}\left[\widehat{\mathbb{G}}\left(\mathbb{Z}\widehat{\delta}^{\,1} + \mathbb{D}\widehat{\alpha}^{\,1}\right)\right] - \frac{1}{n}\left[\mathbb{G}_0\left(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0\right)\right]'\mathbb{Q}\left[\mathbb{G}_0\left(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0\right)\right] = o_p(1).$$

Similarly,

$$\frac{1}{n}\widetilde{\Gamma}'\mathbb{Z}'\mathbb{Q}\widehat{\mathbb{G}}\left(\mathbb{Z}\hat{\delta}^{1}+\mathbb{D}\hat{\alpha}^{1}\right)-\frac{1}{n}\widetilde{\Gamma}'\mathbb{Z}'\mathbb{Q}\mathbb{G}_{0}\left(\mathbb{Z}\delta_{0}+\mathbb{D}\alpha_{0}\right)=o_{p}(1).$$

Since $\|\widehat{\mathbb{G}}-\mathbb{G}_0\|_1=o_p(1)$, $\|\widehat{\mathbb{G}}-\mathbb{G}_0\|_\infty=o_p(1)$, and $\widehat{\mathbb{G}}$ and $\mathbb{G}_0$ are UB, we have

$$\frac{1}{n}\operatorname{tr}\{\widehat{\mathbb{G}}(\widehat{\mathbb{G}}+\widehat{\mathbb{G}}')\}=\frac{1}{n}\operatorname{tr}\{\mathbb{G}_0(\mathbb{G}_0+\mathbb{G}_0')\}+o_p(1). \tag{C.14}$$

Since $\breve{\sigma}^2\xrightarrow{p}\sigma_0^2$ and σ_0^2 is bounded away from zero, the results above imply $\widehat{\Sigma}-\Gamma\Sigma_{\theta_0,n}\Gamma=o_p(1)$. Since $\lim_{n\to\infty}\Gamma\Sigma_{\theta_0,n}\Gamma=\Sigma_{\theta_0}$, we have

$$\widehat{\Sigma}\xrightarrow{p}\Sigma_{\theta_0}.$$

It remains to show the consistency of $\widehat{\Omega}$. Similar to (C.14), the same argument gives

$$\frac{1}{n}\hat{g}'\hat{g}=\frac{1}{n}\sum_{r=1}^{n}(\mathbb{G}_0)_{rr}^2+o_p(1),\qquad \frac{1}{n}\operatorname{tr}(\widehat{\mathbb{G}})=\frac{1}{n}\operatorname{tr}(\mathbb{G}_0)+o_p(1),\qquad \frac{1}{n}\widetilde{\Gamma}'\mathbb{Z}'\hat{g}=\frac{1}{n}\widetilde{\Gamma}'\mathbb{Z}'g_0+o_p(1),$$

where $g_0=((\mathbb{G}_0)_{11},\ldots,(\mathbb{G}_0)_{nn})'$.

Moreover, according to equation (C.13),

$$\begin{aligned}
&\frac{1}{n}\left\|\widehat{\mathbb{G}}\mathbb{Z}\hat{\delta}^{1}+\mathbb{Q}\widehat{\mathbb{G}}\mathbb{D}\hat{\alpha}^{1}-\left(\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right)\right\|^2\\
\leqslant&\frac{2}{n}\left\|\widehat{\mathbb{G}}\mathbb{Z}\hat{\delta}^{1}-\mathbb{G}_0\mathbb{Z}\delta_0\right\|^2+\frac{2}{n}\left\|\mathbb{Q}\left(\widehat{\mathbb{G}}\mathbb{D}\hat{\alpha}^{1}-\mathbb{G}_0\mathbb{D}\alpha_0\right)\right\|^2\\
\leqslant&\frac{2}{n}\left\|\widehat{\mathbb{G}}\mathbb{Z}\hat{\delta}^{1}-\mathbb{G}_0\mathbb{Z}\delta_0\right\|^2+2\|\mathbb{Q}\|_\infty\|\mathbb{Q}\|_1\frac{1}{n}\left\|\widehat{\mathbb{G}}\mathbb{D}\hat{\alpha}^{1}-\mathbb{G}_0\mathbb{D}\alpha_0\right\|^2=o_p(1).
\end{aligned}$$

Since $\frac{1}{n}\hat{g}'\hat{g}=O_p(1)$, and $\frac{1}{n}g_0'g_0=O_p(1)$, the Cauchy–Schwarz inequality gives

$$\frac{1}{n}\left[\widehat{\mathbb{G}}\mathbb{Z}\hat{\delta}^{1}+\mathbb{Q}\widehat{\mathbb{G}}\mathbb{D}\hat{\alpha}^{1}\right]'\hat{g}-\frac{1}{n}\left[\mathbb{G}_0\mathbb{Z}\delta_0+\mathbb{Q}\mathbb{G}_0\mathbb{D}\alpha_0\right]'g_0=o_p(1).$$

Combining the above results with

$$\breve{\mu}_3\xrightarrow{p}\mu_3,\qquad \breve{c}\xrightarrow{p}\frac{\mu_4-3\sigma_0^4}{\sigma_0^4},\qquad \breve{\sigma}^2\xrightarrow{p}\sigma_0^2,$$

we obtain $\widehat{\Omega}-\Gamma\Omega_{\theta_0,n}\Gamma=o_p(1)$. Since $\lim_{n\to\infty}\Gamma\Omega_{\theta_0,n}\Gamma=\Omega_{\theta_0}$, it follows that

$$\widehat{\Omega}\xrightarrow{p}\Omega_{\theta_0}.$$

Finally, Assumption 9 implies that Σ_{θ_0} is nonsingular. Therefore, $\widehat{\Sigma}$ is nonsingular with probability approaching one, and by the continuous mapping theorem,

$$\widehat{\Xi}=\widehat{\Sigma}^{-1}(\widehat{\Sigma}+\widehat{\Omega})\widehat{\Sigma}^{-1}\xrightarrow{p}\Sigma_{\theta_0}^{-1}(\Sigma_{\theta_0}+\Omega_{\theta_0})\Sigma_{\theta_0}^{-1}.$$

Thus, $n^{-1}\Gamma\widehat{\Xi}\Gamma$ is a consistent estimator of the asymptotic covariance matrix of $\widehat{\theta}^{1}$. This completes the proof. □

C.5 Proof of Theorem 5

The proof follows the same route as Theorem 1. We first establish the consistency of $\hat{\theta}^{(-\gamma)}$, and then prove the consistency of $\hat{\gamma}$ from its explicit concentrated expression.

Proof of uniform convergence

The uniform convergence requires showing

$$\sup_{\theta\in\Theta}\left|\frac{1}{n}\left(\bar{L}_n^c(\theta)-\bar{Q}_n(\theta)\right)\right|\xrightarrow{p}0,$$

where $\bar{Q}_n(\theta)\equiv E[\bar{L}_n^c(\theta)]$. As in the proof of Theorem 1, it suffices to show

$$\sup_{\theta\in\Theta}\frac{1}{n}\left|\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta)-E\left[\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta)\right]\right|\xrightarrow{p}0,$$

where $\bar{\mathbb{V}}(\zeta)\equiv\bar{\mathbb{Q}}\left(\mathbb{S}(\rho)\mathbb{Y}-\mathbb{Z}\delta\right)$. Since $\mathbb{Y}=\rho_0\mathbb{W}\mathbb{Y}+\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{J}\tau_0+\mathbb{V}$, and $\bar{\mathbb{Q}}\mathbb{D}=0$, $\bar{\mathbb{Q}}\mathbb{J}=0$, we have $\bar{\mathbb{V}}(\zeta)=\bar{\mathbb{Q}}\left[\mathbb{V}-(\rho-\rho_0)\mathbb{W}\mathbb{Y}-\mathbb{Z}(\delta-\delta_0)\right]$. Thus,

$$\begin{aligned}\frac{1}{n}\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta)=&\frac{1}{n}\mathbb{V}'\bar{\mathbb{Q}}\mathbb{V}+(\rho-\rho_0)^2\frac{1}{n}\mathbb{Y}'\mathbb{W}'\bar{\mathbb{Q}}\mathbb{W}\mathbb{Y}+\frac{1}{n}(\delta-\delta_0)'\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}(\delta-\delta_0)\\&-2(\rho-\rho_0)\frac{1}{n}\mathbb{Y}'\mathbb{W}'\bar{\mathbb{Q}}\mathbb{V}-2\frac{1}{n}(\delta-\delta_0)'\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{V}+2(\rho-\rho_0)\frac{1}{n}(\delta-\delta_0)'\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{W}\mathbb{Y}.\end{aligned}\tag{C.15}$$

Furthermore,

$$\mathbb{W}\mathbb{Y}=\mathbb{G}_0\left(\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{J}\tau_0+\mathbb{V}\right).$$

Compared with Theorem 1, the only additional component is $\mathbb{G}_0\mathbb{J}\tau_0$. Since τ_{t0} is uniformly bounded by Assumption 4′ and $\mathbb{G}_0$ is UB under Assumptions 3 and 5, this term can be treated in the same way as $\mathbb{G}_0\mathbb{D}\alpha_0$. Therefore, applying Lemma 23 to each term in (C.15), all polynomial coefficients in $n^{-1}\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta)$ converge uniformly to their expectations. Since ρ and δ are bounded in Θ, we obtain

$$\sup_{\theta\in\Theta}\frac{1}{n}\left|\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta)-E\left[\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta)\right]\right|\xrightarrow{p}0.$$

Because σ^2 is bounded away from zero, it follows that $\sup_{\theta\in\Theta}\left|\frac{1}{n}\left(\bar{L}_n^c(\theta)-\bar{Q}_n(\theta)\right)\right|\xrightarrow{p}0$.

Proof of equicontinuity

Next we prove that $\frac{1}{n}\bar{Q}_n(\theta)$ is uniformly equicontinuous for $\theta\in\Theta$. The terms involving $-\frac{1}{2}\log\sigma^2$ and $\frac{1}{n}\log|\mathbb{S}(\rho)|$ are identical to those in the proof of Theorem 1, and hence are omitted. Therefore, it remains to verify that replacing the OWFE projection matrix by the TWFE projection matrix $\bar{\mathbb{Q}}$, together with the presence of the time effects $\mathbb{J}\tau_0$, preserves the uniform equicontinuity of the normalized expectation term $\frac{1}{n}E\left[\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta)\right]$.

Substituting $\mathbb{Y}=\mathbb{S}_0^{-1}\left(\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{J}\tau_0+\mathbb{V}\right)$ into $\bar{\mathbb{V}}(\zeta)$ gives

$$\begin{aligned}&E\left[\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta)\right]\\=&E\left[\left\{\mathbb{S}(\rho)\mathbb{S}_0^{-1}\left(\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)-\mathbb{Z}\delta\right\}'\bar{\mathbb{Q}}\left\{\mathbb{S}(\rho)\mathbb{S}_0^{-1}\left(\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)-\mathbb{Z}\delta\right\}\right]\\&+E\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\mathbb{V}\right)'\bar{\mathbb{Q}}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\mathbb{V}\right]+2E\left[\left\{\mathbb{S}(\rho)\mathbb{S}_0^{-1}\left(\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)-\mathbb{Z}\delta\right\}'\bar{\mathbb{Q}}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\mathbb{V}\right].\end{aligned}$$

By Lemma 23 and Assumption 1′, the last term is $o(n)$. Moreover, $\mathbb{S}(\rho)\mathbb{S}_0^{-1}=I_n+(\rho_0-\rho)\mathbb{G}_0$, and $\bar{\mathbb{Q}}\mathbb{D}=0$, $\bar{\mathbb{Q}}\mathbb{J}=0$. Therefore,

$$\begin{aligned}&\frac{1}{n}E\left[\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta)\right]\\=&\frac{(\rho_0-\rho)^2}{n}E\left[\left\{\mathbb{G}_0\left(\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)\right\}'\bar{\mathbb{Q}}\left\{\mathbb{G}_0\left(\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)\right\}\right]\\&+\frac{1}{n}E\left[(\delta_0-\delta)'\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}(\delta_0-\delta)\right]+\frac{2(\rho_0-\rho)}{n}E\left[\left\{\mathbb{G}_0\left(\mathbb{Z}\delta_0+\mathbb{D}\alpha_0+\mathbb{J}\tau_0\right)\right\}'\bar{\mathbb{Q}}\mathbb{Z}(\delta_0-\delta)\right]+\bar{\sigma}_n^2+o(1),\end{aligned}\tag{C.16}$$

where

$$\bar{\sigma}_n^2 = \frac{\sigma_0^2}{n} \operatorname{tr}\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)' \bar{\mathbb{Q}}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right] = \frac{\sigma_0^2}{n} \operatorname{tr}\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)' \mathbb{S}(\rho)\mathbb{S}_0^{-1}\right] - \frac{\sigma_0^2}{n} \operatorname{tr}\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)' \bar{\mathbb{P}}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right].$$

It remains to show that the second term is negligible. Under Assumptions 3 and 5, $\mathbb{S}(\rho)\mathbb{S}_0^{-1}$ is UB uniformly in $\rho \in \Lambda$. Since $\bar{\mathbb{P}}$ is an orthogonal projection matrix with rank $N + T - 1$, we have

$$\begin{aligned}\left|\operatorname{tr}\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)' \bar{\mathbb{P}}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right]\right| &\leqslant \operatorname{rank}\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)' \bar{\mathbb{P}}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right] \left\|\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)' \bar{\mathbb{P}}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right\|_2 \\ &\leqslant \operatorname{rank}\left(\bar{\mathbb{P}}\right) \left\|\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right\|_2^2 \leqslant \operatorname{rank}\left(\bar{\mathbb{P}}\right) \left\|\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right\|_1 \left\|\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right\|_\infty = O(N+T).\end{aligned}$$

The first inequality uses the rank–spectral norm bound for the trace. The second inequality follows from $\operatorname{rank}\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)' \bar{\mathbb{P}}\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right] \leqslant \operatorname{rank}\left(\bar{\mathbb{P}}\right)$ and $\|\bar{\mathbb{P}}\|_2 = 1$. Consequently,

$$\bar{\sigma}_n^2 = \frac{\sigma_0^2}{n} \operatorname{tr}\left[\left(\mathbb{S}(\rho)\mathbb{S}_0^{-1}\right)' \mathbb{S}(\rho)\mathbb{S}_0^{-1}\right] + o(1),$$

uniformly in $\rho \in \Lambda$. The leading term is the same as the corresponding term in Theorem 1 and is therefore uniformly equicontinuous in ρ. Hence, $\bar{\sigma}_n^2$ is uniformly equicontinuous.

It remains to show that the first three terms on the right-hand side of (C.16) are uniformly equicontinuous. These three terms can be combined as $(\rho - \rho_0,\, \delta' - \delta_0')\, E\left[\bar{\mathcal{H}}\right] (\rho - \rho_0,\, \delta' - \delta_0')'$, where $\bar{\mathcal{H}} \equiv \frac{1}{n} \left(\mathbb{G}_0 \left(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{J}\tau_0\right), \mathbb{Z}\right)' \bar{\mathbb{Q}} \left(\mathbb{G}_0 \left(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{J}\tau_0\right), \mathbb{Z}\right)$. Under Assumption 4′, the elements of the regressors, α_0, and τ_0 are uniformly bounded. Together with Assumptions 5–7 and Lemma 23, we have $E\left[\bar{\mathcal{H}}\right] = O(1)$. Therefore, $(\rho - \rho_0,\, \delta' - \delta_0')\, E\left[\bar{\mathcal{H}}\right] (\rho - \rho_0,\, \delta' - \delta_0')'$ is a quadratic function of (ρ, δ) with uniformly bounded coefficients. Therefore it is uniformly equicontinuous.

Combining these results, we obtain that $\frac{1}{n} E\left[\bar{\mathbb{V}}(\zeta)'\bar{\mathbb{V}}(\zeta)\right]$ is uniformly equicontinuous for ζ. Finally, since σ^2 is bounded away from zero, multiplying the normalized expectation term by $-1/(2\sigma^2)$ preserves uniform equicontinuity. Together with the same treatment of $-\frac{1}{2}\log\sigma^2$ and $\frac{1}{n}\log|\mathbb{S}(\rho)|$ as in Theorem 1, this implies that $\frac{1}{n}\bar{Q}_n(\theta)$ is uniformly equicontinuous.

Proof of global identification condition

We now prove that for any $\theta^{(-\gamma)} \neq \theta_0^{(-\gamma)}$,

$$\limsup_{n\to\infty} \frac{1}{n}\left[\bar{Q}_n(\theta) - \bar{Q}_n(\theta_0)\right] < 0.$$

By (C.16), as in equation (C.5) in the proof of Theorem 1, we can write

$$\frac{1}{n}\left[\bar{Q}_n(\theta) - \bar{Q}_n(\theta_0)\right] = \bar{L}_1(\rho, \sigma^2) - \frac{1}{2\sigma^2}\bar{L}_2(\rho, \delta) + o(1),$$

where

$$\bar{L}_1(\rho, \sigma^2) \equiv -\frac{1}{2}\left(\ln\sigma^2 - \ln\sigma_0^2\right) + \frac{1}{n}\left(\ln|\mathbb{S}(\rho)| - \ln|\mathbb{S}_0|\right) - \frac{1}{2\sigma^2}\left(\bar{\sigma}_n^2 - \sigma^2\right),$$

and

$$\begin{aligned}\bar{L}_2(\rho, \delta) \equiv \frac{1}{n} E\Big\{&\left[(\rho_0 - \rho)\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{J}\tau_0) + \mathbb{Z}(\delta_0 - \delta)\right]' \\ &\bar{\mathbb{Q}}\left[(\rho_0 - \rho)\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{J}\tau_0) + \mathbb{Z}(\delta_0 - \delta)\right]\Big\}.\end{aligned}$$

By the information inequality, the same argument as in Theorem 1 gives $\bar{L}_1(\rho, \sigma^2) \leqslant \bar{L}_1(\rho, \bar{\sigma}_n^2) \leqslant 0$. Moreover, the terms involving $\gamma - \gamma_0$ in $\bar{L}_2(\rho, \delta)$ are asymptotically negligible after normalization by n, because $\mathbb{Z}^{(\gamma)}$ is a sparse binary vector with at most N nonzero elements. Hence

$$\lim_{n\to\infty} \bar{L}_2(\rho, \delta) = \left(\rho - \rho_0,\, \delta^{(-\gamma)\prime} - \delta_0^{(-\gamma)\prime}\right) E\left[\bar{\mathcal{H}}^{(-\gamma)}\right] \left(\rho - \rho_0,\, \delta^{(-\gamma)\prime} - \delta_0^{(-\gamma)\prime}\right)'.$$

If Assumption 8′ (i) holds, then $\lim_{n\to\infty} E\left[\bar{\mathcal{H}}^{(-\gamma)}\right]$ exists and is nonsingular. Therefore, $\bar{L}_2(\rho,\delta)$ is strictly positive whenever $(\rho,\delta^{(-\gamma)}) \neq (\rho_0,\delta_0^{(-\gamma)})$, and $(\rho_0,\delta_0^{(-\gamma)})$ is globally identified. If Assumption 8′ (ii) holds, the information inequality implies $\limsup_{n\to\infty} \bar{L}_1(\rho,\sigma^2) < 0$ whenever $\rho \neq \rho_0$. Thus ρ_0 is globally identified. Given $\rho = \rho_0$, Assumption 4′ implies that $\lim_{n\to\infty} \frac{1}{n} E\left(\mathbb{Z}^{(-\gamma)\prime}\bar{\mathbb{Q}}\mathbb{Z}^{(-\gamma)}\right)$ exists and is nonsingular, so $\delta_0^{(-\gamma)}$ is globally identified.

Once $(\rho,\delta^{(-\gamma)})$ is globally identified, σ_0^2 is identified by the same scalar information inequality as in Theorem 1. Hence $\hat{\theta}^{(-\gamma)}$ is consistent.

Finally, given $\hat{\theta}^{(-\gamma)} \xrightarrow{p} \theta_0^{(-\gamma)}$, the concentrated estimator of γ is

$$\begin{aligned}\hat{\gamma} &= \left(\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{Z}^{(\gamma)}\right)^{-1}\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\left(\mathbb{S}(\hat{\rho})\mathbb{Y} - \mathbb{Z}^{(-\gamma)}\hat{\delta}^{(-\gamma)}\right)\\ &= \gamma_0 + \left(\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{Z}^{(\gamma)}\right)^{-1}\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{V} + o_p(1).\end{aligned}$$

The second equality follows from the consistency of $\hat{\theta}^{(-\gamma)}$ and the same order bounds used above. As in the proof of Theorem 1, Assumptions 1′ and 4′ imply that $\frac{1}{N}\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{Z}^{(\gamma)}$ is bounded away from zero and uniformly bounded from above. Therefore,

$$E\left[\left(\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{Z}^{(\gamma)}\right)^{-1}\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{V}\right] = 0,$$

and

$$\mathrm{Var}\left[\left(\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{Z}^{(\gamma)}\right)^{-1}\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{V}\right] = \sigma_0^2\left(\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{Z}^{(\gamma)}\right)^{-1} = O\left(\frac{1}{N}\right).$$

By Chebyshev's inequality, $\left(\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{Z}^{(\gamma)}\right)^{-1}\mathbb{Z}^{(\gamma)\prime}\bar{\mathbb{Q}}\mathbb{V} = o_p(1)$. Thus $\hat{\gamma} \xrightarrow{p} \gamma_0$. Combining this with $\hat{\theta}^{(-\gamma)} \xrightarrow{p} \theta_0^{(-\gamma)}$, we obtain $\hat{\theta} \xrightarrow{p} \theta_0$.

C.6 Proof of Theorem 6

The proof follows the same Taylor expansion argument as in the proof of Theorem 2. We only present the parts that differ from the OWFE case.

By Theorem 5, $\widehat{\theta} \xrightarrow{p} \theta_0$. From the first-order condition $\partial \bar{L}_n^c(\widehat{\theta})/\partial\theta = 0$, the Taylor expansion around θ_0 gives

$$0 = \frac{\partial \bar{L}_n^c(\theta_0)}{\partial\theta} + \frac{\partial^2 \bar{L}_n^c(\bar{\theta})}{\partial\theta\partial\theta'}(\widehat{\theta} - \theta_0),$$

where $\bar{\theta}$ lies between $\widehat{\theta}$ and θ_0. Hence,

$$\sqrt{n}\Gamma^{-1}(\widehat{\theta} - \theta_0) = \left(-\frac{1}{n}\Gamma\frac{\partial^2 \bar{L}_n^c(\bar{\theta})}{\partial\theta\partial\theta'}\Gamma\right)^{-1}\left(\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^c(\theta_0)}{\partial\theta}\right).$$

By the TWFE score decomposition in Section B.5,

$$\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^c(\theta_0)}{\partial\theta} = \frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta} - \bar{\Delta}_n.$$

Moreover, by equation (B.11),

$$\bar{\Delta}_n = \sqrt{\frac{N}{c_u T}}\varphi_n(\theta_0) + \sqrt{\frac{T}{c_u N}}\psi_n(\theta_0) + O_p\left(\frac{1}{\sqrt{T}}\right) + O_p\left(\frac{1}{\sqrt{N}}\right),$$

where $c_u = n/(NT)$, $\varphi_n(\theta_0) = O(1)$, and $\psi_n(\theta_0) = O(1)$. Here $\varphi_n(\theta_0)$ is the OWFE incidental-parameter bias component, while $\psi_n(\theta_0)$ is the additional bias component generated by the time fixed effects.

For the mean-zero score component, the same martingale difference CLT as in the proof of Theorem 2 applies, with $\mathbb{Q}$ replaced by $\bar{\mathbb{Q}}$. Therefore,

$$\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta} \xrightarrow{d} N\left(0, \bar{\Sigma}_{\theta_0} + \bar{\Omega}_{\theta_0}\right),$$

where $\bar{\Sigma}_{\theta_0} = \lim_{n\to\infty}\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma$, $\bar{\Omega}_{\theta_0} = \lim_{n\to\infty}\Gamma\bar{\Omega}_{\theta_0,n}\Gamma$, and $\bar{\Sigma}_{\theta_0,n}$ and $\bar{\Omega}_{\theta_0,n}$ are defined in equation (B.7). By equation (B.7), the corresponding second moment satisfies

$$E\left[\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta}\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta'}\Gamma\frac{1}{\sqrt{n}}\right] = \Gamma\bar{\Sigma}_{\theta_0,n}\Gamma + \Gamma\bar{\Omega}_{\theta_0,n}\Gamma + O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right).$$

For the Hessian matrix, equations (B.8) and (B.9) imply

$$-\frac{1}{n}\Gamma\frac{\partial^2 \bar{L}_n^{c}(\bar{\theta})}{\partial\theta\partial\theta'}\Gamma = \Gamma\bar{\Sigma}_{\theta_0,n}\Gamma + O_p\left(\frac{1}{\sqrt{NT}}\right) + \left\|\Gamma^{-1}(\bar{\theta}-\theta_0)\right\|_2 O_p(1).$$

Since $\widehat{\theta} \xrightarrow{p} \theta_0$, we first have

$$-\frac{1}{n}\Gamma\frac{\partial^2 \bar{L}_n^{c}(\bar{\theta})}{\partial\theta\partial\theta'}\Gamma = \Gamma\bar{\Sigma}_{\theta_0,n}\Gamma + o_p(1).$$

Under Assumption 9′, $\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma$ is nonsingular for sufficiently large n, which has been proved in Section B.5. Hence,

$$\sqrt{n}\Gamma^{-1}(\widehat{\theta}-\theta_0) = \left[\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma + o_p(1)\right]^{-1}\left[\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta} - \bar{\Delta}_n\right].$$

Since the mean-zero score component is $O_p(1)$ and $\bar{\Delta}_n = O_p\left(\sqrt{\frac{N}{T}}\right) + O_p\left(\sqrt{\frac{T}{N}}\right)$, we obtain

$$\Gamma^{-1}(\widehat{\theta}-\theta_0) = O_p\left(\max\left\{\frac{1}{T}, \frac{1}{N}\right\}\right).$$

Therefore, by (B.8) and (B.9),

$$-\frac{1}{n}\Gamma\frac{\partial^2 \bar{L}_n^{c}(\bar{\theta})}{\partial\theta\partial\theta'}\Gamma = \Gamma\bar{\Sigma}_{\theta_0,n}\Gamma + O_p\left(\max\left\{\frac{1}{T}, \frac{1}{N}\right\}\right).$$

Using the standard inverse expansion, we have

$$\left(-\frac{1}{n}\Gamma\frac{\partial^2 \bar{L}_n^{c}(\bar{\theta})}{\partial\theta\partial\theta'}\Gamma\right)^{-1} = \left(\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma\right)^{-1} + O_p\left(\max\left\{\frac{1}{T}, \frac{1}{N}\right\}\right). \tag{C.17}$$

Combining the above results gives

$$\begin{aligned}\sqrt{n}\Gamma^{-1}(\widehat{\theta}-\theta_0) = &\left[\left(\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma\right)^{-1} + O_p\left(\max\left\{\frac{1}{T}, \frac{1}{N}\right\}\right)\right]\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta}\\ &- \left[\left(\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma\right)^{-1} + O_p\left(\max\left\{\frac{1}{T}, \frac{1}{N}\right\}\right)\right]\bar{\Delta}_n.\end{aligned}$$

Since

$$\bar{\Delta}_n = \sqrt{\frac{N}{c_u T}}\varphi_n(\theta_0) + \sqrt{\frac{T}{c_u N}}\psi_n(\theta_0) + O_p\left(\frac{1}{\sqrt{T}}\right) + O_p\left(\frac{1}{\sqrt{N}}\right),$$

we have

$$\begin{aligned}&\sqrt{n}\Gamma^{-1}(\widehat{\theta}-\theta_0) + \sqrt{\frac{N}{c_u T}}\left(\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma\right)^{-1}\varphi_n(\theta_0) + \sqrt{\frac{T}{c_u N}}\left(\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma\right)^{-1}\psi_n(\theta_0)\\ &= \left[\left(\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma\right)^{-1} + o_p(1)\right]\frac{1}{\sqrt{n}}\Gamma\frac{\partial \bar{L}_n^{c*}(\theta_0)}{\partial\theta} + O_p\left(\max\left\{\frac{1}{\sqrt{T}}, \frac{1}{\sqrt{N}}, \sqrt{\frac{N}{T^3}}, \sqrt{\frac{T}{N^3}}\right\}\right).\end{aligned}$$

Because $\lim_{n\to\infty}\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma = \bar{\Sigma}_{\theta_0}$, and the mean-zero score component converges to $N\left(0, \bar{\Sigma}_{\theta_0} + \bar{\Omega}_{\theta_0}\right)$, we obtain

$$\sqrt{n}\Gamma^{-1}(\widehat{\theta} - \theta_0) + \sqrt{\frac{N}{c_u T}}\left(\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma\right)^{-1}\varphi_n(\theta_0) + \sqrt{\frac{T}{c_u N}}\left(\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma\right)^{-1}\psi_n(\theta_0) + O_p\left(\max\left\{\frac{1}{\sqrt{T}}, \frac{1}{\sqrt{N}}, \sqrt{\frac{N}{T^3}}, \sqrt{\frac{T}{N^3}}\right\}\right) \xrightarrow{d} N\left(0, \bar{\Sigma}_{\theta_0}^{-1}\left(\bar{\Sigma}_{\theta_0} + \bar{\Omega}_{\theta_0}\right)\bar{\Sigma}_{\theta_0}^{-1}\right).$$

Since the γ-components of both $\varphi_n(\theta_0)$ and $\psi_n(\theta_0)$ are zero, and Γ only differs from the identity matrix in the γ-component, we have $\Gamma^{-1}\varphi_n(\theta_0) = \varphi_n(\theta_0)$ and $\Gamma^{-1}\psi_n(\theta_0) = \psi_n(\theta_0)$. So equivalently,

$$\sqrt{n}\Gamma^{-1}\left[\widehat{\theta} - \left(\theta_0 - \frac{\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0)}{c_u T} - \frac{\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0)}{c_u N}\right)\right] + O_p\left(\max\left\{\frac{1}{\sqrt{T}}, \frac{1}{\sqrt{N}}, \sqrt{\frac{N}{T^3}}, \sqrt{\frac{T}{N^3}}\right\}\right) \xrightarrow{d} N\left(0, \bar{\Sigma}_{\theta_0}^{-1}\left(\bar{\Sigma}_{\theta_0} + \bar{\Omega}_{\theta_0}\right)\bar{\Sigma}_{\theta_0}^{-1}\right).$$

Furthermore, by Section B.5, $\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0) = O_p(1)$, and $\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0) = O_p(1)$. If v_{it}'s are normally distributed, then $\bar{\Omega}_{\theta_0} = 0$, and the asymptotic variance reduces to $\bar{\Sigma}_{\theta_0}^{-1}$. This completes the proof.

C.7 Proof of Theorem 7

The proof follows directly from Theorem 6 and the same plug-in argument as in the proof of Theorem 3. Define the bias-corrected estimator as

$$\widehat{\theta}^1 = \widehat{\theta} + \left[\left(-\frac{1}{n}\frac{\partial^2 \bar{L}_n^c(\theta)}{\partial\theta\partial\theta'}\right)^{-1}\left\{\frac{\varphi_n(\theta)}{c_u T} + \frac{\psi_n(\theta)}{c_u N}\right\}\right]_{\theta=\widehat{\theta}}.$$

By (C.17), we have

$$\left(-\frac{1}{n}\Gamma\frac{\partial^2 \bar{L}_n^c(\widehat{\theta})}{\partial\theta\partial\theta'}\Gamma\right)^{-1} = \left(\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma\right)^{-1} + O_p\left(\max\left\{\frac{1}{T}, \frac{1}{N}\right\}\right).$$

For the plug-in bias correction term, the proof is the same as in the OWFE case. The continuity of $\varphi_n(\theta)$ follows from Section B.4, and the continuity of $\psi_n(\theta)$ follows similarly. Therefore, by the same mean-value argument as in the OWFE case,

$$\varphi_n(\widehat{\theta}) - \varphi_n(\theta_0) = \left\|\Gamma^{-1}(\widehat{\theta} - \theta_0)\right\|_2 O_p(1), \qquad \psi_n(\widehat{\theta}) - \psi_n(\theta_0) = \left\|\Gamma^{-1}(\widehat{\theta} - \theta_0)\right\|_2 O_p(1).$$

By Theorem 6, $\left\|\Gamma^{-1}(\widehat{\theta} - \theta_0)\right\|_2 = O_p\left(\max\left\{\frac{1}{T}, \frac{1}{N}\right\}\right)$. Therefore,

$$\begin{aligned}
&\widehat{\theta}^1 - \widehat{\theta} - \frac{\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0)}{c_u T} - \frac{\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0)}{c_u N} \\
=&\left\{\left[\left(-\frac{1}{n}\frac{\partial^2 \bar{L}_n^c(\theta)}{\partial\theta\partial\theta'}\right)^{-1}\right]_{\theta=\widehat{\theta}} - \bar{\Sigma}_{\theta_0,n}^{-1}\right\}\left[\frac{\varphi_n(\theta_0)}{c_u T} + \frac{\psi_n(\theta_0)}{c_u N}\right] \\
&+ \left[\left(-\frac{1}{n}\frac{\partial^2 \bar{L}_n^c(\theta)}{\partial\theta\partial\theta'}\right)^{-1}\right]_{\theta=\widehat{\theta}}\left[\frac{\varphi_n(\widehat{\theta}) - \varphi_n(\theta_0)}{c_u T} + \frac{\psi_n(\widehat{\theta}) - \psi_n(\theta_0)}{c_u N}\right] \\
=&O_p\left[\left(\frac{1}{T} + \frac{1}{N}\right)\max\left\{\frac{1}{T}, \frac{1}{N}\right\}\right].
\end{aligned}$$

Multiplying both sides by $\sqrt{n}\Gamma^{-1}$, we obtain

$$\begin{aligned}\sqrt{n}\Gamma^{-1}\left[\widehat{\theta}^1-\widehat{\theta}-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0)}{c_uT}-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0)}{c_uN}\right]=&O_p\left[\sqrt{NT}\left(\frac{1}{T}+\frac{1}{N}\right)\max\left\{\frac{1}{T},\frac{1}{N}\right\}\right]\\=&O_p\left(\frac{1}{\sqrt{NT}}+\sqrt{\frac{N}{T^3}}+\sqrt{\frac{T}{N^3}}\right)=o_p(1),\end{aligned}$$

where the last equality follows from $N/T^3\to 0$ and $T/N^3\to 0$. Therefore,

$$\sqrt{n}\Gamma^{-1}\left(\widehat{\theta}^1-\theta_0\right)=\sqrt{n}\Gamma^{-1}\left[\widehat{\theta}-\left(\theta_0-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0)}{c_uT}-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0)}{c_uN}\right)\right]+o_p(1).$$

By Theorem 6, if $N/T^3\to 0$ and $T/N^3\to 0$, then

$$\sqrt{n}\Gamma^{-1}\left[\widehat{\theta}-\left(\theta_0-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\varphi_n(\theta_0)}{c_uT}-\frac{\bar{\Sigma}_{\theta_0,n}^{-1}\psi_n(\theta_0)}{c_uN}\right)\right]\xrightarrow{d}N\left(0,\bar{\Sigma}_{\theta_0}^{-1}\left(\bar{\Sigma}_{\theta_0}+\bar{\Omega}_{\theta_0}\right)\bar{\Sigma}_{\theta_0}^{-1}\right).$$

Hence,

$$\sqrt{n}\Gamma^{-1}\left(\widehat{\theta}^1-\theta_0\right)\xrightarrow{d}N\left(0,\bar{\Sigma}_{\theta_0}^{-1}\left(\bar{\Sigma}_{\theta_0}+\bar{\Omega}_{\theta_0}\right)\bar{\Sigma}_{\theta_0}^{-1}\right).$$

If v_{it}'s are normally distributed, then $\bar{\Omega}_{\theta_0}=0$, and the asymptotic variance reduces to $\bar{\Sigma}_{\theta_0}^{-1}$. This completes the proof.

C.8 Proof of Theorem 8

Let $\widehat{\theta}^1=(\widehat{\rho}^1,\widehat{\delta}^{1\prime},\widehat{\sigma}^{2,1})'$ be the TWFE bias-corrected estimator. Define

$$(\widehat{\alpha}^{1\prime},\widehat{\tau}^{1\prime})'=[(\mathbb{D},\mathbb{J})'(\mathbb{D},\mathbb{J})]^{-1}(\mathbb{D},\mathbb{J})'\{\mathbb{S}(\widehat{\rho}^1)\mathbb{Y}-\mathbb{Z}\widehat{\delta}^1\},$$

and $\widehat{\mathbb{V}}=\bar{\mathbb{Q}}\{\mathbb{S}(\widehat{\rho}^1)\mathbb{Y}-\mathbb{Z}\widehat{\delta}^1\}$. Let $\check{\sigma}^2=n^{-1}\widehat{\mathbb{V}}'\widehat{\mathbb{V}}$, $\check{\mu}_3=n^{-1}\sum_{r=1}^n\widehat{\mathbb{V}}_r^3$, $\check{\mu}_4=n^{-1}\sum_{r=1}^n\widehat{\mathbb{V}}_r^4$, and $\check{c}=(\check{\mu}_4-3\check{\sigma}^4)/\check{\sigma}^4$, where $\check{\sigma}^4=(\check{\sigma}^2)^2$. Let $\widehat{\mathbb{G}}=\mathbb{W}\mathbb{S}(\widehat{\rho}^1)^{-1}$ and $\widehat{g}=(\widehat{\mathbb{G}}_{11},\ldots,\widehat{\mathbb{G}}_{nn})'$.

Define

$$\widehat{\bar{\Sigma}}=\begin{pmatrix}\left[\begin{array}{c}\frac{1}{n\check{\sigma}^2}\{\widehat{\mathbb{G}}(\mathbb{Z}\widehat{\delta}^1+\mathbb{D}\widehat{\alpha}^1+\mathbb{J}\widehat{\tau}^1)\}'\bar{\mathbb{Q}}\{\widehat{\mathbb{G}}(\mathbb{Z}\widehat{\delta}^1+\mathbb{D}\widehat{\alpha}^1+\mathbb{J}\widehat{\tau}^1)\}\\+\frac{1}{n}\operatorname{tr}\{\widehat{\mathbb{G}}(\widehat{\mathbb{G}}+\widehat{\mathbb{G}}')\}\end{array}\right] & * & *\\ \frac{1}{n\check{\sigma}^2}\widetilde{\Gamma}'\mathbb{Z}'\bar{\mathbb{Q}}\widehat{\mathbb{G}}(\mathbb{Z}\widehat{\delta}^1+\mathbb{D}\widehat{\alpha}^1+\mathbb{J}\widehat{\tau}^1) & \frac{1}{n\check{\sigma}^2}\widetilde{\Gamma}'\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{Z}\widetilde{\Gamma} & *\\ \frac{1}{n\check{\sigma}^2}\operatorname{tr}(\widehat{\mathbb{G}}) & \mathbf{0}_{1\times(k+3)} & \frac{1}{2\check{\sigma}^4}\end{pmatrix}.$$

Next, define

$$\widehat{\bar{\Omega}}=\begin{pmatrix}\frac{\check{c}}{n}\widehat{g}'\widehat{g}+\frac{2\check{\mu}_3}{n\check{\sigma}^4}\left[\widehat{\mathbb{G}}\mathbb{Z}\widehat{\delta}^1+\bar{\mathbb{Q}}\widehat{\mathbb{G}}(\mathbb{D}\widehat{\alpha}^1+\mathbb{J}\widehat{\tau}^1)\right]'\widehat{g} & * & *\\ \frac{\check{\mu}_3}{n\check{\sigma}^4}\widetilde{\Gamma}'\mathbb{Z}'\widehat{g} & \mathbf{0}_{(k+3)\times(k+3)} & *\\ \frac{\check{c}}{2n\check{\sigma}^2}\operatorname{tr}(\widehat{\mathbb{G}}) & \mathbf{0}_{1\times(k+3)} & \frac{\check{c}}{4\check{\sigma}^4}\end{pmatrix}.$$

Set

$$\widehat{\bar{\Xi}}=\widehat{\bar{\Sigma}}^{-1}(\widehat{\bar{\Sigma}}+\widehat{\bar{\Omega}})\widehat{\bar{\Sigma}}^{-1}.$$

Proof. We first show the consistency of the residual moment estimators. Since

$$\widehat{\mathbb{V}} - \bar{\mathbb{Q}}\mathbb{V} = \bar{\mathbb{Q}}\left[-(\widehat{\rho}^{\,1} - \rho_0)\mathbb{W}\mathbb{Y} - \mathbb{Z}(\widehat{\delta}^{\,1} - \delta_0)\right],$$

the same argument as in the proof of Theorem 4, with $\mathbb{Q}$ replaced by $\bar{\mathbb{Q}}$, gives

$$\frac{1}{n}\sum_{r=1}^{n}\left|\widehat{\mathbb{V}}_r - (\bar{\mathbb{Q}}\mathbb{V})_r\right|^q = o_p(1), \qquad q = 2, 3, 4.$$

Here we use that $\bar{\mathbb{Q}}$ is UB, and the moment bounds for $\mathbb{W}\mathbb{Y}$ and the columns of $\mathbb{Z}$ are the same as those used in Theorem 4. Therefore, by the binomial expansion and Hölder's inequality,

$$\frac{1}{n}\sum_{r=1}^{n}\widehat{\mathbb{V}}_r^q - \frac{1}{n}\sum_{r=1}^{n}(\bar{\mathbb{Q}}\mathbb{V})_r^q = o_p(1), \qquad q = 2, 3, 4.$$

It remains to compare the sample moments of $\bar{\mathbb{Q}}\mathbb{V}$ and $\mathbb{V}$. Since $\bar{\mathbb{Q}}\mathbb{V} = \mathbb{V} - \bar{\mathbb{P}}\mathbb{V}$, it suffices to control the average moments of $\bar{\mathbb{P}}\mathbb{V}$. Since $\bar{\mathbb{P}}$ is an orthogonal projection matrix with $\operatorname{rank}(\bar{\mathbb{P}}) = N + T - 1$, for each observation r,

$$E\left|(\bar{\mathbb{P}}\mathbb{V})_r\right|^2 = \sigma_0^2(\bar{\mathbb{P}})_{rr}.$$

Moreover, since $(\bar{\mathbb{P}}\mathbb{V})_r = \sum_{s=1}^{n}(\bar{\mathbb{P}})_{rs}\mathbb{V}_s$, the fourth-moment formula for a linear combination of independent errors gives

$$E\left|(\bar{\mathbb{P}}\mathbb{V})_r\right|^4 = (\mu_4 - 3\sigma_0^4)\sum_{s=1}^{n}(\bar{\mathbb{P}})_{rs}^4 + 3\sigma_0^4\left[\sum_{s=1}^{n}(\bar{\mathbb{P}})_{rs}^2\right]^2 \leqslant c_4\left[\sum_{s=1}^{n}(\bar{\mathbb{P}})_{rs}^2\right]^2 = c_4(\bar{\mathbb{P}})_{rr}^2,$$

where $c_4 = |\mu_4 - 3\sigma_0^4| + 3\sigma_0^4$. The last equality holds because $\bar{\mathbb{P}}$ is symmetric and idempotent,

$$\sum_{s=1}^{n}(\bar{\mathbb{P}})_{rs}^2 = (\bar{\mathbb{P}}\bar{\mathbb{P}})_{rr} = (\bar{\mathbb{P}})_{rr}.$$

By Hölder's inequality,

$$E\left|(\bar{\mathbb{P}}\mathbb{V})_r\right|^3 \leqslant \left\{E\left|(\bar{\mathbb{P}}\mathbb{V})_r\right|^4\right\}^{3/4} \leqslant c_4^{3/4}(\bar{\mathbb{P}})_{rr}^{3/2}.$$

Since $0 \leqslant (\bar{\mathbb{P}})_{rr} \leqslant 1$ and

$$\sum_{r=1}^{n}(\bar{\mathbb{P}})_{rr} = \operatorname{tr}(\bar{\mathbb{P}}) = N + T - 1,$$

we obtain, for $q = 2, 3, 4$,

$$E\left[\frac{1}{n}\sum_{r=1}^{n}|(\bar{\mathbb{P}}\mathbb{V})_r|^q\right] \leqslant c_q\frac{N + T - 1}{n} = O\left(\frac{1}{T}\right) + O\left(\frac{1}{N}\right),$$

where $c_2 = \sigma_0^2$, $c_3 = c_4^{3/4}$, and $c_4 = |\mu_4 - 3\sigma_0^4| + 3\sigma_0^4$. Therefore,

$$\frac{1}{n}\sum_{r=1}^{n}|(\bar{\mathbb{P}}\mathbb{V})_r|^q = o_p(1), \qquad q = 2, 3, 4.$$

Applying the same binomial expansion as in the proof of Theorem 4, we obtain

$$\frac{1}{n}\sum_{r=1}^{n}(\bar{\mathbb{Q}}\mathbb{V})_r^q - \frac{1}{n}\sum_{r=1}^{n}\mathbb{V}_r^q = o_p(1), \qquad q = 2, 3, 4.$$

Consequently,

$$\frac{1}{n}\sum_{r=1}^{n}\widehat{\mathbb{V}}_r^q - \frac{1}{n}\sum_{r=1}^{n}\mathbb{V}_r^q = o_p(1), \qquad q = 2,3,4.$$

By the law of large numbers, it follows that

$$\check{\sigma}^2 \xrightarrow{p} \sigma_0^2, \qquad \check{\mu}_3 \xrightarrow{p} \mu_3, \qquad \check{\mu}_4 \xrightarrow{p} \mu_4, \qquad \check{c} \xrightarrow{p} \frac{\mu_4 - 3\sigma_0^4}{\sigma_0^4}.$$

We next show that $\widehat{\bar{\Sigma}}$ is consistent. By the consistency of the TWFE bias-corrected estimator, $\widehat{\theta}^1 \xrightarrow{p} \theta_0$. In particular, $\widehat{\rho}^1 - \rho_0 = o_p(1)$. As shown in the proof of Theorem 4, for $p = 1, \infty$,

$$\left\|\mathbb{S}(\widehat{\rho}^1)^{-1} - \mathbb{S}(\rho_0)^{-1}\right\|_p = o_p(1), \qquad \left\|\widehat{\mathbb{G}} - \mathbb{G}_0\right\|_p = o_p(1).$$

From the definition of $(\widehat{\alpha}^{1\prime}, \widehat{\tau}^{1\prime})'$,

$$\begin{aligned}\mathbb{D}\widehat{\alpha}^1 + \mathbb{J}\widehat{\tau}^1 - (\mathbb{D}\alpha_0 + \mathbb{J}\tau_0) =& \bar{\mathbb{P}}\left[\mathbb{S}(\widehat{\rho}^1)\mathbb{Y} - \mathbb{Z}\widehat{\delta}^1\right] - (\mathbb{D}\alpha_0 + \mathbb{J}\tau_0) \\ =& \bar{\mathbb{P}}\mathbb{V} - \bar{\mathbb{P}}\left[(\widehat{\rho}^1 - \rho_0)\mathbb{W}\mathbb{Y} + \mathbb{Z}(\widehat{\delta}^1 - \delta_0)\right].\end{aligned}$$

As in the proof of Theorem 4, with $\mathbb{P}$ replaced by $\bar{\mathbb{P}}$, we have

$$\frac{1}{n}\left\|\mathbb{D}\widehat{\alpha}^1 + \mathbb{J}\widehat{\tau}^1 - (\mathbb{D}\alpha_0 + \mathbb{J}\tau_0)\right\|^2 = o_p(1).$$

Since $\widehat{\mathbb{G}}$ and $\mathbb{G}_0$ are UB, and $\|\widehat{\mathbb{G}} - \mathbb{G}_0\|_1 = o_p(1)$ and $\|\widehat{\mathbb{G}} - \mathbb{G}_0\|_\infty = o_p(1)$, we have

$$\begin{aligned}\frac{1}{n}\left\|\widehat{\mathbb{G}}\mathbb{Z}\widehat{\delta}^1 - \mathbb{G}_0\mathbb{Z}\delta_0\right\|^2 &= o_p(1), \\ \frac{1}{n}\left\|\widehat{\mathbb{G}}(\mathbb{D}\widehat{\alpha}^1 + \mathbb{J}\widehat{\tau}^1) - \mathbb{G}_0(\mathbb{D}\alpha_0 + \mathbb{J}\tau_0)\right\|^2 &= o_p(1). \end{aligned} \tag{C.18}$$

Consequently,

$$\frac{1}{n}\left\|\widehat{\mathbb{G}}(\mathbb{Z}\widehat{\delta}^1 + \mathbb{D}\widehat{\alpha}^1 + \mathbb{J}\widehat{\tau}^1) - \mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{J}\tau_0)\right\|^2 = o_p(1).$$

Moreover, $n^{-1}\|\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{J}\tau_0)\|^2 = O_p(1)$, because the additional term $\mathbb{G}_0\mathbb{J}\tau_0$ is handled in the same way as $\mathbb{G}_0\mathbb{D}\alpha_0$. Therefore, using the Cauchy–Schwarz inequality and the UB property of $\bar{\mathbb{Q}}$,

$$\begin{aligned}&\frac{1}{n}\left[\widehat{\mathbb{G}}(\mathbb{Z}\widehat{\delta}^1 + \mathbb{D}\widehat{\alpha}^1 + \mathbb{J}\widehat{\tau}^1)\right]'\bar{\mathbb{Q}}\left[\widehat{\mathbb{G}}(\mathbb{Z}\widehat{\delta}^1 + \mathbb{D}\widehat{\alpha}^1 + \mathbb{J}\widehat{\tau}^1)\right] \\ &- \frac{1}{n}\left[\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{J}\tau_0)\right]'\bar{\mathbb{Q}}\left[\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{J}\tau_0)\right] = o_p(1).\end{aligned}$$

Similarly,

$$\frac{1}{n}\widetilde{\Gamma}'\mathbb{Z}'\bar{\mathbb{Q}}\widehat{\mathbb{G}}(\mathbb{Z}\widehat{\delta}^1 + \mathbb{D}\widehat{\alpha}^1 + \mathbb{J}\widehat{\tau}^1) - \frac{1}{n}\widetilde{\Gamma}'\mathbb{Z}'\bar{\mathbb{Q}}\mathbb{G}_0(\mathbb{Z}\delta_0 + \mathbb{D}\alpha_0 + \mathbb{J}\tau_0) = o_p(1).$$

Also, the trace term satisfies

$$\frac{1}{n}\operatorname{tr}\{\widehat{\mathbb{G}}(\widehat{\mathbb{G}} + \widehat{\mathbb{G}}')\} = \frac{1}{n}\operatorname{tr}\{\mathbb{G}_0(\mathbb{G}_0 + \mathbb{G}_0')\} + o_p(1). \tag{C.19}$$

Similarly, the plug-in argument used in the proof of Theorem 4 gives

$$\widehat{\bar{\Sigma}} - \Gamma\bar{\Sigma}_{\theta_0,n}\Gamma = o_p(1).$$

Since $\lim_{n\to\infty}\Gamma\bar{\Sigma}_{\theta_0,n}\Gamma = \bar{\Sigma}_{\theta_0}$, we obtain

$$\hat{\bar{\Sigma}} \xrightarrow{p} \bar{\Sigma}_{\theta_0}.$$

It remains to show the consistency of $\hat{\bar{\Omega}}$. Similar to (C.19),

$$\frac{1}{n}\hat{g}'\hat{g} = \frac{1}{n}\sum_{r=1}^{n}(\mathbb{G}_0)_{rr}^2 + o_p(1), \qquad \frac{1}{n}\operatorname{tr}(\hat{\mathbb{G}}) = \frac{1}{n}\operatorname{tr}(\mathbb{G}_0) + o_p(1), \qquad \frac{1}{n}\tilde{\Gamma}'\mathbb{Z}'\hat{g} = \frac{1}{n}\tilde{\Gamma}'\mathbb{Z}'g_0 + o_p(1),$$

where $g_0 = ((\mathbb{G}_0)_{11}, \ldots, (\mathbb{G}_0)_{nn})'$. Moreover, by (C.18),

$$\begin{aligned}
&\frac{1}{n}\left\|\hat{\mathbb{G}}\mathbb{Z}\hat{\delta}^1 + \bar{\mathbb{Q}}\hat{\mathbb{G}}(\mathbb{D}\hat{\alpha}^1 + \mathbb{J}\hat{\tau}^1) - \left[\mathbb{G}_0\mathbb{Z}\delta_0 + \bar{\mathbb{Q}}\mathbb{G}_0(\mathbb{D}\alpha_0 + \mathbb{J}\tau_0)\right]\right\|^2 \\
\leqslant &\frac{2}{n}\left\|\hat{\mathbb{G}}\mathbb{Z}\hat{\delta}^1 - \mathbb{G}_0\mathbb{Z}\delta_0\right\|^2 + 2\|\bar{\mathbb{Q}}\|_\infty\|\bar{\mathbb{Q}}\|_1\frac{1}{n}\left\|\hat{\mathbb{G}}(\mathbb{D}\hat{\alpha}^1 + \mathbb{J}\hat{\tau}^1) - \mathbb{G}_0(\mathbb{D}\alpha_0 + \mathbb{J}\tau_0)\right\|^2 = o_p(1).
\end{aligned}$$

Since $n^{-1}\hat{g}'\hat{g} = O_p(1)$ and $n^{-1}g_0'g_0 = O_p(1)$, the Cauchy–Schwarz inequality gives

$$\frac{1}{n}\left[\hat{\mathbb{G}}\mathbb{Z}\hat{\delta}^1 + \bar{\mathbb{Q}}\hat{\mathbb{G}}(\mathbb{D}\hat{\alpha}^1 + \mathbb{J}\hat{\tau}^1)\right]'\hat{g} - \frac{1}{n}\left[\mathbb{G}_0\mathbb{Z}\delta_0 + \bar{\mathbb{Q}}\mathbb{G}_0(\mathbb{D}\alpha_0 + \mathbb{J}\tau_0)\right]'g_0 = o_p(1).$$

Combining the above results with the consistency of $\check{\mu}_3$, $\check{c}$, and $\check{\sigma}^2$, we obtain

$$\hat{\bar{\Omega}} - \Gamma\bar{\Omega}_{\theta_0,n}\Gamma = o_p(1).$$

Since $\lim_{n\to\infty}\Gamma\bar{\Omega}_{\theta_0,n}\Gamma = \bar{\Omega}_{\theta_0}$, it follows that

$$\hat{\bar{\Omega}} \xrightarrow{p} \bar{\Omega}_{\theta_0}.$$

Finally, Assumption 9′ implies that $\bar{\Sigma}_{\theta_0}$ is nonsingular. Therefore, $\hat{\bar{\Sigma}}$ is nonsingular with probability approaching one, and by the continuous mapping theorem,

$$\hat{\bar{\Xi}} = \hat{\bar{\Sigma}}^{-1}(\hat{\bar{\Sigma}} + \hat{\bar{\Omega}})\hat{\bar{\Sigma}}^{-1} \xrightarrow{p} \bar{\Sigma}_{\theta_0}^{-1}(\bar{\Sigma}_{\theta_0} + \bar{\Omega}_{\theta_0})\bar{\Sigma}_{\theta_0}^{-1}.$$

Thus, $n^{-1}\Gamma\hat{\bar{\Xi}}\Gamma$ is a consistent estimator of the asymptotic covariance matrix of $\hat{\theta}^1$. This completes the proof. □

D Detailed Simulation Results

D.1 Supplemental Tables

Tables D.1–D.6 report supplementary simulation results for the baseline OWFE estimator. Tables D.7–D.12 report supplementary simulation results for the TWFE estimator.

Table D.1: Results of OWFE estimator with Normal error, UP=30%, $\theta_0 = (0.1, 0.3, 0.5, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0108	0.0183	-0.0851	-0.0525	-0.0181	-0.1671	$N = 100$	BIAS	0.0005	0.0049	-0.0234	-0.0055	-0.0025	-0.0405
	SD	0.0301	0.0303	0.0261	0.1506	0.0425	0.0498		SD	0.0144	0.0158	0.0120	0.1437	0.0190	0.0269
	RMSE	0.0320	0.0354	0.0890	0.1594	0.0462	0.1744		RMSE	0.0144	0.0165	0.0263	0.1438	0.0191	0.0486
	95%CP	0.9140	0.8950	0.0790	0.9360	0.9200	0.1180		95%CP	0.9590	0.9550	0.4820	0.9480	0.9570	0.6600
	90%CP	0.8360	0.8230	0.0410	0.8730	0.8610	0.0590		90%CP	0.9090	0.8870	0.3490	0.9050	0.9020	0.5310
		After bias correction								After bias correction					
	BIAS	**-0.0050**	**0.0041**	**-0.0214**	**-0.0128**	**-0.0034**	**-0.0485**		BIAS	**0.0004**	**-0.0002**	**-0.0021**	**0.0011**	**-0.0008**	**-0.0061**
	SD	**0.0301**	0.0308	0.0265	**0.1499**	0.0426	0.0565		SD	0.0144	0.0159	0.0121	**0.1431**	**0.0189**	0.0277
	RMSE	**0.0305**	**0.0310**	**0.0341**	**0.1504**	**0.0428**	**0.0744**		RMSE	0.0144	**0.0159**	**0.0123**	**0.1431**	**0.0189**	**0.0284**
	95%CP	**0.9240**	**0.9560**	**0.8700**	**0.9490**	**0.9380**	**0.8380**		95%CP	0.9600	0.9600	**0.9270**	0.9550	0.9600	**0.9350**
	90%CP	**0.8590**	**0.8920**	**0.7740**	**0.8960**	**0.8860**	**0.7460**		90%CP	**0.9070**	0.9140	**0.8790**	**0.8960**	**0.9010**	**0.8860**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0088	0.0211	-0.0844	-0.0501	-0.0206	-0.1594	$N = 400$	BIAS	-0.0003	0.0058	-0.0234	-0.0064	-0.0009	-0.0373
	SD	0.0147	0.0156	0.0133	0.0766	0.0206	0.0253		SD	0.0074	0.0085	0.0059	0.0705	0.0097	0.0139
	RMSE	0.0172	0.0262	0.0854	0.0916	0.0291	0.1613		RMSE	0.0074	0.0102	0.0242	0.0708	0.0098	0.0398
	95%CP	0.8960	0.7090	0.0000	0.8810	0.8260	0.0000		95%CP	0.9520	0.9040	0.0230	0.9600	0.9490	0.2090
	90%CP	0.8180	0.6080	0.0000	0.8050	0.7340	0.0000		90%CP	0.9100	0.8110	0.0060	0.9010	0.8950	0.1420
		After bias correction								After bias correction					
	BIAS	**-0.0031**	**0.0050**	**-0.0205**	**-0.0118**	**-0.0058**	**-0.0398**		BIAS	-0.0005	**0.0005**	**-0.0021**	**-0.0002**	**0.0008**	**-0.0028**
	SD	0.0148	0.0158	**0.0132**	0.0768	**0.0205**	0.0286		SD	0.0074	0.0085	0.0060	0.0709	**0.0097**	0.0143
	RMSE	**0.0151**	**0.0166**	**0.0244**	**0.0777**	**0.0213**	**0.0490**		RMSE	0.0075	**0.0085**	**0.0063**	0.0709	**0.0097**	**0.0146**
	95%CP	**0.9370**	**0.9350**	**0.6270**	**0.9280**	**0.9500**	**0.6610**		95%CP	0.9530	**0.9480**	**0.9250**	**0.9590**	0.9460	**0.9320**
	90%CP	**0.8830**	**0.8850**	**0.5060**	**0.8720**	**0.8940**	**0.5500**		90%CP	*0.9100*	**0.9030**	**0.8630**	**0.9000**	**0.9040**	**0.8540**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0091	0.0211	-0.0842	-0.0504	-0.0192	-0.1598	$N = 1600$	BIAS	0.0000	0.0055	-0.0233	-0.0067	-0.0019	-0.0379
	SD	0.0076	0.0075	0.0069	0.0357	0.0102	0.0124		SD	0.0038	0.0043	0.0028	0.0353	0.0047	0.0066
	RMSE	0.0119	0.0224	0.0845	0.0618	0.0218	0.1603		RMSE	0.0038	0.0070	0.0235	0.0359	0.0051	0.0384
	95%CP	0.7300	0.2210	0.0000	0.7390	0.5500	0.0000		95%CP	0.9520	0.7340	0.0000	0.9460	0.9380	0.0010
	90%CP	0.6220	0.1360	0.0000	0.6300	0.4060	0.0000		90%CP	0.9040	0.6230	0.0000	0.8840	0.8800	0.0000
		After bias correction								After bias correction					
	BIAS	**-0.0034**	**0.0045**	**-0.0204**	**-0.0122**	**-0.0046**	**-0.0402**		BIAS	-0.0002	**0.0001**	**-0.0019**	**-0.0005**	**-0.0003**	**-0.0034**
	SD	**0.0076**	0.0076	0.0069	**0.0357**	**0.0101**	0.0140		SD	**0.0038**	0.0043	0.0029	**0.0352**	**0.0047**	0.0068
	RMSE	**0.0083**	**0.0088**	**0.0215**	**0.0377**	**0.0111**	**0.0426**		RMSE	**0.0038**	**0.0043**	**0.0034**	**0.0352**	**0.0047**	**0.0076**
	95%CP	**0.9030**	**0.9160**	**0.1190**	**0.9350**	**0.9390**	**0.1590**		95%CP	**0.9510**	**0.9500**	**0.8950**	**0.9520**	0.9560	**0.9190**
	90%CP	**0.8380**	**0.8550**	**0.0690**	**0.8920**	**0.8860**	**0.1000**		90%CP	**0.9020**	**0.8920**	**0.8180**	**0.8950**	**0.9080**	**0.8600**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.2: Results of OWFE estimator with Normal error, UP=60%, $\theta_0 = (0.1, 0.3, 0.5, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0161	0.0157	-0.1499	-0.1251	-0.0489	-0.2990	$N = 100$	BIAS	-0.0009	0.0091	-0.0345	-0.0206	-0.0045	-0.0708
	SD	0.0377	0.0418	0.0429	0.1369	0.0606	0.0598		SD	0.0168	0.0182	0.0151	0.1105	0.0260	0.0347
	RMSE	0.0410	0.0446	0.1559	0.1854	0.0779	0.3049		RMSE	0.0168	0.0204	0.0376	0.1124	0.0264	0.0788
	95%CP	0.8990	0.9620	0.0420	0.8450	0.8640	0.0040		95%CP	0.9460	0.9340	0.3760	0.9520	0.9450	0.4690
	90%CP	0.8230	0.8890	0.0190	0.7660	0.7760	0.0010		90%CP	0.8790	0.8570	0.2590	0.8980	0.8900	0.3560
		After bias correction								After bias correction					
	BIAS	**-0.0093**	**0.0056**	**-0.0570**	**-0.0494**	**-0.0183**	**-0.1235**		BIAS	**-0.0003**	**0.0003**	**-0.0044**	**-0.0089**	**-0.0009**	**-0.0122**
	SD	0.0379	0.0423	0.0431	**0.1356**	0.0608	0.0723		SD	0.0168	0.0184	**0.0151**	**0.1104**	**0.0259**	0.0365
	RMSE	**0.0390**	**0.0426**	**0.0715**	**0.1443**	**0.0635**	**0.1431**		RMSE	**0.0168**	**0.0184**	**0.0157**	**0.1107**	**0.0259**	**0.0385**
	95%CP	**0.9100**	**0.9570**	**0.6890**	**0.9300**	**0.9300**	**0.5810**		95%CP	0.9450	**0.9550**	**0.9430**	0.9570	**0.9470**	**0.9290**
	90%CP	**0.8510**	0.9180	**0.5790**	**0.8690**	**0.8680**	**0.4660**		90%CP	**0.8820**	**0.9050**	**0.8880**	0.9090	**0.8970**	**0.8710**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0122	0.0163	-0.1432	-0.1168	-0.0457	-0.2880	$N = 400$	BIAS	-0.0009	0.0099	-0.0338	-0.0119	-0.0040	-0.0673
	SD	0.0183	0.0212	0.0215	0.0649	0.0300	0.0303		SD	0.0080	0.0089	0.0078	0.0576	0.0127	0.0173
	RMSE	0.0220	0.0267	0.1448	0.1336	0.0546	0.2896		RMSE	0.0081	0.0133	0.0347	0.0588	0.0134	0.0695
	95%CP	0.8770	0.8910	0.0000	0.5660	0.6740	0.0000		95%CP	0.9480	0.8060	0.0040	0.9340	0.9430	0.0310
	90%CP	0.7870	0.8090	0.0000	0.4500	0.5510	0.0000		90%CP	0.8870	0.7150	0.0000	0.8850	0.9060	0.0150
		After bias correction								After bias correction					
	BIAS	**-0.0060**	**0.0040**	**-0.0514**	**-0.0425**	**-0.0158**	**-0.1098**		BIAS	**-0.0004**	**0.0007**	**-0.0039**	**-0.0005**	**-0.0006**	**-0.0088**
	SD	**0.0182**	0.0214	**0.0210**	**0.0637**	**0.0299**	0.0368		SD	0.0080	0.0090	0.0078	**0.0571**	0.0127	0.0182
	RMSE	**0.0192**	**0.0218**	**0.0555**	**0.0766**	**0.0338**	**0.1158**		RMSE	**0.0080**	**0.0091**	**0.0087**	**0.0571**	**0.0127**	**0.0202**
	95%CP	**0.9190**	**0.9440**	**0.2790**	**0.9180**	**0.9130**	**0.1440**		95%CP	**0.9500**	**0.9560**	**0.9110**	**0.9410**	**0.9450**	**0.9160**
	90%CP	**0.8530**	**0.8920**	**0.1880**	**0.8360**	**0.8570**	**0.0860**		90%CP	**0.8930**	**0.9020**	**0.8310**	**0.8930**	0.9100	**0.8490**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0118	0.0178	-0.1429	-0.1164	-0.0466	-0.2858	$N = 1600$	BIAS	-0.0009	0.0102	-0.0337	-0.0123	-0.0039	-0.0667
	SD	0.0092	0.0107	0.0105	0.0326	0.0150	0.0149		SD	0.0041	0.0044	0.0038	0.0288	0.0065	0.0087
	RMSE	0.0150	0.0207	0.1432	0.1208	0.0490	0.2862		RMSE	0.0042	0.0111	0.0339	0.0313	0.0076	0.0673
	95%CP	0.7110	0.6220	0.0000	0.0500	0.1350	0.0000		95%CP	0.9390	0.3790	0.0000	0.9180	0.9120	0.0000
	90%CP	0.6060	0.4890	0.0000	0.0260	0.0730	0.0000		90%CP	0.8900	0.2590	0.0000	0.8590	0.8340	0.0000
		After bias correction								After bias correction					
	BIAS	**-0.0056**	**0.0053**	**-0.0510**	**-0.0420**	**-0.0169**	**-0.1070**		BIAS	**-0.0004**	**0.0009**	**-0.0038**	**-0.0010**	**-0.0004**	**-0.0082**
	SD	0.0092	0.0107	0.0105	**0.0322**	**0.0150**	0.0180		SD	**0.0041**	0.0045	**0.0038**	**0.0286**	**0.0065**	0.0092
	RMSE	**0.0108**	**0.0120**	**0.0520**	**0.0529**	**0.0226**	**0.1085**		RMSE	**0.0041**	**0.0046**	**0.0054**	**0.0286**	**0.0065**	**0.0123**
	95%CP	**0.8710**	**0.9180**	*0.0000*	**0.7500**	**0.7820**	*0.0000*		95%CP	0.9350	**0.9530**	**0.8240**	**0.9370**	**0.9510**	**0.8410**
	90%CP	**0.7920**	**0.8570**	*0.0000*	**0.6400**	**0.6790**	*0.0000*		90%CP	**0.8970**	**0.9050**	**0.7230**	**0.8960**	**0.9000**	**0.7650**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.3: Results of OWFE estimator with Exponential error, UP=30%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0038	0.0320	-0.0747	-0.0403	-0.0153	-0.1619	$N = 100$	BIAS	-0.0005	0.0073	-0.0190	-0.0038	-0.0011	-0.0382
	SD	0.0266	0.0376	0.0308	0.1492	0.0426	0.0905		SD	0.0130	0.0177	0.0139	0.1431	0.0190	0.0532
	RMSE	0.0268	0.0494	0.0808	0.1545	0.0452	0.1854		RMSE	0.0130	0.0191	0.0236	0.1432	0.0190	0.0655
	95%CP	0.9260	0.8420	0.3000	0.9410	0.9340	0.6730		95%CP	0.9250	0.9310	0.7170	0.9540	0.9560	0.8910
	90%CP	0.8650	0.7670	0.2170	0.8830	0.8760	0.5290		90%CP	0.8680	0.8740	0.5860	0.9060	0.9060	0.8220
		After bias correction								After bias correction					
	BIAS	**-0.0027**	**0.0049**	**-0.0151**	**-0.0122**	**-0.0059**	**-0.0424**		BIAS	-0.0009	**-0.0002**	**-0.0010**	**-0.0005**	**-0.0002**	**-0.0035**
	SD	**0.0264**	0.0391	0.0321	**0.1489**	**0.0424**	0.1033		SD	0.0130	0.0179	0.0141	0.1432	0.0190	0.0551
	RMSE	**0.0266**	**0.0394**	**0.0355**	**0.1494**	**0.0429**	**0.1117**		RMSE	0.0131	**0.0179**	**0.0142**	0.1432	**0.0190**	**0.0552**
	95%CP	**0.9270**	**0.9340**	**0.8960**	**0.9450**	**0.9430**	**0.9430**		95%CP	**0.9260**	**0.9410**	**0.9520**	0.9550	**0.9530**	**0.9330**
	90%CP	**0.8690**	**0.8610**	**0.8310**	**0.8940**	**0.8920**	**0.8790**		90%CP	**0.8720**	**0.8830**	**0.8880**	0.9070	0.9090	**0.8830**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0029	0.0336	-0.0752	-0.0394	-0.0131	-0.1528	$N = 400$	BIAS	0.0004	0.0083	-0.0195	-0.0010	-0.0009	-0.0376
	SD	0.0140	0.0181	0.0155	0.0719	0.0211	0.0473		SD	0.0066	0.0085	0.0070	0.0677	0.0098	0.0263
	RMSE	0.0143	0.0381	0.0767	0.0820	0.0248	0.1599		RMSE	0.0066	0.0118	0.0207	0.0677	0.0098	0.0459
	95%CP	0.9210	0.5590	0.0010	0.9210	0.8920	0.1590		95%CP	0.9200	0.8640	0.2160	0.9570	0.9540	0.6930
	90%CP	0.8400	0.4330	0.0000	0.8630	0.8300	0.0810		90%CP	0.8610	0.7760	0.1420	0.9030	0.9110	0.5740
		After bias correction								After bias correction					
	BIAS	**-0.0020**	**0.0063**	**-0.0155**	**-0.0117**	**-0.0037**	**-0.0322**		BIAS	**0.0000**	**0.0007**	**-0.0014**	**0.0022**	**0.0000**	**-0.0029**
	SD	**0.0139**	0.0186	0.0160	**0.0713**	**0.0210**	0.0540		SD	**0.0066**	0.0086	0.0071	**0.0676**	**0.0097**	0.0272
	RMSE	**0.0140**	**0.0196**	**0.0223**	**0.0722**	**0.0213**	**0.0629**		RMSE	**0.0066**	**0.0086**	**0.0072**	**0.0677**	**0.0097**	**0.0274**
	95%CP	**0.9270**	**0.9350**	**0.8130**	**0.9410**	**0.9420**	**0.9050**		95%CP	**0.9250**	**0.9550**	**0.9440**	**0.9510**	**0.9530**	**0.9460**
	90%CP	**0.8470**	**0.8810**	**0.7050**	**0.9010**	**0.8940**	**0.8160**		90%CP	**0.8640**	**0.9170**	**0.8880**	*0.9030*	0.9130	**0.8860**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0020	0.0337	-0.0748	-0.0352	-0.0113	-0.1518	$N = 1600$	BIAS	0.0004	0.0078	-0.0191	-0.0032	-0.0009	-0.0373
	SD	0.0068	0.0090	0.0076	0.0363	0.0107	0.0237		SD	0.0033	0.0043	0.0034	0.0352	0.0047	0.0128
	RMSE	0.0071	0.0348	0.0752	0.0506	0.0156	0.1536		RMSE	0.0033	0.0090	0.0194	0.0353	0.0048	0.0394
	95%CP	0.9230	0.0490	0.0000	0.8310	0.7960	0.0000		95%CP	0.9250	0.5980	0.0000	0.9600	0.9540	0.1740
	90%CP	0.8640	0.0250	0.0000	0.7440	0.7040	0.0000		90%CP	0.8780	0.4690	0.0000	0.9080	0.8930	0.1060
		After bias correction								After bias correction					
	BIAS	**-0.0011**	**0.0061**	**-0.0153**	**-0.0079**	**-0.0021**	**-0.0312**		BIAS	**0.0000**	**0.0003**	**-0.0011**	**-0.0001**	**0.0000**	**-0.0026**
	SD	0.0068	0.0093	0.0079	**0.0361**	**0.0107**	0.0270		SD	0.0033	0.0044	0.0034	**0.0351**	**0.0047**	0.0132
	RMSE	**0.0069**	**0.0111**	**0.0172**	**0.0370**	**0.0109**	**0.0413**		RMSE	**0.0033**	**0.0044**	**0.0036**	**0.0351**	**0.0047**	**0.0135**
	95%CP	**0.9280**	**0.9060**	**0.4760**	**0.9540**	**0.9360**	**0.7800**		95%CP	**0.9270**	**0.9580**	**0.9440**	0.9610	0.9610	**0.9360**
	90%CP	**0.8750**	**0.8350**	**0.3540**	**0.9090**	**0.8730**	**0.6650**		90%CP	0.8720	**0.9270**	**0.8920**	**0.9070**	**0.9050**	**0.8810**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.4: Results of OWFE estimator with Exponential error, UP=60%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0083	0.0575	-0.1318	-0.0966	-0.0382	-0.2907	$N = 100$	BIAS	-0.0008	0.0140	-0.0311	-0.0062	-0.0023	-0.0669
	SD	0.0311	0.0470	0.0463	0.1326	0.0596	0.1097		SD	0.0138	0.0203	0.0184	0.1096	0.0261	0.0683
	RMSE	0.0322	0.0743	0.1397	0.1640	0.0708	0.3107		RMSE	0.0138	0.0247	0.0361	0.1097	0.0262	0.0956
	95%CP	0.9280	0.7690	0.1570	0.8720	0.9100	0.4080		95%CP	0.9380	0.9040	0.6420	0.9550	0.9430	0.8540
	90%CP	0.8730	0.6840	0.0980	0.7960	0.8340	0.2730		90%CP	0.8800	0.8150	0.4950	0.9180	0.9010	0.7510
		After bias correction								After bias correction					
	BIAS	**-0.0053**	**0.0180**	**-0.0440**	**-0.0379**	**-0.0154**	**-0.1133**		BIAS	-0.0012	**0.0006**	**-0.0022**	**0.0013**	**0.0000**	**-0.0080**
	SD	**0.0309**	0.0484	0.0472	**0.1294**	**0.0594**	0.1357		SD	**0.0138**	0.0206	0.0186	**0.1090**	0.0262	0.0723
	RMSE	**0.0313**	**0.0516**	**0.0645**	**0.1348**	**0.0613**	**0.1768**		RMSE	0.0138	**0.0206**	**0.0187**	**0.1090**	**0.0262**	**0.0728**
	95%CP	**0.9330**	**0.9310**	**0.8100**	**0.9290**	**0.9440**	**0.8900**		95%CP	0.9360	**0.9520**	**0.9510**	0.9600	**0.9480**	**0.9450**
	90%CP	**0.8870**	**0.8750**	**0.7220**	**0.8870**	**0.8950**	**0.7940**		90%CP	*0.8800*	**0.9050**	**0.8880**	**0.9160**	0.9030	**0.8860**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0062	0.0591	-0.1309	-0.0848	-0.0333	-0.2760	$N = 400$	BIAS	0.0003	0.0151	-0.0322	-0.0072	-0.0030	-0.0670
	SD	0.0159	0.0240	0.0236	0.0645	0.0300	0.0542		SD	0.0071	0.0102	0.0091	0.0562	0.0131	0.0339
	RMSE	0.0171	0.0638	0.1330	0.1065	0.0448	0.2813		RMSE	0.0071	0.0182	0.0334	0.0566	0.0134	0.0751
	95%CP	0.9170	0.2860	0.0000	0.7260	0.7920	0.0080		95%CP	0.9360	0.7050	0.0660	0.9440	0.9530	0.5290
	90%CP	0.8560	0.1950	0.0000	0.6160	0.6920	0.0030		90%CP	0.8790	0.5850	0.0290	0.9030	0.8910	0.3920
		After bias correction								After bias correction					
	BIAS	**-0.0034**	**0.0187**	**-0.0429**	**-0.0257**	**-0.0111**	**-0.0952**		BIAS	**-0.0002**	**0.0015**	**-0.0033**	**0.0001**	**-0.0007**	**-0.0081**
	SD	**0.0159**	0.0250	0.0245	**0.0635**	**0.0299**	0.0672		SD	0.0071	0.0104	0.0093	**0.0559**	**0.0131**	0.0359
	RMSE	**0.0162**	**0.0312**	**0.0494**	**0.0685**	**0.0319**	**0.1165**		RMSE	0.0071	**0.0105**	**0.0098**	**0.0559**	**0.0131**	**0.0368**
	95%CP	**0.9260**	**0.8750**	**0.5030**	**0.9340**	**0.9350**	**0.7400**		95%CP	**0.9380**	**0.9490**	**0.9420**	*0.9440*	0.9540	**0.9350**
	90%CP	**0.8710**	**0.7710**	**0.3890**	**0.8700**	**0.8720**	**0.6160**		90%CP	**0.8890**	**0.8950**	**0.8850**	0.9060	**0.9040**	**0.8780**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1,600$	BIAS	-0.0054	0.0583	-0.1300	-0.0870	-0.0326	-0.2746	$N = 1600$	BIAS	0.0002	0.0148	-0.0318	-0.0085	-0.0024	-0.0661
	SD	0.0076	0.0119	0.0113	0.0324	0.0149	0.0274		SD	0.0037	0.0050	0.0045	0.0271	0.0065	0.0164
	RMSE	0.0093	0.0595	0.1305	0.0928	0.0358	0.2759		RMSE	0.0037	0.0157	0.0321	0.0284	0.0070	0.0681
	95%CP	0.8840	0.0010	0.0000	0.2270	0.3970	0.0000		95%CP	0.9290	0.1800	0.0000	0.9400	0.9310	0.0270
	90%CP	0.8140	0.0000	0.0000	0.1420	0.2790	0.0000		90%CP	0.8760	0.1000	0.0000	0.8980	0.8800	0.0110
		After bias correction								After bias correction					
	BIAS	**-0.0026**	**0.0177**	**-0.0422**	**-0.0288**	**-0.0106**	**-0.0936**		BIAS	**-0.0002**	**0.0013**	**-0.0030**	**-0.0013**	**-0.0001**	**-0.0072**
	SD	0.0076	0.0122	0.0116	**0.0321**	0.0150	0.0339		SD	**0.0036**	0.0050	0.0046	**0.0271**	**0.0065**	0.0174
	RMSE	**0.0081**	**0.0215**	**0.0438**	**0.0431**	**0.0183**	**0.0996**		RMSE	**0.0037**	**0.0052**	**0.0055**	**0.0271**	**0.0065**	**0.0188**
	95%CP	**0.9290**	**0.6820**	**0.0360**	**0.8560**	**0.8980**	**0.2240**		95%CP	**0.9320**	**0.9450**	**0.8960**	**0.9500**	**0.9530**	**0.9340**
	90%CP	**0.8730**	**0.5700**	**0.0150**	**0.7670**	**0.8110**	**0.1390**		90%CP	0.8750	**0.8910**	**0.8320**	0.9090	**0.8990**	**0.8760**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.5: Results of OWFE estimator with Laplace error, UP=30%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0041	0.0333	-0.0760	-0.0364	-0.0100	-0.1630	$N = 100$	BIAS	-0.0001	0.0077	-0.0192	-0.0055	-0.0017	-0.0377
	SD	0.0281	0.0351	0.0303	0.1491	0.0418	0.0760		SD	0.0125	0.0171	0.0135	0.1450	0.0199	0.0410
	RMSE	0.0284	0.0484	0.0818	0.1535	0.0430	0.1798		RMSE	0.0125	0.0187	0.0235	0.1451	0.0200	0.0557
	95%CP	0.8990	0.8660	0.2880	0.9390	0.9530	0.4900		95%CP	0.9430	0.9350	0.7310	0.9480	0.9450	0.8510
	90%CP	0.8270	0.7890	0.1950	0.8860	0.8850	0.3650		90%CP	0.8860	0.8850	0.6040	0.8900	0.8910	0.7740
		After bias correction								After bias correction					
	BIAS	**-0.0030**	**0.0066**	**-0.0167**	**-0.0081**	**-0.0008**	**-0.0439**		BIAS	-0.0004	**0.0001**	**-0.0011**	**-0.0025**	**-0.0008**	**-0.0030**
	SD	**0.0280**	0.0363	0.0316	0.1501	**0.0417**	0.0865		SD	0.0125	0.0173	0.0137	0.1451	0.0199	0.0425
	RMSE	**0.0282**	**0.0369**	**0.0358**	**0.1503**	**0.0417**	**0.0970**		RMSE	0.0125	**0.0173**	**0.0137**	0.1452	**0.0199**	**0.0426**
	95%CP	*0.8990*	**0.9490**	**0.9060**	**0.9480**	0.9560	**0.9120**		95%CP	0.9420	**0.9530**	**0.9540**	0.9470	**0.9480**	**0.9500**
	90%CP	**0.8280**	**0.8980**	**0.8430**	**0.8950**	**0.9060**	**0.8380**		90%CP	0.8850	**0.9120**	**0.9020**	0.8850	*0.8910*	**0.8960**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0026	0.0335	-0.0747	-0.0364	-0.0118	-0.1537	$N = 400$	BIAS	0.0002	0.0077	-0.0193	-0.0023	-0.0010	-0.0372
	SD	0.0132	0.0184	0.0147	0.0750	0.0209	0.0372		SD	0.0068	0.0087	0.0071	0.0710	0.0100	0.0209
	RMSE	0.0134	0.0382	0.0762	0.0834	0.0240	0.1582		RMSE	0.0068	0.0116	0.0206	0.0711	0.0100	0.0427
	95%CP	0.9200	0.5530	0.0020	0.9060	0.9020	0.0330		95%CP	0.9360	0.8670	0.2160	0.9540	0.9250	0.5730
	90%CP	0.8820	0.4330	0.0000	0.8480	0.8500	0.0160		90%CP	0.8600	0.7950	0.1240	0.8980	0.8900	0.4410
		After bias correction								After bias correction					
	BIAS	**-0.0016**	**0.0062**	**-0.0152**	**-0.0086**	**-0.0026**	**-0.0334**		BIAS	**-0.0002**	**0.0001**	**-0.0013**	**0.0008**	**-0.0001**	**-0.0025**
	SD	0.0132	0.0190	0.0153	0.0753	**0.0209**	0.0423		SD	**0.0068**	0.0088	0.0071	0.0711	0.0100	0.0217
	RMSE	**0.0133**	**0.0200**	**0.0216**	**0.0758**	**0.0210**	**0.0539**		RMSE	**0.0068**	**0.0088**	**0.0072**	0.0711	**0.0100**	**0.0218**
	95%CP	**0.9250**	**0.9350**	**0.8230**	**0.9400**	**0.9580**	**0.8690**		95%CP	0.9320	**0.9550**	**0.9370**	**0.9530**	*0.9250*	**0.9490**
	90%CP	0.8800	**0.8810**	**0.7190**	**0.8910**	**0.8910**	**0.7810**		90%CP	0.8550	**0.9020**	**0.8820**	0.8950	**0.8930**	**0.8980**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0019	0.0339	-0.0743	-0.0345	-0.0114	-0.1526	$N = 1600$	BIAS	0.0003	0.0078	-0.0191	-0.0041	-0.0009	-0.0369
	SD	0.0068	0.0091	0.0072	0.0369	0.0106	0.0183		SD	0.0034	0.0046	0.0034	0.0355	0.0048	0.0101
	RMSE	0.0070	0.0351	0.0747	0.0505	0.0155	0.1537		RMSE	0.0034	0.0090	0.0194	0.0358	0.0049	0.0383
	95%CP	0.9120	0.0490	0.0000	0.8360	0.8010	0.0000		95%CP	0.9290	0.6110	0.0010	0.9480	0.9440	0.0480
	90%CP	0.8540	0.0270	0.0000	0.7410	0.6850	0.0000		90%CP	0.8750	0.4900	0.0000	0.9050	0.9000	0.0250
		After bias correction								After bias correction					
	BIAS	**-0.0010**	**0.0063**	**-0.0148**	**-0.0070**	**-0.0021**	**-0.0322**		BIAS	**-0.0001**	**0.0002**	**-0.0010**	**-0.0011**	**0.0000**	**-0.0022**
	SD	**0.0067**	0.0093	0.0075	**0.0368**	**0.0105**	0.0208		SD	0.0034	0.0047	0.0035	**0.0355**	**0.0048**	0.0105
	RMSE	**0.0068**	**0.0113**	**0.0166**	**0.0375**	**0.0107**	**0.0383**		RMSE	**0.0034**	**0.0047**	**0.0036**	**0.0355**	**0.0048**	**0.0107**
	95%CP	**0.9260**	**0.9040**	**0.4930**	**0.9510**	**0.9400**	**0.6610**		95%CP	**0.9300**	**0.9480**	**0.9480**	**0.9500**	**0.9480**	**0.9500**
	90%CP	**0.8700**	**0.8250**	**0.3610**	**0.8890**	**0.8900**	**0.5450**		90%CP	**0.8770**	**0.9020**	**0.8820**	0.9120	0.9010	**0.8960**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.6: Results of OWFE estimator with Laplace error, UP=60%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0079	0.0594	-0.1331	-0.0947	-0.0325	-0.2905	$N = 100$	BIAS	-0.0007	0.0144	-0.0319	-0.0030	-0.0035	-0.0690
	SD	0.0292	0.0468	0.0453	0.1316	0.0599	0.0870		SD	0.0132	0.0204	0.0188	0.1144	0.0261	0.0516
	RMSE	0.0303	0.0756	0.1406	0.1622	0.0682	0.3033		RMSE	0.0132	0.0249	0.0370	0.1145	0.0263	0.0862
	95%CP	0.9270	0.7620	0.1490	0.8790	0.9240	0.1930		95%CP	0.9500	0.8980	0.6120	0.9450	0.9490	0.7830
	90%CP	0.8580	0.6660	0.0790	0.7980	0.8570	0.1180		90%CP	0.9070	0.8320	0.4940	0.8830	0.9000	0.6710
		After bias correction								After bias correction					
	BIAS	**-0.0048**	**0.0204**	**-0.0461**	**-0.0357**	**-0.0103**	**-0.1135**		BIAS	-0.0011	**0.0009**	**-0.0028**	0.0043	**-0.0011**	**-0.0099**
	SD	**0.0290**	0.0482	0.0462	**0.1306**	**0.0597**	0.1073		SD	0.0133	0.0207	0.0191	0.1147	0.0260	0.0546
	RMSE	**0.0294**	**0.0523**	**0.0653**	**0.1354**	**0.0606**	**0.1562**		RMSE	0.0133	**0.0207**	**0.0193**	0.1148	**0.0260**	**0.0555**
	95%CP	**0.9330**	**0.9230**	**0.8200**	**0.9510**	**0.9510**	**0.8370**		95%CP	*0.9500*	**0.9530**	**0.9540**	0.9440	0.9480	**0.9510**
	90%CP	**0.8730**	**0.8630**	**0.7160**	**0.8820**	**0.8990**	**0.7240**		90%CP	0.9090	**0.9050**	**0.8880**	*0.8830*	0.9060	**0.9070**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0060	0.0587	-0.1305	-0.0871	-0.0319	-0.2762	$N = 400$	BIAS	0.0004	0.0148	-0.0323	-0.0063	-0.0031	-0.0663
	SD	0.0153	0.0227	0.0221	0.0656	0.0304	0.0456		SD	0.0071	0.0106	0.0095	0.0555	0.0129	0.0262
	RMSE	0.0164	0.0630	0.1323	0.1090	0.0441	0.2799		RMSE	0.0071	0.0182	0.0337	0.0558	0.0132	0.0713
	95%CP	0.8980	0.2980	0.0000	0.7270	0.7980	0.0010		95%CP	0.9360	0.7120	0.0730	0.9540	0.9550	0.3250
	90%CP	0.8340	0.1980	0.0000	0.6070	0.6950	0.0000		90%CP	0.8860	0.6100	0.0430	0.9060	0.8870	0.2190
		After bias correction								After bias correction					
	BIAS	**-0.0031**	**0.0184**	**-0.0427**	**-0.0288**	**-0.0099**	**-0.0956**		BIAS	**0.0000**	**0.0012**	**-0.0034**	**0.0010**	**-0.0008**	**-0.0074**
	SD	**0.0153**	0.0233	0.0229	**0.0644**	**0.0302**	0.0562		SD	0.0071	0.0108	0.0096	0.0555	**0.0129**	0.0277
	RMSE	**0.0156**	**0.0297**	**0.0485**	**0.0705**	**0.0317**	**0.1108**		RMSE	0.0071	**0.0109**	**0.0102**	**0.0555**	**0.0129**	**0.0287**
	95%CP	**0.9150**	**0.8840**	**0.4980**	**0.9380**	**0.9370**	**0.5800**		95%CP	**0.9370**	**0.9420**	**0.9360**	0.9580	**0.9490**	**0.9530**
	90%CP	**0.8590**	**0.8050**	**0.3850**	**0.8820**	**0.8880**	**0.4600**		90%CP	**0.8880**	**0.8930**	**0.8660**	0.9070	**0.9060**	**0.8790**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0054	0.0594	-0.1309	-0.0877	-0.0330	-0.2755	$N = 1600$	BIAS	0.0003	0.0149	-0.0321	-0.0078	-0.0023	-0.0649
	SD	0.0076	0.0119	0.0114	0.0327	0.0151	0.0223		SD	0.0035	0.0049	0.0046	0.0282	0.0066	0.0136
	RMSE	0.0093	0.0605	0.1314	0.0936	0.0363	0.2764		RMSE	0.0036	0.0157	0.0324	0.0292	0.0070	0.0663
	95%CP	0.8510	0.0010	0.0000	0.2320	0.4110	0.0000		95%CP	0.9340	0.1610	0.0000	0.9440	0.9300	0.0020
	90%CP	0.7530	0.0010	0.0000	0.1480	0.2850	0.0000		90%CP	0.8780	0.1020	0.0000	0.8870	0.8800	0.0010
		After bias correction								After bias correction					
	BIAS	**-0.0025**	**0.0188**	**-0.0432**	**-0.0297**	**-0.0110**	**-0.0949**		BIAS	**-0.0001**	**0.0013**	**-0.0032**	**-0.0006**	**0.0000**	**-0.0060**
	SD	**0.0076**	0.0123	0.0119	0.0328	**0.0150**	0.0276		SD	**0.0035**	0.0050	0.0047	**0.0280**	**0.0066**	0.0145
	RMSE	**0.0080**	**0.0224**	**0.0448**	**0.0442**	**0.0186**	**0.0988**		RMSE	**0.0035**	**0.0051**	**0.0057**	**0.0280**	**0.0066**	**0.0156**
	95%CP	**0.9100**	**0.6510**	**0.0340**	**0.8390**	**0.8870**	**0.0780**		95%CP	**0.9380**	**0.9470**	**0.8870**	**0.9500**	**0.9480**	**0.9170**
	90%CP	**0.8310**	**0.5410**	**0.0210**	**0.7520**	**0.7910**	**0.0420**		90%CP	**0.8820**	**0.9140**	**0.8220**	**0.9040**	**0.8970**	**0.8540**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.7: Results of TWFE estimator with Normal error, UP=30%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0139	0.0383	-0.0739	-0.0365	-0.0094	-0.1661	$N = 100$	BIAS	-0.0096	0.0119	-0.0181	-0.0025	0.0008	-0.0489
	SD	0.0251	0.0349	0.0287	0.1644	0.0420	0.0501		SD	0.0125	0.0178	0.0144	0.1429	0.0198	0.0268
	RMSE	0.0287	0.0519	0.0793	0.1684	0.0431	0.1735		RMSE	0.0158	0.0214	0.0231	0.1429	0.0198	0.0557
	95%CP	0.8920	0.8090	0.2940	0.9380	0.9380	0.1150		95%CP	0.8540	0.8910	0.7370	0.9480	0.9490	0.5740
	90%CP	0.8290	0.7130	0.1910	0.8870	0.8990	0.0570		90%CP	0.7710	0.8200	0.6280	0.9020	0.8890	0.4390
		After bias correction								After bias correction					
	BIAS	**-0.0042**	**0.0083**	**-0.0166**	**-0.0096**	**-0.0020**	**-0.0399**		BIAS	**-0.0006**	**0.0004**	**-0.0011**	**-0.0016**	**-0.0003**	**-0.0053**
	SD	**0.0249**	0.0360	0.0295	**0.1607**	0.0421	0.0567		SD	**0.0124**	0.0179	0.0146	**0.1426**	**0.0197**	0.0277
	RMSE	**0.0253**	**0.0369**	**0.0339**	**0.1610**	**0.0422**	**0.0693**		RMSE	**0.0125**	**0.0179**	**0.0146**	**0.1426**	**0.0197**	**0.0282**
	95%CP	**0.9340**	**0.9400**	**0.9150**	**0.9500**	**0.9420**	**0.8840**		95%CP	**0.9320**	**0.9490**	**0.9450**	0.9470	*0.9510*	**0.9460**
	90%CP	**0.8760**	**0.9000**	**0.8550**	**0.9020**	0.8930	**0.8010**		90%CP	**0.8780**	**0.9000**	**0.8890**	**0.9000**	**0.8940**	**0.8940**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0052	0.0354	-0.0732	-0.0398	-0.0111	-0.1553	$N = 400$	BIAS	-0.0026	0.0092	-0.0187	-0.0008	-0.0004	-0.0397
	SD	0.0128	0.0178	0.0146	0.0877	0.0217	0.0253		SD	0.0061	0.0088	0.0070	0.0729	0.0096	0.0128
	RMSE	0.0138	0.0397	0.0746	0.0964	0.0244	0.1574		RMSE	0.0066	0.0128	0.0200	0.0729	0.0096	0.0417
	95%CP	0.9160	0.5130	0.0010	0.8960	0.9070	0.0000		95%CP	0.9270	0.8330	0.2370	0.9390	0.9480	0.1610
	90%CP	0.8530	0.3740	0.0000	0.8400	0.8490	0.0000		90%CP	0.8710	0.7290	0.1560	0.8870	0.9090	0.0880
		After bias correction								After bias correction					
	BIAS	**-0.0015**	**0.0059**	**-0.0148**	**-0.0095**	**-0.0027**	**-0.0336**		BIAS	**-0.0003**	**0.0003**	**-0.0010**	0.0018	**-0.0000**	**-0.0029**
	SD	**0.0127**	0.0184	0.0152	**0.0786**	0.0218	0.0286		SD	*0.0061*	0.0089	0.0071	**0.0726**	*0.0096*	0.0133
	RMSE	**0.0128**	**0.0193**	**0.0213**	**0.0791**	**0.0219**	**0.0441**		RMSE	**0.0061**	**0.0089**	**0.0072**	**0.0727**	*0.0096*	**0.0136**
	95%CP	**0.9360**	**0.9340**	**0.8250**	**0.9550**	**0.9310**	**0.7500**		95%CP	**0.9400**	**0.9500**	**0.9390**	0.9380	*0.9480*	**0.9510**
	90%CP	**0.8830**	**0.8730**	**0.7380**	**0.9000**	**0.8780**	**0.6320**		90%CP	**0.8910**	**0.8870**	**0.8960**	0.8860	0.9140	**0.9010**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0032	0.0365	-0.0739	-0.0393	-0.0112	-0.1527	$N = 1600$	BIAS	-0.0005	0.0086	-0.0189	-0.0031	-0.0007	-0.0372
	SD	0.0067	0.0093	0.0077	0.0602	0.0105	0.0130		SD	0.0032	0.0045	0.0035	0.0366	0.0049	0.0066
	RMSE	0.0074	0.0377	0.0743	0.0719	0.0153	0.1533		RMSE	0.0032	0.0097	0.0192	0.0368	0.0049	0.0378
	95%CP	0.8860	0.0250	0.0000	0.7060	0.8240	0.0000		95%CP	0.9390	0.5180	0.0000	0.9450	0.9470	0.0000
	90%CP	0.8350	0.0130	0.0000	0.6210	0.7090	0.0000		90%CP	0.8850	0.3990	0.0000	0.9010	0.8950	0.0000
		After bias correction								After bias correction					
	BIAS	**-0.0011**	**0.0071**	**-0.0154**	**-0.0073**	**-0.0024**	**-0.0320**		BIAS	**-0.0001**	**0.0003**	**-0.0010**	**0.0001**	**-0.0000**	**-0.0021**
	SD	*0.0067*	0.0094	0.0078	**0.0421**	*0.0105*	0.0148		SD	*0.0032*	0.0046	*0.0035*	**0.0356**	*0.0049*	0.0068
	RMSE	**0.0068**	**0.0118**	**0.0172**	**0.0427**	**0.0108**	**0.0353**		RMSE	*0.0032*	**0.0046**	**0.0037**	**0.0356**	*0.0049*	**0.0071**
	95%CP	**0.9280**	**0.8770**	**0.4460**	**0.9410**	**0.9350**	**0.3460**		95%CP	**0.9430**	**0.9480**	**0.9430**	**0.9520**	**0.9490**	**0.9390**
	90%CP	**0.8560**	**0.7960**	**0.3460**	**0.8760**	**0.8870**	**0.2510**		90%CP	**0.8870**	**0.8890**	**0.8880**	0.9050	0.8940	**0.8850**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.8: Results of TWFE estimator with Normal error, UP=60%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0171	0.0619	-0.1294	-0.1117	-0.0287	-0.3051	$N = 100$	BIAS	-0.0083	0.0197	-0.0318	0.0002	-0.0005	-0.0887
	SD	0.0297	0.0434	0.0432	0.1790	0.0623	0.0601		SD	0.0133	0.0196	0.0190	0.1203	0.0264	0.0348
	RMSE	0.0342	0.0756	0.1364	0.2110	0.0686	0.3110		RMSE	0.0157	0.0278	0.0370	0.1203	0.0264	0.0953
	95%CP	0.8630	0.7280	0.1530	0.8780	0.9210	0.0030		95%CP	0.8750	0.8350	0.5840	0.9440	0.9480	0.2980
	90%CP	0.7890	0.6200	0.0800	0.7950	0.8550	0.0010		90%CP	0.7990	0.7290	0.4760	0.8940	0.8990	0.2030
		After bias correction								After bias correction					
	BIAS	**-0.0079**	**0.0201**	**-0.0450**	**-0.0397**	**-0.0083**	**-0.1182**		BIAS	**-0.0011**	**0.0026**	**-0.0046**	0.0064	**-0.0004**	**-0.0128**
	SD	**0.0293**	0.0448	0.0442	**0.1702**	**0.0620**	0.0739		SD	**0.0132**	0.0198	0.0193	**0.1199**	*0.0263*	0.0371
	RMSE	**0.0303**	**0.0491**	**0.0631**	**0.1748**	**0.0626**	**0.1394**		RMSE	**0.0132**	**0.0200**	**0.0198**	**0.1200**	*0.0263*	**0.0392**
	95%CP	**0.9060**	**0.9390**	**0.8280**	**0.9390**	**0.9520**	**0.6070**		95%CP	**0.9390**	**0.9540**	**0.9310**	**0.9460**	0.9470	**0.9400**
	90%CP	**0.8500**	**0.8860**	**0.7150**	**0.8750**	**0.8960**	**0.4850**		90%CP	**0.8710**	**0.9020**	**0.8780**	*0.8940*	0.8980	**0.8810**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0085	0.0632	-0.1292	-0.1090	-0.0321	-0.2815	$N = 400$	BIAS	-0.0020	0.0165	-0.0312	-0.0115	-0.0023	-0.0714
	SD	0.0142	0.0213	0.0216	0.0934	0.0300	0.0305		SD	0.0064	0.0096	0.0090	0.0583	0.0134	0.0181
	RMSE	0.0165	0.0667	0.1310	0.1436	0.0439	0.2831		RMSE	0.0068	0.0191	0.0325	0.0594	0.0136	0.0737
	95%CP	0.8720	0.1930	0.0000	0.7110	0.8000	0.0000		95%CP	0.9280	0.5950	0.0690	0.9460	0.9440	0.0240
	90%CP	0.7880	0.1060	0.0000	0.5900	0.7000	0.0000		90%CP	0.8700	0.4860	0.0340	0.8880	0.8860	0.0150
		After bias correction								After bias correction					
	BIAS	**-0.0036**	**0.0206**	**-0.0435**	**-0.0364**	**-0.0112**	**-0.0991**		BIAS	**-0.0002**	**0.0015**	**-0.0033**	**-0.0036**	**-0.0007**	**-0.0084**
	SD	*0.0141*	0.0219	0.0222	**0.0797**	0.0301	0.0376		SD	*0.0064*	0.0097	0.0090	**0.0574**	*0.0134*	0.0191
	RMSE	**0.0146**	**0.0301**	**0.0488**	**0.0876**	**0.0321**	**0.1060**		RMSE	**0.0064**	**0.0098**	**0.0096**	**0.0575**	**0.0134**	**0.0209**
	95%CP	**0.9110**	**0.8500**	**0.4810**	**0.9270**	**0.9390**	**0.2170**		95%CP	**0.9430**	**0.9570**	**0.9400**	**0.9530**	**0.9500**	**0.9070**
	90%CP	**0.8420**	**0.7690**	**0.3540**	**0.8670**	**0.8690**	**0.1570**		90%CP	**0.8890**	**0.9110**	**0.8890**	**0.8980**	**0.8890**	**0.8410**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0063	0.0623	-0.1272	-0.1058	-0.0306	-0.2750	$N = 1600$	BIAS	-0.0007	0.0158	-0.0309	-0.0097	-0.0019	-0.0662
	SD	0.0071	0.0112	0.0113	0.0647	0.0147	0.0153		SD	0.0030	0.0048	0.0046	0.0312	0.0067	0.0086
	RMSE	0.0095	0.0633	0.1277	0.1239	0.0340	0.2755		RMSE	0.0031	0.0166	0.0312	0.0326	0.0070	0.0667
	95%CP	0.8110	0.0000	0.0000	0.3100	0.4330	0.0000		95%CP	0.9500	0.1110	0.0000	0.9260	0.9300	0.0000
	90%CP	0.7040	0.0000	0.0000	0.2550	0.3240	0.0000		90%CP	0.9000	0.0530	0.0000	0.8600	0.8740	0.0000
		After bias correction								After bias correction					
	BIAS	**-0.0025**	**0.0195**	**-0.0413**	**-0.0342**	**-0.0096**	**-0.0937**		BIAS	**-0.0003**	**0.0014**	**-0.0028**	**-0.0014**	**-0.0000**	**-0.0064**
	SD	**0.0070**	0.0114	**0.0112**	**0.0437**	*0.0147*	0.0188		SD	*0.0030*	0.0049	0.0046	**0.0300**	*0.0067*	0.0090
	RMSE	**0.0074**	**0.0226**	**0.0428**	**0.0555**	**0.0176**	**0.0956**		RMSE	**0.0030**	**0.0051**	**0.0054**	**0.0300**	**0.0067**	**0.0111**
	95%CP	**0.9320**	**0.5750**	**0.0330**	**0.8400**	**0.9110**	**0.0010**		95%CP	0.9550	**0.9430**	**0.9150**	**0.9480**	**0.9410**	**0.8800**
	90%CP	**0.8520**	**0.4570**	**0.0130**	**0.7420**	**0.8380**	**0.0010**		90%CP	0.9110	**0.8940**	**0.8370**	**0.8770**	**0.8830**	**0.8140**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.9: Results of TWFE estimator with Exponential error, UP=30%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0131	0.0371	-0.0746	-0.0490	-0.0089	-0.1709	$N = 100$	BIAS	-0.0094	0.0116	-0.0175	-0.0030	0.0011	-0.0490
	SD	0.0265	0.0356	0.0302	0.1673	0.0425	0.0934		SD	0.0118	0.0176	0.0139	0.1443	0.0193	0.0511
	RMSE	0.0295	0.0514	0.0805	0.1743	0.0434	0.1948		RMSE	0.0151	0.0210	0.0223	0.1443	0.0193	0.0708
	95%CP	0.8970	0.8110	0.3000	0.9330	0.9330	0.6450		95%CP	0.8830	0.8960	0.7650	0.9520	0.9550	0.8710
	90%CP	0.8150	0.7270	0.2030	0.8670	0.8880	0.4900		90%CP	0.7890	0.8230	0.6530	0.9100	0.9040	0.7600
		After bias correction								After bias correction					
	BIAS	**-0.0035**	**0.0071**	**-0.0174**	**-0.0211**	**-0.0016**	**-0.0453**		BIAS	**-0.0005**	**0.0000**	**-0.0005**	**-0.0018**	**-0.0000**	**-0.0054**
	SD	**0.0263**	0.0365	0.0312	**0.1622**	**0.0423**	0.1068		SD	*0.0118*	0.0177	0.0141	**0.1434**	*0.0192*	0.0533
	RMSE	**0.0265**	**0.0372**	**0.0357**	**0.1636**	**0.0423**	**0.1160**		RMSE	**0.0118**	**0.0177**	**0.0141**	**0.1434**	**0.0192**	**0.0536**
	95%CP	**0.9280**	**0.9460**	**0.9020**	**0.9480**	**0.9510**	**0.9340**		95%CP	**0.9490**	**0.9490**	**0.9560**	**0.9510**	*0.9550*	**0.9600**
	90%CP	**0.8690**	**0.8880**	**0.8350**	**0.8910**	**0.9020**	**0.8670**		90%CP	**0.9090**	**0.8940**	**0.9070**	0.9120	0.9080	**0.9060**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0059	0.0358	-0.0728	-0.0362	-0.0098	-0.1526	$N = 400$	BIAS	-0.0026	0.0095	-0.0191	-0.0051	0.0001	-0.0397
	SD	0.0132	0.0176	0.0148	0.0891	0.0212	0.0464		SD	0.0062	0.0090	0.0068	0.0748	0.0098	0.0264
	RMSE	0.0144	0.0399	0.0743	0.0961	0.0233	0.1595		RMSE	0.0067	0.0131	0.0203	0.0749	0.0098	0.0476
	95%CP	0.8900	0.4870	0.0010	0.9010	0.9130	0.1390		95%CP	0.9330	0.8150	0.2080	0.9380	0.9500	0.6950
	90%CP	0.8300	0.3570	0.0010	0.8360	0.8620	0.0800		90%CP	0.8590	0.7000	0.1170	0.8870	0.9040	0.5620
		After bias correction								After bias correction					
	BIAS	**-0.0021**	**0.0063**	**-0.0144**	**-0.0076**	**-0.0014**	**-0.0304**		BIAS	**-0.0004**	**0.0006**	**-0.0014**	**-0.0023**	0.0004	**-0.0028**
	SD	*0.0132*	0.0181	0.0152	**0.0802**	**0.0211**	0.0529		SD	*0.0062*	0.0091	0.0070	**0.0744**	*0.0098*	0.0274
	RMSE	**0.0133**	**0.0192**	**0.0210**	**0.0806**	**0.0212**	**0.0610**		RMSE	**0.0062**	**0.0091**	**0.0071**	**0.0744**	*0.0098*	**0.0275**
	95%CP	**0.9270**	**0.9360**	**0.8260**	**0.9470**	**0.9440**	**0.9160**		95%CP	**0.9470**	**0.9500**	**0.9390**	**0.9410**	0.9480	**0.9460**
	90%CP	**0.8590**	**0.8790**	**0.7320**	**0.8940**	**0.8900**	**0.8400**		90%CP	**0.9020**	**0.8910**	**0.8870**	*0.8870*	**0.9030**	**0.8960**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0038	0.0364	-0.0735	-0.0372	-0.0107	-0.1517	$N = 1600$	BIAS	-0.0006	0.0088	-0.0190	-0.0047	-0.0004	-0.0371
	SD	0.0068	0.0090	0.0076	0.0564	0.0106	0.0242		SD	0.0032	0.0044	0.0035	0.0376	0.0048	0.0132
	RMSE	0.0077	0.0375	0.0739	0.0676	0.0150	0.1537		RMSE	0.0033	0.0098	0.0194	0.0379	0.0048	0.0394
	95%CP	0.8770	0.0190	0.0000	0.7450	0.8200	0.0000		95%CP	0.9310	0.4990	0.0010	0.9380	0.9560	0.2090
	90%CP	0.8130	0.0090	0.0000	0.6570	0.7230	0.0000		90%CP	0.8810	0.3910	0.0010	0.8810	0.8940	0.1260
		After bias correction								After bias correction					
	BIAS	**-0.0016**	**0.0070**	**-0.0150**	**-0.0080**	**-0.0019**	**-0.0309**		BIAS	**-0.0002**	**0.0005**	**-0.0012**	**-0.0010**	**0.0003**	**-0.0020**
	SD	*0.0068*	*0.0090*	0.0077	**0.0406**	**0.0106**	0.0276		SD	*0.0032*	0.0044	*0.0035*	**0.0363**	*0.0048*	0.0136
	RMSE	**0.0070**	**0.0114**	**0.0169**	**0.0413**	**0.0107**	**0.0414**		RMSE	**0.0032**	**0.0045**	**0.0037**	**0.0363**	*0.0048*	**0.0138**
	95%CP	**0.9220**	**0.8880**	**0.4710**	**0.9440**	**0.9390**	**0.7630**		95%CP	*0.9310*	**0.9530**	**0.9330**	**0.9490**	*0.9560*	**0.9440**
	90%CP	**0.8560**	**0.8030**	**0.3730**	**0.8950**	**0.8850**	**0.6720**		90%CP	**0.8950**	**0.9060**	**0.8740**	**0.8950**	**0.8960**	**0.8820**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.10: Results of TWFE estimator with Exponential error, UP=60%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0152	0.0623	-0.1315	-0.1107	-0.0325	-0.3065	$N = 100$	BIAS	-0.0081	0.0192	-0.0318	-0.0073	-0.0012	-0.0891
	SD	0.0298	0.0452	0.0457	0.1660	0.0642	0.1017		SD	0.0133	0.0190	0.0184	0.1216	0.0269	0.0670
	RMSE	0.0335	0.0769	0.1392	0.1995	0.0719	0.3229		RMSE	0.0156	0.0270	0.0367	0.1219	0.0269	0.1114
	95%CP	0.8900	0.7340	0.1520	0.8890	0.9090	0.3540		95%CP	0.9020	0.8540	0.5910	0.9400	0.9440	0.7670
	90%CP	0.8180	0.6140	0.0960	0.8250	0.8470	0.2010		90%CP	0.8190	0.7700	0.4640	0.8900	0.8870	0.6330
		After bias correction								After bias correction					
	BIAS	**-0.0064**	**0.0203**	**-0.0472**	**-0.0404**	**-0.0122**	**-0.1202**		BIAS	**-0.0008**	**0.0020**	**-0.0045**	**-0.0006**	**-0.0010**	**-0.0127**
	SD	**0.0298**	0.0462	0.0466	**0.1546**	**0.0635**	0.1274		SD	**0.0131**	0.0193	0.0187	**0.1210**	**0.0268**	0.0720
	RMSE	**0.0304**	**0.0505**	**0.0663**	**0.1598**	**0.0647**	**0.1752**		RMSE	**0.0132**	**0.0194**	**0.0192**	**0.1210**	**0.0268**	**0.0731**
	95%CP	**0.9300**	**0.9230**	**0.8080**	**0.9530**	**0.9430**	**0.8900**		95%CP	**0.9460**	**0.9540**	**0.9470**	**0.9410**	**0.9460**	**0.9500**
	90%CP	**0.8590**	**0.8560**	**0.6970**	**0.9100**	**0.8890**	**0.7880**		90%CP	**0.9010**	**0.9010**	**0.8870**	**0.8920**	**0.8890**	**0.9000**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0078	0.0641	-0.1294	-0.1099	-0.0301	-0.2828	$N = 400$	BIAS	-0.0025	0.0170	-0.0314	-0.0150	-0.0021	-0.0698
	SD	0.0146	0.0227	0.0234	0.0972	0.0303	0.0539		SD	0.0065	0.0101	0.0092	0.0594	0.0128	0.0347
	RMSE	0.0166	0.0680	0.1315	0.1468	0.0427	0.2879		RMSE	0.0070	0.0198	0.0328	0.0613	0.0130	0.0779
	95%CP	0.8750	0.1940	0.0000	0.6970	0.8180	0.0060		95%CP	0.9320	0.5910	0.0730	0.9330	0.9570	0.4880
	90%CP	0.8070	0.1220	0.0000	0.6050	0.7350	0.0040		90%CP	0.8640	0.4750	0.0400	0.8870	0.9110	0.3800
		After bias correction								After bias correction					
	BIAS	**-0.0029**	**0.0218**	**-0.0440**	**-0.0373**	**-0.0095**	**-0.1008**		BIAS	**-0.0006**	**0.0020**	**-0.0035**	**-0.0065**	**-0.0006**	**-0.0067**
	SD	**0.0146**	0.0232	0.0238	**0.0810**	**0.0301**	0.0669		SD	*0.0065*	0.0102	0.0093	**0.0585**	*0.0128*	0.0369
	RMSE	**0.0148**	**0.0318**	**0.0500**	**0.0891**	**0.0316**	**0.1210**		RMSE	**0.0065**	**0.0104**	**0.0099**	**0.0589**	**0.0128**	**0.0375**
	95%CP	**0.9310**	**0.8370**	**0.4650**	**0.9300**	**0.9410**	**0.6920**		95%CP	**0.9470**	**0.9390**	**0.9380**	**0.9410**	**0.9560**	**0.9400**
	90%CP	**0.8640**	**0.7270**	**0.3530**	**0.8670**	**0.8900**	**0.5620**		90%CP	**0.8910**	**0.8850**	**0.8750**	**0.9020**	0.9160	**0.8900**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0066	0.0621	-0.1278	-0.1106	-0.0314	-0.2743	$N = 1600$	BIAS	-0.0005	0.0158	-0.0310	-0.0108	-0.0020	-0.0657
	SD	0.0072	0.0110	0.0114	0.0669	0.0147	0.0276		SD	0.0033	0.0049	0.0046	0.0321	0.0067	0.0168
	RMSE	0.0098	0.0630	0.1283	0.1293	0.0346	0.2757		RMSE	0.0033	0.0165	0.0313	0.0339	0.0070	0.0678
	95%CP	0.8210	0.0000	0.0000	0.3000	0.4310	0.0000		95%CP	0.9380	0.0990	0.0000	0.9110	0.9330	0.0340
	90%CP	0.7320	0.0000	0.0000	0.2410	0.3150	0.0000		90%CP	0.8810	0.0600	0.0000	0.8440	0.8780	0.0140
		After bias correction								After bias correction					
	BIAS	**-0.0028**	**0.0193**	**-0.0416**	**-0.0381**	**-0.0103**	**-0.0927**		BIAS	**-0.0000**	**0.0013**	**-0.0029**	**-0.0022**	**-0.0001**	**-0.0058**
	SD	*0.0072*	0.0112	0.0115	**0.0442**	**0.0146**	0.0343		SD	*0.0033*	0.0050	*0.0046*	**0.0307**	*0.0067*	0.0178
	RMSE	**0.0077**	**0.0224**	**0.0432**	**0.0583**	**0.0179**	**0.0989**		RMSE	*0.0033*	**0.0051**	**0.0054**	**0.0307**	**0.0067**	**0.0187**
	95%CP	**0.9070**	**0.5850**	**0.0340**	**0.8190**	**0.9000**	**0.2370**		95%CP	**0.9410**	**0.9300**	**0.9020**	**0.9410**	**0.9400**	**0.9440**
	90%CP	**0.8520**	**0.4670**	**0.0170**	**0.7310**	**0.8320**	**0.1510**		90%CP	*0.8810*	**0.8790**	**0.8350**	**0.8880**	**0.8880**	**0.8830**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.11: Results of TWFE estimator with Laplace error, UP=30%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0126	0.0346	-0.0736	-0.0342	-0.0100	-0.1660	$N = 100$	BIAS	-0.0092	0.0118	-0.0186	-0.0019	0.0016	-0.0479
	SD	0.0265	0.0362	0.0310	0.1621	0.0424	0.0775		SD	0.0119	0.0176	0.0139	0.1420	0.0198	0.0404
	RMSE	0.0294	0.0501	0.0799	0.1656	0.0436	0.1832		RMSE	0.0150	0.0212	0.0232	0.1420	0.0199	0.0626
	95%CP	0.8930	0.8290	0.3080	0.9440	0.9350	0.5020		95%CP	0.8810	0.8980	0.7510	0.9580	0.9420	0.8150
	90%CP	0.8290	0.7330	0.1980	0.8960	0.8910	0.3740		90%CP	0.7860	0.8150	0.6290	0.9090	0.8980	0.7160
		After bias correction								After bias correction					
	BIAS	**-0.0030**	**0.0044**	**-0.0164**	**-0.0079**	**-0.0027**	**-0.0401**		BIAS	**-0.0002**	**0.0003**	**-0.0016**	**-0.0006**	**0.0005**	**-0.0043**
	SD	**0.0262**	0.0371	0.0318	**0.1570**	**0.0422**	0.0888		SD	**0.0118**	0.0177	0.0140	**0.1419**	*0.0198*	0.0421
	RMSE	**0.0264**	**0.0373**	**0.0358**	**0.1572**	**0.0423**	**0.0974**		RMSE	**0.0118**	**0.0177**	**0.0141**	**0.1419**	**0.0198**	**0.0423**
	95%CP	**0.9310**	**0.9450**	**0.8970**	**0.9520**	**0.9460**	**0.9190**		95%CP	**0.9530**	**0.9460**	**0.9470**	0.9600	**0.9450**	**0.9500**
	90%CP	**0.8710**	**0.8840**	**0.8420**	0.9140	**0.8960**	**0.8430**		90%CP	**0.9080**	**0.8870**	**0.8910**	*0.9090*	**0.9000**	**0.9070**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0062	0.0365	-0.0735	-0.0382	-0.0109	-0.1566	$N = 400$	BIAS	-0.0025	0.0093	-0.0188	-0.0019	-0.0007	-0.0401
	SD	0.0134	0.0176	0.0153	0.0877	0.0210	0.0375		SD	0.0063	0.0088	0.0071	0.0711	0.0097	0.0209
	RMSE	0.0147	0.0405	0.0751	0.0956	0.0237	0.1610		RMSE	0.0067	0.0128	0.0201	0.0711	0.0097	0.0452
	95%CP	0.9070	0.4540	0.0030	0.9090	0.9120	0.0280		95%CP	0.9130	0.8210	0.2240	0.9530	0.9480	0.5310
	90%CP	0.8230	0.3410	0.0000	0.8350	0.8460	0.0130		90%CP	0.8560	0.7060	0.1460	0.9120	0.9010	0.3940
		After bias correction								After bias correction					
	BIAS	**-0.0024**	**0.0070**	**-0.0153**	**-0.0082**	**-0.0024**	**-0.0347**		BIAS	**-0.0002**	**0.0003**	**-0.0011**	**0.0010**	**-0.0004**	**-0.0032**
	SD	**0.0133**	0.0180	0.0158	**0.0799**	**0.0210**	0.0429		SD	*0.0063*	0.0089	0.0073	**0.0706**	*0.0097*	0.0217
	RMSE	**0.0135**	**0.0194**	**0.0220**	**0.0803**	**0.0211**	**0.0552**		RMSE	**0.0063**	**0.0089**	**0.0073**	**0.0706**	*0.0097*	**0.0219**
	95%CP	**0.9360**	**0.9260**	**0.7940**	**0.9600**	**0.9420**	**0.8610**		95%CP	**0.9450**	**0.9510**	**0.9340**	0.9560	**0.9490**	**0.9450**
	90%CP	**0.8670**	**0.8740**	**0.7020**	**0.8930**	**0.8900**	**0.7780**		90%CP	**0.8860**	**0.9060**	**0.8810**	0.9140	0.8970	**0.8930**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0036	0.0363	-0.0735	-0.0395	-0.0106	-0.1521	$N = 1600$	BIAS	-0.0005	0.0084	-0.0189	-0.0048	-0.0005	-0.0374
	SD	0.0069	0.0088	0.0071	0.0562	0.0102	0.0186		SD	0.0031	0.0044	0.0035	0.0376	0.0049	0.0106
	RMSE	0.0078	0.0374	0.0738	0.0687	0.0147	0.1532		RMSE	0.0032	0.0095	0.0192	0.0379	0.0049	0.0389
	95%CP	0.8840	0.0180	0.0000	0.7490	0.8180	0.0000		95%CP	0.9450	0.5330	0.0000	0.9310	0.9500	0.0620
	90%CP	0.8060	0.0080	0.0000	0.6620	0.7230	0.0000		90%CP	0.9000	0.4110	0.0000	0.8840	0.8960	0.0350
		After bias correction								After bias correction					
	BIAS	**-0.0014**	**0.0070**	**-0.0149**	**-0.0081**	**-0.0019**	**-0.0313**		BIAS	**-0.0001**	**0.0002**	**-0.0010**	**-0.0015**	**0.0002**	**-0.0023**
	SD	*0.0069*	0.0090	0.0073	**0.0396**	*0.0102*	0.0212		SD	*0.0031*	*0.0045*	*0.0036*	**0.0364**	*0.0049*	0.0110
	RMSE	**0.0071**	**0.0114**	**0.0166**	**0.0404**	**0.0104**	**0.0378**		RMSE	*0.0031*	**0.0045**	**0.0037**	**0.0365**	*0.0049*	**0.0112**
	95%CP	**0.9170**	**0.8810**	**0.4920**	**0.9460**	**0.9530**	**0.6900**		95%CP	**0.9490**	**0.9530**	**0.9350**	**0.9480**	0.9560	**0.9310**
	90%CP	**0.8510**	**0.8120**	**0.3600**	**0.9050**	**0.8960**	**0.5740**		90%CP	0.8960	**0.8990**	**0.8780**	**0.9000**	**0.8990**	**0.8800**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

Table D.12: Results of TWFE estimator with Laplace error, UP=60%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$.

		ρ	λ	ν	γ	β	σ^2			ρ	λ	ν	γ	β	σ^2
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 100$	BIAS	-0.0139	0.0649	-0.1310	-0.1187	-0.0349	-0.3011	$N = 100$	BIAS	-0.0080	0.0186	-0.0306	-0.0068	-0.0005	-0.0910
	SD	0.0285	0.0430	0.0451	0.1704	0.0616	0.0874		SD	0.0132	0.0193	0.0185	0.1181	0.0273	0.0516
	RMSE	0.0317	0.0779	0.1386	0.2076	0.0708	0.3136		RMSE	0.0154	0.0268	0.0357	0.1183	0.0273	0.1046
	95%CP	0.8980	0.7040	0.1570	0.8810	0.9120	0.1690		95%CP	0.8920	0.8540	0.6220	0.9510	0.9490	0.6500
	90%CP	0.8410	0.5940	0.0810	0.8140	0.8380	0.1010		90%CP	0.8280	0.7600	0.5020	0.8940	0.8850	0.5240
		After bias correction								After bias correction					
	BIAS	**-0.0048**	**0.0225**	**-0.0456**	**-0.0446**	**-0.0147**	**-0.1129**		BIAS	**-0.0007**	**0.0014**	**-0.0033**	**-0.0004**	**-0.0003**	**-0.0150**
	SD	**0.0284**	0.0444	0.0462	**0.1630**	*0.0615*	0.1096		SD	*0.0132*	0.0197	0.0188	**0.1175**	**0.0272**	0.0556
	RMSE	**0.0288**	**0.0498**	**0.0649**	**0.1690**	**0.0633**	**0.1573**		RMSE	**0.0132**	**0.0197**	**0.0191**	**0.1175**	**0.0272**	**0.0576**
	95%CP	**0.9270**	**0.9240**	**0.8020**	**0.9390**	**0.9540**	**0.8430**		95%CP	**0.9350**	**0.9550**	**0.9370**	0.9520	**0.9500**	**0.9400**
	90%CP	**0.8720**	**0.8570**	**0.7170**	**0.8830**	**0.8960**	**0.7080**		90%CP	**0.8860**	**0.9040**	**0.8890**	*0.8940*	**0.8920**	**0.8860**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 400$	BIAS	-0.0076	0.0625	-0.1286	-0.1135	-0.0316	-0.2796	$N = 400$	BIAS	-0.0020	0.0158	-0.0305	-0.0103	-0.0019	-0.0701
	SD	0.0143	0.0227	0.0223	0.0961	0.0296	0.0449		SD	0.0065	0.0098	0.0092	0.0589	0.0135	0.0266
	RMSE	0.0162	0.0665	0.1305	0.1487	0.0433	0.2832		RMSE	0.0068	0.0186	0.0318	0.0598	0.0136	0.0749
	95%CP	0.8780	0.2070	0.0000	0.6760	0.8110	0.0000		95%CP	0.9280	0.6460	0.0810	0.9450	0.9410	0.2910
	90%CP	0.8110	0.1260	0.0000	0.5880	0.7140	0.0000		90%CP	0.8600	0.5190	0.0460	0.8970	0.8870	0.1730
		After bias correction								After bias correction					
	BIAS	**-0.0027**	**0.0199**	**-0.0427**	**-0.0401**	**-0.0107**	**-0.0969**		BIAS	**-0.0001**	**0.0008**	**-0.0026**	**-0.0025**	**-0.0004**	**-0.0070**
	SD	*0.0143*	0.0234	0.0229	**0.0802**	**0.0294**	0.0555		SD	*0.0065*	0.0099	0.0094	**0.0581**	*0.0135*	0.0283
	RMSE	**0.0145**	**0.0307**	**0.0484**	**0.0897**	**0.0313**	**0.1117**		RMSE	**0.0065**	**0.0100**	**0.0097**	**0.0582**	**0.0135**	**0.0292**
	95%CP	**0.9200**	**0.8450**	**0.5010**	**0.9260**	**0.9380**	**0.5770**		95%CP	**0.9350**	**0.9560**	**0.9410**	**0.9510**	*0.9410*	**0.9350**
	90%CP	**0.8600**	**0.7630**	**0.3880**	**0.8520**	**0.8750**	**0.4560**		90%CP	**0.8920**	**0.8950**	**0.8760**	**0.9010**	**0.8960**	**0.8890**
$T = 10$		Before bias correction						$T = 40$		Before bias correction					
$N = 1600$	BIAS	-0.0064	0.0622	-0.1274	-0.1106	-0.0322	-0.2745	$N = 1600$	BIAS	-0.0005	0.0160	-0.0311	-0.0097	-0.0021	-0.0656
	SD	0.0073	0.0114	0.0121	0.0666	0.0154	0.0220		SD	0.0032	0.0050	0.0048	0.0302	0.0063	0.0138
	RMSE	0.0097	0.0633	0.1280	0.1291	0.0356	0.2754		RMSE	0.0032	0.0167	0.0315	0.0317	0.0066	0.0670
	95%CP	0.8030	0.0020	0.0000	0.2920	0.4190	0.0000		95%CP	0.9440	0.1110	0.0000	0.9280	0.9440	0.0020
	90%CP	0.7000	0.0000	0.0000	0.2380	0.3070	0.0000		90%CP	0.8860	0.0580	0.0000	0.8720	0.8880	0.0000
		After bias correction								After bias correction					
	BIAS	**-0.0026**	**0.0194**	**-0.0412**	**-0.0367**	**-0.0110**	**-0.0931**		BIAS	**-0.0001**	**0.0015**	**-0.0030**	**-0.0009**	**-0.0001**	**-0.0058**
	SD	**0.0072**	0.0116	**0.0119**	**0.0437**	**0.0153**	0.0271		SD	*0.0032*	0.0050	0.0049	**0.0291**	*0.0063*	0.0146
	RMSE	**0.0077**	**0.0226**	**0.0429**	**0.0571**	**0.0188**	**0.0970**		RMSE	*0.0032*	**0.0053**	**0.0058**	**0.0291**	**0.0063**	**0.0157**
	95%CP	**0.9000**	**0.5870**	**0.0560**	**0.8210**	**0.8890**	**0.0810**		95%CP	**0.9470**	**0.9330**	**0.8880**	**0.9470**	**0.9550**	**0.9190**
	90%CP	**0.8320**	**0.4650**	**0.0330**	**0.7300**	**0.8100**	**0.0510**		90%CP	**0.8940**	**0.8800**	**0.8020**	**0.8870**	0.9130	**0.8600**

Notes: UP $= 1 - \frac{n}{NT}$. In the table, **bold** numbers indicate that the bias-corrected results have improved, while *italicized* numbers indicate that the bias-corrected results perform equally well as before. The 95%CP and 90%CP indicate the frequencies that the true parameter value is contained within the 95% and 90% confidence intervals of the estimator, respectively.

D.2 Supplemental Figures

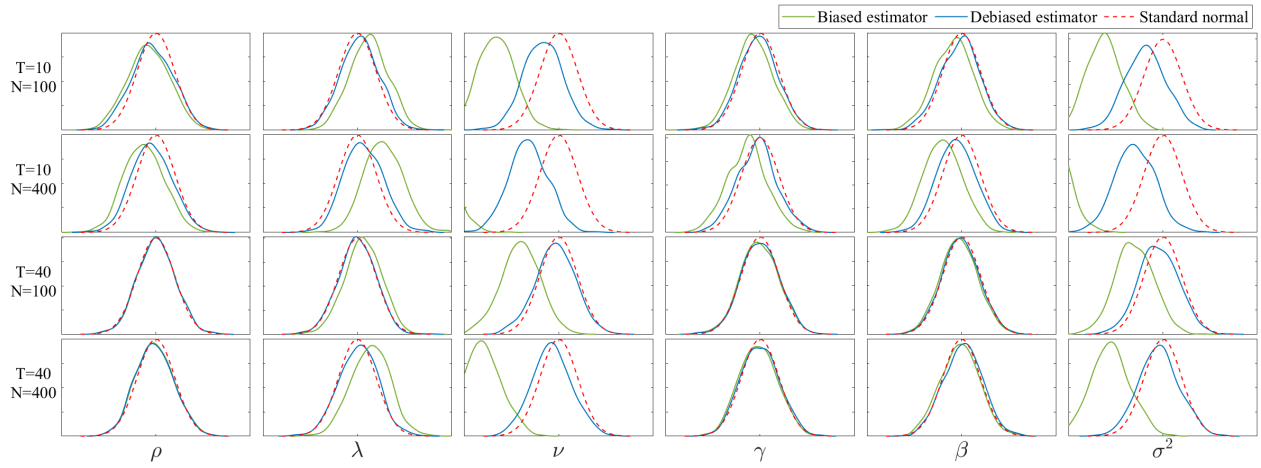

Figure D.1: PDF of OWFE standardized estimator with normal error, UP=30%, $\theta_0 = (0.1, 0.3, 0.5, 1, 1, 1)$

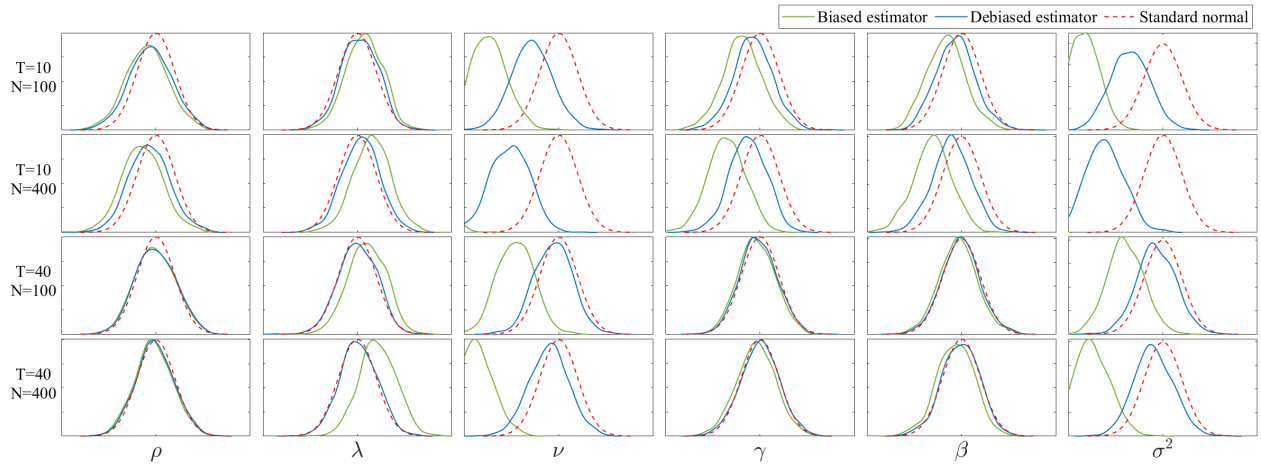

Figure D.2: PDF of OWFE standardized estimator with normal error, UP=60%, $\theta_0 = (0.1, 0.3, 0.5, 1, 1, 1)$

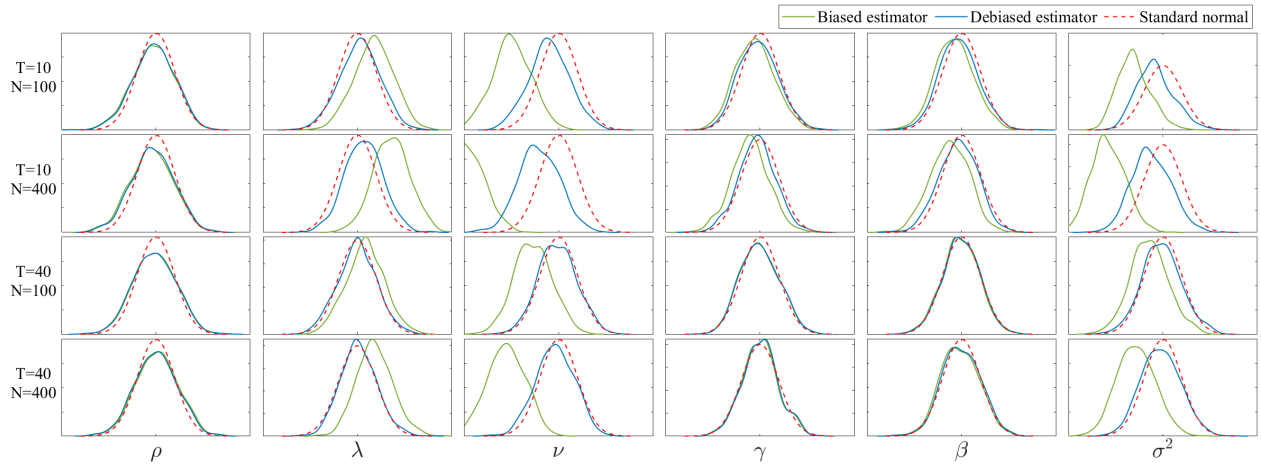

Figure D.3: PDF of OWFE standardized estimator with Exponential error, UP=30%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$

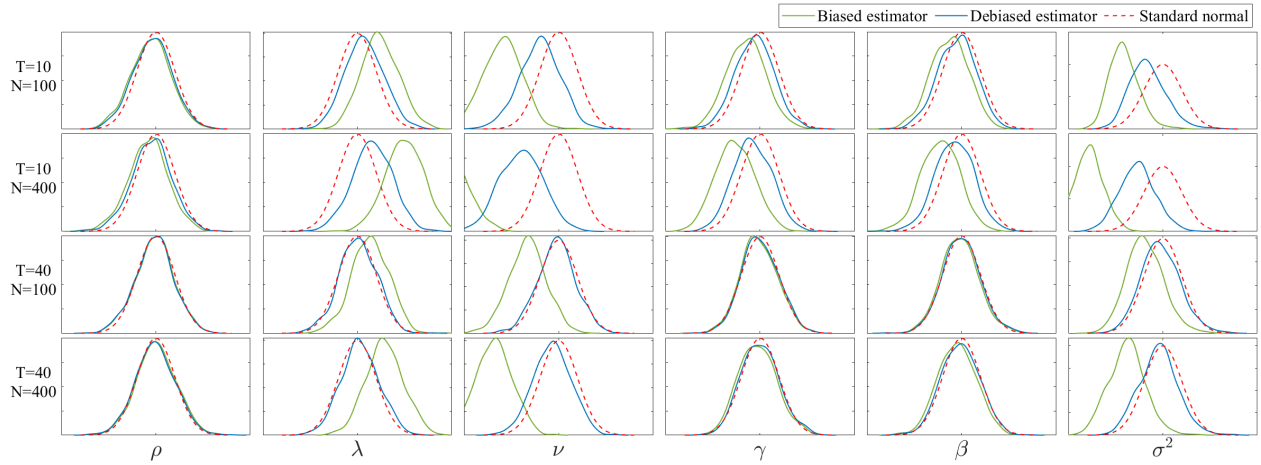

Figure D.4: PDF of OWFE standardized estimator with Exponential error, UP=60%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$

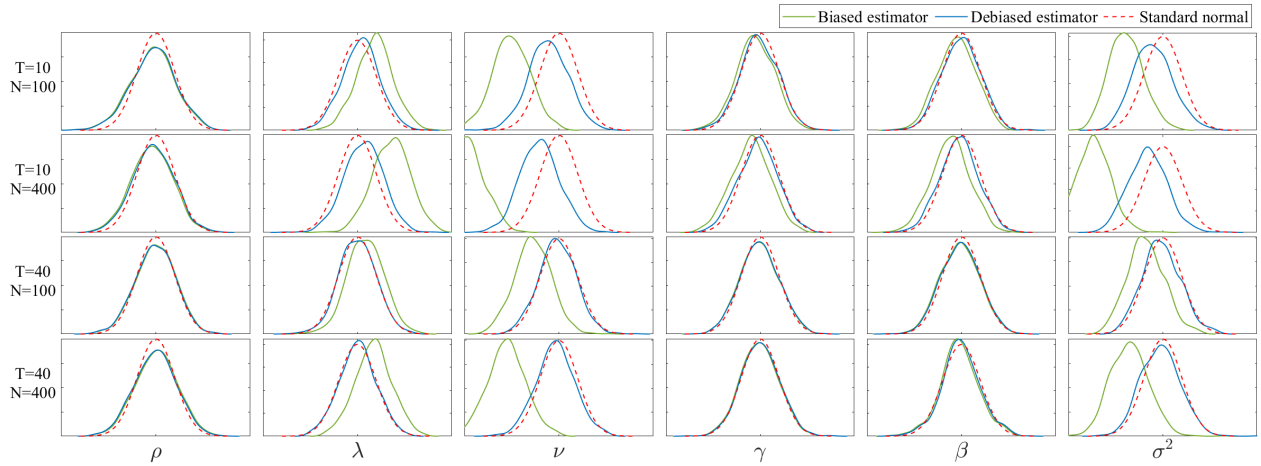

Figure D.5: PDF of OWFE standardized estimator with Laplace error, UP=30%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$

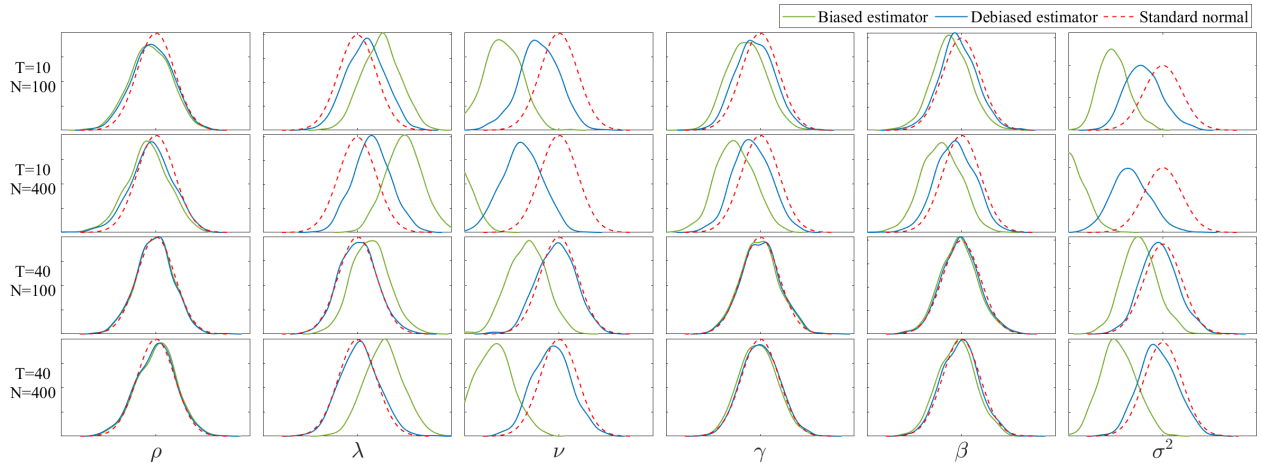

Figure D.6: PDF of OWFE standardized estimator with Laplace error, UP=60%, $\theta_0 = (0.5, 0.2, 0.1, 1, 1, 1)$

E Detailed Empirical Results

E.1 Descriptive Statistics of the Data

Table E.1: Descriptive Statistics

	Sample Size	Mean	Std	Min	Q25	Median	Q75	Max
				Auckland Dataset				
log rental price	80716	5.421	0.8183	2.7726	4.8363	5.3279	5.8972	11.5129
avail30d	80716	14.0899	10.5291	0	4	14	24	30
avgAvail30d1km	80716	13.9605	3.8352	0	11.127	13.8058	16.0676	30
				Queenstown Dataset				
log rental price	95455	5.9705	0.7675	2.9957	5.4467	5.9789	6.4425	10.6968
avail30d	95455	10.7122	9.4054	0	2	9	18	30
avgAvail30d1km	95455	10.6611	3.4676	0	8.3091	10.0317	12.7324	30
				Wellington Dataset				
log rental price	18723	5.1562	0.701	3.091	4.7707	5.2095	5.5947	8.5172
avail30d	18723	13.5583	10.3087	0	3	14	23	30
avgAvail30d1km	18723	13.782	3.9405	3.9861	10.318	14.401	17.4	22.0946
	Sample Size	Mean	Std	Min	Q25	Median	Q75	Max
				Williamsburg Dataset				
log rental price	14567	5.1221	0.7946	2.3026	4.6052	5.0752	5.5759	10.5966
avail30d	14567	13.1242	12.5974	0	0	9	28	30
avgAvail30d1km	14567	12.9429	2.5332	4.7143	10.9963	12.5299	14.7712	23
				Midtown Dataset				
log rental price	23672	5.7745	0.8484	2.0794	5.2933	5.717	6.1431	10.8208
avail30d	23672	14.0729	12.0784	0	1	13	28	30
avgAvail30d1km	23672	14.3105	2.09	8.0087	12.7817	14.3718	15.613	20.0807
				Hell's Kitchen Dataset				
log rental price	16862	5.357	0.868	3.4657	4.8283	5.2832	5.7746	10.8235
avail30d	16862	12.8441	12.4652	0	0	9	28	30
avgAvail30d1km	16862	12.5631	2.434	6.0616	10.6333	12.6123	14.4707	18.6227

E.2 Estimation Results

Table E.2: Estimation Results for Auckland Dataset

	d=0.4km	d=0.7km	d=1.1km
$\rho \quad (W_t Y_t)$	0.0725(0.0034)***	0.1335(0.0047)***	0.1612(0.0053)***
$\lambda \quad (M_t Y_{t-1})$	-0.0697(0.0034)***	-0.1419(0.0046)***	-0.1640(0.0053)***
$\nu \quad (Y_{t-1})$	0.3867(0.0033)***	0.3985(0.0031)***	0.3982(0.0031)***
avail30d	-0.0007(0.0001)***	-0.0006(0.0001)***	-0.0006(0.0001)***
avgAvail30d1km	-0.0056(0.0004)***	-0.0052(0.0004)***	-0.0053(0.0004)***
Listing effect	0.2994(0.0046)***	0.2430(0.0053)***	0.2221(0.0059)***

*In this dataset, $N = 5896$, $T = 30$, $n = 80716$, UP $= 1 - \frac{n}{NT} = 54.37\%$. Estimates are presented with their standard errors in parentheses. Statistical significance is indicated by stars: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.

Table E.3: Estimation Results for Queenstown Dataset

	d=0.4km	d=0.7km	d=1.1km
ρ (W_tY_t)	0.0350(0.0024)***	0.0669(0.0040)***	0.0960(0.0055)***
λ (M_tY_{t-1})	-0.0298(0.0024)***	-0.0623(0.0039)***	-0.0791(0.0052)***
ν (Y_{t-1})	0.2984(0.0030)***	0.3003(0.0029)***	0.3006(0.0028)***
avail30d	-0.0005(0.0001)***	-0.0005(0.0001)***	-0.0005(0.0001)***
avgAvail30d1km	-0.0059(0.0004)***	-0.0059(0.0004)***	-0.0058(0.0004)***
Listing effect	0.2822(0.0043)***	0.2507(0.0054)***	0.2335(0.0064)***

*In this dataset, $N = 6095$, $T = 30$, $n = 95455$, UP $= 1 - \frac{n}{NT} = 47.80\%$. Estimates are presented with their standard errors in parentheses. Statistical significance is indicated by stars: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.

Table E.4: Estimation Results for Wellington Dataset

	d=0.4km	d=0.7km	d=1.1km
ρ (W_tY_t)	0.0674(0.0083)***	0.1329(0.0210)***	0.1591(0.0333)***
λ (M_tY_{t-1})	-0.0661(0.0077)***	-0.0919(0.0199)***	-0.0735(0.0301)**
ν (Y_{t-1})	0.2109(0.0050)***	0.2089(0.0049)***	0.2082(0.0049)***
avail30d	-0.0003(0.0002)**	-0.0003(0.0002)**	-0.0003(0.0002)**
avgAvail30d1km	0.0011(0.0018)	0.0008(0.0016)	0.0015(0.0016)
Listing effect	0.1301(0.0082)***	0.1049(0.0184)***	0.1207(0.0274)***

*In this dataset, $N = 1549$, $T = 30$, $n = 18723$, UP $= 1 - \frac{n}{NT} = 59.71\%$. Estimates are presented with their standard errors in parentheses. Statistical significance is indicated by stars: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.

Table E.5: Estimation Results for Williamsburg Dataset

	d=0.4km	d=0.6km	d=1.1km
ρ (W_tY_t)	0.5475(0.0182)***	0.7018(0.0234)***	0.7690(0.0310)***
λ (M_tY_{t-1})	-0.1517(0.0158)***	-0.1818(0.0211)***	-0.5242(0.0397)***
ν (Y_{t-1})	0.4306(0.0064)***	0.4369(0.0063)***	0.4638(0.0064)***
avail30d	0.0006(0.0002)**	0.0006(0.0002)**	0.0006(0.0002)***
avgAvail30d1km	0.0031(0.0033)	0.0017(0.0031)	0.0035(0.0031)
Listing effect	0.3199(0.0176)***	0.2925(0.0234)***	-0.0687(0.0438)

*In this dataset, $N = 1566$, $T = 17$, $n = 14567$, UP $= 1 - \frac{n}{NT} = 45.28\%$. Estimates are presented with their standard errors in parentheses. Statistical significance is indicated by stars: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.

Table E.6: Estimation Results for Midtown Dataset

	d=0.1km	d=0.2km	d=0.4km
ρ (W_tY_t)	0.2338(0.0073)***	0.4837(0.0138)***	0.4812(0.0238)***
λ (M_tY_{t-1})	-0.0978(0.0085)***	-0.2489(0.0207)***	-0.2740(0.0319)***
ν (Y_{t-1})	0.4422(0.0086)***	0.4597(0.0071)***	0.4597(0.0068)***
avail30d	-0.0013(0.0003)***	-0.0010(0.0003)***	-0.0012(0.0003)***
avgAvail30d1km	-0.0105(0.0053)**	-0.0036(0.0038)	-0.0020(0.0034)
Listing effect	0.4586(0.0169)***	0.2852(0.0284)***	0.2528(0.0420)***

*In this dataset, $N = 2390$, $T = 17$, $n = 23672$, UP $= 1 - \frac{n}{NT} = 41.74\%$. Estimates are presented with their standard errors in parentheses. Statistical significance is indicated by stars: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.

Table E.7: Estimation Results for Hell's Kitchen Dataset

	d=0.2km	d=0.3km	d=0.5km
$\rho \quad (W_t Y_t)$	0.3932(0.0129)***	0.5211(0.0164)***	0.5414(0.0267)***
$\lambda \quad (M_t Y_{t-1})$	-0.2657(0.0153)***	-0.3238(0.0206)***	-0.4066(0.0328)***
$\nu \quad (Y_{t-1})$	0.3989(0.0065)***	0.3948(0.0064)***	0.3969(0.0063)***
avail30d	-0.0002(0.0003)	-0.0002(0.0003)	-0.0002(0.0003)
avgAvail30d1km	0.0022(0.0116)	0.0019(0.0108)	0.0013(0.0093)
Listing effect	0.1662(0.0179)***	0.0926(0.0237)***	-0.0022(0.0380)

*In this dataset, $N = 1943$, $T = 17$, $n = 16862$, UP $= 1 - \frac{n}{NT} = 48.95\%$. Estimates are presented with their standard errors in parentheses. Statistical significance is indicated by stars: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.